# MECHANICAL ABBREVIATIONS NOTE: NOT ALL ABBREVIATIONS MAY BE USED ON THE PROJECT.

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|---|---|
| (A) | ABANDONED |
| A/E | ARCHITECT / ENGINEER |
| AB | AIR BLENDER |
| ACC | AIR COOLED CONDENSER |
| ACCH | AIR COOLED CHILLER |
| ACCU | AIR-COOLED CONDENSING UNIT |
| ACU | AIR CONDITIONING UNIT |
| AD | ACCESS DOOR |
| AF | AFTER FILTER |
| AFCV | AIR FLOW CONTROL VALVE |
| AFF | ABOVE FINISHED FLOOR |
| AFMD | AIR FLOW MEASURING DEVICE |
| AHU | AIR-HANDLING UNIT |
| AMP | AMPERAGE |
| AP | ACCESS PANEL |
| APD | AIR PRESSURE DROP |
| ARI | AIR CONDITIONING AND REFRIGERATION INSTITUTE |
| AS | AIR SEPARATOR |
| ASME | AMERICAN SOCIETY OF MECHANICAL ENGINEERS |
| AV | AUTOMATIC AIR VENT |
| AXF | AXIAL FLOW |
| B | BOILER |
| BAS | BUILDING AUTOMATION SYSTEM |
| BD | BLOWDOWN |
| BDD | BACKDRAFT DAMPER |
| BBR | BASE BOARD RADIATOR |
| BFP | BACKFLOW PREVENTER |
| BHP | BRAKE HORSEPOWER |
| BIW | BACKWARD INCLINED WHEEL (FAN) |
| BOD | BOTTOM OF DUCT |
| BOP | BOTTOM OF PIPE |
| BSC | BIOLOGICAL SAFETY CABINETS |
| BT | BLOWOFF TANK |
| BTU | BRITISH THERMAL UNIT |
| BTUH | BRITISH THERMAL UNIT PER HOUR |
| C | CONVECTOR |
| CC | COOLING COIL |
| CD | CEILING DIFFUSER |
| CENT | CENTRIFUGAL |
| CFH | CUBIC FEET PER HOUR |
| CFM | CUBIC FEET PER MINUTE |
| CU FT | CUBIC FEET |
| CFP | CHEMICAL FEED PUMP |
| CG | CEILING GRILLE |
| CH | CHILLER |
| CHP | CHILLED WATER PUMP |
| CHR | CHILLED WATER RETURN |
| CHS | CHILLED WATER SUPPLY |
| CI | CAST IRON |
| CO | CARBON MONOXIDE |
| CO | CLEAN OUT |
| $CO_2$ | CARBON DIOXIDE |
| COMP | COMPRESSOR UNIT |
| COR | CONTRACTING OFFICER REPRESENTATIVE |
| COP | COEFFICIENT OF PERFORMANCE |
| CP | CONDENSATE PUMP |
| CS | CONDENSATE STORAGE TANK |
| CSG | CLEAN STEAM GENERATOR |
| CT | COOLING TOWER |
| CU | CONDENSING UNIT |
| CUH | CABINET UNIT HEATER |
| CV | CONSTANT VOLUME |
| CW | COLD WATER (POTABLE) |
| CWCC | CHILLED WATER COOLING COIL |
| CWP | CONDENSER WATER PUMP |
| CWR | CONDENSER WATER RETURN (TO COOLING TOWER) |
| CWS | CONDENSER WATER SUPPLY (FROM COOLING TOWER) |
| (D) | DEMOLISH |
| Db | DECIBELS |
| DB | DRY-BULB TEMPERATURE |
| DDC | DIRECT DIGITAL CONTROLS |
| DEG | DEGREE |
| DIA | DIAMETER |
| DMPR | DAMPER |
| DP | DEW POINT TEMPERATURE |
| DPS | DIFFERENTIAL PRESSURE SENSOR |
| DX | DIRECT EXPANSION |
| DXCC | DIRECT EXPANSION COOLING COIL |
| (E) | EXISTING |
| EA | EXHAUST AIR |
| EAT | ENTERING AIR TEMPERATURE |
| EAV | EXHAUST AIR VALVE |
| EC | EVAPORATIVE COOLER |
| ECC | ENGINEERING CONTROL CENTER |
| ECM | ELECTRICALLY COMMUTATED MOTOR |
| ECU | EVAPORATIVE CONDENSER UNIT |
| EDH | ELECTRIC DUCT HEATER |
| EER | ENERGY EFFICIENCY RATIO |
| EF | EXHAUST FAN |
| EG | EXHAUST GRILLE |
| EGS | EMERGENCY GAS SHUTOFF |
| EGT | ENTERING GLYCOL TEMPERATURE |
| EH | EXHAUST HOOD |
| EJ | EXPANSION JOINT |
| ENT | ENTERING |
| ER | EXHAUST REGISTER |
| ERC | ELECTRIC REHEAT COIL |
| ERP | ELECTRIC RADIANT PANEL |
| ESP | EXTERNAL STATIC PRESSURE |
| ET | EXPANSION TANK |
| ETO | ETHYLENE OXIDE |
| EUH | ELECTRIC UNIT HEATER |
| EWC | EVAPORATIVE WATER COOLER |
| EWT | ENTERING WATER TEMPERATURE |
| EX | EXISTING |
| EXH | EXHAUST |
| F | FAHRENHEIT |
| F&T | FLOAT AND THERMOSTATIC |
| F/SD | COMBINATION FIRE SMOKE DAMPER |
| FA | FIRE ALARM |
| FA | FREE AREA |
| FC | FLEXIBLE CONNECTION |
| FCU | FAN COIL UNIT (4 PIPE) |
| FCW | FORWARD CURVED WHEEL (FAN) |
| FD | FLOOR DRAIN |
| FD | FIRE DAMPER |
| FT | FINAL FILTER |
| FHX | FLUE GAS/FEEDWATER HEAT EXCHANGER |
| FM | FLOW METER |
| FOP | FUEL OIL PUMP |
| FOT | FUEL OIL TANK |
| FPM | FEET PER MINUTE |
| FPS | FEET PER SECOND |
| FPTU | FAN POWERED TERMINAL UNIT |
| FRP | FIBER REINFORCED POLYESTER |
| FS | FLOW SWITCH |
| FSTAT | FREEZESTAT |
| FT | FEET |
| FT-LB | FOOT-POUND |
| FTR | FIN TUBE RADIATION |
| FV | FACE VELOCITY |
| GA | GAUGE |
| GAL | GALLONS |
| GPD | GALLONS PER DAY |
| GPH | GALLONS PER HOUR |
| GPM | GALLONS PER MINUTE |
| GPR | GAS PRESSURE REGULATOR |
| GS | GALVANIZED STEEL |
| H | HUMIDIFIER |
| H&CW | HOT & COLD WATER |
| HAC | HOUSEKEEPING AID CLOSET |
| HB | HOSE BIBB |
| HC | HEATING COIL |
| HD | HEAD |
| HOA | HAND/OFF/AUTOMATIC |
| HP | HEAT PUMP |
| HP | HORSEPOWER |
| HPR | HIGH PRESSURE RETURN (STEAM CONDENSATE) |
| HPS | HIGH PRESSURE SUPPLY (STEAM) |
| HRC | HEAT RECOVERY COIL |
| HRP | HYDRONIC RADIANT (CEILING) PANEL |
| HRW | HEAT RECOVERY WHEEL |
| HSTAT | HUMIDISTAT |
| HUM | HUMIDIFIER |
| HVU | HEATING AND VENTILATING UNIT |
| HW | HOT WATER |
| HWC | HOT WATER COIL |
| HWP | HEATING HOT WATER PUMP |
| HWR | HEATING HOT WATER RETURN |
| HWS | HEATING HOT WATER SUPPLY |
| HWUH | HOT WATER UNIT HEATER |
| HX | HEAT EXCHANGER |
| HZ | HERTZ |
| I/O | INPUT/OUTPUT |
| IAQ | INDOOR AIR QUALITY |
| IBT | INVERTED BUCKET TRAP |
| ICF | IN-LINE CENTRIFUGAL FAN |
| ICU | INTENSIVE CARE UNIT |
| ID | INSIDE DIAMETER |
| IFB | INTEGRAL FACE AND BYPASS |
| IN | INCHES |
| IN HG | HG INCHES OF MERCURY |
| IN WC | INCH WATER COLUMN |
| IN WG | INCH WATER GAUGE |
| IN-LB | INCH-POUND |
| IPLV | INTEGRATED PART LOAD VALUE |
| IRH | INFRARED HEATER |
| IS | INSECT SCREEN |
| IU | INDUCTION UNIT |
| kW | KILOWATT |
| kWH | KILOWATT HOUR |
| LAT | LEAVING AIR TEMPERATURE |
| LBS/HR | POUNDS PER HOUR |
| LF | LINEAR FOOT (FEET) |
| LGT | LEAVING GLYCOL TEMPERATURE |
| LH | LATENT HEAT |
| LPG | LIQUID PROPANE GAS |
| LPC | LOW PRESSURE RETURN (STEAM CONDENSATE) |
| LPRC | LOW PRESSURE STEAM RETURN (CLEAN) |
| LPS | LOW PRESSURE STEAM |
| LPSC | LOW PRESSURE STEAM (CLEAN) |
| LSD | LINEAR SLOT DIFFUSER |
| LVG | LEAVING |
| LVR | LOUVER |
| LWT | LEAVING WATER TEMPERATURE |
| MA | MIXED AIR |
| MAT | MIXED AIR TEMPERATURE |
| MAU | MAKE-UP AIR UNIT |
| MAX | MAXIMUM |
| MB | MIXING BOX |
| MBH | 1000 BTUH |
| MCA | MINIMUM BRANCH CIRCUIT AMPACITY |
| MER | MECHANICAL EQUIPMENT ROOM |
| MERV | MINIMUM EFFICIENCY REPORTING VALUE |
| MHP | MOTOR HORSEPOWER |
| MIN | MINIMUM |
| MOD | MOTOR OPERATED DAMPER |
| MOV | MOTOR OPERATED VALVE |
| MPR | MEDIUM PRESSURE RETURN (STEAM CONDENSATE) |
| MPS | MEDIUM PRESSURE STEAM |
| MRI | MAGNETIC RESONANCE IMAGING |
| MTD | MEAN TEMPERATURE DIFFERENCE |
| MV | MANUAL AIR VENT |
| MVD | MANUAL VOLUME DAMPER |
| (N) | NEW |
| NA | NOT APPLICABLE |
| NC | NOISE CRITERIA |
| NC | NORMALLY CLOSED |
| NG | NATURAL GAS |
| NO | NORMALLY OPEN |
| NOM | NOMINAL |
| NPLV | NON-STANDARD PART LOAD VALUE |
| NPSH | NET POSITIVE SUCTION HEAD |
| NTS | NOT TO SCALE |
| OA | OUTDOOR AIR |
| OAG | OUTDOOR AIR GRILLE |
| OAI | OUTDOOR AIR INTAKE |
| OD | OUTSIDE DIAMETER |
| OED | OPEN END DUCT |
| OFM | OIL FLOWMETER |
| OR | OPERATING ROOM |
| P | PUMP |
| PC | PUMPED CONDENSATE |
| PCF | POUNDS PER CUBIC FOOT (FEET) |
| PD | PRESSURE DROP |
| PF | PRE-FILTER |
| PG | PRESSURE GAGE |
| PGW | PROPYLENE GLYCOL-WATER (SOLUTION) |
| PHC | PREHEAT COIL |
| PPM | PARTS PER MILLION |
| PRV | PRESSURE REDUCING VALVE |
| PSI | POUNDS PER SQUARE INCH |
| PSIA | POUNDS PER SQUARE INCH – ABSOLUTE |
| PSIG | POUNDS PER SQUARE INCH – GAGE |
| PTAC | PACKAGED TERMINAL AIR CONDITIONER |
| (R) | RELOCATE(D) |
| RA | RETURN AIR |
| RAD | REFRIGERANT AIR DRYER |
| RAHX | ROTARY AIR HEAT EXCHANGER |
| RAT | RETURN AIR TEMPERATURE |
| RAV | RETURN AIR VALVE |
| RDS | ROOM DATA SHEETS |
| REA | RELIEF AIR |
| RF | RETURN FAN |
| RG | RETURN GRILLE |
| RH | RELATIVE HUMIDITY |
| RHC | REHEAT COIL |
| RHG | REFRIGERANT HOT GAS |
| RL | REFRIGERANT LIQUID LINE |
| RLA | RUN LOAD AMPERE |
| RO | REVERSE OSMOSIS |
| RPM | REVOLUTIONS PER MINUTE |
| RR | RETURN REGISTER |
| RS | REFRIGERANT SUCTION |
| RTU | ROOF TOP UNIT |
| RV | RELIEF VALVE |
| SA | SUPPLY AIR |
| SAD | SOUND ATTENUATING DEVICE |
| SAT | SUPPLY AIR TEMPERATURE |
| SAV | SUPPLY AIR VALVE |
| SC | SHADING COEFFICIENT |
| SCFM | STANDARD CUBIC FEET PER MINUTE |
| SCI | SPINAL CODE INJURY |
| SCR | SILICON CONTROLLED RECTIFIER |
| SD | SMOKE DETECTOR |
| SD | SUPPLY AIR DIFFUSER |
| SDPR | SMOKE DAMPER |
| SEN | SENSIBLE HEAT |
| SF | SUPPLY FAN |
| SG | SUPPLY AIR GRILLE |
| SH | STEAM HUMIDIFIER |
| SHC | STEAM HEATING COIL |
| SP | STATIC PRESSURE |
| SP GR | SPECIFIC GRAVITY |
| SPD | STERILE PROCESS AND DISTRIBUTION |
| SPS | STATIC PRESSURE SENSOR |
| SQ FT | SQUARE FOOT (FEET) |
| SQ IN | SQUARE INCH(ES) |
| SR | SUPPLY AIR REGISTER |
| SRV | SAFETY RELIEF VALVE |
| SS | STAINLESS STEEL |
| ST | STEAM TRAP |
| SUH | STEAM UNIT HEATER |
| SVS | STEAM VENT SILENCER |
| SWHX | STEAM TO WATER HEAT EXCHANGER |
| TAB | TESTING, ADJUSTING & BALANCING |
| TD | TEMPERATURE DIFFERENCE |
| TDH | TOTAL DYNAMIC HEAD |
| TDS | TOTAL DISSOLVED SOLIDS |
| TG | TRANSFER GRILLE |
| TP | TRAP |
| TSP | TOTAL STATIC PRESSURE |
| TSTAT | THERMOSTAT |
| TU | TERMINAL UNIT |
| TWU | THRU-WALL UNIT |
| UC | UNDER CUT |
| UH | UNIT HEATER |
| UL | UNDERWRITERS LABORATORY |
| V | VALVE |
| VAF | VANE-AXIAL FAN |
| VAV | VARIABLE AIR VOLUME |
| VD | VOLUME DAMPER (MANUAL BALANCING) |
| VFD | VARIABLE FREQUENCY DRIVE |
| VHA | VETERANS HEALTH ADMINISTRATION |
| VI | VIBRATION ISOLATOR |
| VIF | VERIFY IN FIELD |
| VP | VACUUM PUMP |
| VPS | VARIABLE PRIMARY SYSTEM |
| VUH | VERTICAL UNIT HEATER |
| W | WATTS |
| WAG | WASTE ANESTHESIA GAS |
| WB | WET-BULB (TEMPERATURE) |
| WC | WATER COLUMN |
| WCCH | WATER COOLED CHILLER |
| WCCU | WATER COOLED CONDENSING UNIT |
| WCHP | WATER COOLED HEAT PUMPS |
| WCPU | WATER COOLED PACKAGED UNIT |
| WEF | WALL EXHAUST FAN |
| WF | WATER FILTER |
| WFM | WATER FLOWMETER |
| WG | WATER GAGE |
| WMS | WIRE MESH SCREEN |
| WPD | WATER SIDE PRESSURE DROP |

# MECHANICAL SYMBOLS

- UP / DN — SUPPLY DUCT (UP & DOWN)
- UP / DN — EXHAUST DUCT (UP & DOWN)
- UP / DN — RETURN DUCT (UP & DOWN)
- SQUARE 1-WAY CEILING DIFFUSERS
- SQUARE 2-WAY CEILING DIFFUSERS
- SQUARE 3-WAY CEILING DIFFUSERS
- ROUND AND SQUARE 4-WAY CEILING DIFFUSERS
- LINEAR SLOT DIFFUSER
- SUPPLY REGISTER OR GRILLE (WALL TYPE)
- EXHAUST/RETURN CEILING REGISTER OR GRILLE
- EXHAUST/RETURN REGISTER OR GRILLE (WALL TYPE)
- VANED ELBOW & AIR SPLIT TYPE DUCT TAKE-OFF
- CONNECT NEW DUCT TO EXISTING DUCT
- INCLINED RISE, IN DIRECTION OF AIR FLOW (R)
- INCLINED DROP, IN DIRECTION OF AIR FLOW (D)
- LIMIT OF DEMOLITION
- FLEXIBLE CONNECTION, EQUIPMENT, VIBRATION, OR SEISMIC (FC)
- VANED ELBOW (PROVIDE ALL SQUARE OR RECTANGULAR ELBOWS WITH VANES, EVEN IF SYMBOL IS MISSING)
- VANED ELBOW (SHORT RADIUS)
- STANDARD RADIUS ELBOW (LONG RADIUS)
- NEW DUCT (INSIDE DIMENSIONS: WIDTH x DEPTH) (10x8)
- EXISTING DUCT TO REMAIN
- EXISTING DUCT TO BE REMOVED
- LOUVER (LOUVER SPECIFIED IN ARCHITECTURAL SECTION.)
- FLEXIBLE DUCTWORK (INSULATED)
- DUCT WITH SOUND LINING
- MANUAL VOLUME DAMPER (VD)

# MECHANICAL SYMBOLS (CONTINUED)

- FIRE DAMPER (FD)
- BACK DRAFT DAMPER (BDD)
- COMBINATION FIRE/SMOKE DAMPER (F/SD)
- POINT OF CHANGE IN DUCT CONSTRUCTION BY STATIC PRESSURE CLASS. THE NUMBER ASSIGNS PRESSURE CLASS (IN. OF WATER) WHICH WILL ACCOMMODATE MAXIMUM OPERATING PRESSURE IN THE DUCT SUBSECTION. THE SYMBOL CONTINUES THE ASSIGNMENT UNTIL THE DUCT TERMINATES OR ANOTHER SYMBOL APPEARS. A "N" SUPERSCRIPT INDICATES NEGATIVE PRESSURE. (2 N; 4; 1/2; 10)
- AUTOMATIC CONTROL DAMPER MODULATING
- AUTOMATIC CONTROL DAMPER TWO POSITION
- STAINLESS STEEL DUCT
- MANUAL SPLITTER DAMPER
- STANDARD BRANCH SUPPLY, NO SPLITTER (45° TAP) (45° SUPPLY)
- STANDARD BRANCH RETURN, NO SPLITTER (45° TAP) (45° RETURN)
- DUCT MOUNTED COIL (HOT WATER OR STEAM COIL)
- DUCT MOUNTED COIL (ELECTRIC)
- TERMINAL UNIT
- TERMINAL UNIT WITH REHEAT COIL
- DOUBLE DUCT MIXING BOX
- FAN POWERED VARIABLE VOLUME TERMINAL UNIT WITH HEATING COIL
- CONVECTOR OR RADIATOR (RECESSED)
- CONVECTOR OR RADIATOR (WALL HUNG)
- (A) FCU — FLOOR MOUNTED VERTICAL RECESSED FAN COIL UNIT. LETTER INDICATES UNIT SIZE.
- (A) FCU — FLOOR MOUNTED VERTICAL CABINET FAN COIL UNIT. LETTER INDICATES UNIT SIZE.
- (A) TWU — THRU WALL AIR CONDITIONING UNIT. LETTER INDICATES UNIT SIZE.
- (A) PTAC — WINDOW TYPE AIR CONDITIONING UNIT. LETTER INDICATES UNIT SIZE.
- (A) PTAC — FLOOR MOUNTED HEAT PUMP. LETTER INDICATES UNIT SIZE.
- AIR CURTAIN
- UNIT HEATER (HORIZONTAL)
- UNIT HEATER (VERTICAL)
- 2'x2' RADIANT CEILING PANEL
- 2'x4' RADIANT CEILING PANEL

# PIPING SYMBOLS

- DIRECTION OF PIPE PITCH (DOWN)
- DIRECTION OF FLOW
- ANCHOR
- REDUCER OR INCREASER
- ECCENTRIC REDUCER
- TOP CONNECTION, 45° OR 90°
- BOTTOM CONNECTION, 45° OR 90°
- SIDE CONNECTION
- CAPPED OUTLET
- RISE OR DROP IN PIPE
- UNION
- PIPE UP OR RISE
- PIPE DOWN OR DROP
- QUICK-COUPLE HOSE CONNECTOR
- INVERTED BUCKET TRAP SET INCLUDING PIPING ACCESSORIES SEE DETAIL
- FLOAT & THERMOSTATIC TRAP SET INCLUDING PIPING ACCESSORIES SEE DETAIL
- THERMOSTATIC TRAP SET INCLUDING PIPING ACCESSORIES SEE DETAIL
- THERMODYNAMIC TRAP SET INCLUDING PIPING ACCESSORIES SEE DETAIL
- THERMOMETER
- PRESSURE GAGE
- FLOW ELEMENT (FE)
- REFRIGERANT SIGHT GLASS
- TEST PLUG (PRESSURE/TEMPERATURE) (TP)
- AUTOMATIC AIR VENT (AV)
- MANUAL AIR VENT (MV)
- GATE VALVE – THREADED/FLANGED
- GLOBE VALVE – THREADED/FLANGED
- GATE VALVE WITH 3/4" HOSE ADAPTER (C)
- CHECK VALVE
- WYE STRAINER W/VALVED DRAIN AND QUICK-COUPLE HOSE CONNECTOR
- FLEXIBLE CONNECTION
- ANGLE GLOBE VALVE
- BUTTERFLY VALVE
- BALL VALVE
- MODULATING CONTROL VALVE
- MODULATING CONTROL BUTTERFLY VALVE
- TWO POSITION CONTROL VALVE
- THREE-WAY MODULATING CONTROL VALVE
- THREE-WAY TWO POSITION CONTROL VALVE
- PRESSURE REGULATING VALVE
- PRESSURE SAFETY VALVE
- AUTOMATIC BALANCING CONTROL VALVE
- BALANCING VALVE
- CIRCUIT SETTER VALVE
- GATE VALVE WITH GLOBE-VALVED BYPASS
- PLUG VALVE
- CONTROL VALVE (CV) – FLOAT-OPERATED
- PRESSURE REDUCING VALVE (PRV)
- A — COMPRESSED AIR
- BO — BOILER BLOWOFF
- BWS — BOILER WATER SAMPLE
- CBD — CONTINUOUS BLOWDOWN
- CF — CHEMICAL FEED
- CHR — CHILLED WATER RETURN
- CHS — CHILLED WATER SUPPLY
- CTPD — CONDENSATE TRANSFER PUMP DISCHARGE
- CTPS — CONDENSATE TRANSFER PUMP SUCTION
- CW — COLD WATER (CITY WATER)
- CWS — CONDENSER WATER SUPPLY (FROM TOWER)
- CWR — CONDENSER WATER RETURN (TO TOWER)
- D — DRAIN LINE
- FOS — FUEL OIL SUPPLY
- FOR — FUEL OIL RETURN
- FWPD — FEEDWATER PUMP DISCHARGE
- FWPS — FEEDWATER PUMP SUCTION
- FWS — FEEDWATER SAMPLE (FROM DEAERATOR)
- G — NATURAL GAS MAIN FUEL
- GCI — NATURAL GAS IGNITER FUEL
- GCR — CHILLED GLYCOL-WATER RETURN
- GCS — CHILLED GLYCOL-WATER SUPPLY

# PIPING SYMBOLS (CONTINUED)

- GHR — GLYCOL-WATER HEATING RETURN
- GHS — GLYCOL-WATER HEATING SUPPLY
- GRR — GLYCOL-WATER RUN AROUND RETURN
- GRS — GLYCOL-WATER RUN AROUND SUPPLY
- HPR — HIGH PRESSURE STEAM CONDENSATE RETURN
- HPS — HIGH PRESSURE STEAM (60 PSIG AND ABOVE)
- HWR — HOT WATER HEATING RETURN
- HWS — HOT WATER HEATING SUPPLY
- LPC — LOW PRESSURE STEAM CONDENSATE RETURN
- LPS — LOW PRESSURE STEAM (15 PSIG AND BELOW)
- MPR — MEDIUM PRESSURE STEAM CONDENSATE RETURN
- MPS — MEDIUM PRESSURE STEAM (16 PSIG THRU 59 PSIG)
- MW — MAKE-UP WATER
- OFL — OVERFLOW
- PC — PUMPED CONDENSATE
- RHG — REFRIGERANT HOT GAS
- RL — REFRIGERANT LIQUID
- RS — REFRIGERANT SUCTION
- SW — SOFTENED WATER
- V — VENT LINE
- NEW PIPE
- EXISTING PIPE TO REMAIN
- EXISTING PIPE TO BE DEMOLISHED

# DRAWING SYMBOLS

- 2 / H4 — DETAIL NUMBER / DRAWING NUMBER WHERE SHOWN
- A / H7 — SECTION LETTER / DRAWING NUMBER WHERE SHOWN
- 26-TU-1D1 — BUILDING NO. WHERE EQUIPMENT IS LOCATED; ITEM (TERMINAL UNIT SHOWN); ITEM NUMBER (TERMINAL UNIT NO. 1); AREA OF BUILDING (PHASE 1 SHOWN); FLOOR OF BUILDING

# HVAC NOTES AND CONDITIONS:

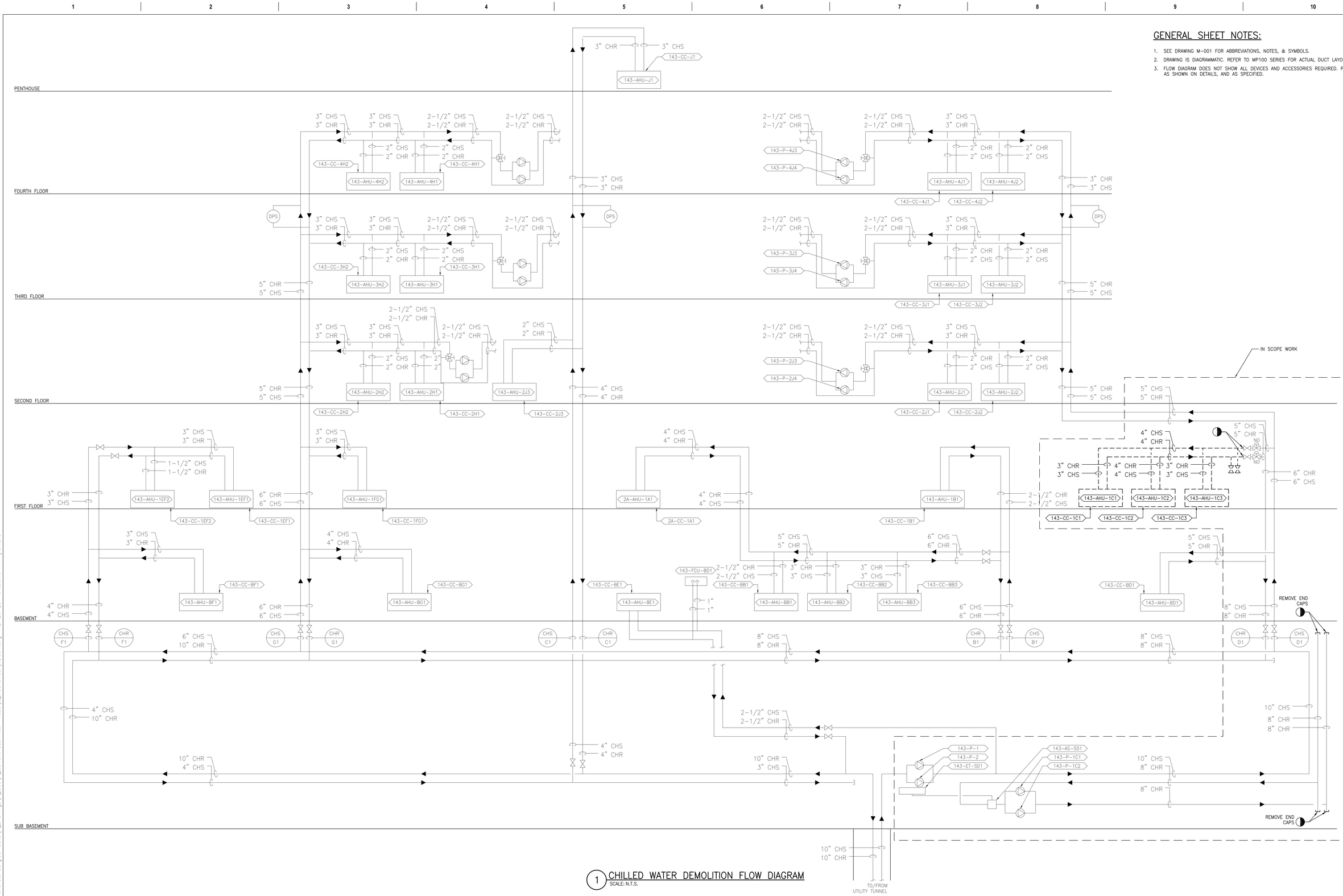
1. ALL DUCTWORK DIMENSIONS, AS SHOWN ON THE DRAWINGS, ARE INTERNAL CLEAR DIMENSIONS. CONTRACTOR SHALL INCREASE THE DUCT SIZE ACCORDINGLY TO COMPENSATE FOR ACOUSTICAL DUCT LINING THICKNESS (UNLESS OTHERWISE NOTED). CONTRACTOR MAY ADJUST DUCT SIZES ACCORDINGLY TO SUIT FIELD CONDITIONS AS LONG AS THE DUCT FREE AREA AND STATIC PRESSURE DROP ARE EQUAL TO THE PROJECT DOCUMENTS.
2. THE FIRST FIGURE OF DUCT SIZE INDICATES DIMENSION OF FACE SHOWN OR INDICATED. DUCT SIZES ARE NET INSIDE DIMENSIONS
3. ACCESS PANELS IN HARD SUSPENDED CEILINGS ARE REQUIRED FOR ALL VALVES, TRAPS, DAMPERS, CLEANOUTS, CONTROLS, ETC. AND ANY OTHER ITEMS THAT REQUIRE MECHANICAL MAINTENANCE. ACCESS PANELS SHALL BE FURNISHED AND INSTALLED UNDER THE ARCHITECTURAL SPECIFICATIONS.
4. DIFFUSER, REGISTER AND GRILLE SIZES SHOWN ON FLOOR PLANS ARE NECK SIZES.
5. WATER PIPE CONNECTIONS TO AIR HEATING AND COOLING COILS SHALL BE MADE TO PROVIDE COUNTER FLOW BETWEEN WATER AND AIR.
6. CONTRACTOR SHALL PROVIDE ACCESS DOORS IN DUCTWORK FOR OPERATION, ADJUSTMENT AND MAINTENANCE OF ALL FANS, VALVES, COILS AND MECHANICAL EQUIPMENT.
7. ADEQUATELY SIZED ACCESS PANELS SHOULD BE PROVIDED AT SMOKE DETECTORS, FIRE DAMPERS, SMOKE DAMPERS, HUMIDIFIERS, COILS AND OTHER DEVICES LOCATED IN THE DUCTWORK WHICH REQUIRE INSPECTION AND/OR MAINTENANCE.
8. CONTRACTOR SHALL PROVIDE FLEXIBLE CONNECTIONS IN ALL DUCTWORK SYSTEMS (SUPPLY, RETURN, VENTILATION AIR AND EXHAUST) CONNECTED TO ALL AIR HANDLING UNITS, FANS AND OTHER EQUIPMENT WHICH REQUIRE VIBRATION ISOLATION. FLEXIBLE CONNECTIONS SHALL BE PROVIDED AT THE POINT OF CONNECTION TO THE EQUIPMENT (UNLESS NOTED OTHERWISE).
9. SMOOTH TURN RADIUS DUCTWORK OR TURNING VANES AND 45 DEGREE ENTRY OR CONICAL DUCT BRANCH CONNECTIONS SHALL BE USED THROUGHOUT.
10. ALL DUCTWORK SHALL BE COORDINATED WITH ALL TRADES INVOLVED. OFFSETS IN DUCTS, INCLUDING TRANSITIONS AROUND OBSTRUCTIONS, SHALL BE PROVIDED AT NO ADDITIONAL COST.
11. COORDINATE DIFFUSER, REGISTER, AND GRILLE LOCATIONS WITH ARCHITECTURAL REFLECTED CEILING PLANS. LIGHTING AND OTHER CEILING ITEMS AND MAKE MINOR DUCT MODIFICATIONS TO SUIT.
12. ALL INTERIOR AND EXTERIOR WALL, CEILING AND ROOF OPENINGS AND DUCT AND PIPE THROUGH WALL PENETRATIONS, FRAMING, SLEEVES, WEATHERPROOFING AND FIRE SEALING SHALL BE PROVIDED AND INSTALLED BY THE CONTRACTOR, UNLESS NOTED OTHERWISE.
13. CONTRACTOR SHALL PROVIDE AND INSTALL ALL FIRE DAMPERS IN ACCORDANCE WITH UL 555 AND MANUFACTURER'S REQUIREMENTS WHERE DUCTS PENETRATE FIRE RATED WALLS OR FLOORS. ALL FIRE DAMPERS SHALL BE UL LISTED AND LABELED CONSISTENT WITH THE WALLS OR FLOORS THAT ARE PENETRATED. PROVIDE ACCESS DOORS SIZED AND LOCATED TO FACILITATE TESTING AND RESETTING OF ALL FIRE DAMPERS.
14. CONTRACTOR SHALL TEST, ADJUST AND BALANCE THE COMPLETED SYSTEM. ALL BALANCING WORK SHALL BE DONE UNDER THE DIRECT SUPERVISION OF AN AABC OR NEBB-QUALIFIED SUPERVISOR WHO SHALL CERTIFY THE RESULTS.
15. TEST, ADJUST AND BALANCE WORK WILL DEMONSTRATE THAT ALL SYSTEMS OPERATE AS THEY WERE INTENDED TO. UPON COMPLETION OF THE WORK, A WRITTEN TEST & BALANCE REPORT WILL BE SUBMITTED TO THE OWNER. FORMS AND PROCEDURES UTILIZED FOR THIS WORK WILL BE AS PROPOSED BY THE AABC OR OTHER NATIONALLY RECOGNIZED ORGANIZATION ACCEPTABLE TO THE OWNER.
16. ALL DUCTWORK, PIPING AND EQUIPMENT SHOWN ON NEW WORK PLANS AND DETAILS SHALL BE CONSIDERED AS NEW UNLESS IDENTIFIED AS EXISTING [IDENTIFIED AS LIGHT LINEWORK ON THE DRAWINGS] TO REMAIN.

# GENERAL NOTES AND CONDITIONS

1. THESE DRAWINGS WERE PREPARED FROM INFORMATION TAKEN FROM THE AVAILABLE BUILDING DRAWINGS, ARCHITECTURAL BACKGROUNDS PROVIDED BY THE OWNER AND FIELD SURVEY INFORMATION COMPILED BY THE ENGINEERING DESIGN TEAM FOR THE PURPOSE OF ENGINEERING DESIGN. EXISTING CONDITIONS ARE SHOWN AS ACCURATELY AS POSSIBLE. THERE IS THE POSSIBILITY THAT CONDITIONS SHOWN ARE NOT EXACTLY AS EXISTING. CONTRACTOR SHALL VERIFY ALL DIMENSIONS, ELEVATIONS, LOCATIONS, SIZES AND CONDITIONS AT THE SITE AND REPORT ANY DISCREPANCIES TO THE ENGINEER PRIOR TO BEGINNING INSTALLATION OR FABRICATION WORK.
   1.1. IT IS THE RESPONSIBILITY OF THE CONTRACTOR TO REVIEW THE PLANS AND SPECIFICATIONS, AS WELL AS ALL RELATED PROJECT PLANS AND SPECIFICATIONS FROM OTHER DISCIPLINES, TO BECOME FAMILIAR WITH THE FULL PROJECT SCOPE AND COORDINATED RESPONSIBILITIES.
   1.2. SHOULD IT APPEAR THAT THE WORK INTENDED TO BE DESCRIBED OR RELATED WORK ARE NOT SUFFICIENTLY DETAILED OR EXPLAINED ON THE DRAWINGS, OR IN THE SPECIFICATIONS, CONSULT THE ENGINEER FOR NECESSARY CLARIFICATIONS, AND CONFORM TO THOSE CLARIFICATIONS INSOFAR AS THEY ARE CONSISTENT WITH THE ORIGINAL DRAWINGS AND SPECIFICATIONS. IN NO CASE SHALL WORK PROCEED IN UNCERTAINTY.
   1.3. EQUIPMENT ARRANGEMENTS ARE DESIGNED TO SHOW PREFERRED CONFIGURATIONS TO SUIT KNOWN CONDITIONS. ACTUAL INSTALLATION BY CONTRACTOR MAY BE ALTERED AS REQUIRED TO SUIT FIELD CONDITIONS ENCOUNTERED DURING CONSTRUCTION WITHOUT COMPROMISING THE INTENT OF THE ORIGINAL DESIGN. COORDINATE ANY MODIFICATIONS WITH THE ENGINEER AND OWNER PRIOR TO STARTING WORK IN AREAS AFFECTED.
   1.4. THE CONTRACTOR SHALL CHECK AND VERIFY ALL DIMENSIONS AND CONDITIONS AT SITE PRIOR TO STARTING WORK.
   1.5. DO NOT SCALE DRAWINGS.
2. BIDDERS ARE TO VISIT THE SITE AND FAMILIARIZE THEMSELVES AS THE NATURE AND SCOPE OF THE WORK. THE SUBMISSION OF A BID WILL BE EVIDENCE THAT SUCH AN EXAMINATION HAS BEEN MADE AND ENSURE THAT ALL ALTERNATE PRICING INFORMATION NOTED WITHIN THE CONSTRUCTION DOCUMENTS HAVE BEEN INCLUDED WITHIN THE ORIGINAL BID SUBMISSION. LATER CLAIMS FOR LABOR, EQUIPMENT OR MATERIALS REQUIRED, OR FOR ANY DIFFICULTIES ENCOUNTERED WHICH COULD HAVE BEEN FORESEEN HAD AN EXAMINATION BEEN MADE, WILL NOT BE ACCEPTED.
3. THE CONTRACTOR WILL BE RESPONSIBLE FOR ENSURING THAT ALL RULES AND REGULATIONS, INCLUDING THOSE WHICH MAY BE ISSUED BY THE OWNER, ARE BEING OBSERVED, PARTICULARLY WORKPLACE SAFETY AND THE CONDUCT OF ALL THOSE EMPLOYED DIRECTLY AND INDIRECTLY ON THE PREMISES, AND THE OWNER'S EMPLOYEES WHO MAY BE IMPACTED OR AFFECTED BY CONSTRUCTION ACTIVITIES. THE CONTRACTOR WILL INSTALL SIGNAGE, BARRIERS, AND OTHER MEANS TO PROVIDE WARNING AND PERSONNEL SAFETY. PLACEMENT OF THESE ITEMS WILL BE COORDINATED WITH THE OWNER AND HIS ONGOING OPERATIONS AND WILL PROMPTLY BE REVISED WHEN WORK IN A PARTICULAR AREA HAS BEEN COMPLETED.
   3.1. DURING PERFORMANCE OF WORK, THE CONTRACTOR SHALL BE RESPONSIBLE FOR PROVISION AND MAINTENANCE OF WARNING SIGNS, LIGHT SIGNAL DEVICES, GUARD LIGHTS, BARRICADES, GUARD RAILS, FENCES AND OTHER DEVICES, APPROPRIATELY LOCATED ON AND AROUND THE JOB SITE WHICH GIVE PROPER AND UNDERSTANDABLE WARNING TO PERSONS WITH REGARD TO HAZARDOUS CONDITIONS, EQUIPMENT AND OPERATIONS BEING PERFORMED IN CONJUNCTION WITH THE WORK.
4. THE CONTRACTOR SHALL BE RESPONSIBLE TO COORDINATE WITH OTHER TRADES FOR ITEMS IN THEIR SCOPE OF WORK WHICH COULD REQUIRE ADDITIONAL MECHANICAL/PLUMBING WORK, AND ITEMS THAT ARE NOT INDICATED ON THE DRAWINGS.
5. THIS INSTALLATION WILL CONFORM TO ALL CODES AND THE REQUIREMENTS OF FEDERAL, STATE, AND LOCAL REGULATORY AGENCIES HAVING JURISDICTION. IN PARTICULAR, THE WORK WILL BE IN ACCORDANCE WITH THE LATEST EDITION OF NFPA 90A, NFPA 96, NFPA 99 AND NFPA 101.
6. ALL WORK WILL BE LAWFULLY EXECUTED IN A NEAT AND WORKMANLIKE MANNER AND WILL BE COMPLETED IN ACCORDANCE WITH THE GOVERNING CODES (ABOVE), INDUSTRY STANDARDS, AND IN CONFORMANCE WITH THE MANUFACTURERS RECOMMENDATIONS AND REQUIREMENTS.
7. WORK UNDER THIS CONTRACT SHALL CONSIST OF THE CONTRACTOR PROVIDING ALL LABOR, MATERIALS, AND SERVICES, INCLUDING WORK NOT SPECIFICALLY SHOWN BUT REASONABLY IMPLIED. THIS SHALL INCLUDE CUTTING, PATCHING AND RESTORATION OF EXISTING SURFACES DAMAGED DURING THE CONSTRUCTION. CONTRACTOR SHALL ALSO PROVIDE ALL EQUIPMENT SHOWN OR SPECIFIED OR AN APPROVED EQUIVALENT. SUBSTITUTED EQUIPMENT OR MATERIALS SHALL NOT BE INSTALLED UNTIL GIVEN WRITTEN APPROVAL BY THE OWNER.
8. CONTRACTOR SHALL BE RESPONSIBLE FOR COSTS INCURRED FOR NONCOMPLIANCE WITH THESE CONTRACT DOCUMENTS. CONTRACTOR WILL NOT BE ALLOWED CHANGE

## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MP100 SERIES FOR ACTUAL DUCT LAYOUT.
3. FLOW DIAGRAM DOES NOT SHOW ALL DEVICES AND ACCESSORIES REQUIRED. PROVIDE AS SHOWN ON DETAILS, AND AS SPECIFIED.




1 CHILLED WATER DEMOLITION FLOW DIAGRAM
SCALE: N.T.S.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

## CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

## ENGINEER OF RECORD

MILLER•REMICK
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MILLER-REMICK LLC
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ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL DEMOLITION CHILLED WATER FLOW DIAGRAM | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Project Number** 658-19-103 | **Building Number** 143 |
| | | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** MD601 |
| | | **Issue Date** 2025-12-30 / **Checked** BH / **Drawn** CS | |

VA FORM 08-6231

three inches = one foot | one and one half inches = one foot | one inch = one foot | three quarters inch = one foot | one half inch = one foot | three eighths inch = one foot | one quarter inch = one foot | one eighth inch = one foot
P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\MD601.dwg 12-29-25 01:36:16 PM jromano

three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
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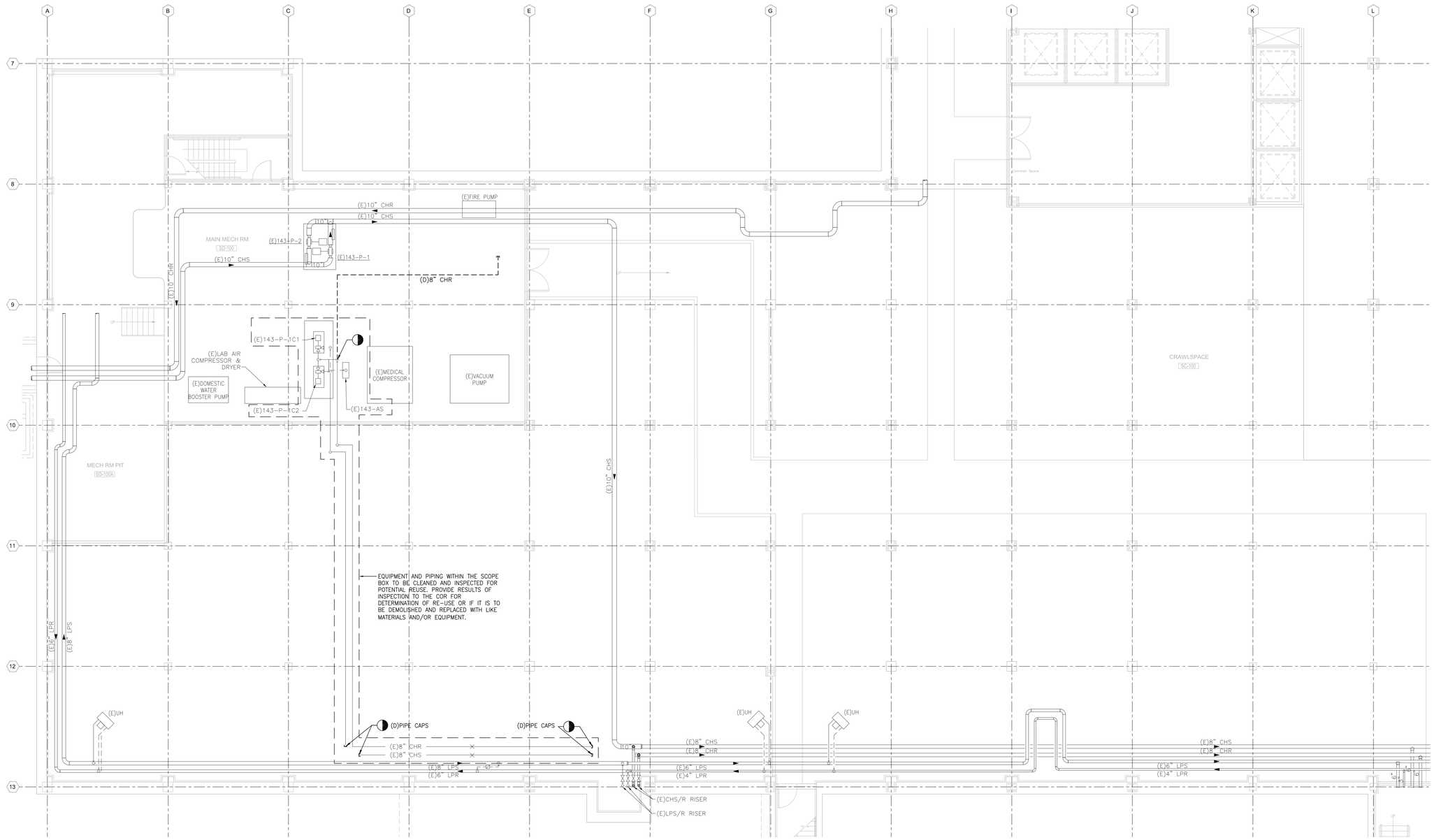
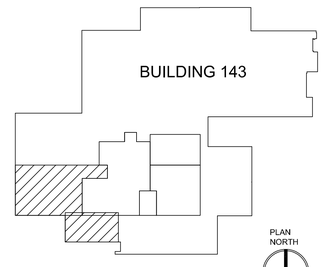
## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.




1 SUB-BASEMENT DEMOLITION PLAN
SCALE: 1/8" = 1'-0"




KEYPLAN

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| NO. | DESCRIPTION | DATE |
|---|---|---|

CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

ENGINEER OF RECORD

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12.30.2025

Office of Construction and Facilities Management

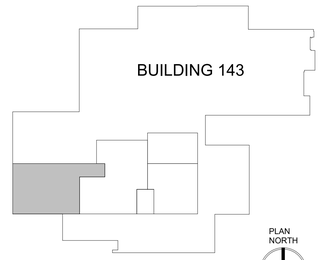
VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL PIPING DEMOLITION PLAN SUB-BASEMENT, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | BH |
| Drawn | PNL |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MD100 |

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

1 DEMOLITION PLAN - BASEMENT, PHASE 1
SCALE: 1/4" = 1'-0"

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| Revisions: | Date: |
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CONSULTANT

| Architecture: | Civil Engineering: | Fire Protection: |
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12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DEMOLITION PLAN BASEMENT PHASE 1 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| | | Project Number: 658-19-103 | Building Number: 143 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: P1-MD101 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

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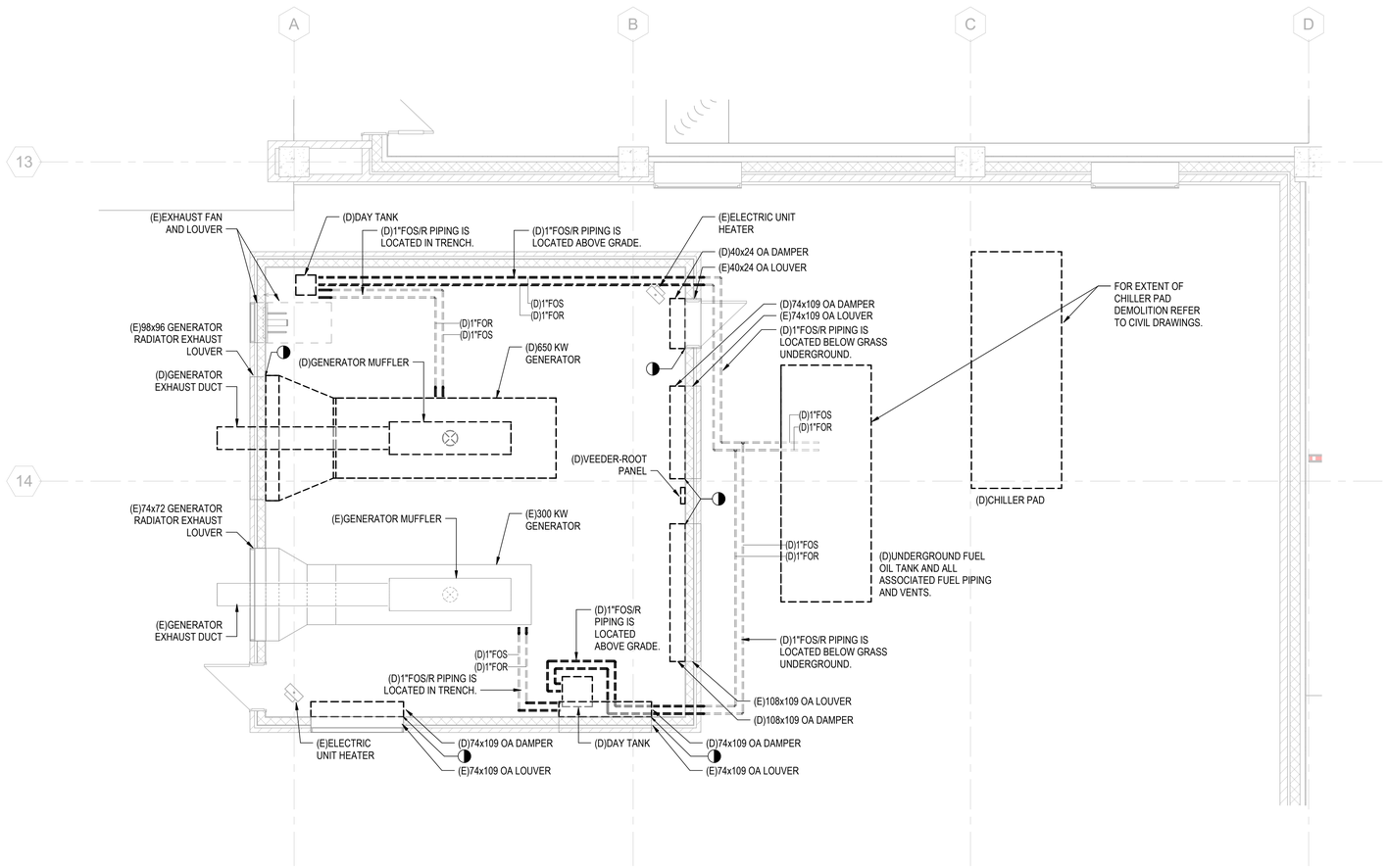
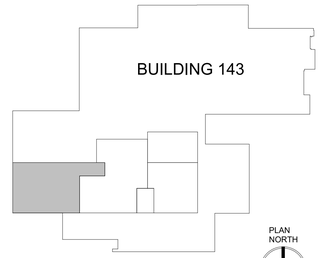
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.
2. COORDINATE ALL UNDERGROUND UTILITY DEMOLITION WITH CIVIL DRAWINGS.
3. COORDINATE ALL DEMOLITION WORK IN GENERATOR ROOM WITH STRUCTURAL DRAWINGS.




## 1 DEMOLITION PLAN - BASEMENT GENERATOR, PHASE 1

SCALE: 1/4" = 1'-0"

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12-30-2025

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|---|---|
| Drawing Title | MECHANICAL DEMOLITION PLAN BASEMENT GENERATOR, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MD102 |

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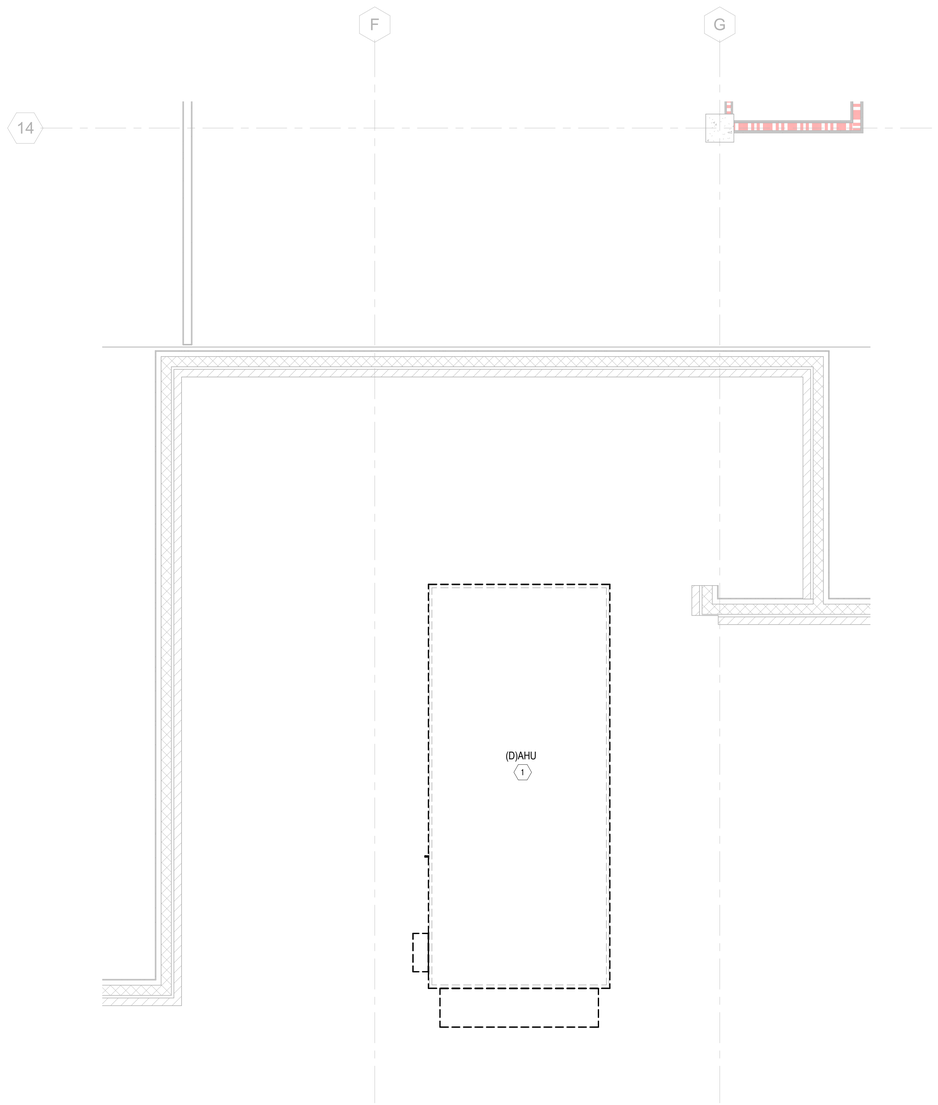
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

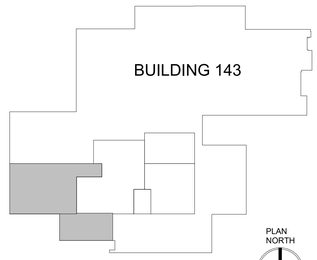
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

## SHEET KEYNOTES:

1 EXISTING AHU IS ABANDONED. CONTRACTOR TO DISCONNECT FROM ANY UTILITIES, DRAIN THE UNIT OF ANY FLUIDS AND DISPOSE OF THEM IN A LEGAL MANNER. MOVE THE EQUIPMENT TO A LOCATION ON THE SALEM VAMC CAMPUS AS DIRECTED BY THE COR. EC TO REMOVAL ELECTRICAL CONNECTIONS BACK TO SOURCE.




1 DEMOLITION PLAN - BASEMENT, PHASE 1
SCALE: 1/4" = 1'-0"

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Lic. No. 402055079
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12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DEMOLITION PLAN BASEMENT, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MD103 |

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VA FORM 08 - 6231

## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

## SHEET KEYNOTES:

1. DEMOLISH EXISTING DUCT WORK, TERMINAL UNITS, CONTROLS, CONTROL WIRING, HANGERS, REGISTERS, GRILLES AND ALL APPURTENANCES.
2. DEMOLISHED DUCTWORK AND PIPING CONTINUES INTO MECHANICAL ROOM. REFER TO DRAWING P1-MD112.

1 MECHANICAL DEMOLITION DUCTWORK PLAN – 1ST FLOOR, PHASE 1
SCALE: 1/4" = 1'-0"




KEYPLAN

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**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

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|---|---|---|---|
| **Drawing Title** MECHANICAL DEMOLITION DUCTWORK PLAN 1ST FLOOR, PHASE 1 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Project Number** 658-19-103 | **Building Number** 143 |
| | | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P1-MD111 |
| | | **Issue Date** 2025-12-30 / **Checked** BH / **Drawn** CS | |

VA FORM 08-6231

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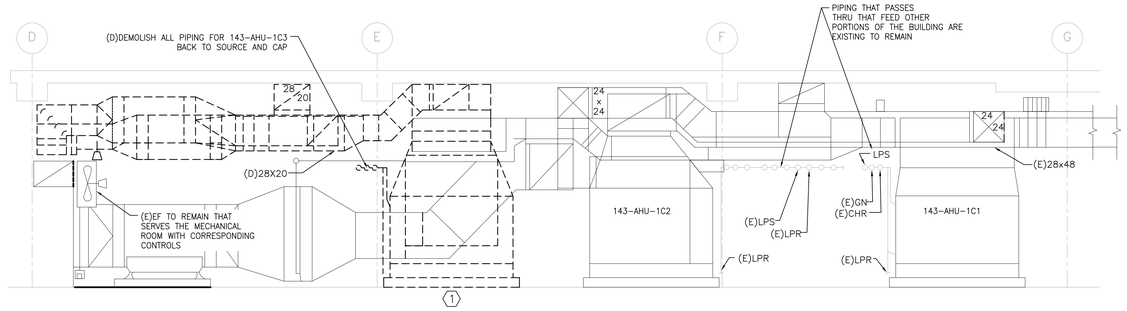
## GENERAL SHEET NOTES:

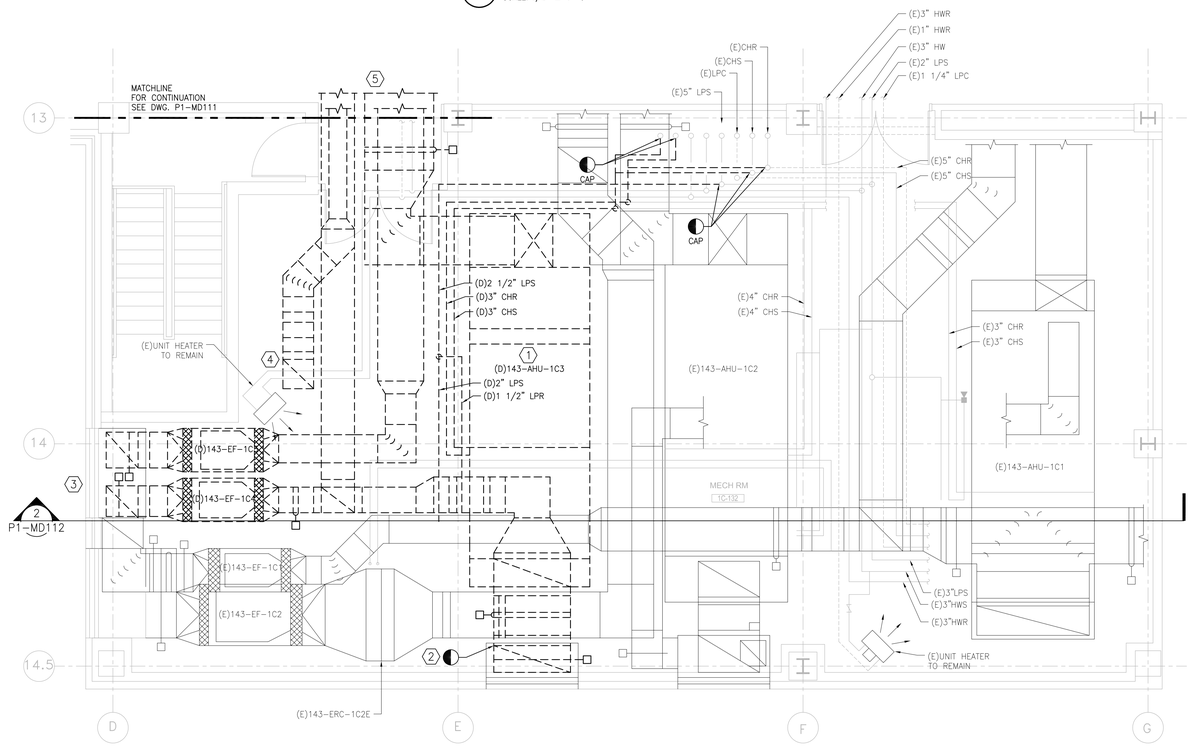
1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

## ⬡ SHEET KEYNOTES:

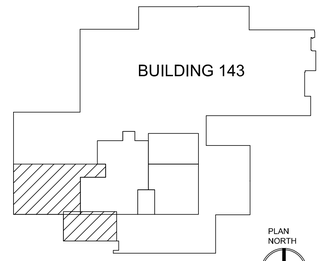
1. DEMOLISH EXISTING AHU, DUCT WORK, PIPING, CONTROLS, CONTROL WIRING, HANGERS, SUPPORTS AND ALL APPURTENANCES ASSOCIATED WITH 143-AHU-1C3 UNDER PHASE 1.
2. DEMOLISH OA DUCTWORK AND DISCONNECT OA DUCTWORK FROM OA PLENUM. OA PLENUM EXISTING TO REMAIN.
3. DEMOLISH EA DUCTWORK AND DISCONNECT EA DUCTWORK FROM EA PLENUM. EA PLENUM EXISTING TO REMAIN.
4. DEMOLISH 24x24 DUCTWORK UP THROUGH ROOF. PATCH ROOF OPENING TO MATCH EXISTING ROOF.
5. DEMOLISH DUCTWORK OUT TO FIRST FLOOR. REFER TO DRAWING P1-MD111.




2 SECTION AT MECHANICAL ROOM 1C-132
SCALE: 1/4" = 1'-0"




1 MECHANICAL DEMOLITION PLAN – MECHANICAL ROOM, PHASE 1
SCALE: 1/4" = 1'-0"




KEYPLAN

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**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DEMOLITION PLAN MECHANICAL ROOM, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MD112 |

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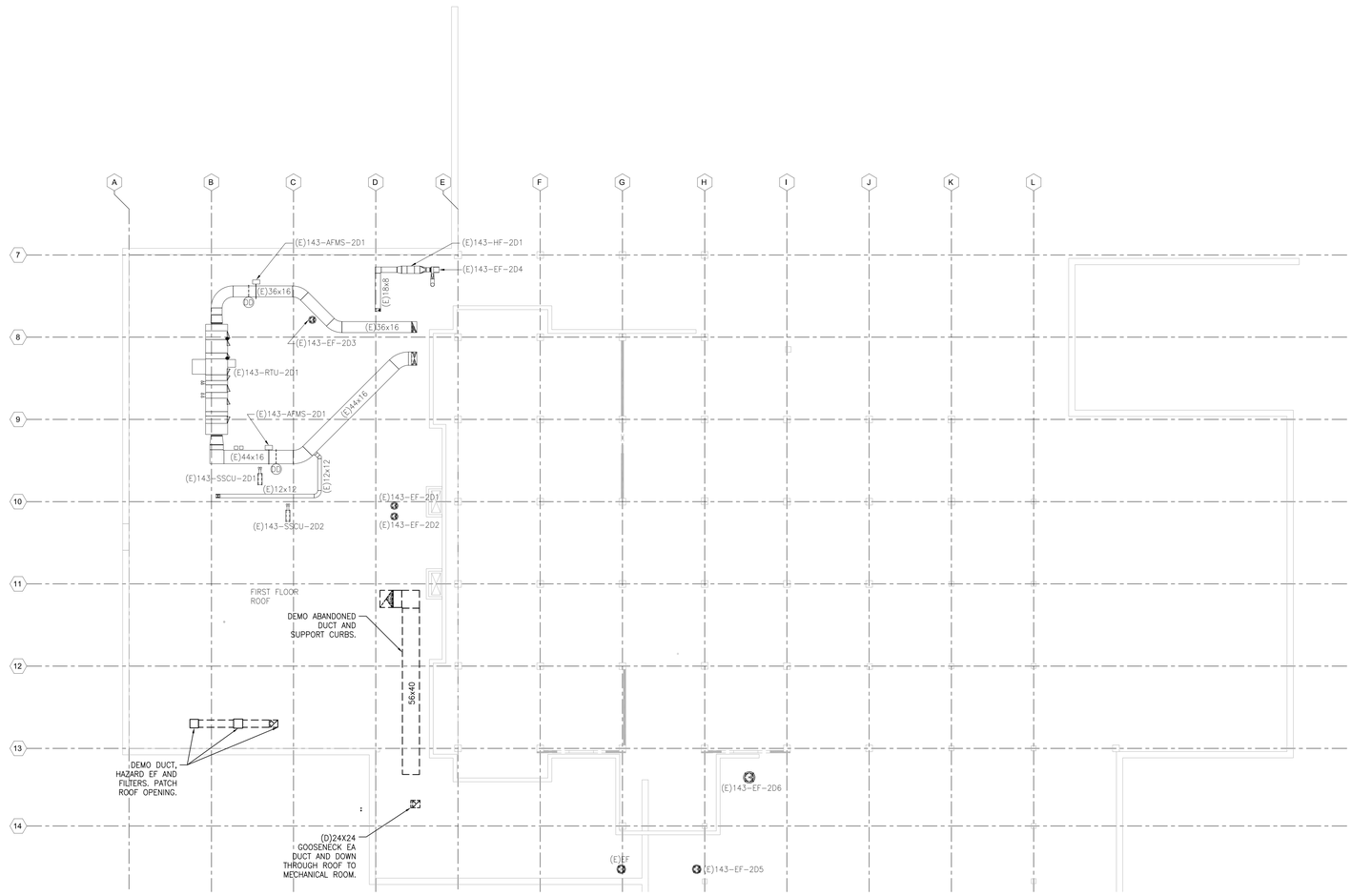
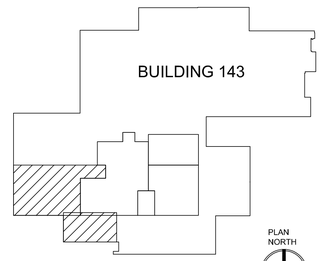
P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P1-MD112.dwg 12-29-25 01:27:50 PM jromano

## GENERAL SHEET NOTES:

1. MAINTAIN EXISTING ROOF WARRANTY. PATCH ALL ROOF OPENINGS FROM DEMOLISHED EQUIPMENT, DUCTWORK AND PIPING.




1 MECHANICAL DEMOLITION ROOF PLAN – PHASE 1
SCALE: 1/16" = 1'-0"

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|---|---|---|
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**CONSULTANTS**

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MILR-PROJECT-NO-0499-0136

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PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 0402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

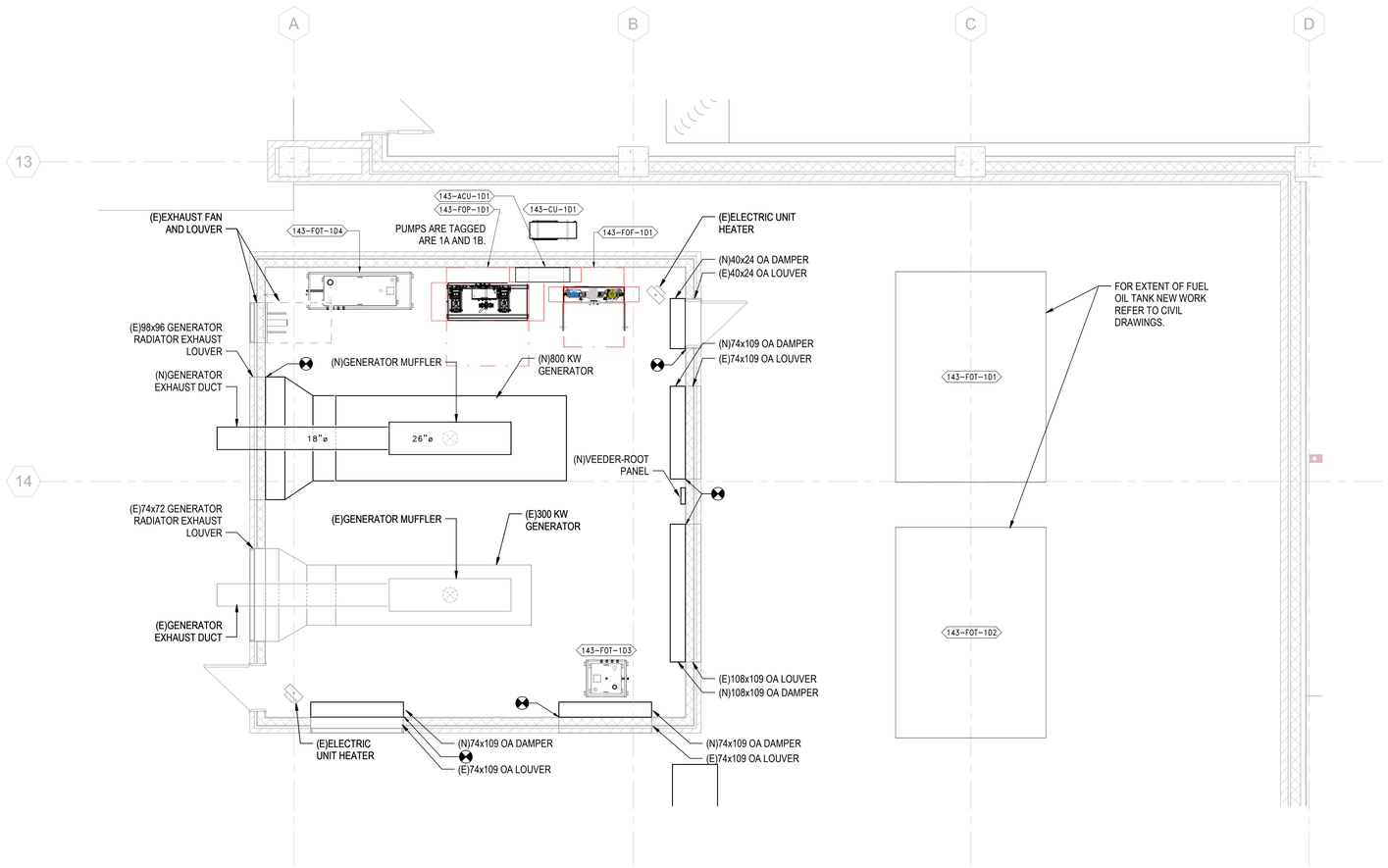
| | |
|---|---|
| Drawing Title | MECHANICAL DEMOLITION ROOF PLAN, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | BH |
| Drawn | PNL |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MD121 |



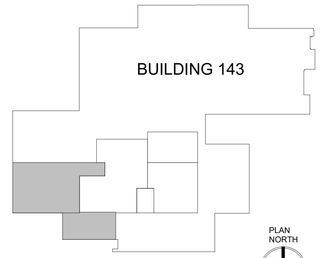
P:\0499_OWA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P1-MD121.dwg 12-29-25 01:28:24 PM jromano

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.
2. COORDINATE ALL UNDERGROUND UTILITY NEW WORK WITH CIVIL DRAWINGS.




1 DUCTWORK PLAN - 1ST FLOOR, PHASE 1 - GENERATOR
SCALE: 1/4" = 1'-0"

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**CONSULTANT**

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Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DUCTWORK PLAN BASEMENT, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MH101 |

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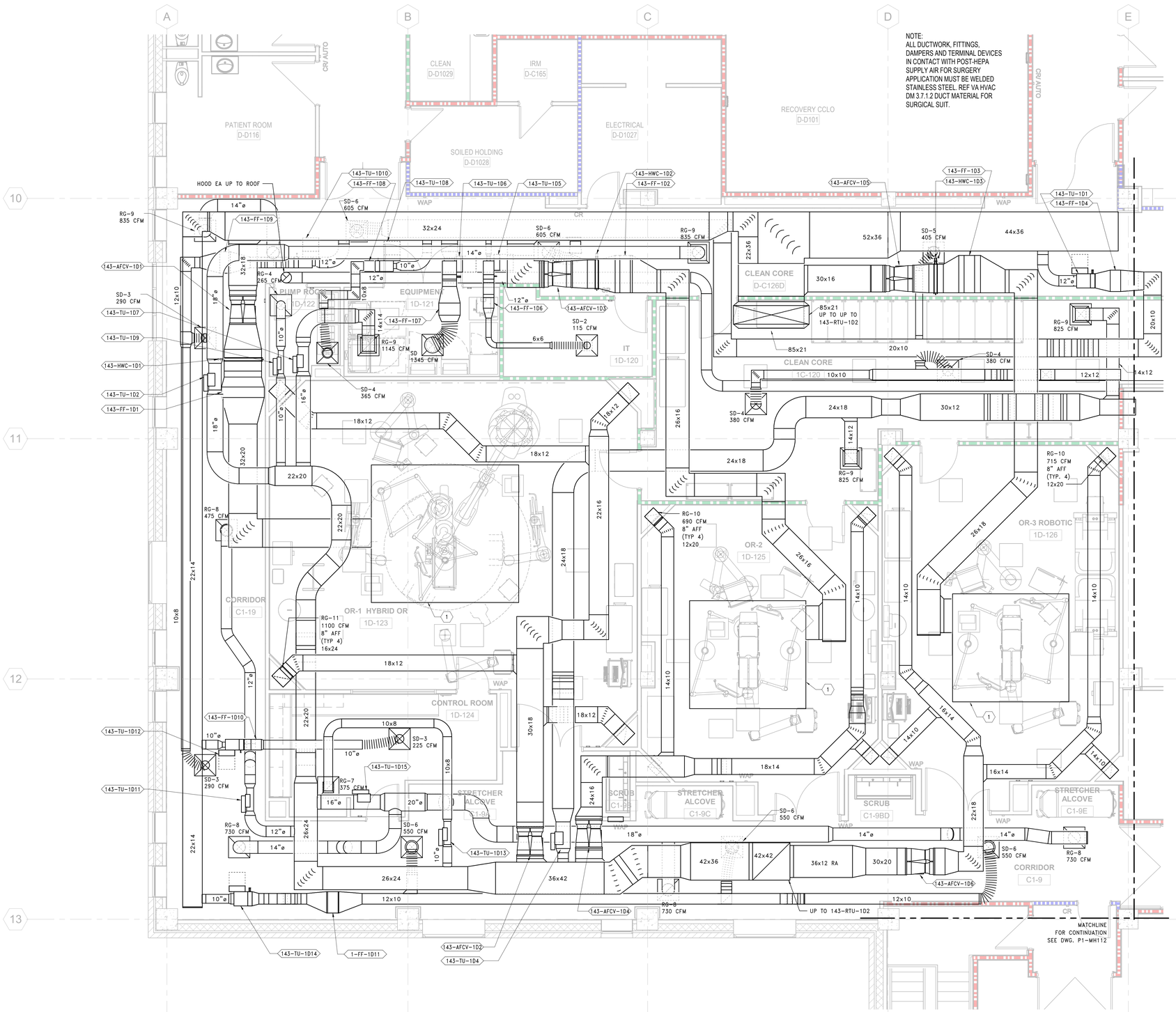



## GENERAL SHEET NOTES:

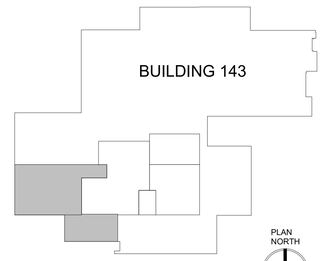
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.
2. ALL DUCTWORK, FITTINGS, DAMPERS AND TERMINAL DEVICES IN CONTACT WITH POST-HEPA SUPPLY AIR FOR SURGERY APPLICATION MUST BE WELDED STAINLESS STEEL. REF VA HVAC DM 3.7.1.2 DUCT MATERIAL FOR SURGICAL SUIT.

## SHEET KEYNOTES:

1 DIFFUSER FIELD IN CEILING. REFER TO ARCHITECTURAL DRAWINGS.

## 1 DUCTWORK PLAN - 1ST FLOOR, PHASE 1

SCALE: 1/4" = 1'-0"

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ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DUCTWORK PLAN 1ST FLOOR, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | CS |
| Drawn | SP |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MH111 |

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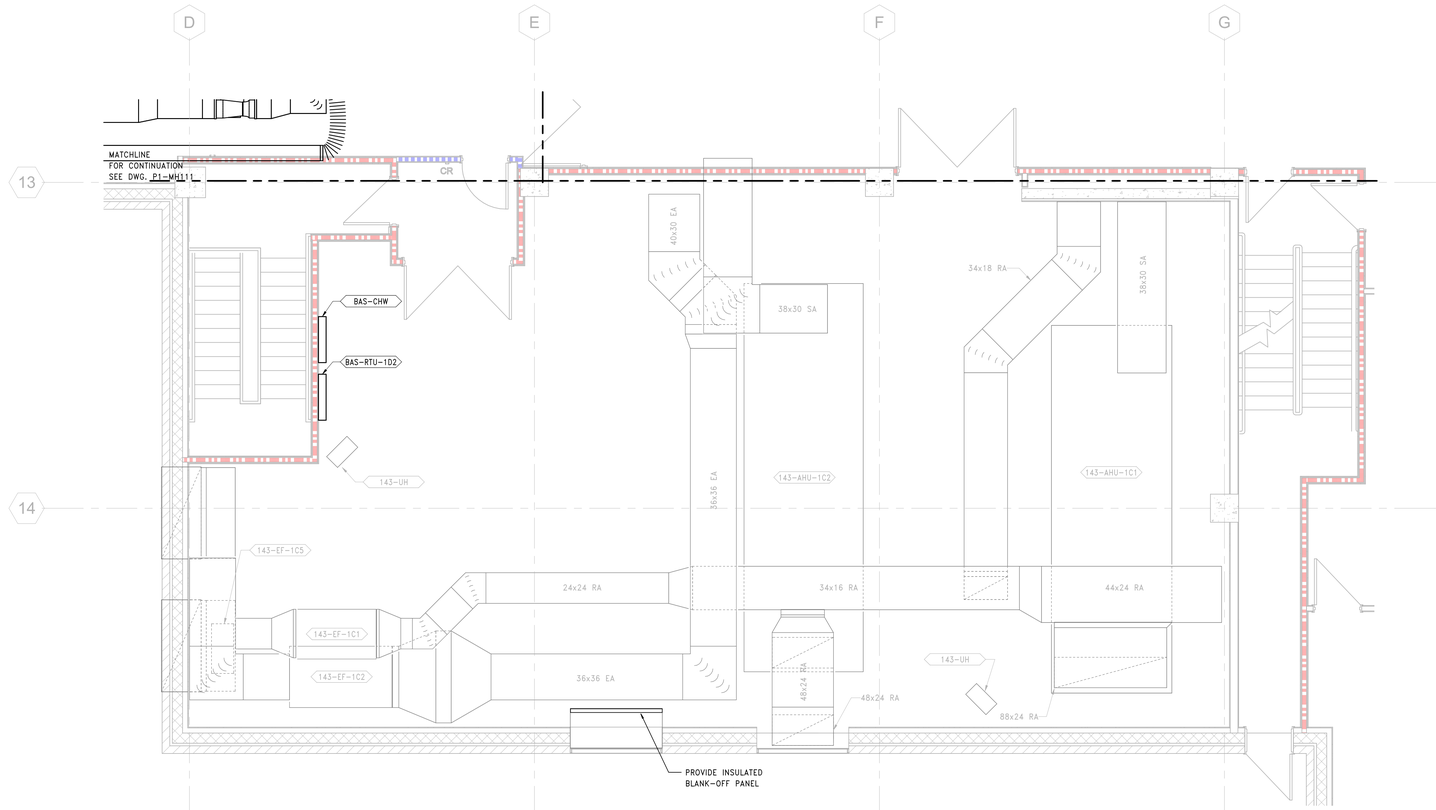
## GENERAL SHEET NOTES:

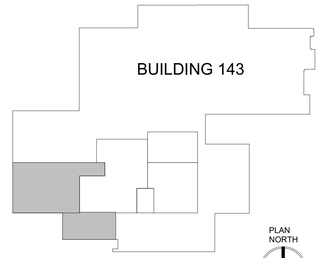
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.
2. ALL DUCTWORK, FITTINGS, DAMPERS AND TERMINAL DEVICES IN CONTACT WITH POST-HEPA SUPPLY AIR FOR SURGERY APPLICATION MUST BE WELDED STAINLESS STEEL. REF VA HVAC DM 3.7.1.2 DUCT MATERIAL FOR SURGICAL SUIT.

GENERAL NOTE:
NO NEW WORK TO BE PERFORMED IN THIS AREA OTHER THAN THE INSTALLATION OF BAS CONTROL PANELS FOR ROOFTOP AHU D2 AND CHILLED WATER SYSTEM.
DURING PHASE I, EXISTING AHU HAS BEEN DEMOLISHED AND REPLACED BY ROOFTOP UNIT 143-RTU-D2.
INSULATED BLANKOFF PANEL TO BE PROVIDED ON THE OA PLENUM FROM DEMOLISHED AHU.




## 1 DUCTWORK PLAN - MECHANICAL ROOM, PHASE 1

SCALE: 1/4" = 1'-0"

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

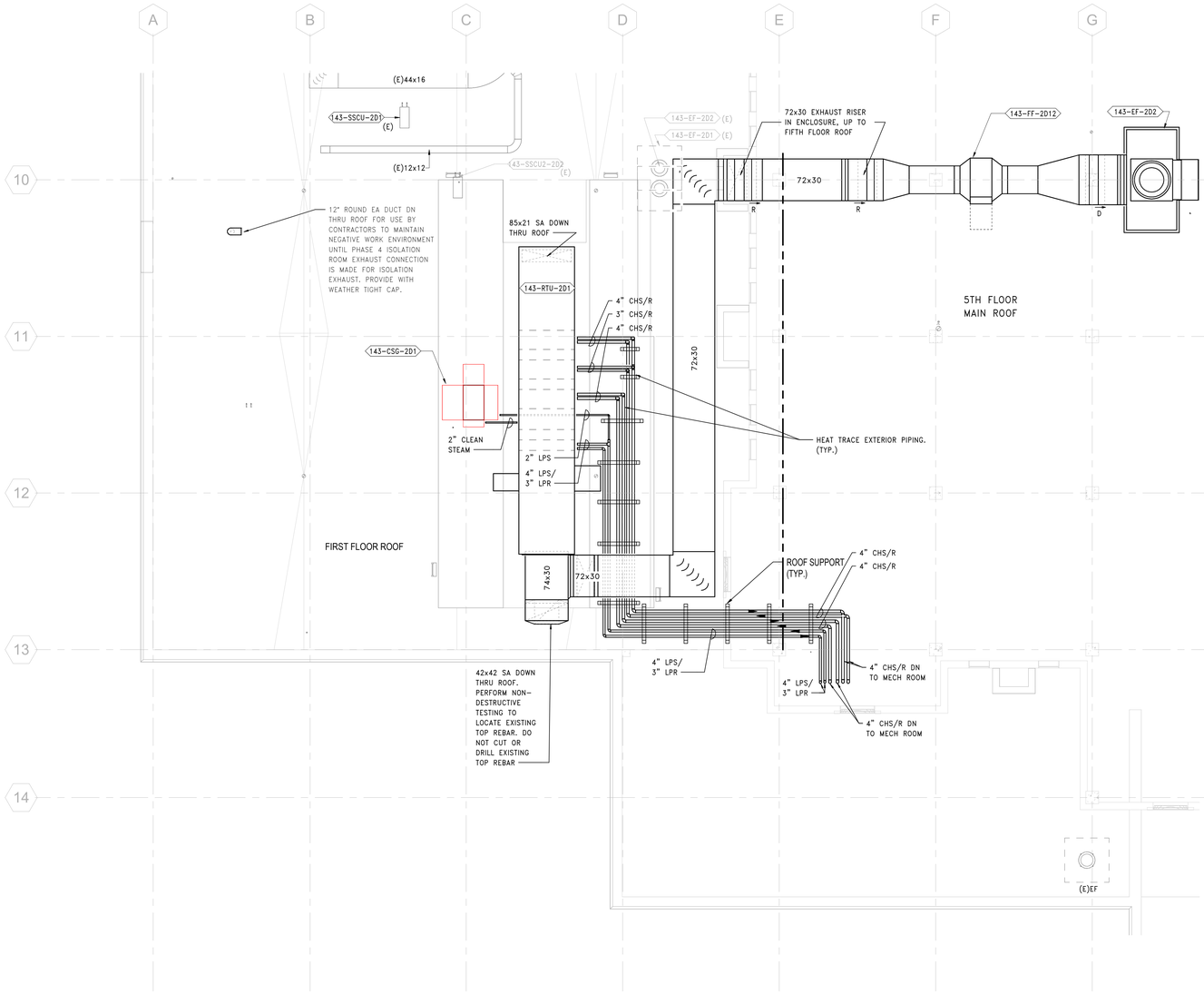
| | |
|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN MECHANICAL ROOM, PHASE 1 | Phase: BID DOCUMENTS |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED |

| | | |
|---|---|---|
| Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 | |
| Project Number: 658-19-103 | Building Number: 143 | |
| Location: SALEM VA MEDICAL CENTER | Drawing Number: P1-MH112 | |
| Issue Date: 2025-12-30 | Checked: CS | Drawn: SP |

VA FORM 08 - 6231

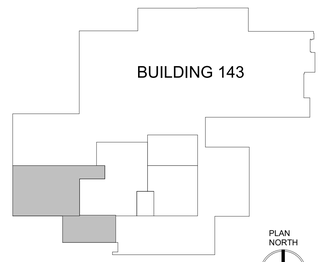
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




## 1 DUCTWORK AND PIPING PLAN - ROOF, PHASE 1

SCALE: 1/8" = 1'-0"

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M-R PROJECT NO. 0499-006

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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER

12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| **Drawing Title** | MECHANICAL DUCTWORK PLAN ROOF, PHASE 1 |
| **Approved by:** | Jon Manninen, COR |
| **Phase** | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| **Project Title** | RENOVATE SURGICAL SUITE PHASE II |
| **Project Number** | 658-19-103 |
| **Location** | SALEM VA MEDICAL CENTER |
| **Issue Date** | 2025-12-30 |
| **Checked** | DJ |
| **Drawn** | JAK |
| **VA Drawing Number** | 2617 |
| **Building Number** | 143 |
| **Drawing Number** | P1-MH121 |

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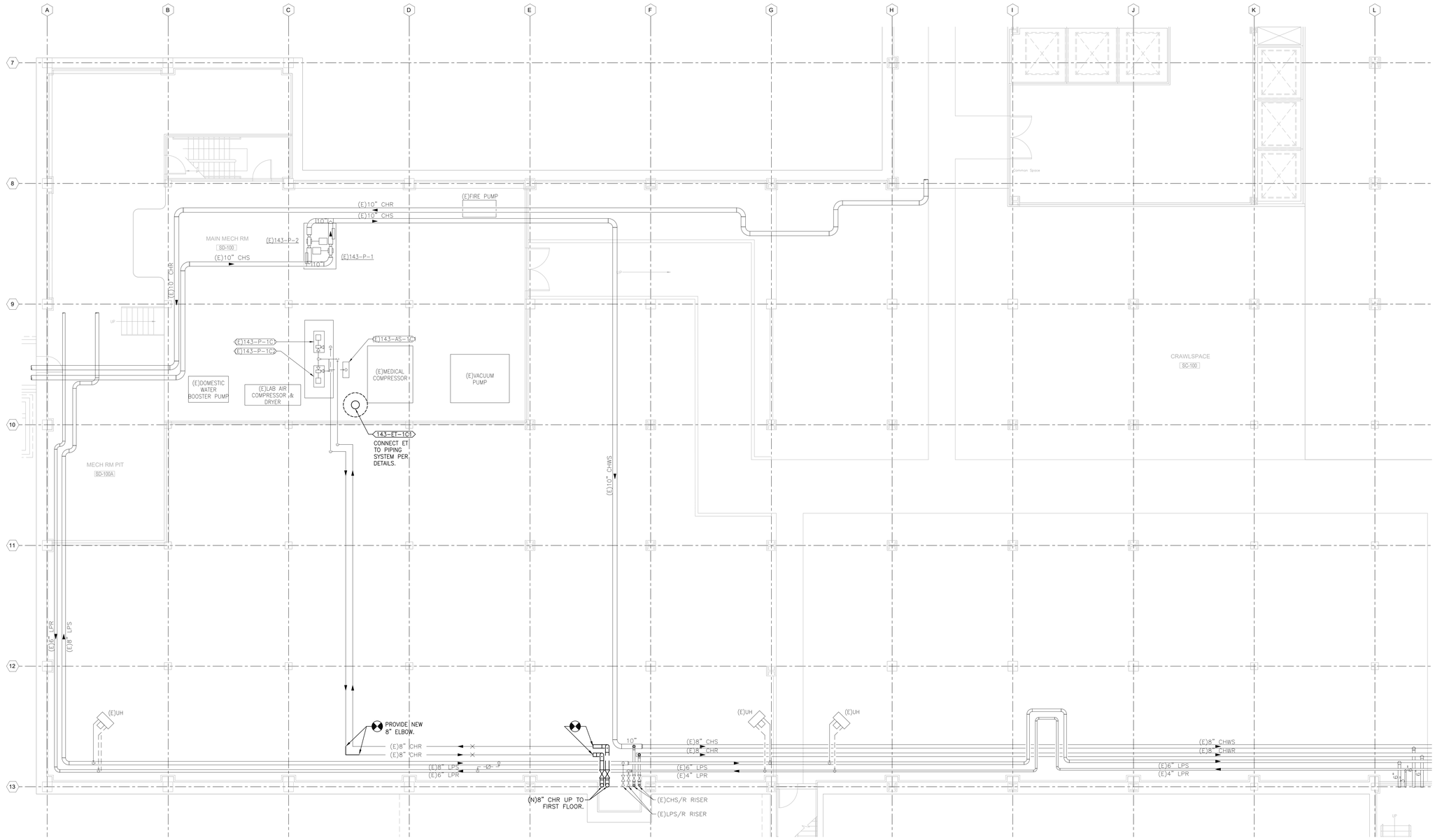
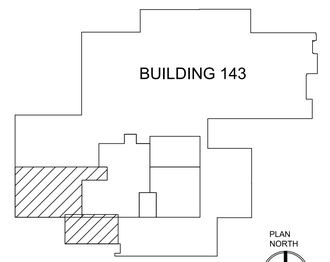
## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.




1 SUB-BASEMENT DEMOLITION PLAN
SCALE: 1/8" = 1'-0"

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

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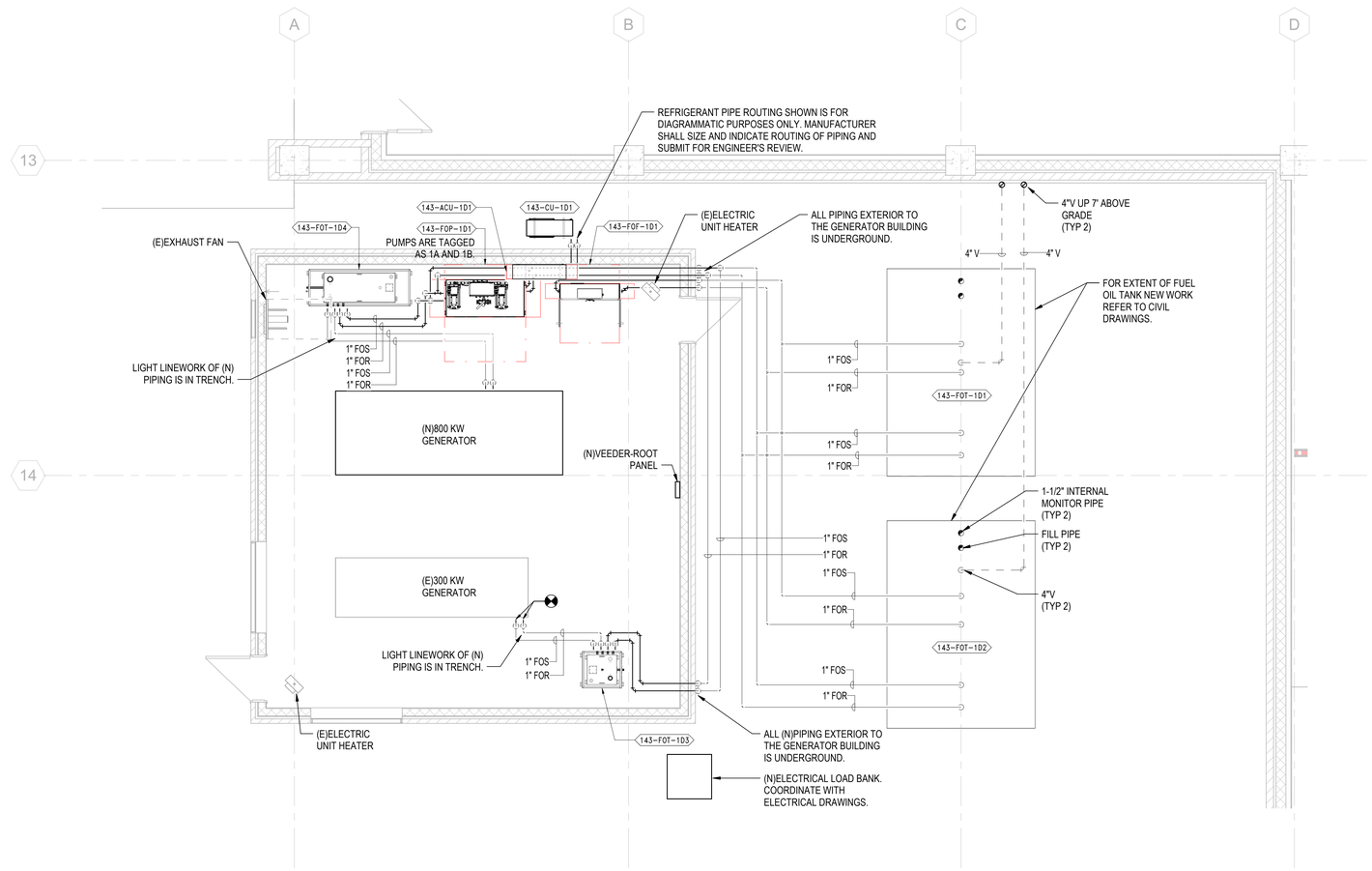
|---|---|
| Drawing Title | MECHANICAL PIPING PLAN SUB-BASEMENT |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | BH |
| Drawn | PNL |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MP100 |

VA FORM 08-6231

three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P1-MP100.dwg 12-30-25 11:11:52 AM mcifelli

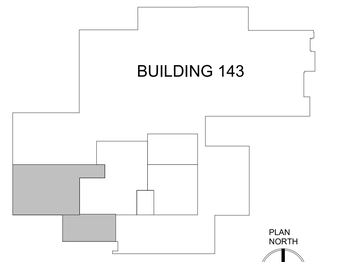
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.
2. COORDINATE ALL UNDERGROUND UTILITY NEW WORK WITH CIVIL DRAWINGS.




## 1 PIPING PLAN - BASEMENT GENERATOR, PHASE 1

SCALE: 1/4" = 1'-0"

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ANTHONY DANIEL PACARO JR
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12.30.2025

Office of Construction and Facilities Management

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| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL PIPING PLAN BASEMENT GENERATOR, PHASE 1 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P1-MP102 |
| | | Issue Date: 2025-12-30 / Checked: JAK / Drawn: JLR | |

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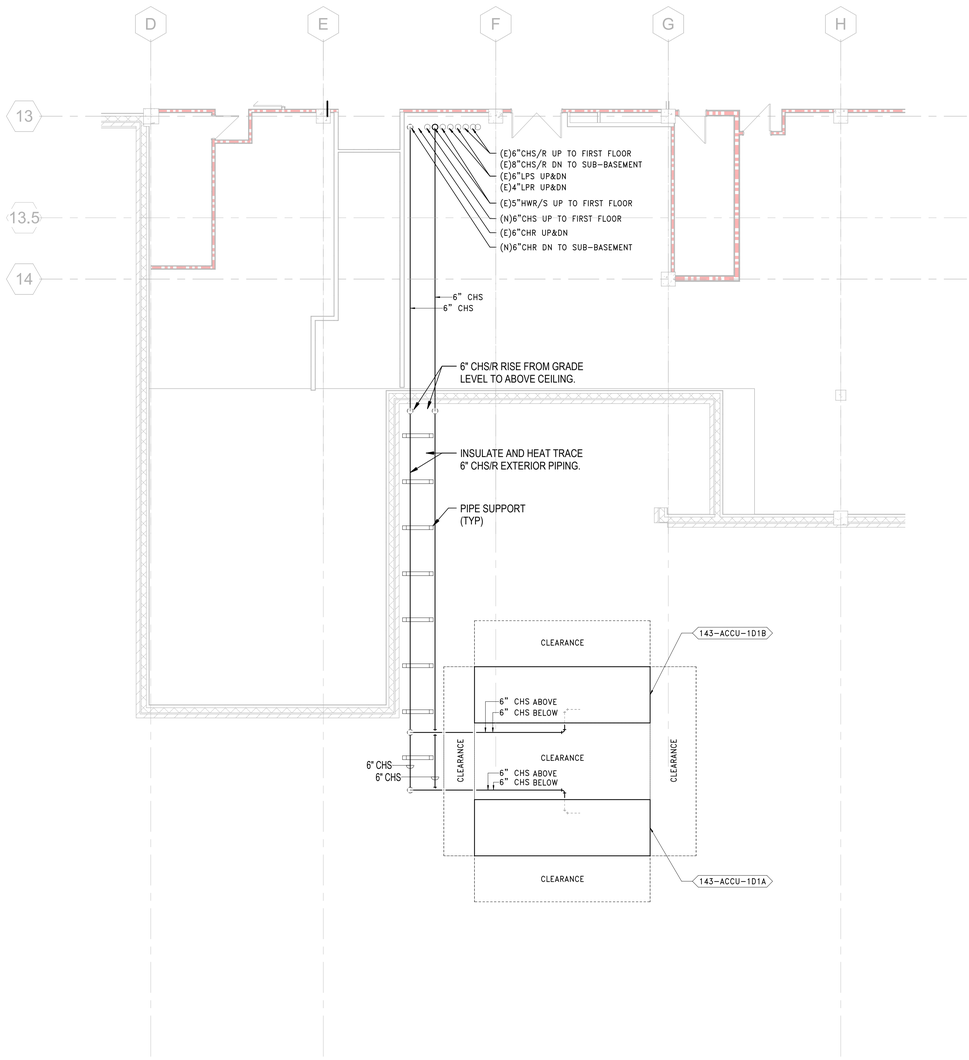
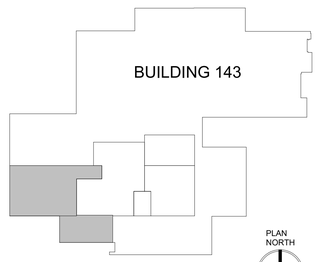
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 PIPING PLAN - BASEMENT CHILLER, PHASE 1
SCALE: 1/8" = 1'-0"

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Lic. No. 402055070
PROFESSIONAL ENGINEER
12-30-2025

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|---|---|---|---|
| **Drawing Title** MECHANICAL PIPING PLAN BASEMENT CHILLER, PHASE 1 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| | | **Project Number** 658-19-103 | **Building Number** 143 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P1-MP103 |
| | | **Issue Date** 2025-12-30 / **Checked** JAK / **Drawn** JLR | |

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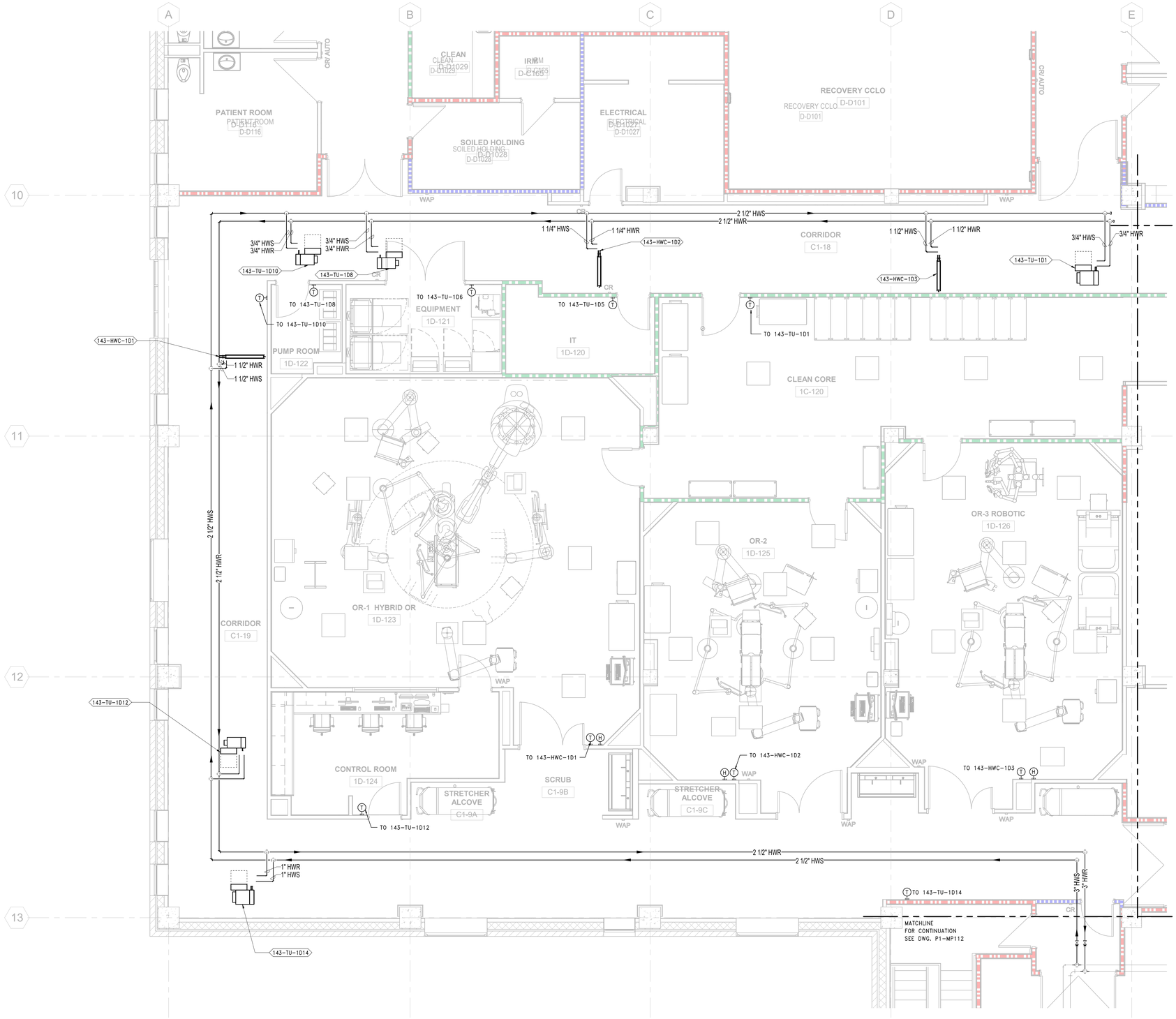
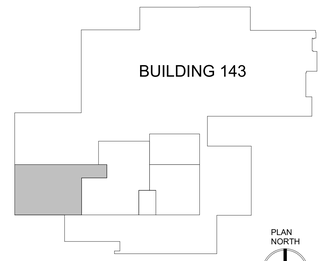



## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

## 1 PIPING PLAN - 1ST FLOOR, PHASE 1

SCALE: 1/4" = 1'-0"

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M-R PROJECT NO. 5009-0036

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL PIPING PLAN 1ST FLOOR, PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | JAK |
| Drawn | JLR |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P1-MP111 |

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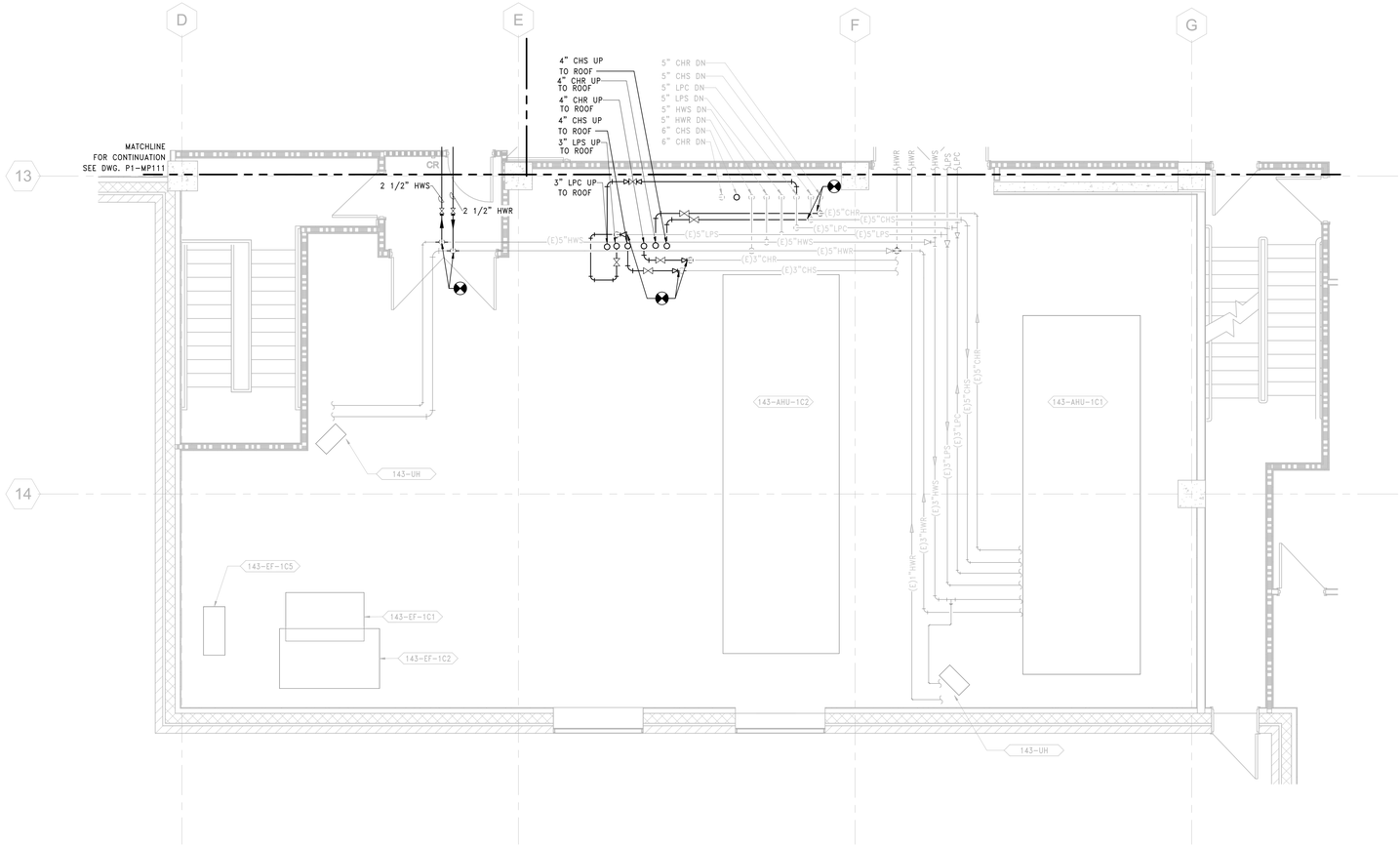
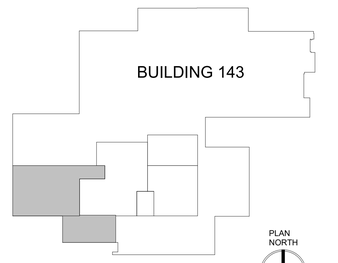
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 PIPING PLAN - MECHANICAL ROOM, PHASE 1
SCALE: 1/4" = 1'-0"




KEYPLAN

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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055070
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL PIPING PLAN MECHANICAL ROOM, PHASE 1 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| | | **Project Number** 658-19-103 | **Building Number** 143 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P1-MP112 |
| | | **Issue Date** 2025-12-30 / **Checked** JAK / **Drawn** JLR | |

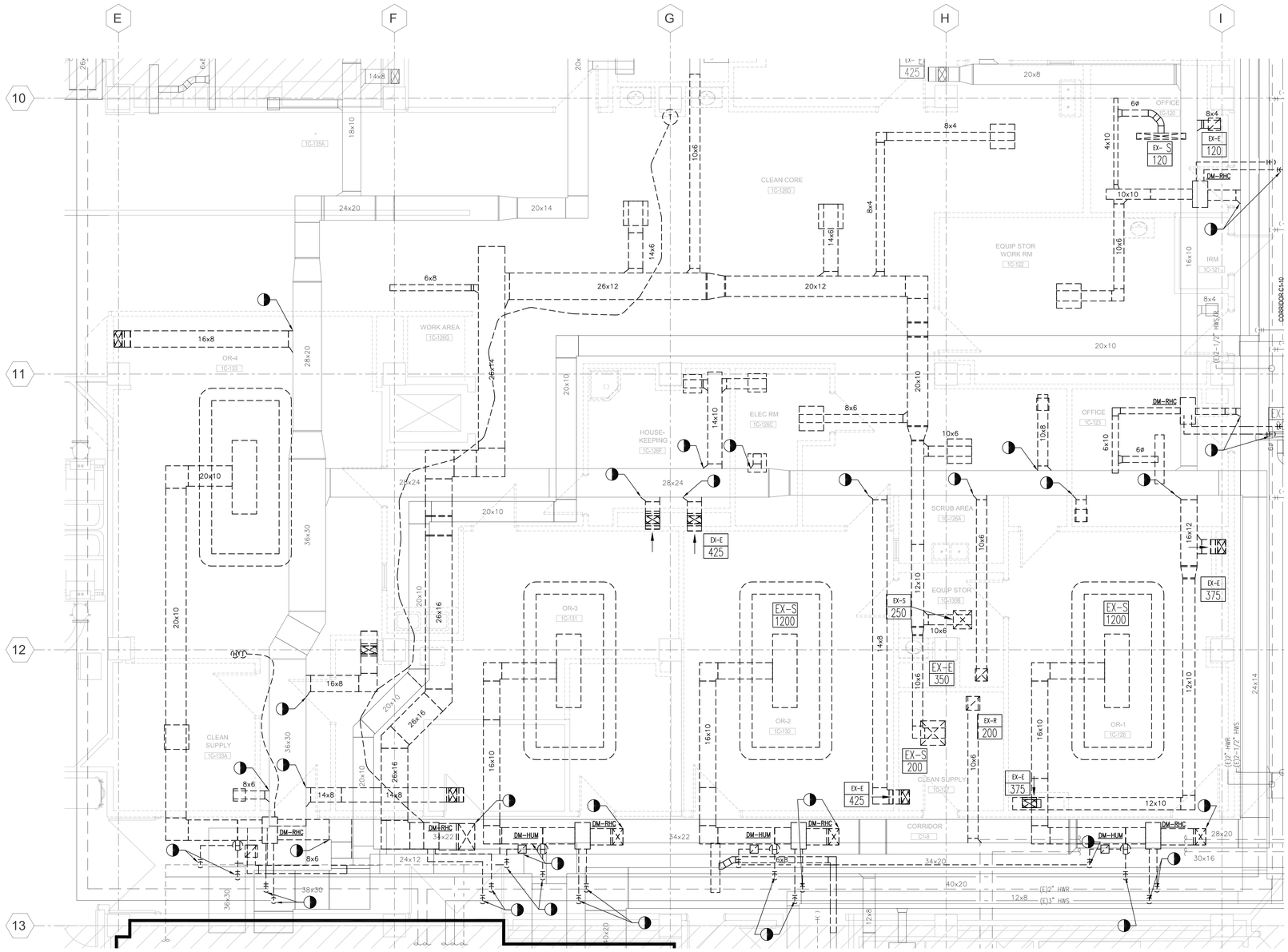
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

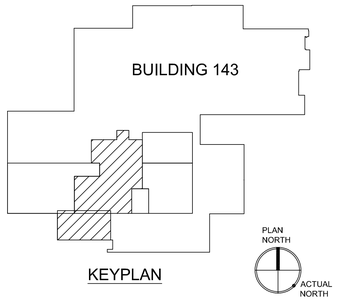
1. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

## SHEET KEYNOTES:

1. DEMOLISH EXISTING DUCT WORK, TERMINAL UNITS, CONTROLS, CONTROL WIRING, HANGERS, REGISTERS, GRILLES AND ALL APPURTENANCES.




1 MECHANICAL DEMOLITION DUCTWORK PLAN – 1ST FLOOR, PHASE 2 (SOUTH)
SCALE: 1/4" = 1'-0"

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DEMOLITION DUCTWORK PLAN 1ST FLOOR, PHASE 2 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P2-MD111 |
| | | Issue Date: 2025-12-30 / Checked: PK / Drawn: SP | |

VA FORM 08-6231

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## GENERAL SHEET NOTES:

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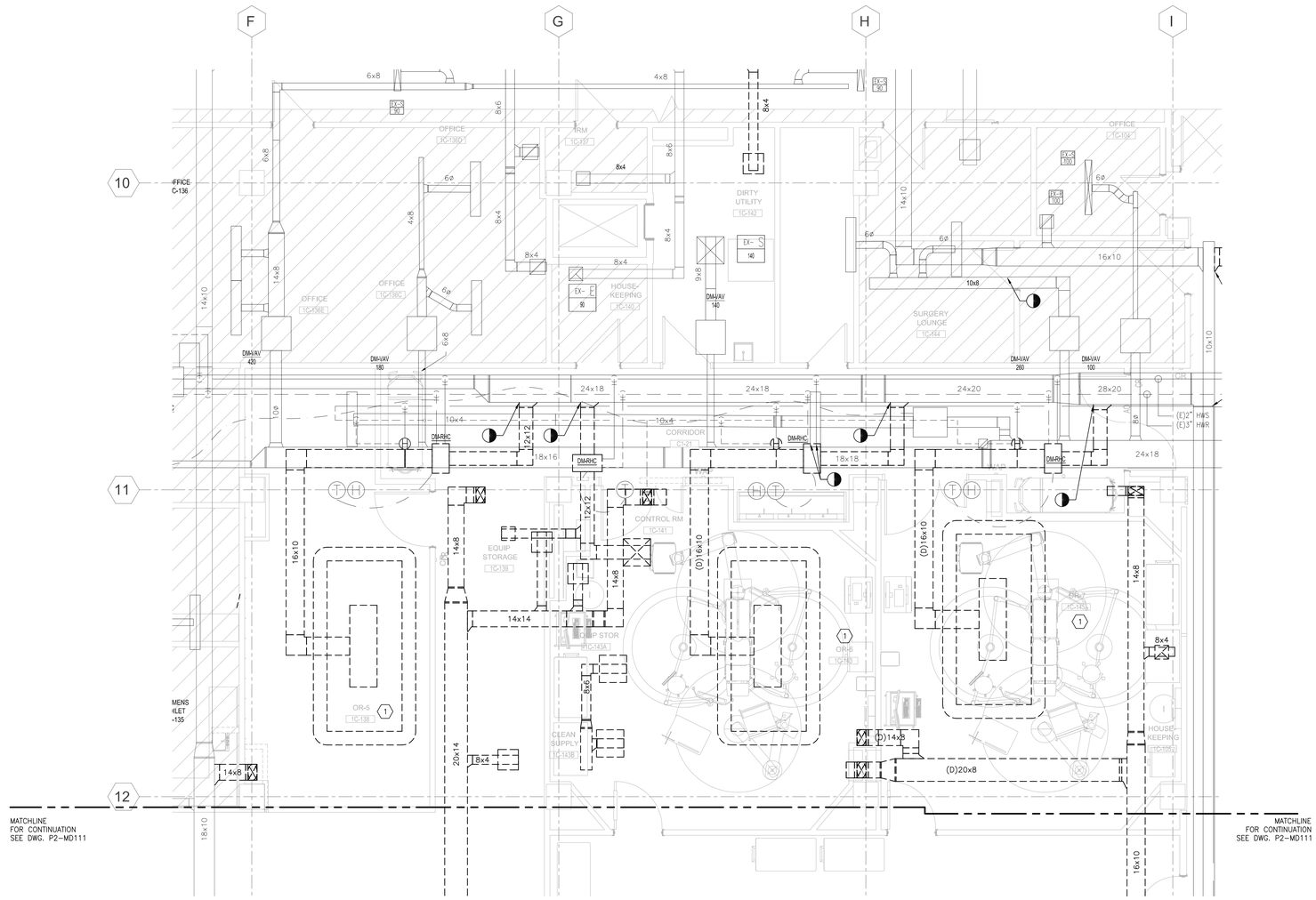
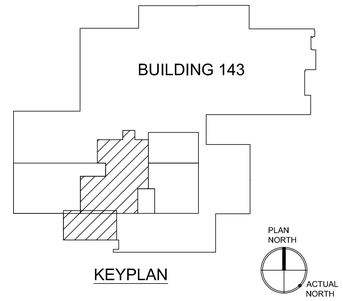



1 MECHANICAL DEMOLITION DUCTWORK PLAN – 1ST FLOOR, PHASE 2 (NORTH)
SCALE: 1/4" = 1'-0"

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**CONSULTANTS**

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SAM.GOV/CAGE: NO: 0AR60/DUNS: 079-

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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12.30.2025

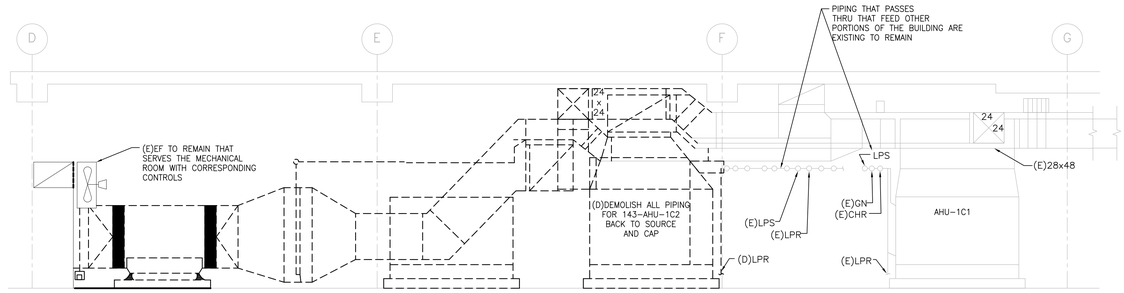
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL DEMOLITION DUCTWORK PLAN 1ST FLOOR, PHASE 2 (NORTH) | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| | | **Project Number** 658-19-103 | **Building Number** 143 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P2-MD112 |
| | | **Issue Date** 2025-12-30 / **Checked** BH / **Drawn** PNL | |

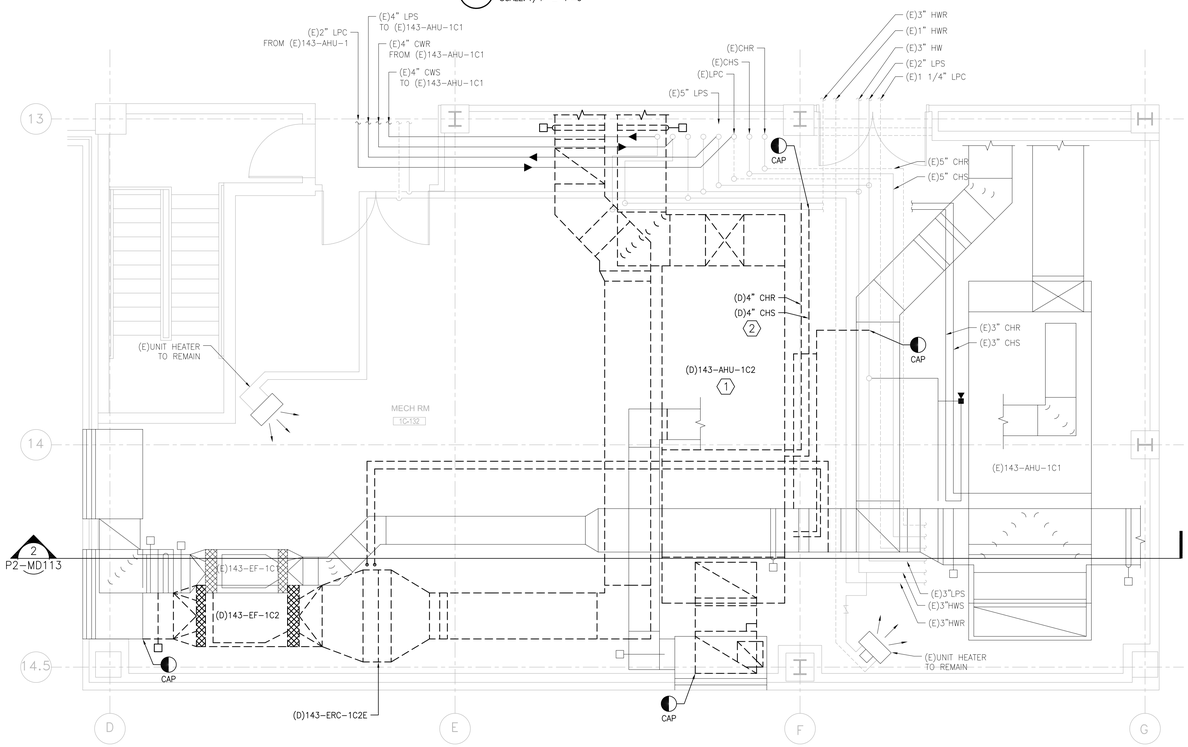
VA FORM 08-6231

three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P2-MD112.dwg 12-29-25 01:30:57 PM jromano




## 2 SECTION AT MECHANICAL ROOM 1C-132

SCALE: 1/4" = 1'-0"




## 1 MECHANICAL DEMOLITION PLAN – MECHANICAL ROOM, PHASE 2

SCALE: 1/4" = 1'-0"

## GENERAL SHEET NOTES:

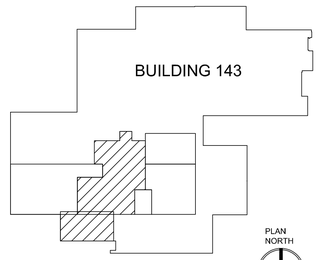
1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

## SHEET KEYNOTES:

1. DEMOLISH EXISTING AHU's, DUCT WORK, CONTROLS, CONTROL WIRING, HANGERS, SUPPORTS AND ALL APPURTENANCES.
2. DEMOLISH EXISTING PIPING, HANGERS, SUPPORTS AND ALL APPURTENANCES.

PHASING

1. PHASE II HVAC WORK: DEMOLISH 143-AHU-1C2 AND ALL ASSOCIATED DUCTWORK AND MECHANICAL PIPING THROUGHOUT THE PHASE II AREA.
2. INSTALL NEW AHU 143-AHU-1C2 AND ASSOCIATED EQUIPMENT AS SHOWN IN SHEET P2-MH113.
3. INSTALL ALL ASSOCIATED DUCTWORK IN ACCORDANCE WITH SHEET P2-MH113.
4. INSTALL ALL NEW MECHANICAL PIPING IN ACCORDANCE WITH SHEET P2-MP113.

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CONSULTANTS

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M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DEMOLITION PLAN MECHANICAL ROOM, PHASE 2 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P2-MD113 |
| | | Issue Date: 2025-12-30 / Checked: BH / Drawn: CS | |

P:\0499_OVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P2-MD113.dwg 12-29-25 01:31:40 PM jromano

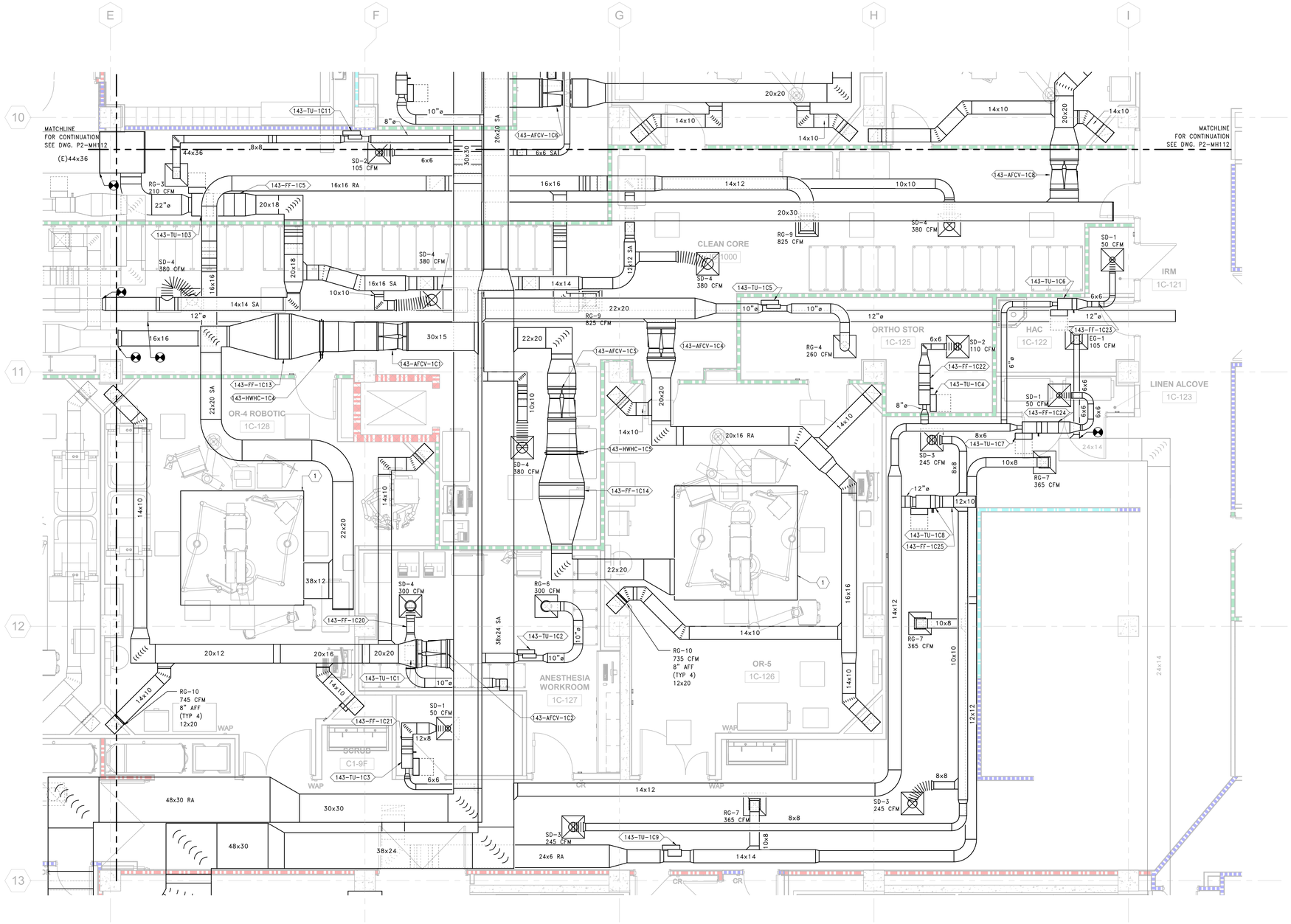
VA FORM 08-6231

## GENERAL SHEET NOTES:

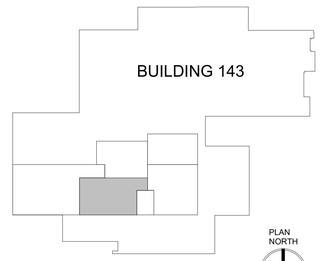
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

## SHEET KEYNOTES:

1 DIFFUSER FIELD IN CEILING. REFER TO ARCHITECTURAL DRAWINGS.




1 DUCTWORK PLAN - 1ST FLOOR, PHASE 2 (SOUTH)
SCALE: 1/4" = 1'-0"

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Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

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VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DUCTWORK PLAN 1ST FLOOR, PHASE 2 (SOUTH) |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | CS |
| Drawn | JLR |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P2-MH111 |

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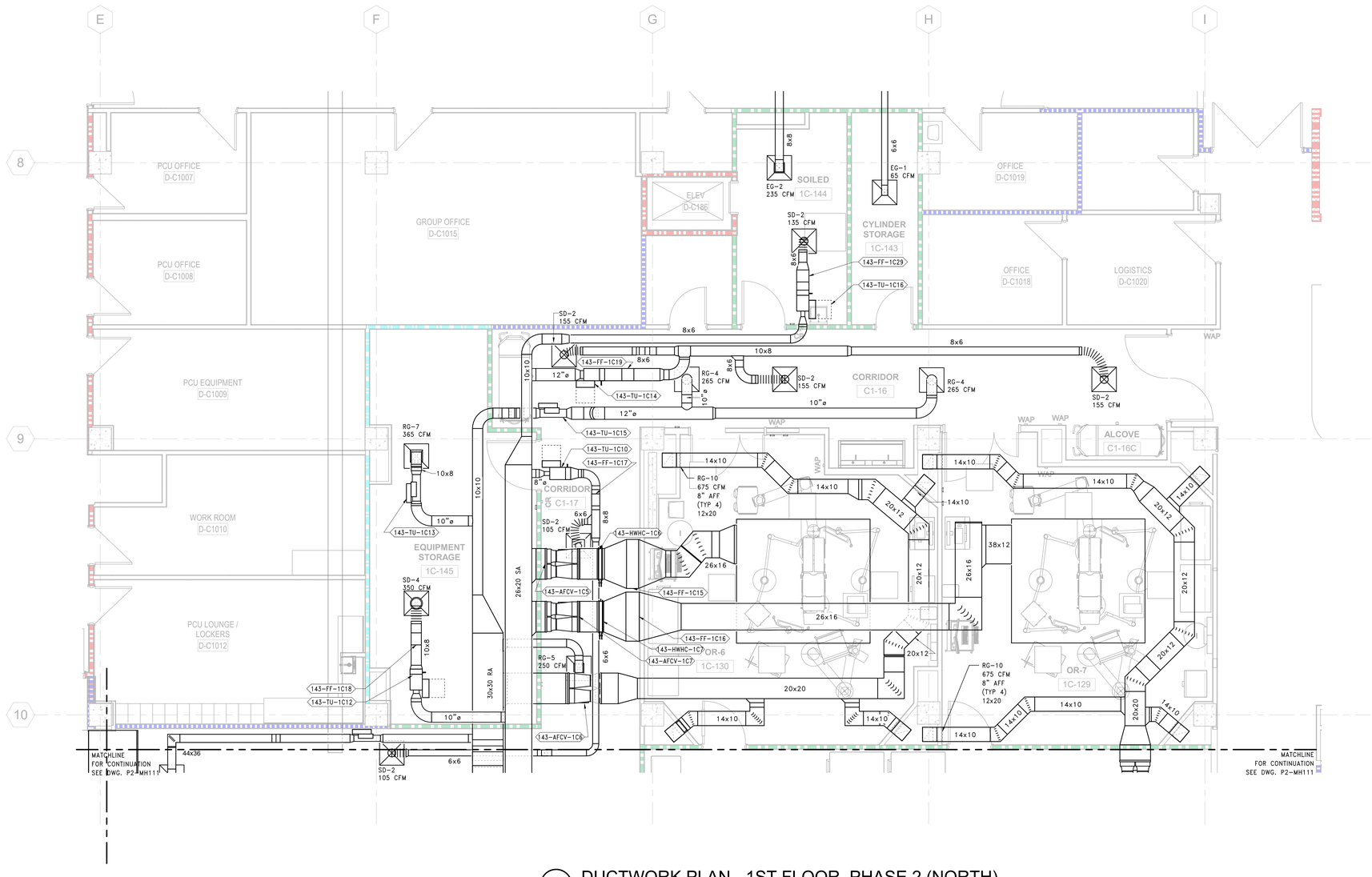
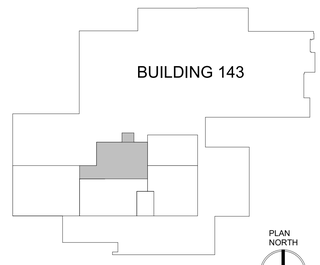
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 DUCTWORK PLAN - 1ST FLOOR, PHASE 2 (NORTH)
SCALE: 1/4" = 1'-0"

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PROFESSIONAL ENGINEER
12.30.2025

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|---|---|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN 1ST FLOOR, PHASE 2 (NORTH) | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| | | Project Number: 658-19-103 | Building Number: 143 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: P2-MH112 |
| | | Issue Date: 2025-12-30 / Checked: CS / Drawn: SP | |

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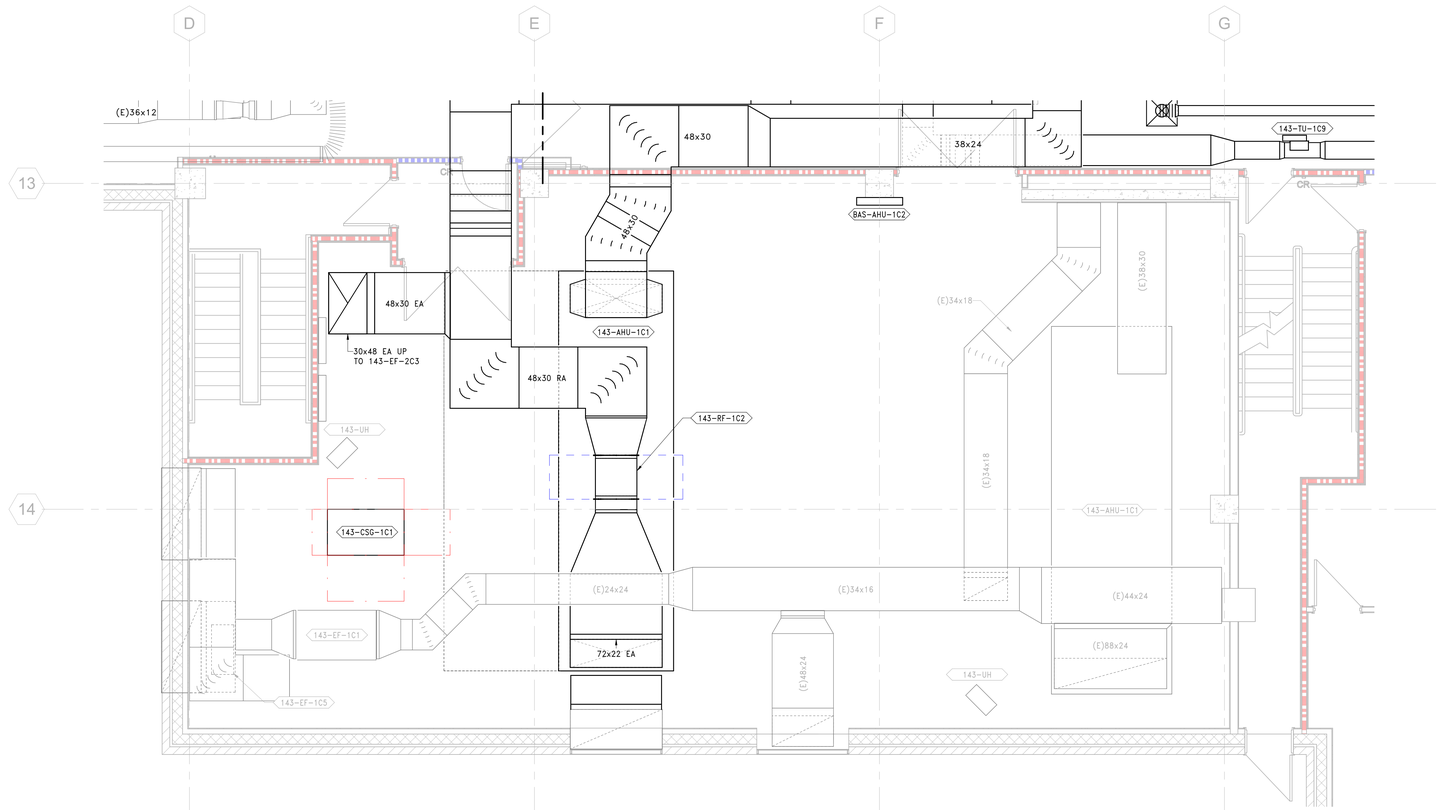
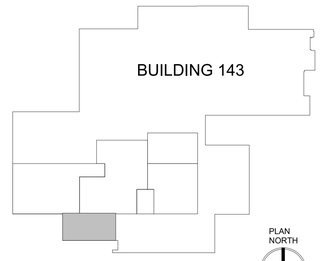
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 DUCTWORK PLAN - MECHANICAL ROOM, PHASE 2
SCALE: 1/4" = 1'-0"

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**MILLER-REMICK LLC PROFESSIONAL ENGINEER**

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ANTHONY DANIEL PACARO JR.
Lic. No. 0402055075
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DUCTWORK PLAN MECHANICAL ROOM, PHASE 2 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | CS |
| Drawn | SP |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P2-MH113 |

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## GENERAL SHEET NOTES:

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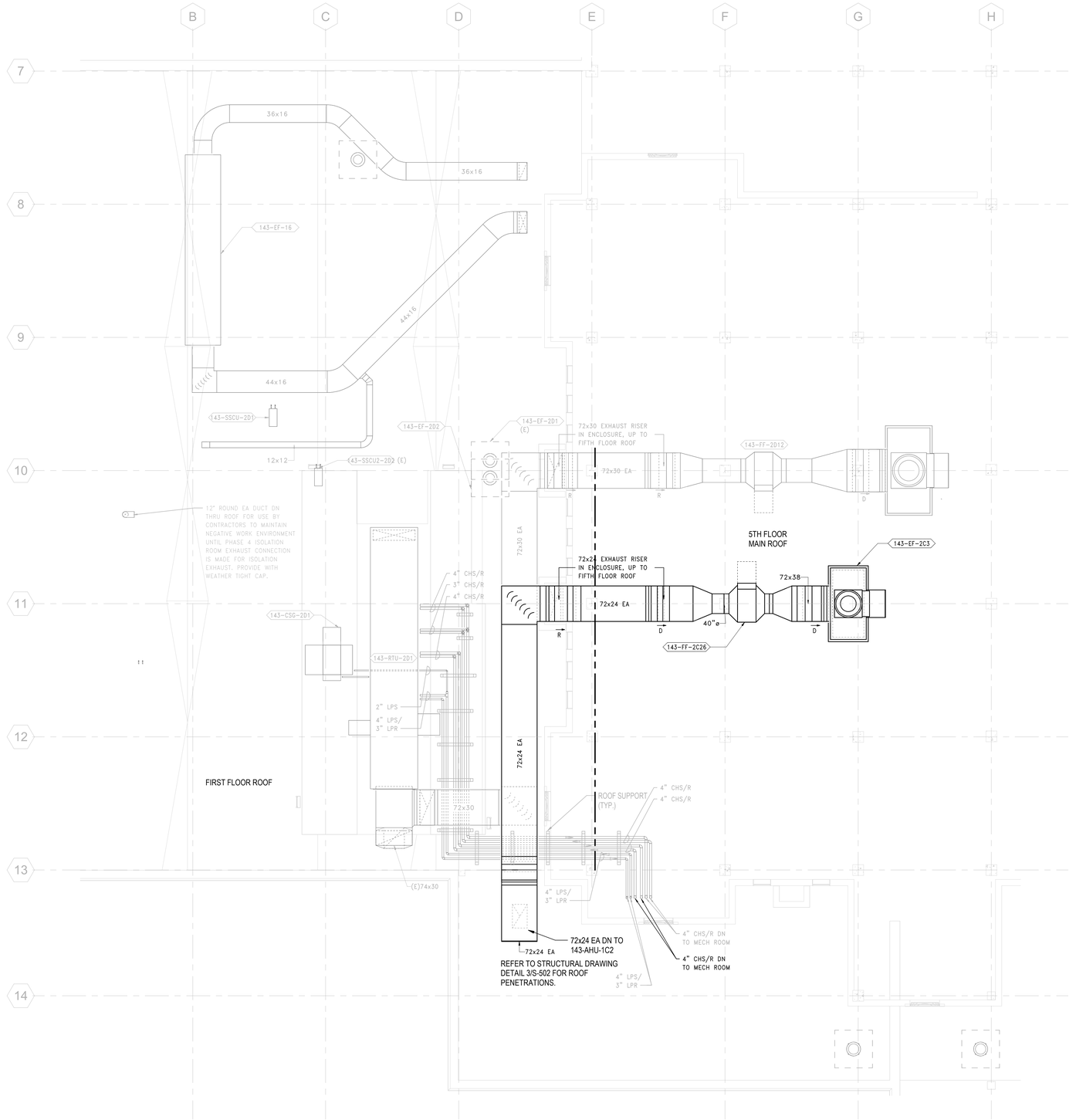
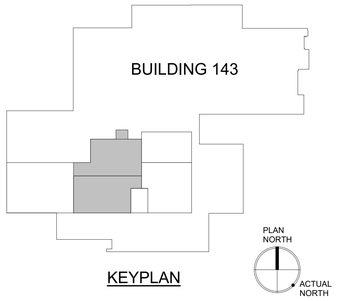



1 DUCTWORK PLAN - ROOF, PHASE 2
SCALE: 1/8" = 1'-0"

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12.30.2025

Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN ROOF, PHASE 2 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P2-MH121 |
| | | Issue Date: 2025-12-30 / Checked: JAK / Drawn: JLR | |

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## GENERAL SHEET NOTES:

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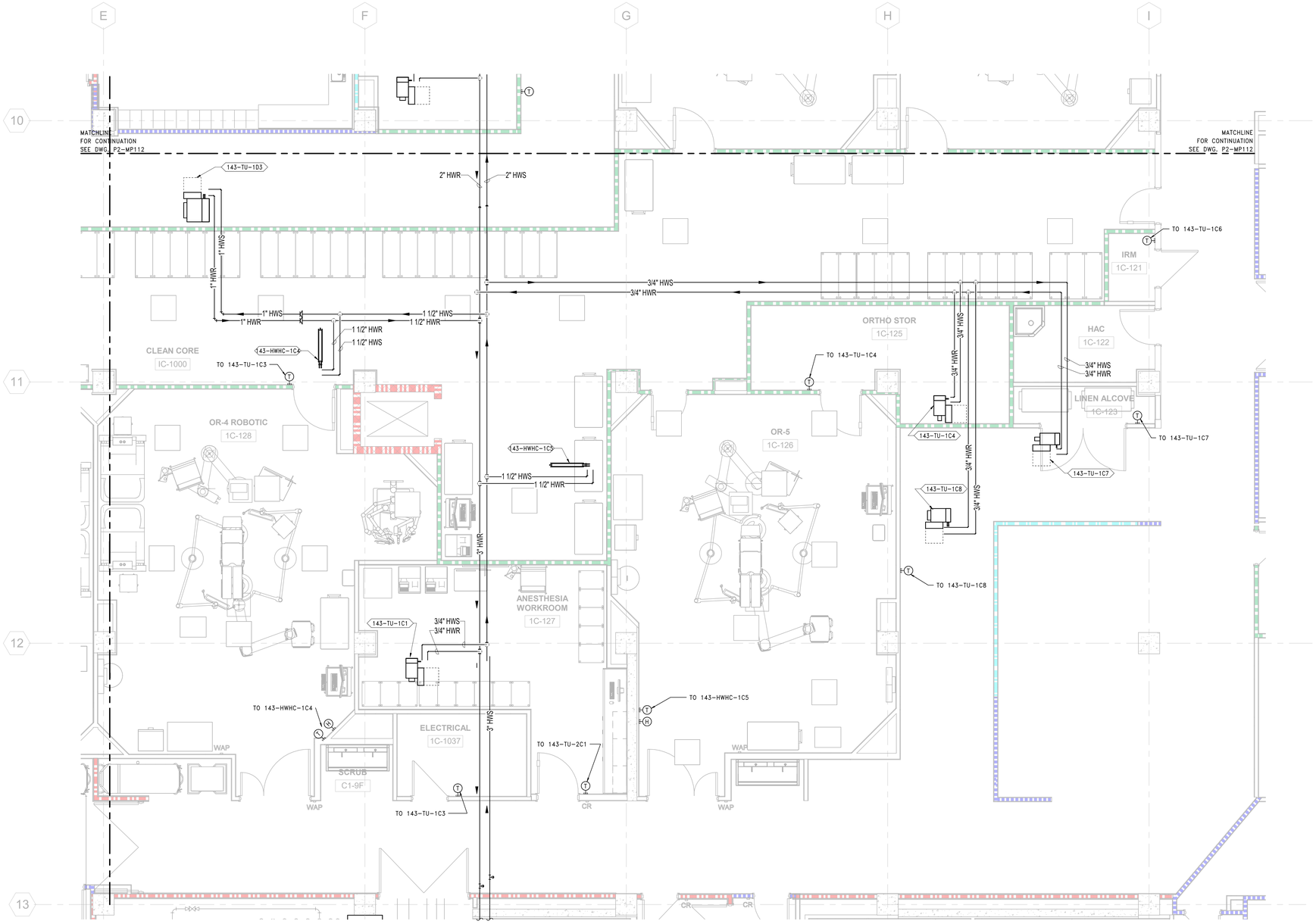
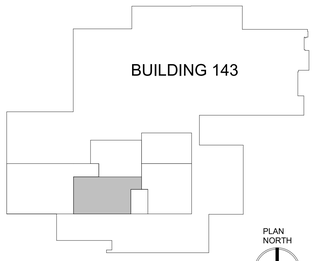



## 1 PIPING PLAN - 1ST FLOOR, PHASE 2 (SOUTH)

SCALE: 1/4" = 1'-0"

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M-R PROJECT NO. 0909-0036

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ANTHONY DANIEL PACARO JR
Lic. No. 0402055070
PROFESSIONAL ENGINEER

12-30-2025

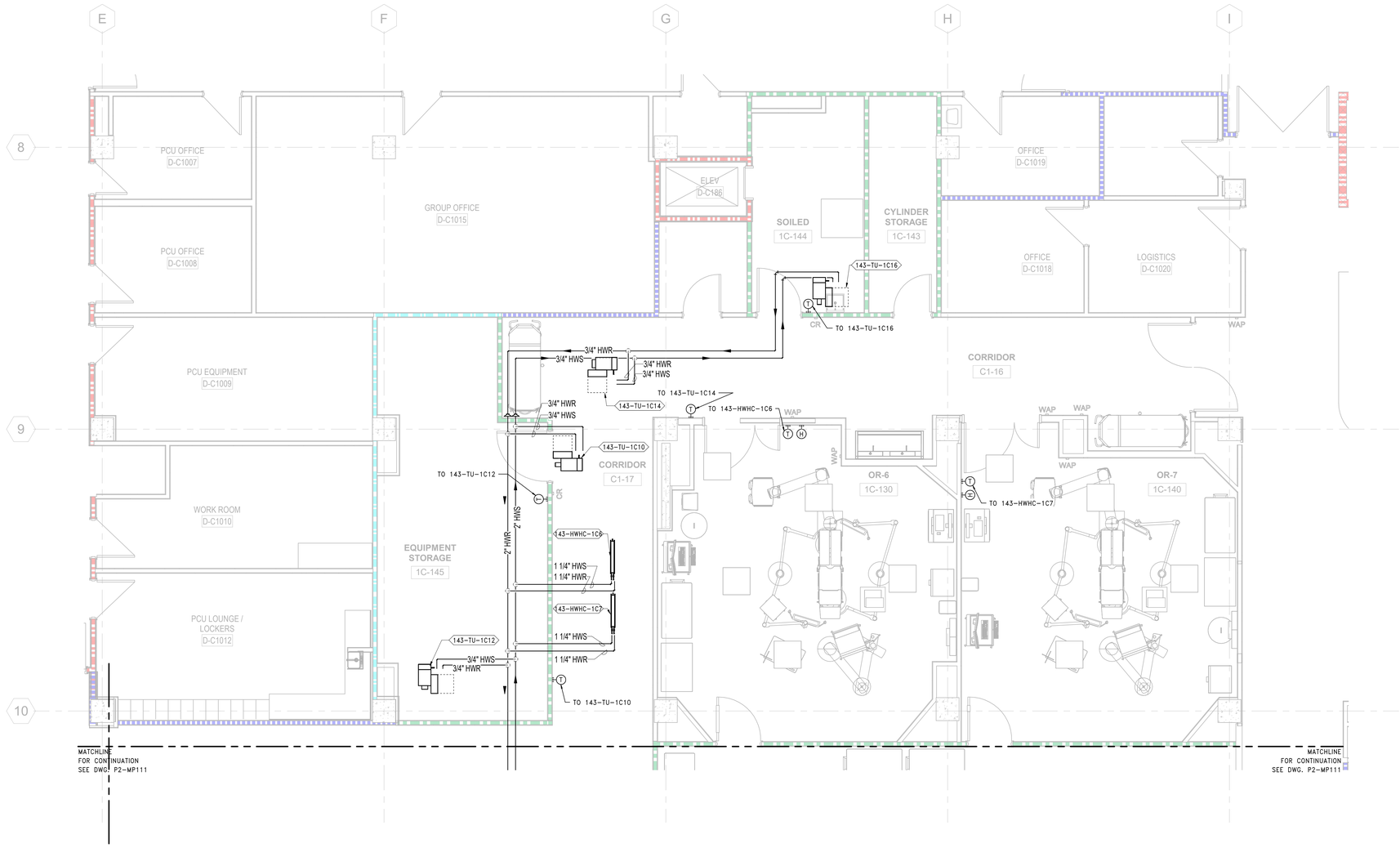
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

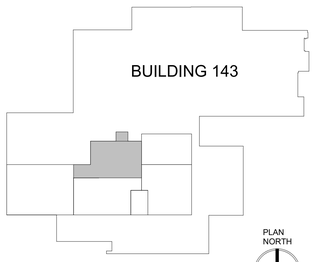
| | |
|---|---|
| Drawing Title | MECHANICAL<br>PIPING PLAN<br>1ST FLOOR, PHASE 2 (SOUTH) |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE<br>PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | JAK |
| Drawn | JLR |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P2-MP111 |

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 PIPING PLAN - 1ST FLOOR, PHASE 2 (NORTH)
SCALE: 1/4" = 1'-0"




KEYPLAN

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| Revisions: | Date: |
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CONSULTANT

| Architecture: | Civil Engineering: | Fire Protection: |
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PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL<br>PIPING PLAN<br>1ST FLOOR, PHASE 2 (NORTH) |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE<br>PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | JAK |
| Drawn | JLR |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P2-MP112 |

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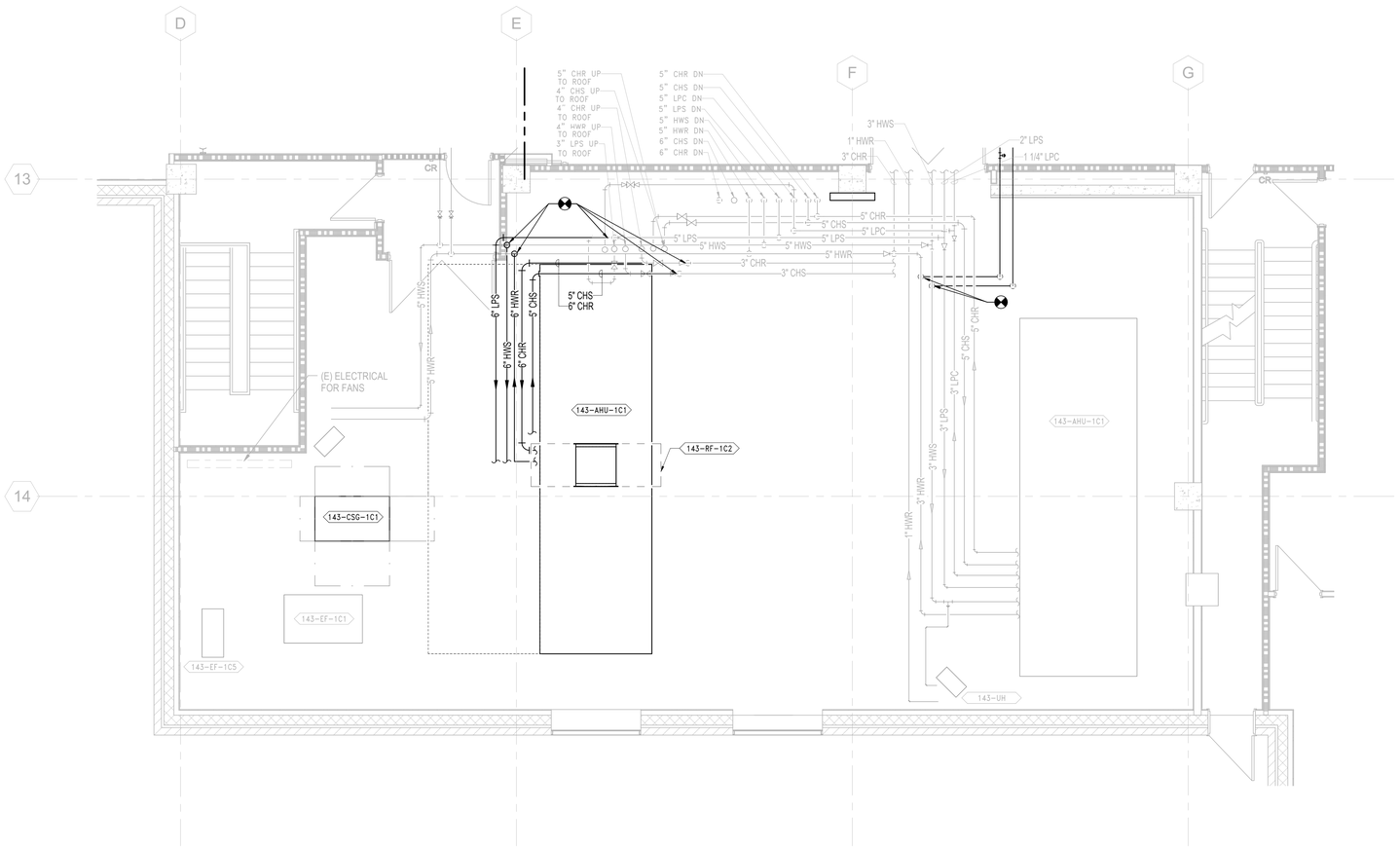
VA FORM 08 - 6231

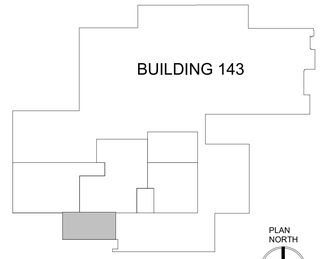
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




## 1 PIPING PLAN - MECHANICAL ROOM, PHASE 2

SCALE: 1/4" = 1'-0"

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**CONSULTANT**

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**MILLER-REMICK LLC PROFESSIONAL ENGINEER**

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL PIPING PLAN MECHANICAL ROOM, PHASE 2 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| | | **Project Number** 658-19-103 | **Building Number** 143 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P2-MP113 |
| | | **Issue Date** 2025-12-30 / **Checked** JAK / **Drawn** JLR | |

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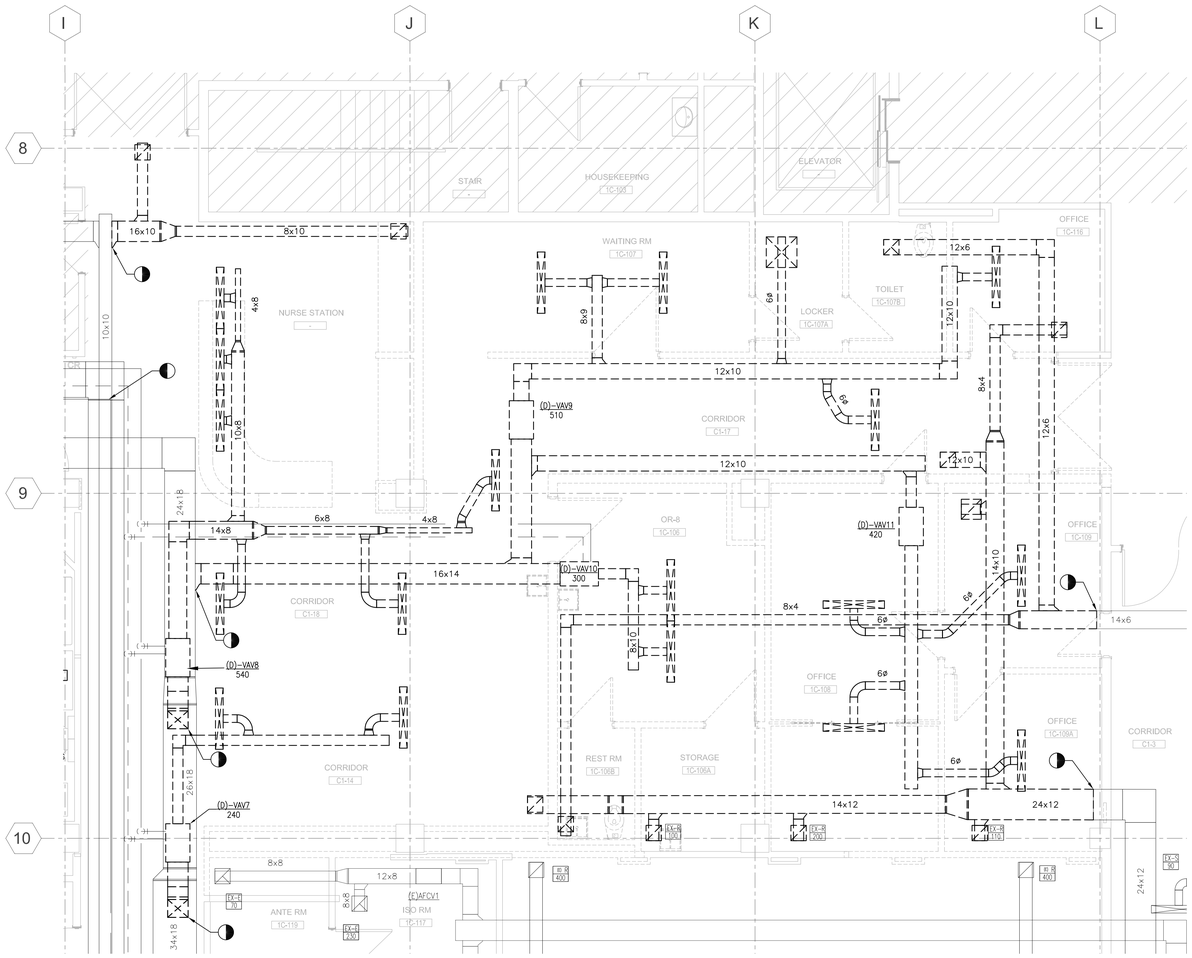
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

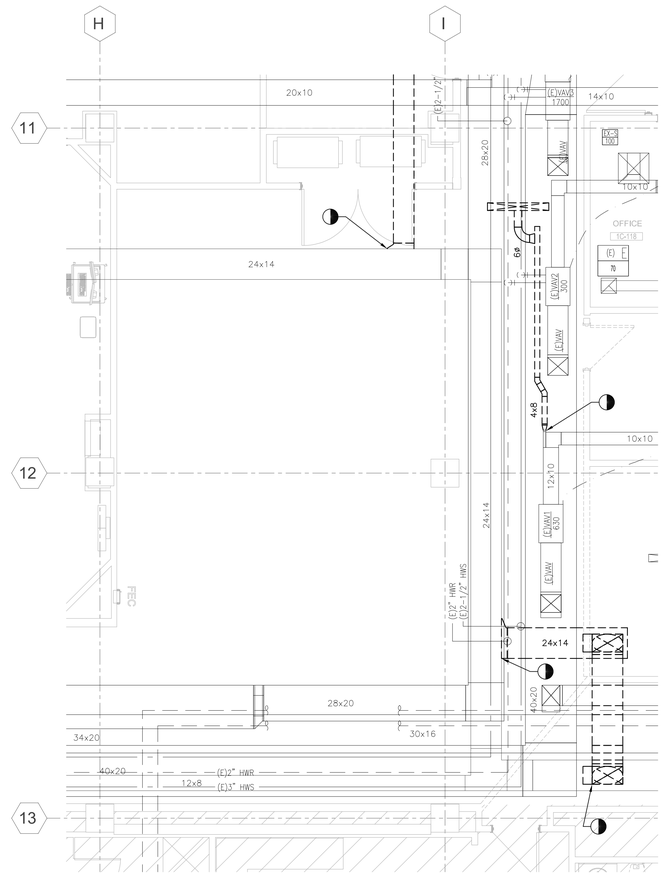
1. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.
2. PATCH OPENINGS CREATED IN REMAINING DUCTWORK THAT WERE CREATED BY DEMOLITION OF BRANCH DUCTWORK.

## SHEET KEYNOTES:

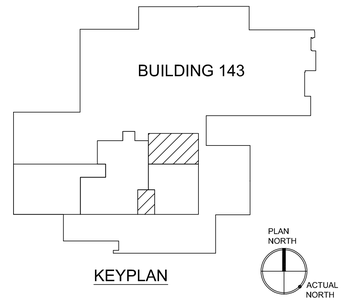
1. DEMOLISH EXISTING DUCT WORK, TERMINAL UNITS, CONTROLS, CONTROL WIRING, HANGERS, REGISTERS, GRILLES AND ALL APPURTENANCES.




1 MECHANICAL DEMOLITION DUCTWORK PLAN – 1ST FLOOR, PHASE 3, NORTH
SCALE: 1/4" = 1'-0"




2 MECHANICAL DEMOLITION DUCTWORK PLAN – 1ST FLOOR, PHASE 3, SOUTH
SCALE: 1/4" = 1'-0"

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CONSULTANTS

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MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

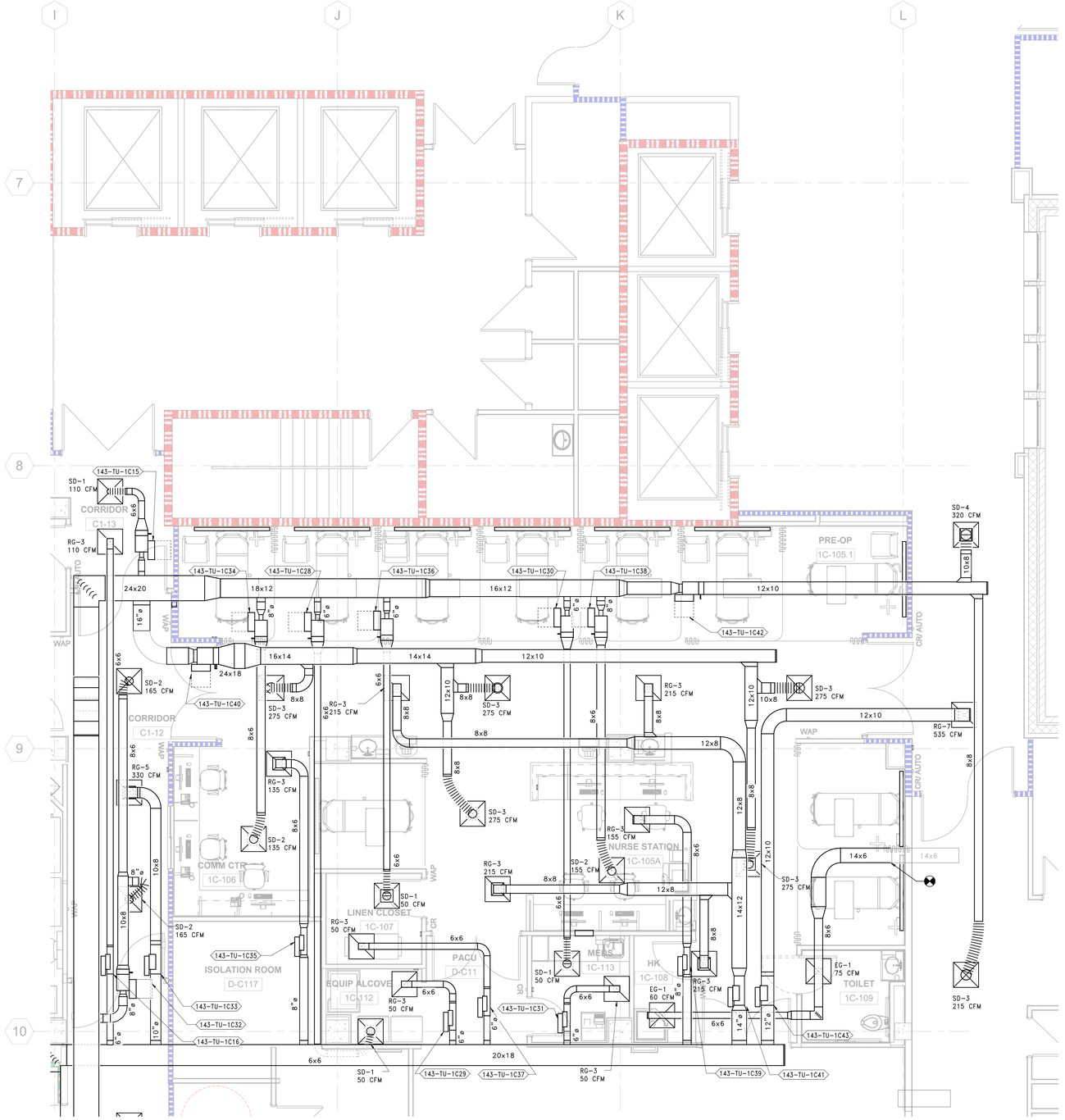
| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DEMOLITION DUCTWORK PLAN 1ST FLOOR, PHASE 3 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P3-MD111 |
| | | Issue Date: 2025-12-30 / Checked: BH / Drawn: PNL | |

VA FORM 08-6231

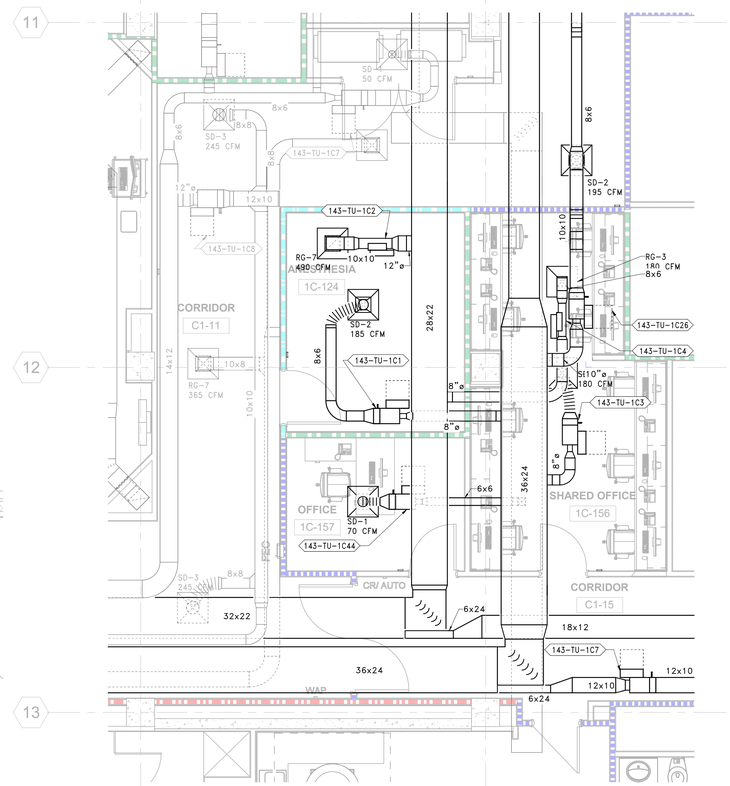
P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P3-MD111.dwg 12-29-25 01:32:58 PM jromano

## GENERAL SHEET NOTES:

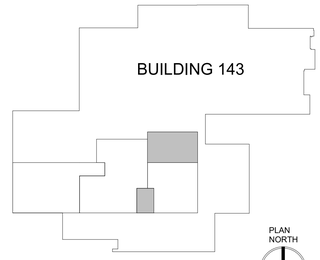
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 DUCTWORK PLAN - 1ST FLOOR, PHASE 3 (NORTH)
SCALE: 1/4" = 1'-0"




2 DUCTWORK PLAN - 1ST FLOOR, PHASE 3 (SOUTH)
SCALE: 1/4" = 1'-0"

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**CONSULTANT**

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STATE REGISTER NO. P0027919

**MILLER-REMICK LLC PROFESSIONAL ENGINEER**

COMMONWEALTH OF VIRGINIA — ANTHONY DANIEL PACARO JR. — Lic. No. 402055079 — PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN 1ST FLOOR, PHASE 3 | Phase: BID DOCUMENTS |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED |

| | |
|---|---|
| Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Project Number: 658-19-103 | Building Number: 143 |
| Location: SALEM VA MEDICAL CENTER | Drawing Number: P3-MH111 |
| Issue Date: 2025-12-30 / Checked: CS / Drawn: SP | |

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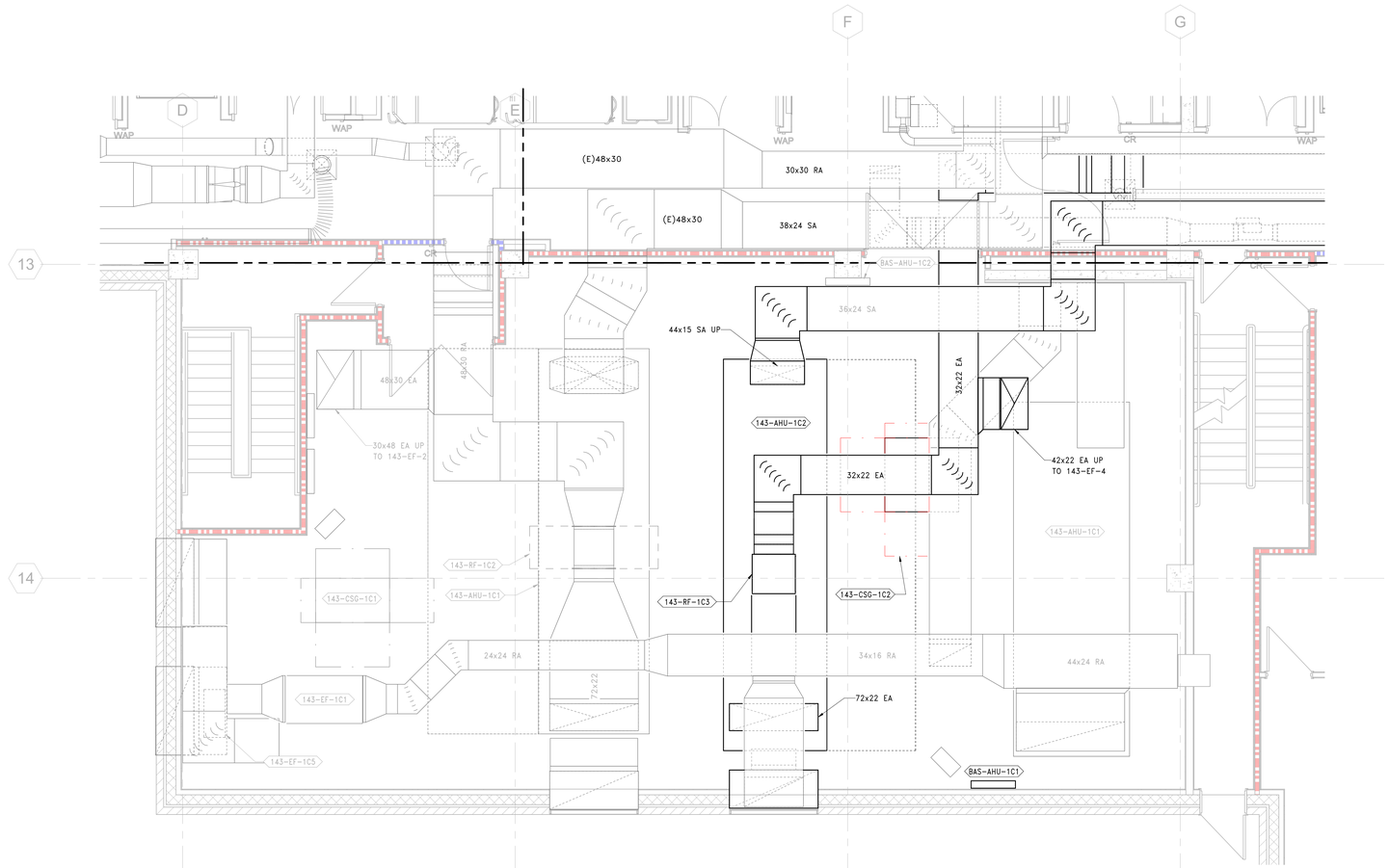
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

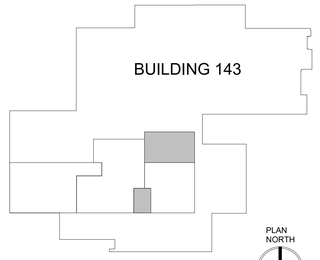
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

**PHASING:**

1. **PHASE III HVAC WORK: REPLACE PHASE III DUCTWORK.**
2. **INSTALL NEW AHU TO CONNECT TO NEW DUCTWORK.**
3. **AHU-1C1 TO RMAIN DURING THIS PHASE TO PROVIDE HVAC TO PHASE 4 AREA.**




1 DUCTWORK PLAN - MECHANICAL ROOM, PHASE 3
SCALE: 1/4" = 1'-0"

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CONSULTANT

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ENGINEER OF RECORD

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M-R PROJECT NO. 0499-0036

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 0402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL DUCTWORK PLAN MECHANICAL ROOM, PHASE 3 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | CS |
| Drawn | SP |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P3-MH112 |

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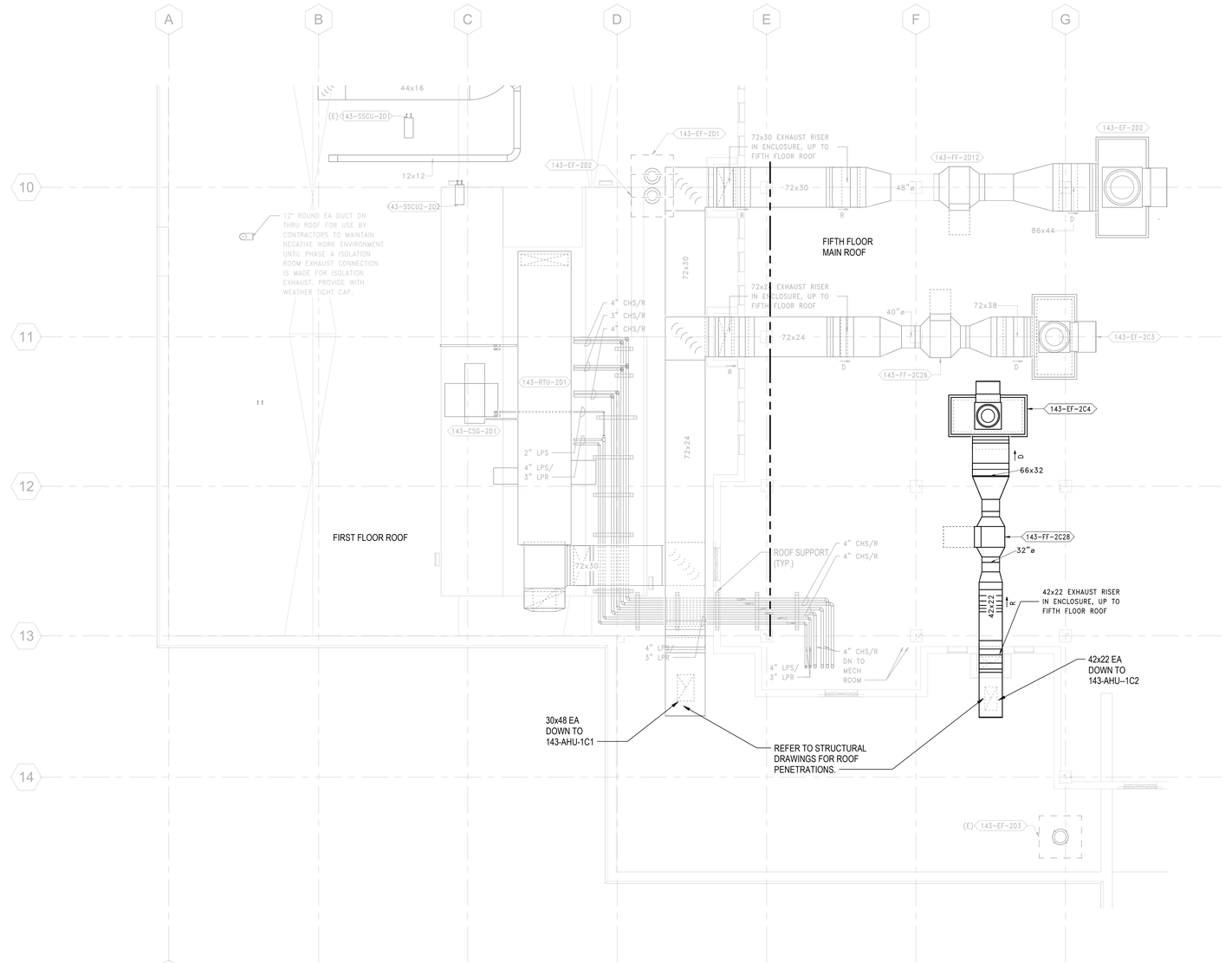
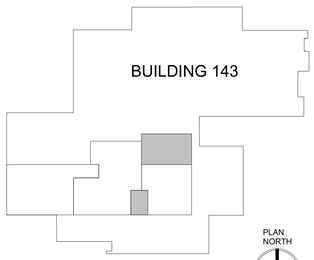
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 DUCTWORK PLAN - ROOF, PHASE 3
SCALE: 1/8" = 1'-0"

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MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN ROOF, PHASE 3 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P3-MH121 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

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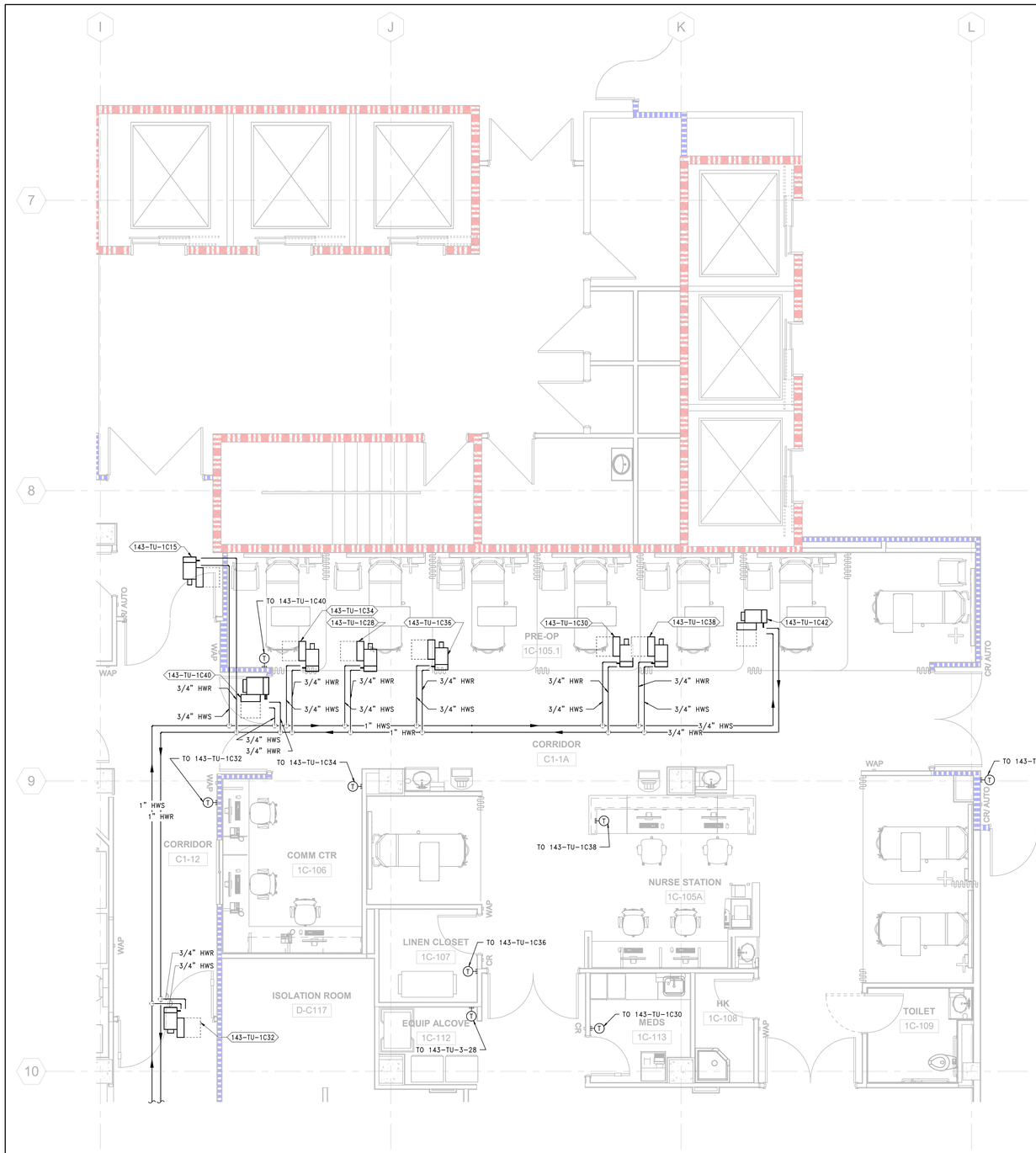
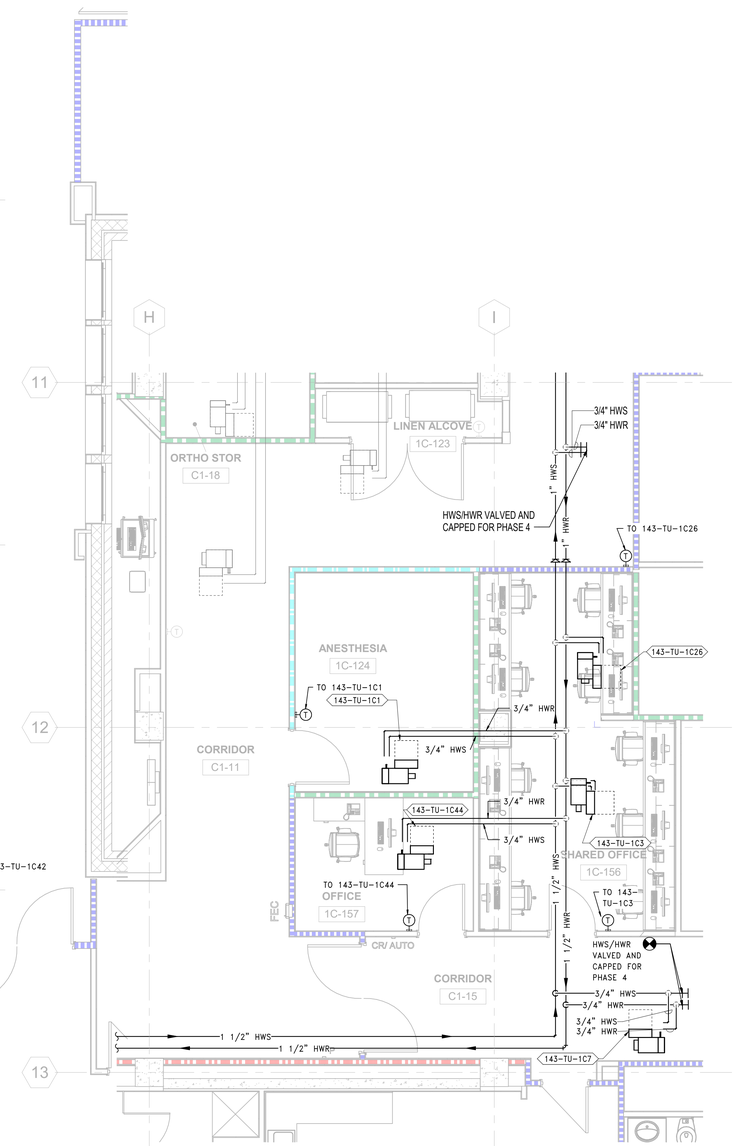
VA FORM 08 - 6231

## GENERAL SHEET NOTES:

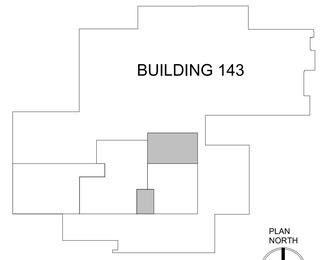
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 PIPING PLAN - 1ST FLOOR, PHASE 3, (NORTH)
SCALE: 1/4" = 1'-0"




2 PIPING PLAN - 1ST FLOOR, PHASE 3 (SOUTH)
SCALE: 1/4" = 1'-0"

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ANTHONY DANIEL PACARDO JR
Lic. No. 0402055074
PROFESSIONAL ENGINEER

12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

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|---|---|
| Drawing Title | MECHANICAL PIPING PLAN 1ST FLOOR, PHASE 3 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | CS |
| Drawn | SP |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | P3-MP111 |

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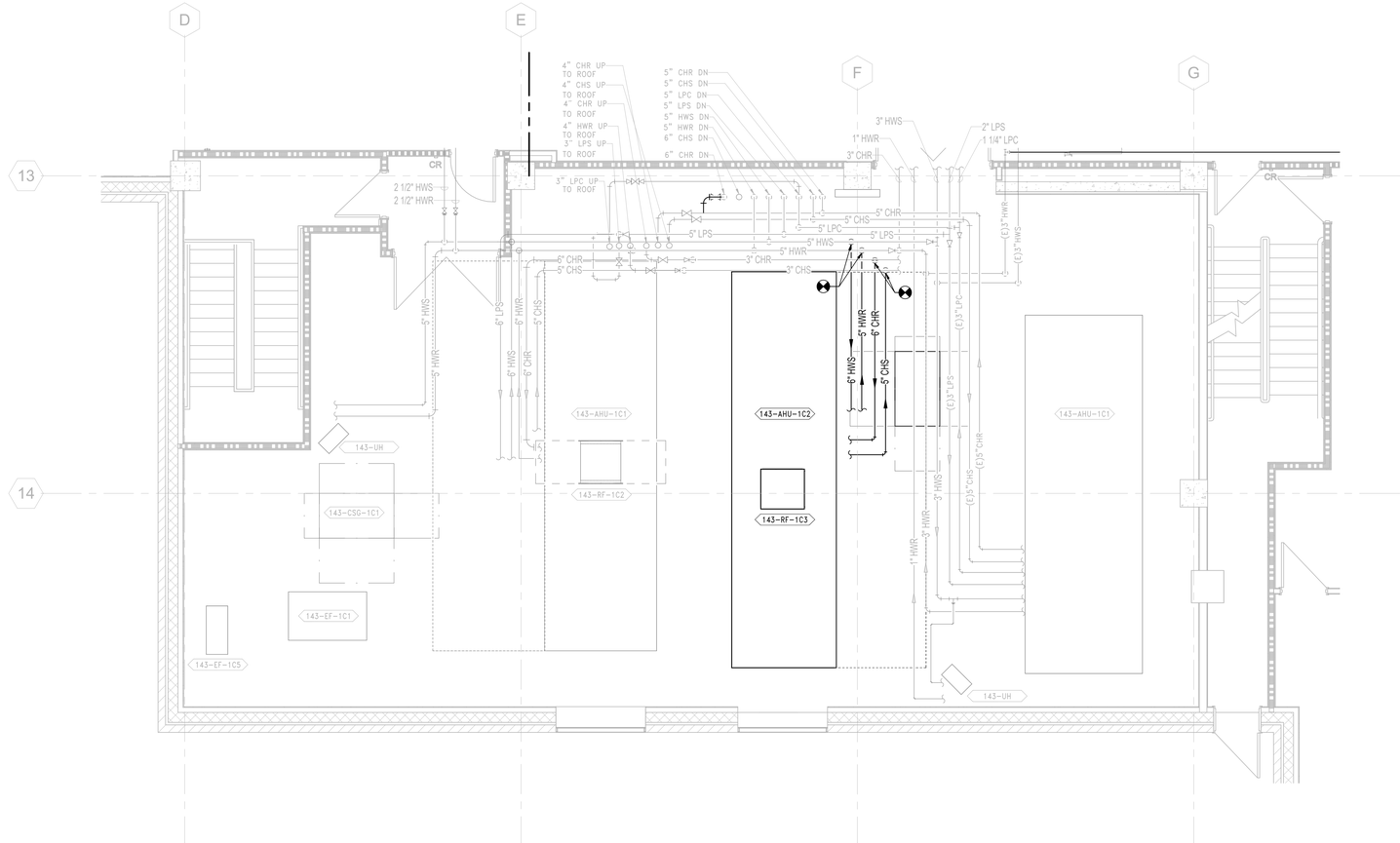
## GENERAL SHEET NOTES:

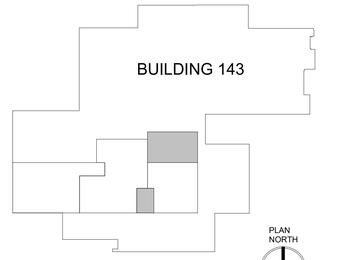
1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.

## ⬡ SHEET KEYNOTES:




## 1 PIPING PLAN - MECHANICAL ROOM, PHASE 3

SCALE: 1/4" = 1'-0"




KEYPLAN

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**CONSULTANT**

| Architecture: | Civil Engineering: | Fire Protection: |
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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| **Drawing Title** | MECHANICAL<br>PIPING PLAN<br>MECHANICAL ROOM, PHASE 3 |
| **Approved by:** | Jon Manninen, COR |
| **Phase** | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| **Project Title** | RENOVATE SURGICAL SUITE<br>PHASE II |
| **Project Number** | 658-19-103 |
| **Location** | SALEM VA MEDICAL CENTER |
| **Issue Date** | 2025-12-30 |
| **Checked** | CS |
| **Drawn** | SP |
| **VA Drawing Number** | 2617 |
| **Building Number** | 143 |
| **Drawing Number** | P3-MP112 |

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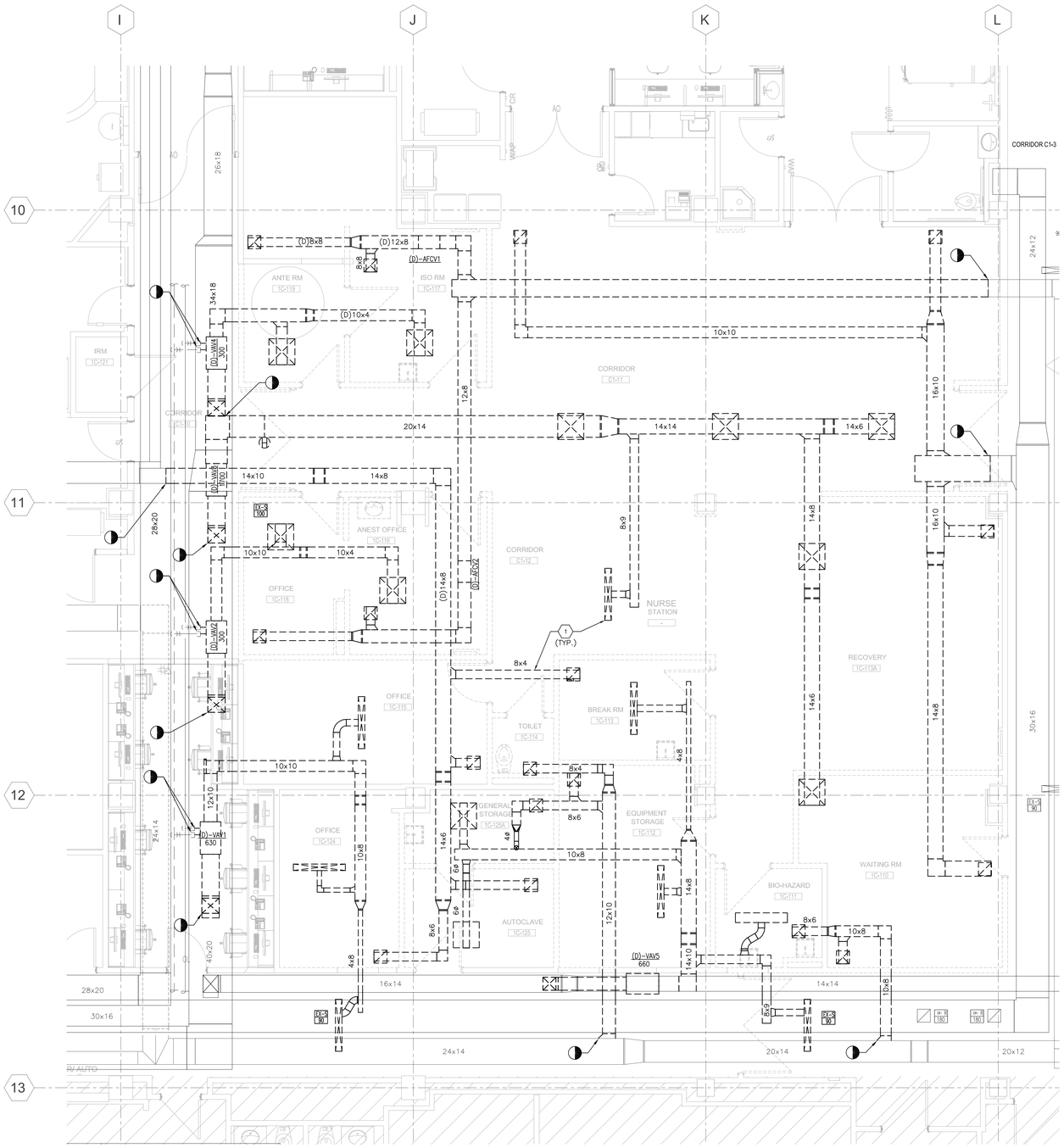
## GENERAL SHEET NOTES:

1. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

## SHEET KEYNOTES:

1. DEMOLISH EXISTING DUCT WORK, TERMINAL UNITS, CONTROLS, CONTROL WIRING, HANGERS, REGISTERS, GRILLES AND ALL APPURTENANCES.



## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. THE CONTRACTOR MAY BE REQUIRED TO WORK ACROSS THE CONSTRUCTION BOUNDARY LINE TO COMPLETE NEW WORK OR TERMINATE EXISTING. THE CONTRACTOR IS REQUIRED TO ADHERE TO ALL ICRA REQUIREMENTS.

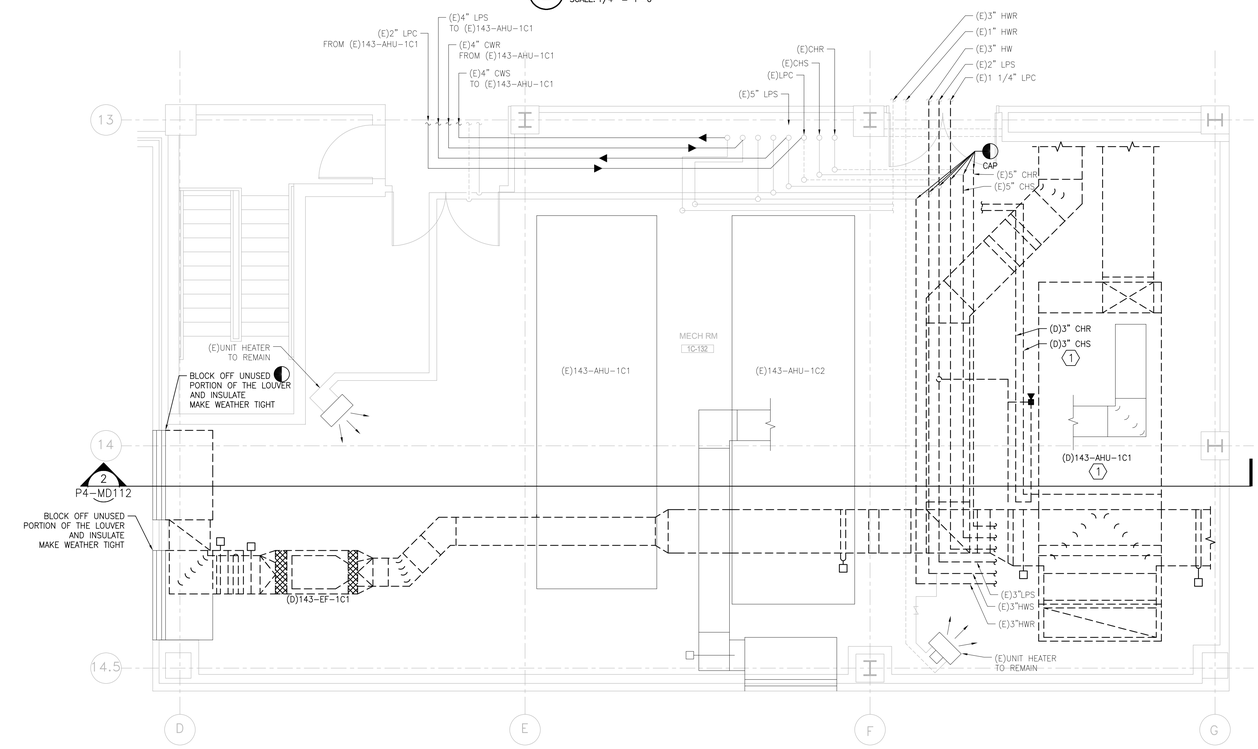
## SHEET KEYNOTES:

1. DEMOLISH EXISTING AHU's, DUCT WORK, CONTROLS, CONTROL WIRING, HANGERS, SUPPORTS AND ALL APPURTENANCES.
2. DEMOLISH EXISTING PIPING, HANGERS, SUPPORTS AND ALL APPURTENANCES.

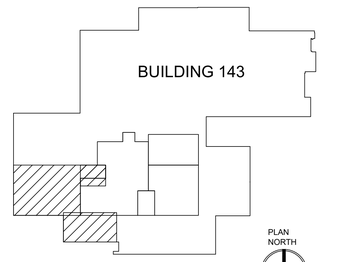
PHASING

1. PHASE IV HVAC WORK: DEMOLISH REMAINING UNIT AND DUCTWORK IN PHASE IV AREA.
2. CONNECT NEW DUCTWORK IN PHASE IV AREA TO NEW AHU-1C2.

2 SECTION AT MECHANICAL ROOM 1C-132
SCALE: 1/4" = 1'-0"




1 MECHANICAL DEMOLITION PLAN – MECHANICAL ROOM, PHASE 4
SCALE: 1/4" = 1'-0"

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| NO. | DESCRIPTION | DATE |
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**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

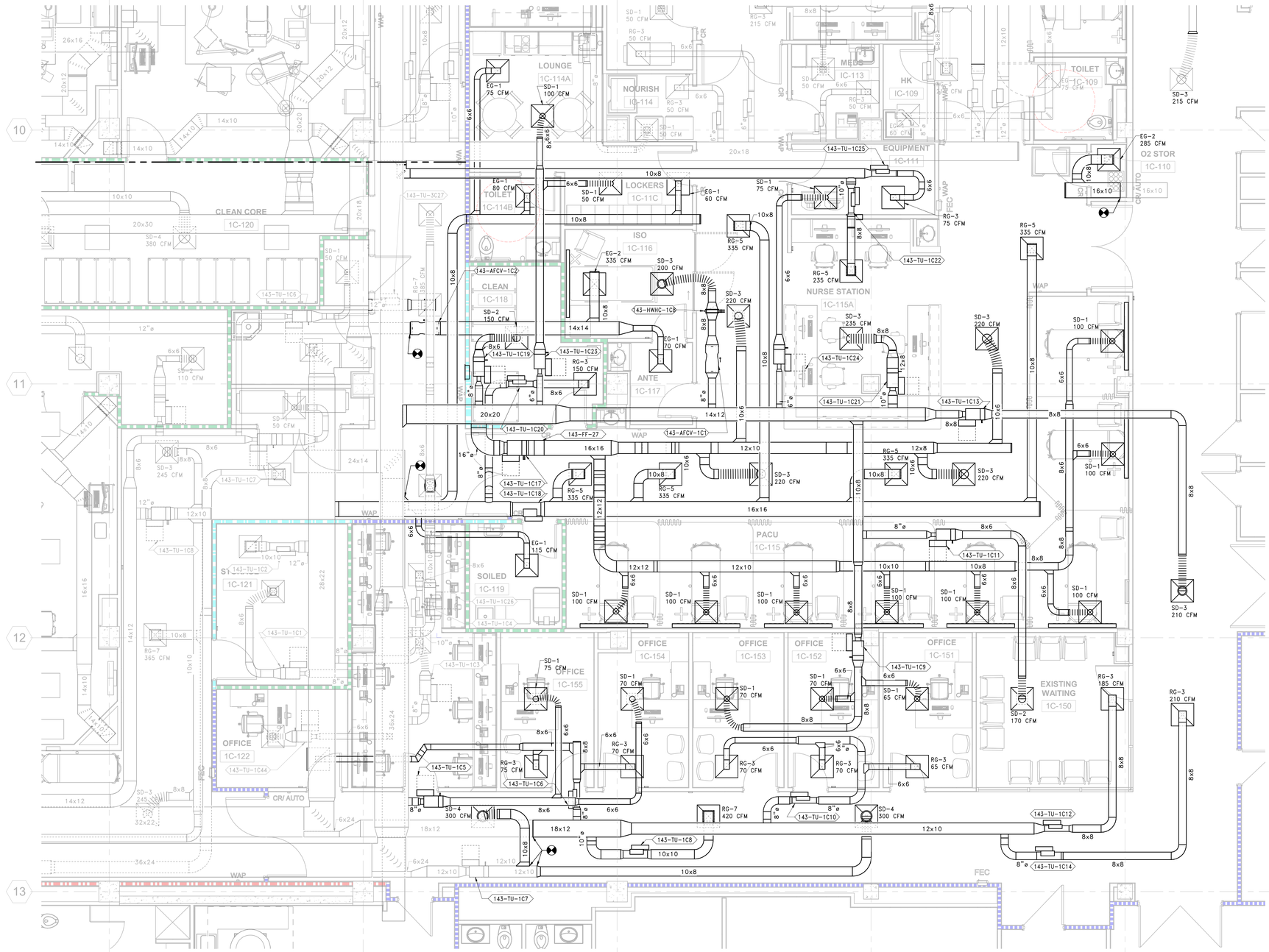
| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL DEMOLITION PLAN MECHANICAL ROOM, PHASE 4 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| | | **Project Number** 658-19-103 | **Building Number** 143 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** P4-MD112 |
| | | **Issue Date** 2025-12-30 / **Checked** BH / **Drawn** CS | |

VA FORM 08-6231

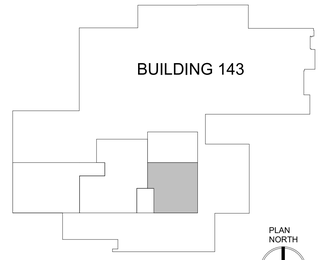
three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
P:\0499_0VA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\P4-MD112.dwg 12-29-25 04:05:04 PM JAlopsis

## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




1 DUCTWORK PLAN - 1ST FLOOR, PHASE 4
SCALE: 1/4" = 1'-0"




KEYPLAN

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12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN 1ST FLOOR, PHASE 4 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: P4-MH111 |
| | | Issue Date: 2025-12-30 / Checked: CS / Drawn: SP | |

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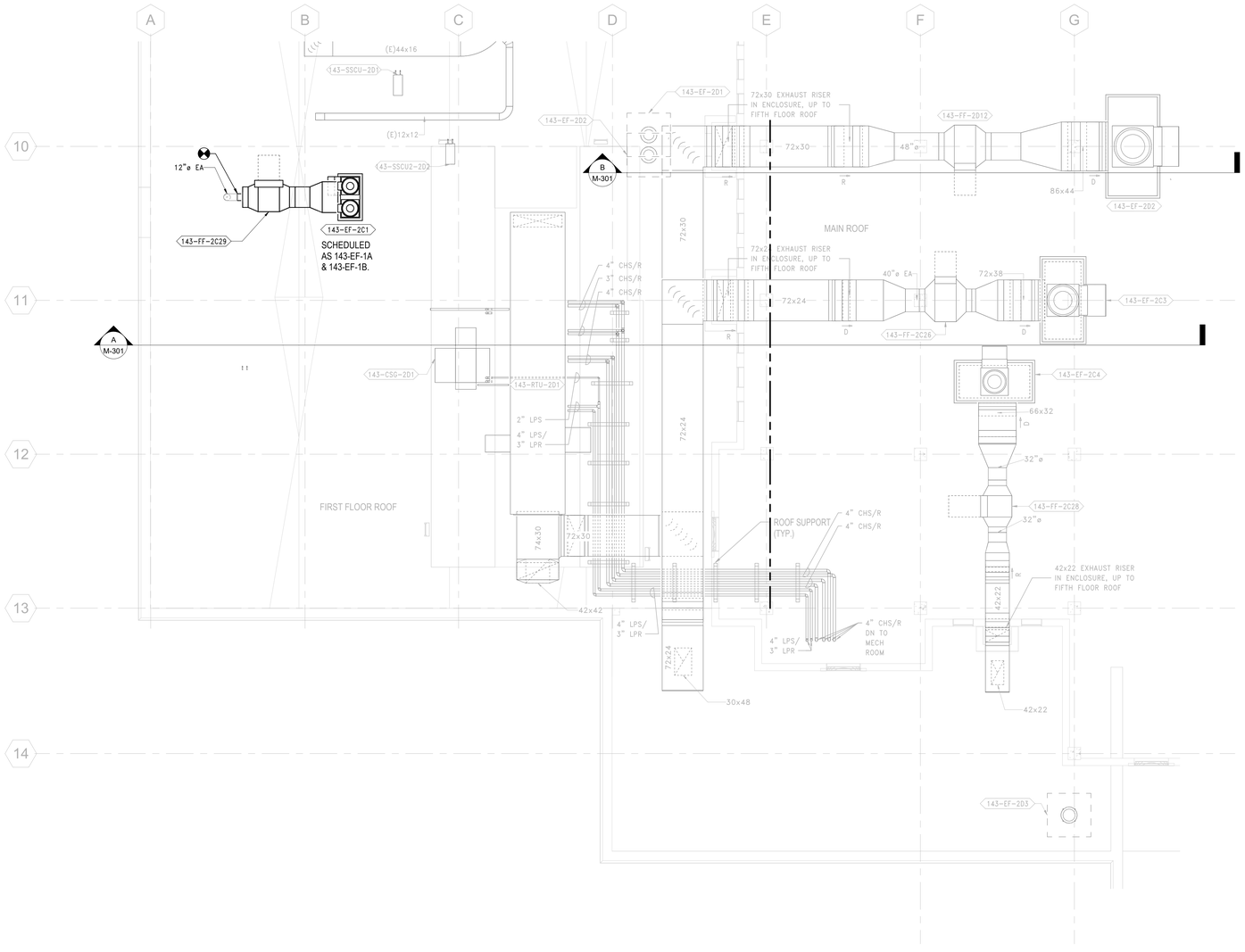
VA FORM 08 - 6231

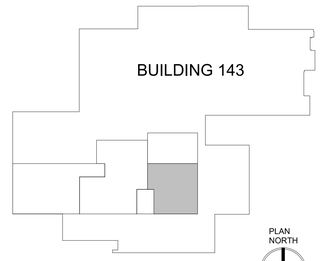
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




## 1 DUCTWORK PLAN - ROOF, PHASE 4

SCALE: 1/8" = 1'-0"

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| | |
|---|---|
| Drawing Title: MECHANICAL DUCTWORK PLAN ROOF, PHASE 4 | Phase: BID DOCUMENTS |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED |

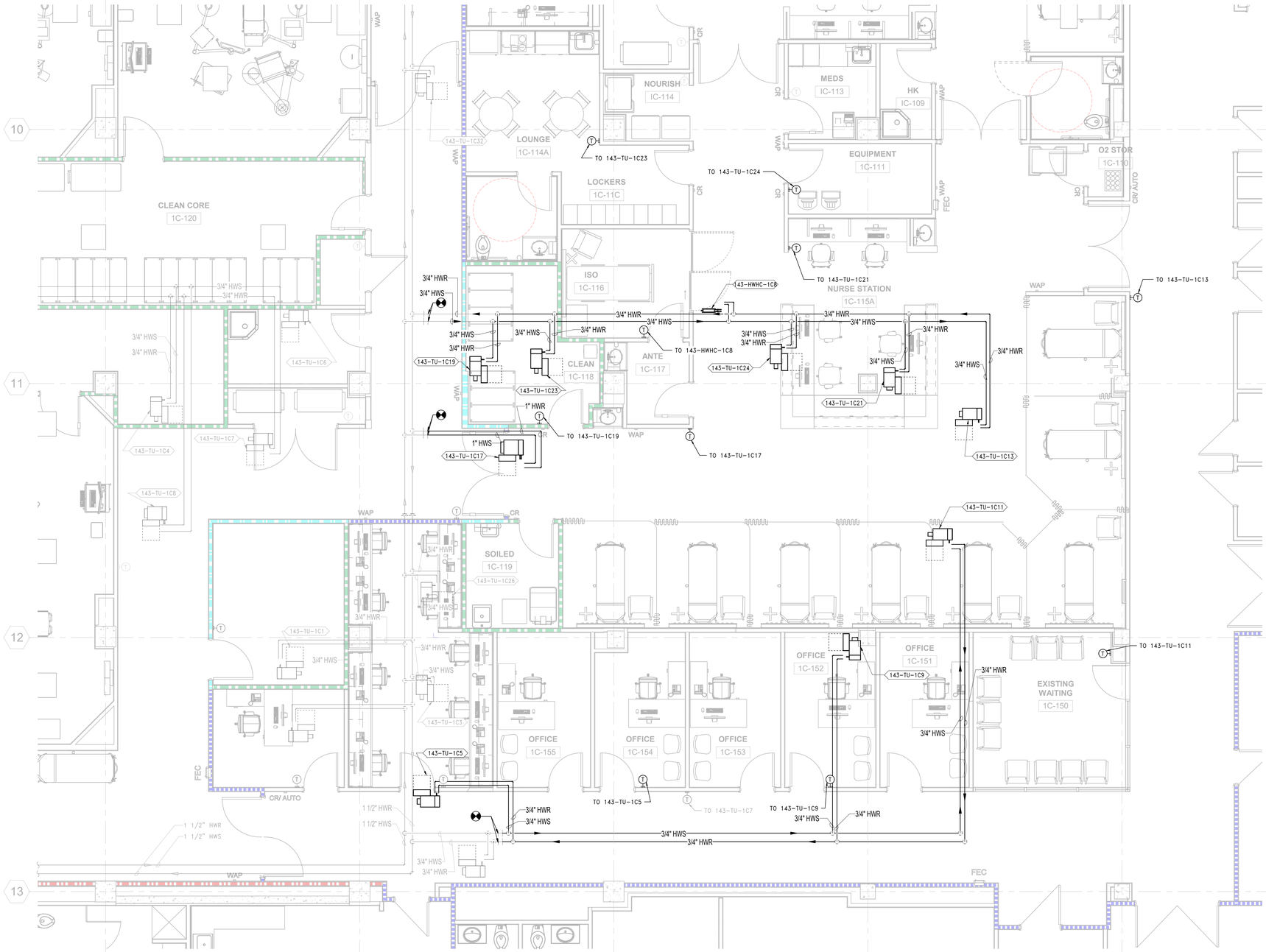
| | | | |
|---|---|---|---|
| Project Title: RENOVATE SURGICAL SUITE PHASE II | | | VA Drawing Number: 2617 |
| Project Number: 658-19-103 | | | Building Number: 143 |
| Location: SALEM VA MEDICAL CENTER | | | Drawing Number: P4-MH121 |
| Issue Date: 2025-12-30 | Checked: DJ | Drawn: JAK | |

12/29/2025 1:56:14 PM

VA FORM 08 - 6231

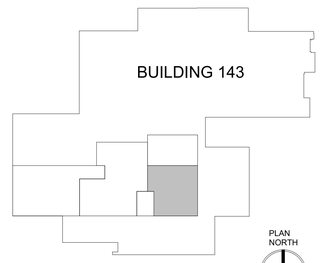
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL GENERAL NOTES, ABBREVIATIONS & SYMBOLS.




## 1 PIPING PLAN - 1ST FLOOR, PHASE 4

SCALE: 1/4" = 1'-0"

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Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL PIPING PLAN 1ST FLOOR, PHASE 4 | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| | | Project Number: 658-19-103 | Building Number: 143 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: P4-MP111 |
| | | Issue Date: 2025-12-30 / Checked: CS / Drawn: SP | |

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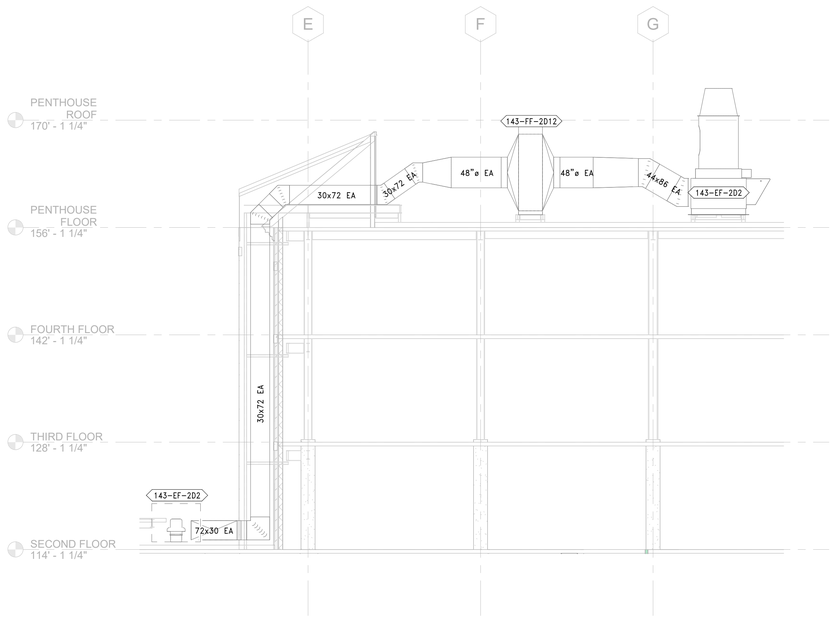
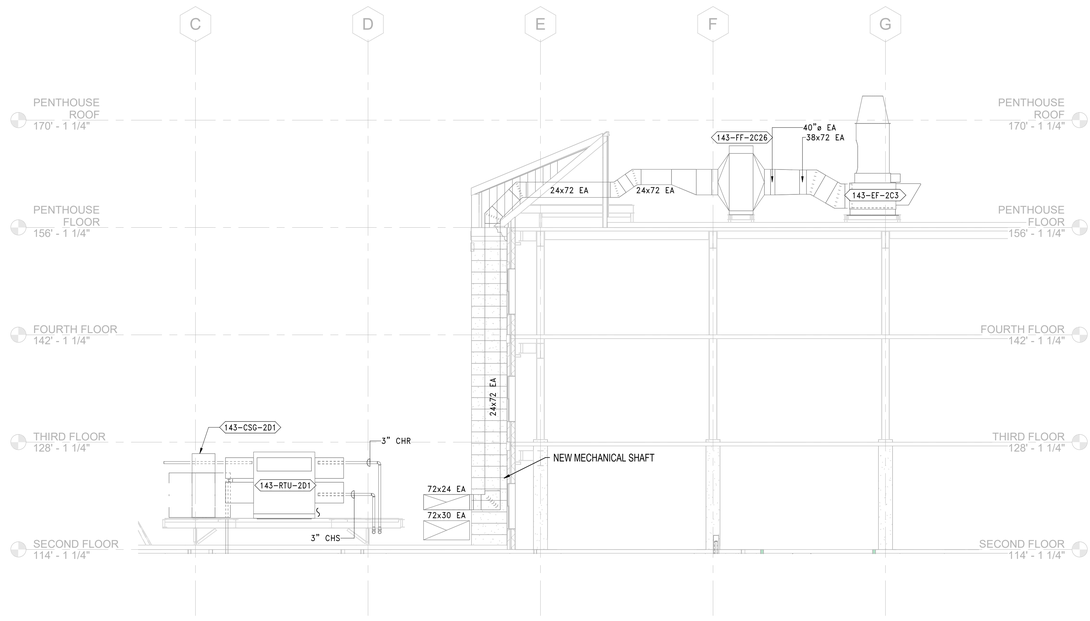


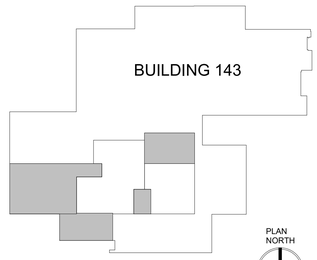

B SECTION AT 143-EF-2D2 DUCT CHASE

SCALE: 1/8" = 1'-0"




A SECTION AT 143-EF-2C3 DUCT CHASE

SCALE: 1/8" = 1'-0"

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12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| Drawing Title | Phase | Project Title | VA Drawing Number |
|---|---|---|---|
| MECHANICAL SECTIONS | BID DOCUMENTS | RENOVATE SURGICAL SUITE PHASE II | 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number 658-19-103 | Building Number 143 |
| | | Location SALEM VA MEDICAL CENTER | Drawing Number M-301 |
| | | Issue Date 2025-12-30 / Checked DJ / Drawn JAK | |

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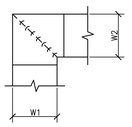
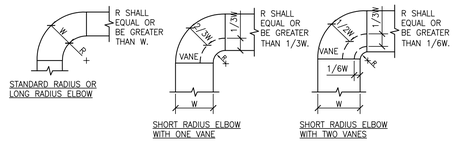
VA FORM 08 - 6231




NOTE:

1. ALL VANE ELBOWS SHALL BE CONSTRUCTED AND INSTALLED AS DETAILED BY SMACNA.
2. WHEN W1 DOES NOT EQUAL W2, VANE SHALL BE SINGLE THICKNESS VANE TYPE REGARDLESS OF W DIMENSION.
3. ALL SINGLE THICKNESS VANES SHALL HAVE A 2" RADIUS, 1-1/2" MAXIMUM SPACE BETWEEN VANES AND A 3/4" TRAILING EDGE.
4. WHEN W EQUALS W2 AND W1 IS GREATER THAN 20" VANES SHALL BE DOUBLE VANE TYPE.

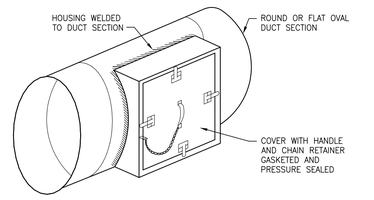
## 1 RECTANGULAR DUCT TURNING VANE DETAIL
SCALE: NTS



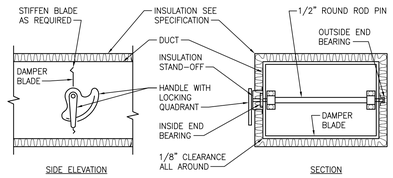

NOTE:

1. THE INTERIOR SURFACE OF ALL RADIUS ELBOWS SHALL BE MADE ROUND.
2. ALL STANDARD RADIUS ELBOWS CAN BE SUBSTITUTED WITH SHORT RADIUS ELBOWS. ALL SHORT RADIUS ELBOWS SHALL HAVE VANES. VANES SHALL BE CONSTRUCTED, SUPPORTED AND FASTENED AS RECOMMENDED BY SMACNA.
3. MAX NUMBER OF MITERED PIECES TO COMPLY WITH SMACNA TABLE 3-1 MITERED ELBOWS CHART.

## 2 DUCT RADIUS ELBOW DETAIL
SCALE: NTS



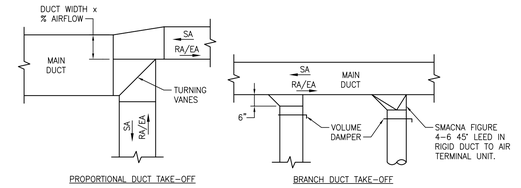

## 3 ACCESS DOOR FOR ROUND DUCT DETAIL
SCALE: NTS



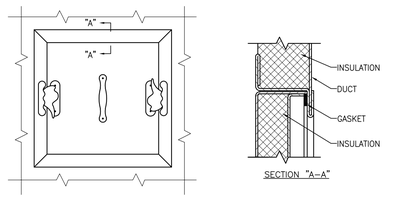

NOTE:

1. DELETE INSULATION STAND-OFF ON DUCTWORK WITHOUT EXTERIOR INSULATION.
2. DETAIL SHOWS SINGLE BLADE DAMPER. DAMPER INSTALLATION SHALL BE SIMILAR FOR MULTI-BLADE DAMPERS & ROUND DAMPERS.

## 4 RECTANGULAR VOLUME DAMPER DETAIL
SCALE: NTS

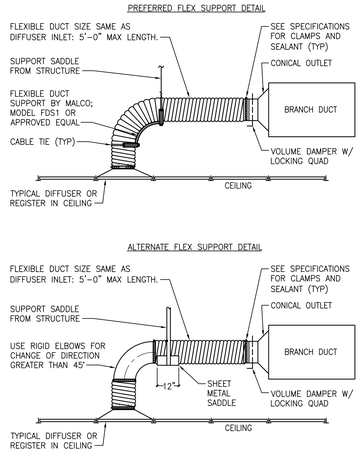



## 5 DUCT TAKE-OFF DETAIL
SCALE: NTS

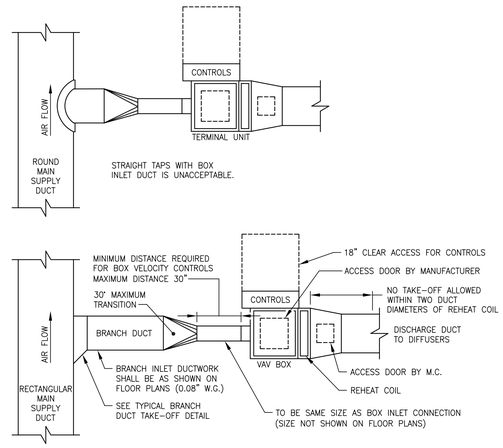



NOTES:
1. LATCHES SHALL BE OF THE WEDGE TYPE TO CLOSE DOORS TIGHTLY.
2. HINGES ON THE ACCESS DOORS SHALL HAVE NON-CORROSIVE PINS.
3. SEE SMACNA 2005, FIGURE 9-15

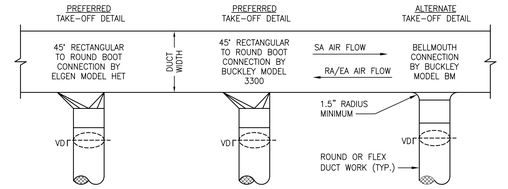
## 6 ACCESS PANEL AND DOOR DETAIL
SCALE: NTS




## 7 FLEXIBLE AIR DUCT CONNECTOR
SCALE: NTS




## 8 AIR VOLUME BOX DUCT CONNECTION DETAIL
SCALE: NTS



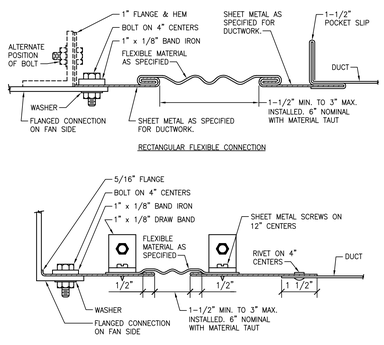

| ROUND BRANCH DUCT SIZE | DUCT OPENING SIZE | MINIMUM TRUNK LINE DUCT HEIGHT |
|---|---|---|
| 5"ø | 5" x 10" | 8" |
| 6"ø | 6" x 12" | 8" |
| 7"ø | 6" x 12" | 8" |
| 8"ø | 6" x 12" | 8" |
| 9"ø | 6-3/4" x 14" | 10" |
| 10"ø | 6-3/4" x 15" | 10" |
| 12"ø | 8-1/2" x 18" | 12" |
| 14"ø | 9-1/2" x 20" | 12" |
| 16"ø | 12" x 24" | 14" |

| ROUND BRANCH DUCT SIZE | DUCT OPENING SIZE | MINIMUM TRUNK LINE DUCT HEIGHT |
|---|---|---|
| 5"ø | 5" x 10" | 8" |
| 6"ø | 7" x 9" | 10" |
| 7"ø | 7" x 9-1/4" | 10" |
| 8"ø | 7" x 10-1/4" | 10" |
| 9"ø | 5-3/4" x 15" | 8" |
| 10"ø | 8" x 12-1/2" | 10" |
| 12"ø | 10" x 14-1/2" | 12" |
| 14"ø | 11-1/4" x 16" | 14" |
| 16"ø | 11-1/4" x 17-3/4" | 14" |

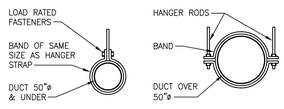
| ROUND BRANCH DUCT SIZE | DUCT OPENING SIZE | MINIMUM TRUNK LINE DUCT HEIGHT |
|---|---|---|
| 5"ø | 8"ø | 10" |
| 6"ø | 9"ø | 10" |
| 7"ø | 10"ø | 12" |
| 8"ø | 11"ø | 12" |
| 9"ø | 12"ø | 14" |
| 10"ø | 13"ø | 14" |
| 12"ø | 15"ø | 16" |
| 14"ø | 17"ø | 18" |
| 16"ø | 19"ø | 20" |

NOTES:
1. ALL PRESSURE CLASSES: USE GASKETED CONNECTION ONLY. SPIN-IN CONNECTIONS ARE UNACCEPTABLE.
2. ROUND TO ROUND (ALL PRESSURES): PROVIDE CONICAL TEE.

## 9 ROUND BRANCH DUCT CONNECTION DETAIL
SCALE: NTS



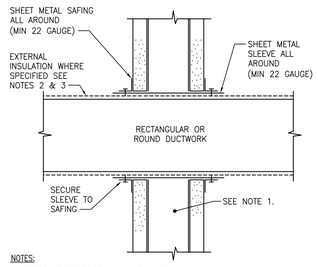

## 10 FLEXIBLE CONNECTION DETAIL
SCALE: NTS




| HANGER STRAPS OR RODS | | | |
|---|---|---|---|
| MAX. DUCT ø IN. [mm] | QUANTITY/SIZE IN. [mm] | MAX. LOAD LBS. [kg] | MAX. SPACING IN. [mm] |
| 26 [650] | ONE 1 [25] x 22 GA STRAP | 260 [119] | 144 [3600] |
| 36 [900] | ONE 1 [25] x 18 GA STRAP | 420 [190] | 144 [3600] |
| 50 [1250] | ONE 1 [25] x 16 GA STRAP | 700 [317] | 144 [3600] |
| 60 [1500] | TWO 3/8 [10]ø. RODS | 1320 [598] | 144 [3600] |
| 84 [2100] | TWO 1/2 [13]ø RODS | 2500 [1133] | 144 [3600] |

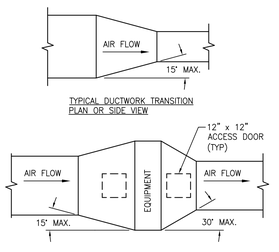
NOTE:
TABULATED DATA FROM SMACNA ALLOWS FOR DUCT REINFORCING AND INSULATION, BUT NO EXTERNAL LOAD.

## 11 ROUND DUCT HANGER DETAIL
SCALE: NTS



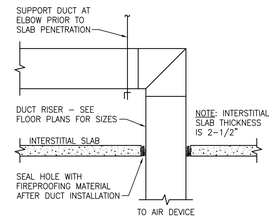

NOTES:
1. CONCRETE OR BLOCK WALL SIMILAR.
2. PROVIDE UL LISTED THROUGH PENETRATION FIRE STOP SEAL FOR RATED WALL WHERE DUCT DOES NOT REQUIRE A FIRE DAMPER.
3. INSULATION TIGHT TO SLEEVE FOR INSULATED DUCT. CAULK CLEAR SPACE AIR TIGHT FOR INSULATED AND NON-INSULATED DUCT.

## 12 DUCT THROUGH WALL DETAIL
SCALE: NTS




NOTE:
UNLESS OTHERWISE INDICATED ON PLANS, MAXIMUM ANGLES SHOWN SHALL APPLY.

## 13 DUCT TRANSITION DETAIL
SCALE: NTS




## 14 DUCT RISER SUPPORTS
SCALE: NTS

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| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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A Service-Disabled Veteran-Owned Small Business

(856) 429-4000 ★ Miller-Remick.com

M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR
Lic. No. 0402055070
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DETAILS | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | | Project Number: 658-19-103 | Building Number: 143 |
| | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: M-501 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

VA FORM 08-6231

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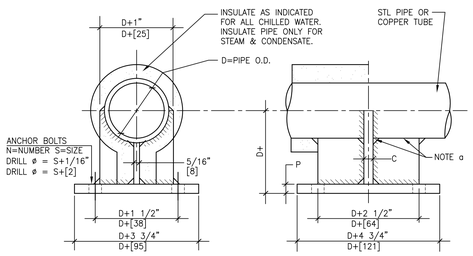
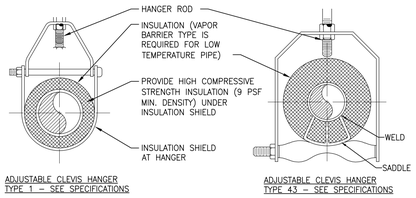
| PIPE ANCHOR SCHEDULE | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| D | | P | | C | | N | | S | | BOLT PATTERN |
| IN | MM | IN | MM | IN | MM | IN | MM | IN | MM | |
| 4" | 102 | ⅜" | 16 | ¾" | 19 | 4" | 102 | ¾" | 19 | |
| 3" | 76 | ½" | 13 | ½" | 13 | 4" | 102 | ⅜" | 16 | |
| 2½" | 64 | ⅜" | 10 | ⅜" | 10 | 4" | 102 | ⅜" | 16 | |
| 2" | 51 | ⅜" | 10 | ⅜" | 10 | 4" | 102 | ⅜" | 16 | |
| 1½" | 38 | ⅜" | 10 | ¼" | 6 | 4" | 102 | ½" | 13 | |



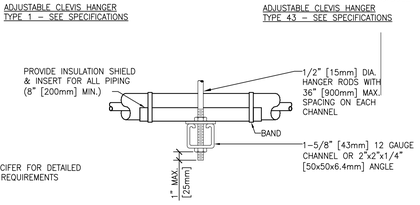

NOTE:
a. WHERE USED FOR COPPER TUBE OR PIPE, BRAZE TO FABRICATED STEEL ANCHOR.

## 1 SMALL PIPE ANCHOR (1 1/2"–4")
SCALE: NTS




ADJUSTABLE CLEVIS HANGER
TYPE 1 – SEE SPECIFICATIONS

ADJUSTABLE CLEVIS HANGER
TYPE 43 – SEE SPECIFICATIONS

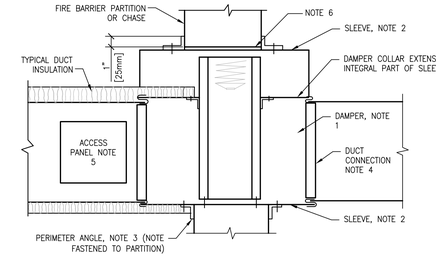



NOTES:
SEE SPECIFER FOR DETAILED HANGER REQUIREMENTS

SIDE VIEW TRAPEZE HANGER
FOR UP TO 1000 LB. [453KG] UNIFORM LOAD

| MAXIMUM PIPE/TUBING SUPPORT SPACING | | | | | | | | | | | | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| NOM. SIZE | IN. | THRU 3/4 | 1 | 1 1/4 | 1 1/2 | 2 | 2 1/2 | 3 | 4 | 5 | 6 | 8 | 10 | 12 | 14 | 16 | 18 | 20 | 24 |
| | [mm] | THRU [20] | [25] | [32] | [40] | [50] | [65] | [75] | [100] | [125] | [150] | [200] | [250] | [300] | [350] | [400] | [450] | [500] | [600] |
| PIPE | FT. | 7 | 7 | 7 | 9 | 10 | 11 | 12 | 14 | 16 | 17 | 19 | 22 | 23 | 25 | 27 | 28 | 30 | 32 |
| | [mm] | [2100] | [2100] | [2100] | [2700] | [3000] | [3400] | [3700] | [4100] | [4900] | [5200] | [5800] | [6700] | [7000] | [7600] | [8200] | [8500] | [9100] | [9600] |
| TUBING | FT. | 5 FT | 6 | 7 | 8 | 8 | 9 | 10 | 12 | 13 | 14 | 16 | – | – | – | – | – | – | – |
| | [mm] | [1500] | [1800] | [2100] | [2400] | [2400] | [2700] | [3000] | [3700] | [4000] | [4100] | [4900] | – | – | – | – | – | – | – |
| NOTE: FOR TRAPEZE HANGER TAKE SPACING OF SMALLEST SIZE ON TRAPEZE. | | | | | | | | | | | | | | | | | | | | | |

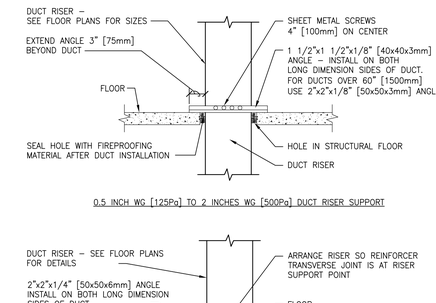
## 2 PIPE HANGERS
SCALE: NTS



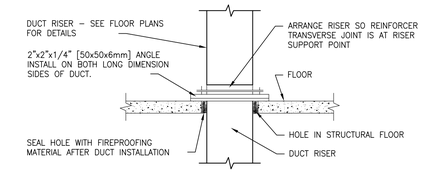

NOTE:

1. A VERTICAL DAMPER IS SHOWN. HORIZONTAL DAMPER INSTALLATION, IS SIMILAR. FOLLOW DAMPER MANUFACTURER'S INSTRUCTIONS, INCLUDING FASTENER OPTIONS AND GAGES FOR SLEEVE AND PERIMETER ANGLES. FIRE DAMPERS MUST BE INSTALLED IN THE PARTITION OR FLOOR AND NOT OUTSIDE THE PENETRATION.
2. GALVANIZED SLEEVE: GAGE NOT LESS THAN CONNECTING DUCT. FASTEN SLEEVE TO DAMPER FRAME AND TO PERIMETER ANGLES.
3. PERIMETER ANGLES: GALVANIZED STEEL, NOT LESS THAN 1 1/2"x1 1/2" [40x40mm], 14 GAGE, TO PROVIDE 1" [25mm] MINIMUM OVERLAP OF OPENING ON ALL 4 SIDES.
4. BREAKAWAY DUCT CONNECTION: CONTRACTOR'S OPTION OF TYPES SHOWN IN SMACNA.
5. ACCESS PANELS: SIZE AND LOCATION TO PERMIT SERVICING THE FUSIBLE LINK OR LIMS.
6. PROVIDE 1/4" TO 1/2" [6 TO 15mm] CLEARANCE ON HEIGHT AND WIDTH.

## 3 SECTION THRU FIRE DAMPER INSTALLATION
SCALE: NTS




0.5 INCH WG [125Pa] TO 2 INCHES WG [500Pa] DUCT RISER SUPPORT

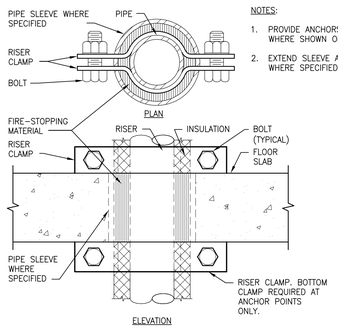



2 INCHES WG [500Pa] TO 4 INCHES WG [1000Pa] DUCT RISER SUPPORT

NOTE:

ALL DUCT WORK RISERS WHICH ARE RUN EXPOSED, SUCH AS THRU ATTIC FLOORS AND FAN ROOM FLOORS SHALL BE PROVIDED WITH A 3" [75mm] HIGH CONCRETE CURB AROUND OPENING FOR DUCT.

## 4 DUCT RISER SUPPORTS
SCALE: NTS

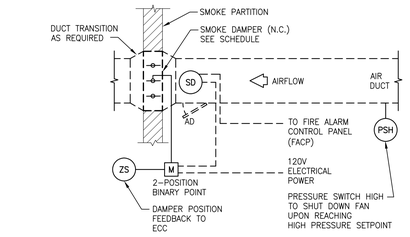



ELEVATION

NOTES:

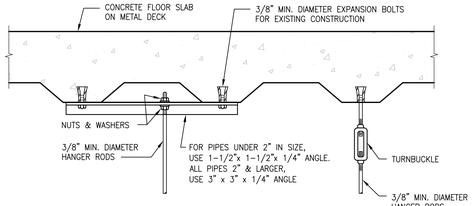
1. PROVIDE ANCHORS ONLY WHERE SHOWN ON DRAWINGS.
2. EXTEND SLEEVE ABOVE FLOOR WHERE SPECIFIED.

## 5 SUPPORT/ANCHOR FOR PIPE RISERS
SCALE: NTS

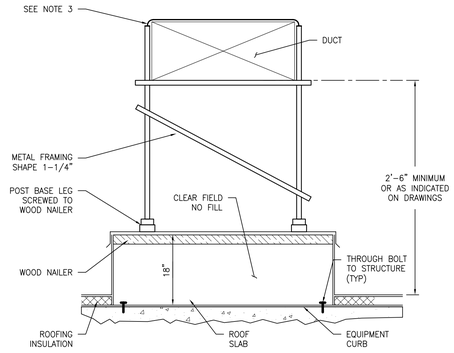



NOTE:
UPON DETECTION OF SMOKE BY THE SMOKE DETECTOR, THE SMOKE DAMPER SHALL CLOSE & SEND AN ALARM TO THE ECC.

## 6 SMOKE DAMPER CONTROL DIAGRAM
SCALE: NTS



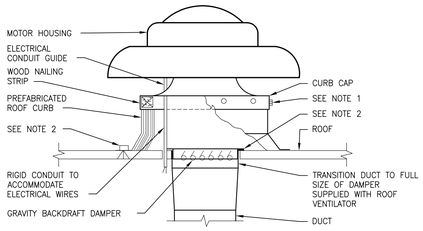

## 7 SECURING HANGER RODS IN CONCRETE
SCALE: NTS




NOTES:

1. MINIMUM CURB HEIGHT IS TO BE 12 INCHES.
2. METAL FRAMING SHAPE TO BE 1–1/4" UNISTRUT OR EQUAL FOR INSTALLATIONS WITH TOP OF DUCT LESS THAN 60". HIGHER TOP OF DUCT INSTALLATIONS TO BE 1 5/8".
3. STAINLESS STEEL STRAP DOWN DUCT TO SUPPORT, PER SMACNA.

## 8 ROOFTOP DUCT SUPPORT DETAIL
SCALE: NTS




NOTES:

1. SECURE CURB CAP TO WOOD NAILING STRIP WITH 3/8" [10mm] CADMIUM PLATED LAG BOLTS NOT OVER 12" [300mm] ON CENTER.
2. SECURE ROOF CURB, DUCTWORK AND DAMPER TO ROOF WITH EXPANSION BOLTS (CONCRETE ROOF) OR RUST RESISTANT BOLTS (METAL DECK AND BAR JOIST ROOF).
3. RUN ELECTRICAL LINES THROUGH CLEARANCE HOLE PROVIDED IN GRAVITY DAMPER, THEN THROUGH VENTILATOR ELECTRICAL CONDUIT GUIDE.

## 9 POWER ROOF VENTILATOR
SCALE: NTS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

ENGINEER OF RECORD

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A Service-Disabled Veteran-Owned Small Business
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M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
CHRISTOPHER ANTHONY PANEL PACARO JR
Lic. No. 0402055070
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DETAILS | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | | Project Number: 658-19-103 | Building Number: 143 |
| | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: M-502 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

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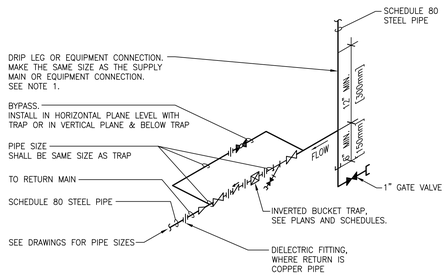
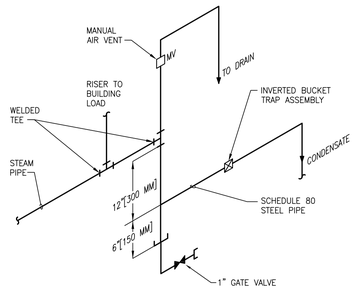


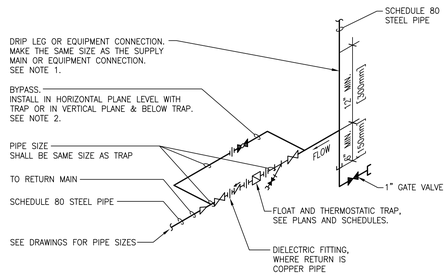

NOTES:

1. ALL DRIP POINTS ON STEAM MAINS SHALL BE PROVIDED WITH A 12" MINIMUM HIGH DRIP LEG FROM BOTTOM OF STEAM MAIN TO TRAP INLET. DRIP LEG SHALL HAVE 6" SCALE POCKET BELOW TRAP INLET.
2. PROVIDE BYPASS PIPING.

## 1 INVERTED BUCKET STEAM TRAP ASSEMBLY
SCALE: NTS



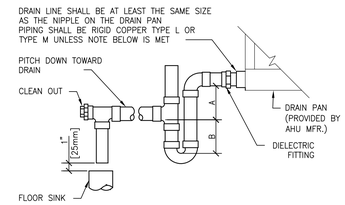

## 2 STEAM LINE DRIP TRAP AT RISER
SCALE: NTS




NOTES:

1. ALL DRIP POINTS ON STEAM MAINS SHALL BE PROVIDED WITH A 12" MINIMUM HIGH DRIP LEG FROM BOTTOM OF STEAM MAIN TO TRAP INLET. DRIP LEG SHALL HAVE 6" SCALE POCKET BELOW TRAP INLET.
2. PROVIDE BYPASS PIPING FOR ALL TRAPS 1" AND LARGER.

## 3 FLOAT AND THERMOSTATIC STEAM TRAP ASSEMBLY
SCALE: NTS



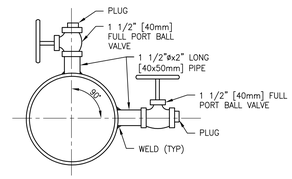

NOTE:

1. CPVC PIPE MAY BE USED ONLY IF APPROVED BY LOCAL VA AND IS INDOORS AND DOES NOT PASS THROUGH RATED BARRIERS.
2. DIELECTRIC FITTING TO BE USED WHEN TWO DISSIMILAR METALS ARE TO BE CONNECTED.

| UNIT TYPE | A | B |
|---|---|---|
| DRAW THRU | 2" [50mm] PLUS X | X |
| BLOW THRU | 1" [25mm] MINIMUM | 2X |

WHERE X = STATIC PRESSURE IN PAN
(PROVIDED BY AHU MANUFACTURER)

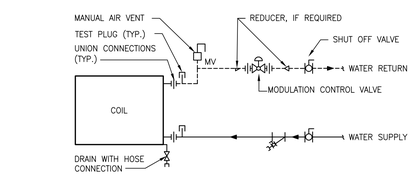
## 4 AIR HANDLING UNIT DRAIN TRAP DETAIL
SCALE: NTS



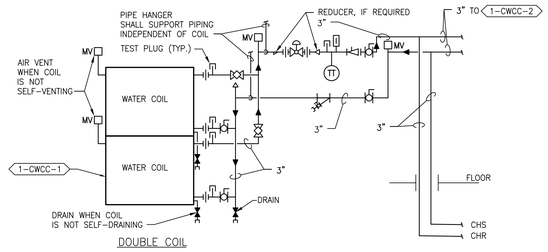

NOTES:

1. PROVIDE IN CHILLED WATER MAIN.
2. LOCATE PILOT TUBE TAPS 20 PIPE DIAMETERS DOWNSTREAM AND 10 PIPE DIAMETERS UPSTREAM FROM THE NEAREST PIPE FITTING. EITHER TOP OR SIDE LOCATION. BOTH ARE NOT REQUIRED AT SAME LOCATION.

## 5 FLOW ELEMENT CONNECTIONS
SCALE: NTS




## 6 TERMINAL UNIT WATER COILS PIPING CONNECTIONS
SCALE: NTS

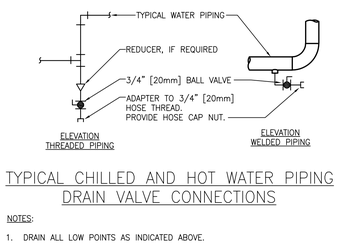



DOUBLE COIL

NOTES:

1. PIPING SHALL BE INSTALLED IN SUCH MANNER THAT IT WILL NOT BLOCK THE SWING OR USE OF ACCESS DOORS OR PANELS; NEITHER SHALL IT BLOCK THE SERVICING OF FILTERS, VALES, OR EQUIPMENT.
2. THE FLOW ELEMENT MAY BE INSTALLED IN THE SUPPLY PIPING IF THE REQUIRED MINIMUM UPSTREAM AND DOWNSTREAM DIMENSIONS CANNOT BE OBTAINED IN THE RETURN PIPING.

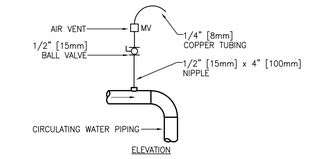
## 7 WATER COILS – PIPING CONNECTIONS
SCALE: NTS




## TYPICAL CHILLED AND HOT WATER PIPING DRAIN VALVE CONNECTIONS

NOTES:

1. DRAIN ALL LOW POINTS AS INDICATED ABOVE.
2. WHERE SCALE POCKETS ARE SHOWN ON PIPE RISER DIAGRAMS AND/OR PLANS LOCATE DRAIN AT BOTTOM OF SCALE POCKET.

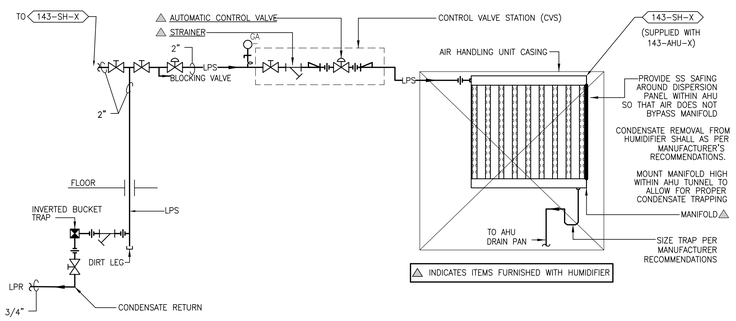



## TYPICAL MANUAL AIR VENT

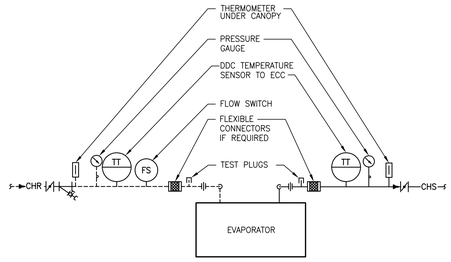
NOTES:

1. VENT ALL HIGH POINTS INDICATED ABOVE.

## 8 DRAIN VALVE AND AIR VENT CONNECTIONS (HYDRONIC SYSTEMS)
SCALE: NTS




## 9 AHU STEAM HUMIDIFIER PIPING CONNECTIONS (DISPERSION TYPE)
SCALE: NTS




## 10 AIR COOLED CHILLER PIPING CONNECTIONS
SCALE: NTS

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| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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A Service-Disabled Veteran-Owned Small Business
(856) 429-4000 ✧ Miller-Remick.com
M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL DETAILS | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | | Project Number: 658-19-103 | Building Number: 143 |
| | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER | Drawing Number: M-503 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

VA FORM 08-6231

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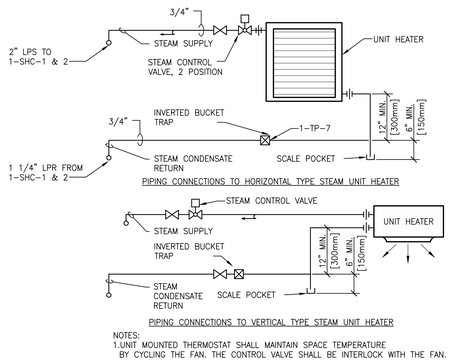
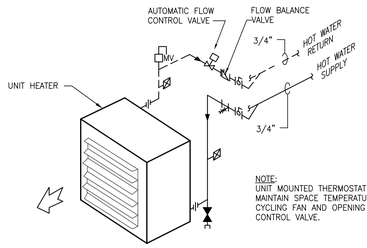



NOTES:
1. UNIT MOUNTED THERMOSTAT SHALL MAINTAIN SPACE TEMPERATURE BY CYCLING THE FAN. THE CONTROL VALVE SHALL BE INTERLOCK WITH THE FAN.

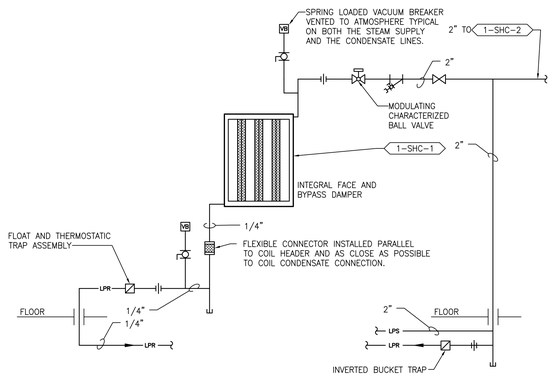
## 1 UNIT HEATERS (STEAM) PIPING CONNECTIONS
SCALE: NTS



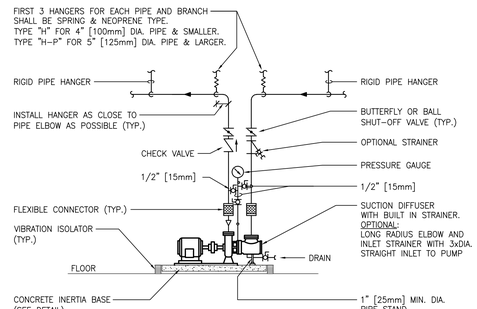

NOTE:
UNIT MOUNTED THERMOSTAT SHALL MAINTAIN SPACE TEMPERATURE BY CYCLING FAN AND OPENING CONTROL VALVE.

TYPICAL CONNECTIONS TO HORIZONTAL HOT WATER UNIT HEATER

## 2 UNIT HEATERS (HOT WATER) PIPING CONNECTIONS
SCALE: NTS

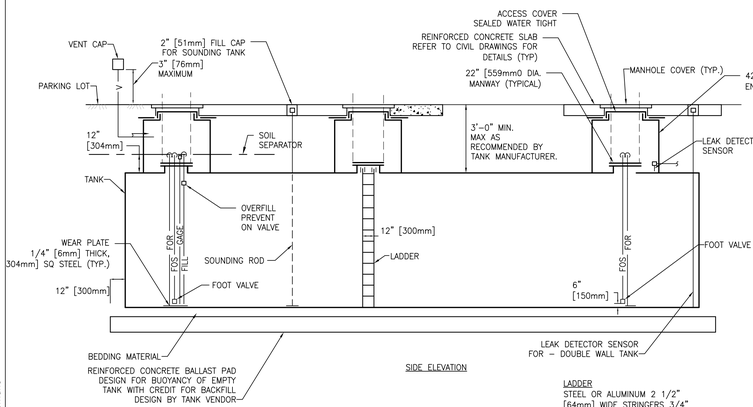



## 3 INTEGRAL FACE AND BYPASS STEAM COIL DETAIL
SCALE: NTS

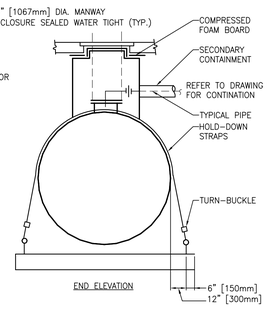



NOTES:
1. SEE SPECIFICATION SECTION "PUMPS" FOR Y STRAINER OPTION.

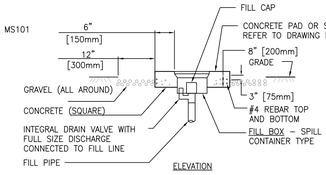
## 4 SINGLE SUCTION FLOOR-MOUNTED PUMPS CONNECTIONS WITH FLEXIBLE CONNECTORS
SCALE: NTS




SIDE ELEVATION




END ELEVATION

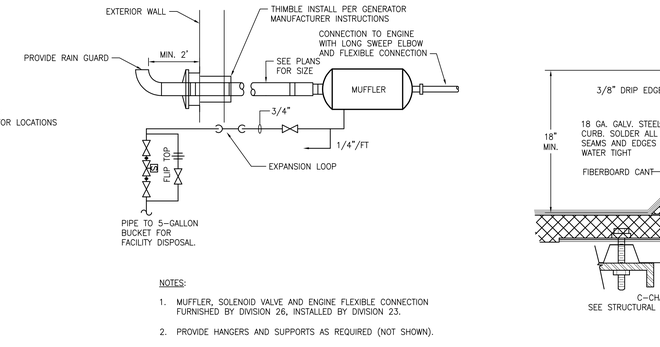



ELEVATION

LADDER
STEEL OR ALUMINUM 2 1/2" [64mm] WIDE STRINGERS 3/4" [19mm] DIA. RUNGS ANCHOR AT BOTTOM GUIDE AT TOP

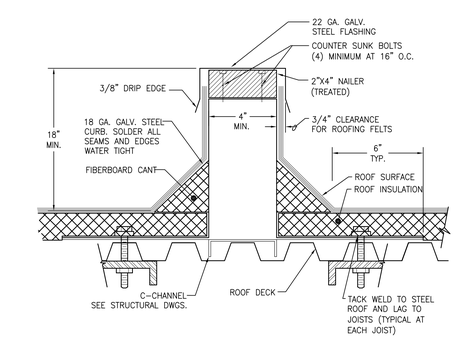
NOTE:
1. INSTALL TANKS PER MANUFACTURERS WRITTEN INSTRUCTIONS.
2. PROVIDE DOUBLE WALLED INTERSTITIAL SPACE MONITORING SYSTEM. BASIS OF DESIGN: VEEDER-ROOT.
3. INSTALLATION MUST COMPLY WITH PA ENVIRONMENTAL REGULATIONS. INSTALLERS MUST BE CERTIFIED WITH THE DEP.
4. PIPING CONNECTIONS ARE DIAGRAMMATIC. REFER TO DRAWING M-101 FOR PIPE CONNECTION LOCATIONS.

## 5 UNDERGROUND DOUBLE WALL FUEL OIL STORAGE TANK
SCALE: NTS

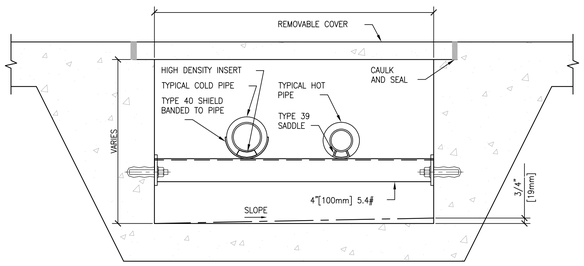



NOTES:
1. MUFFLER, SOLENOID VALVE AND ENGINE FLEXIBLE CONNECTION FURNISHED BY DIVISION 26, INSTALLED BY DIVISION 23.
2. PROVIDE HANGERS AND SUPPORTS AS REQUIRED (NOT SHOWN).

## 6 EMERGENCY GENERATOR ENGINE EXHAUST THROUGH WALL PREFABRICATED DOUBLE WALL EXHAUST SYSTEM
SCALE: NTS

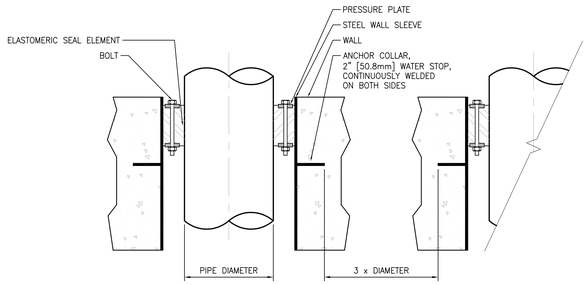



## 7 EQUIPMENT SUPPORT RAIL
SCALE: NTS




NOTES:
1. COORDINATE TRENCH DETAIL WITH ARCHITECTURAL DETAIL 2/29-AE400 & STRUCTURAL DETAILS 6/S-501 AND 8/S-501.
2. REFER TO SPECIFICATION, SEALING & CAULKING.

## 8 PIPE TRENCH SUPPORT DETAIL – FUEL OIL PIPING
SCALE: NTS




PLAN VIEW SECTION

NOTES:
1. ALL PIPING PENETRATIONS THROUGH BELOW-GRADE WALL SHALL BE CAST-IN-PLACE WATERSTOP SLEEVES. CORED HOLED ARE PROHIBITED.
2. COORDINATE SLEEVE LOCATION WITH GC PRIOR TO FORMING OF WALL.
3. PROVIDE WATERSTOP SLEEVES TO GC, FOR PLACEMENT IN FORMS, PRIOR TO POURING OF CONCRETE.
4. SLEEVES SHALL BE HOT DIPPED GALVANIZED.
5. LINK SEAL ELEMENTS SHALL BE APPROPRIATE FOR SERVICE TYPE OF THE PIPE.

## 9 PIPE PENETRATION THROUGH BELOW-GRADE WALL DETAIL
SCALE: NTS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title: MECHANICAL DETAILS | Phase: BID DOCUMENTS |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED |

| | |
|---|---|
| Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Project Number: 658-19-103 | Building Number: 143 |
| Location: SALEM VA MEDICAL CENTER | Drawing Number: M-504 |
| Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

VA FORM 08-6231

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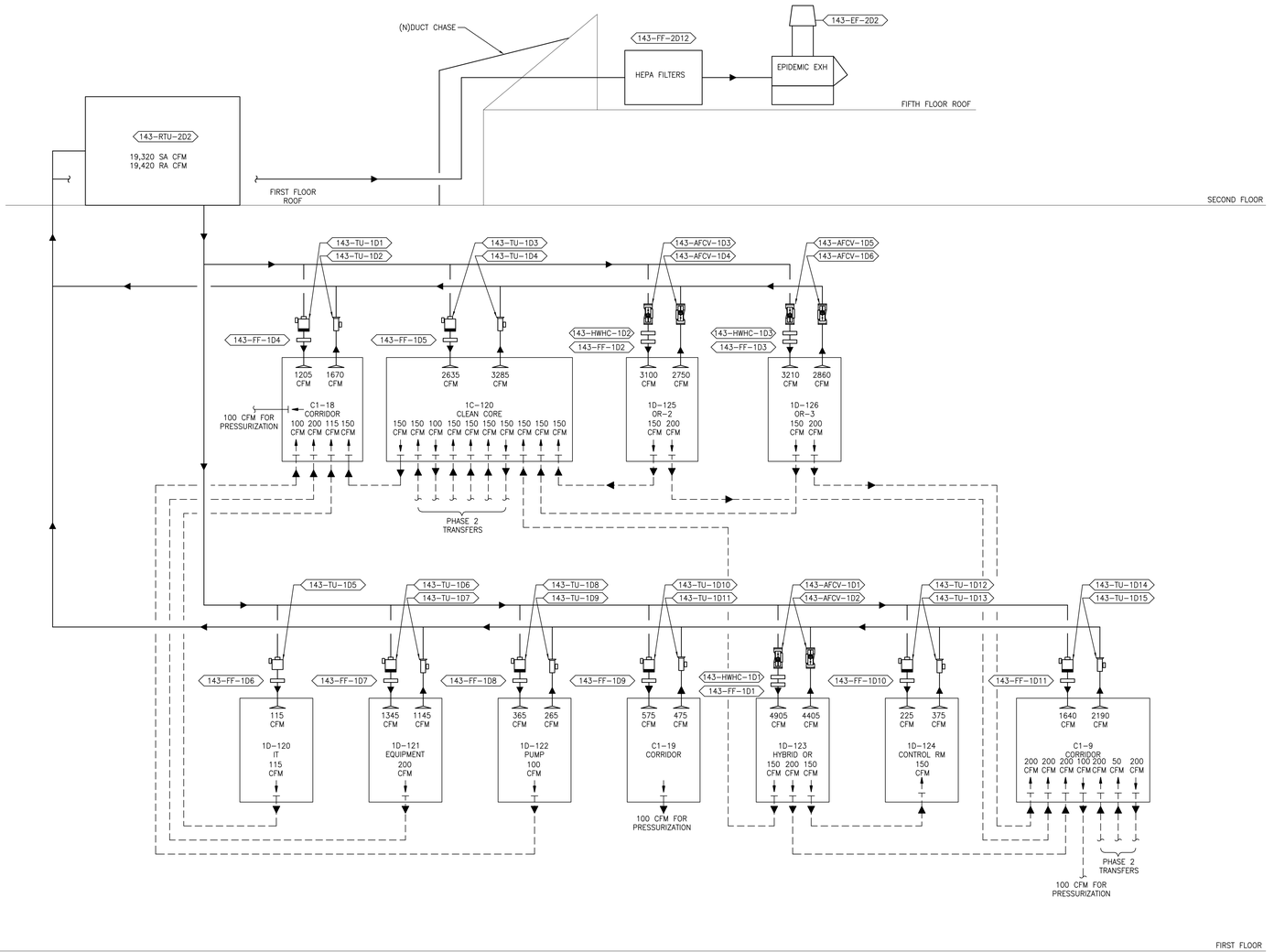
## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MH100 SERIES FOR ACTUAL DUCT LAYOUT.




1 AIRFLOW DIAGRAM – PHASE 1
SCALE: N.T.S.

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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL AIR FLOW DIAGRAM PHASE 1 |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | BH |
| Drawn | CS |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-601 |

VA FORM 08-6231

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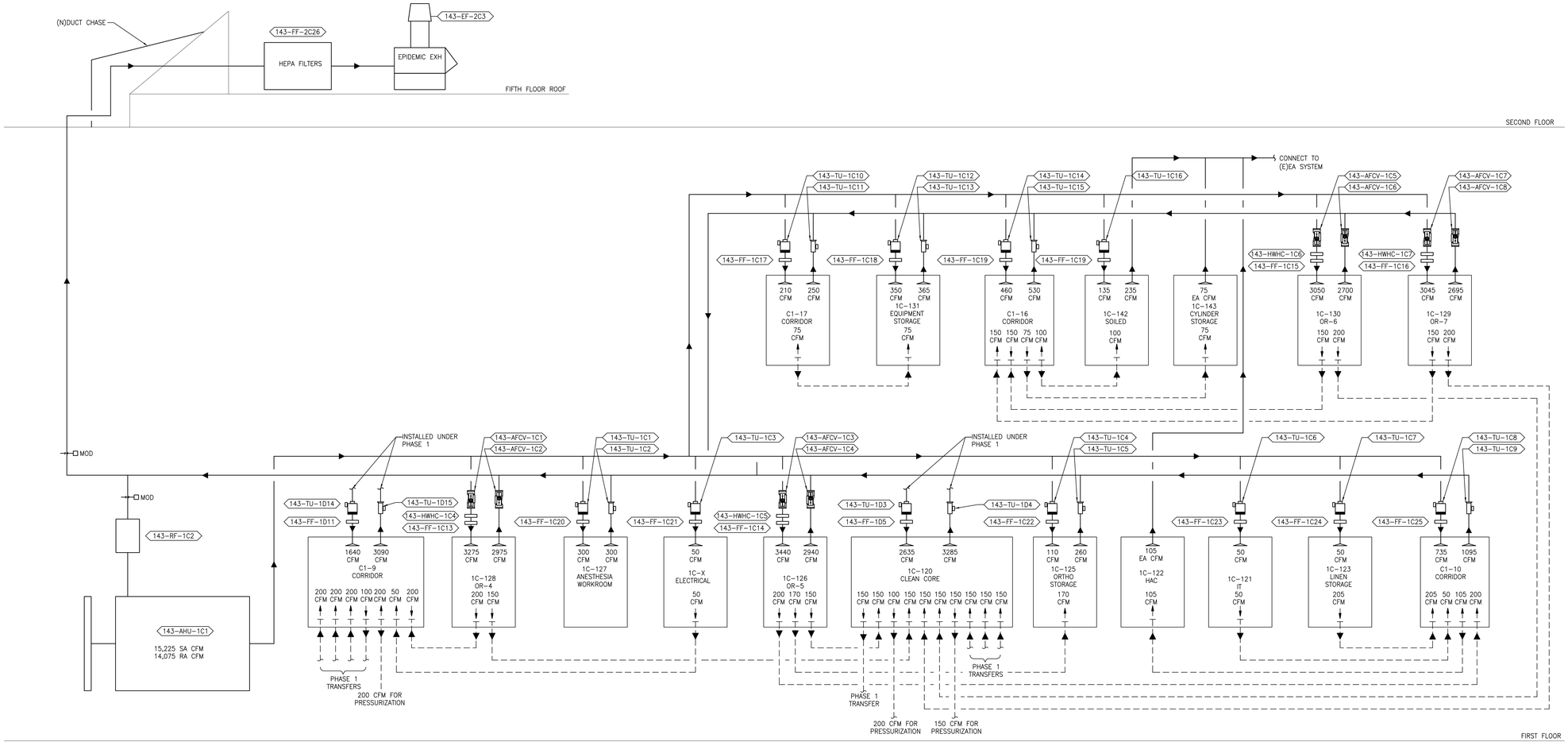
## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MH100 SERIES FOR ACTUAL DUCT LAYOUT.




1 AIRFLOW DIAGRAM – PHASES 2
SCALE: N.T.S.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

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**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL AIR FLOW DIAGRAM PHASES 2 & 3 |
| Phase | BID DOCUMENTS |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Approved by: | Jon Manninen, COR |
| | FULLY SPRINKLERED |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | BH |
| Drawn | CS |
| Drawing Number | M-602 |

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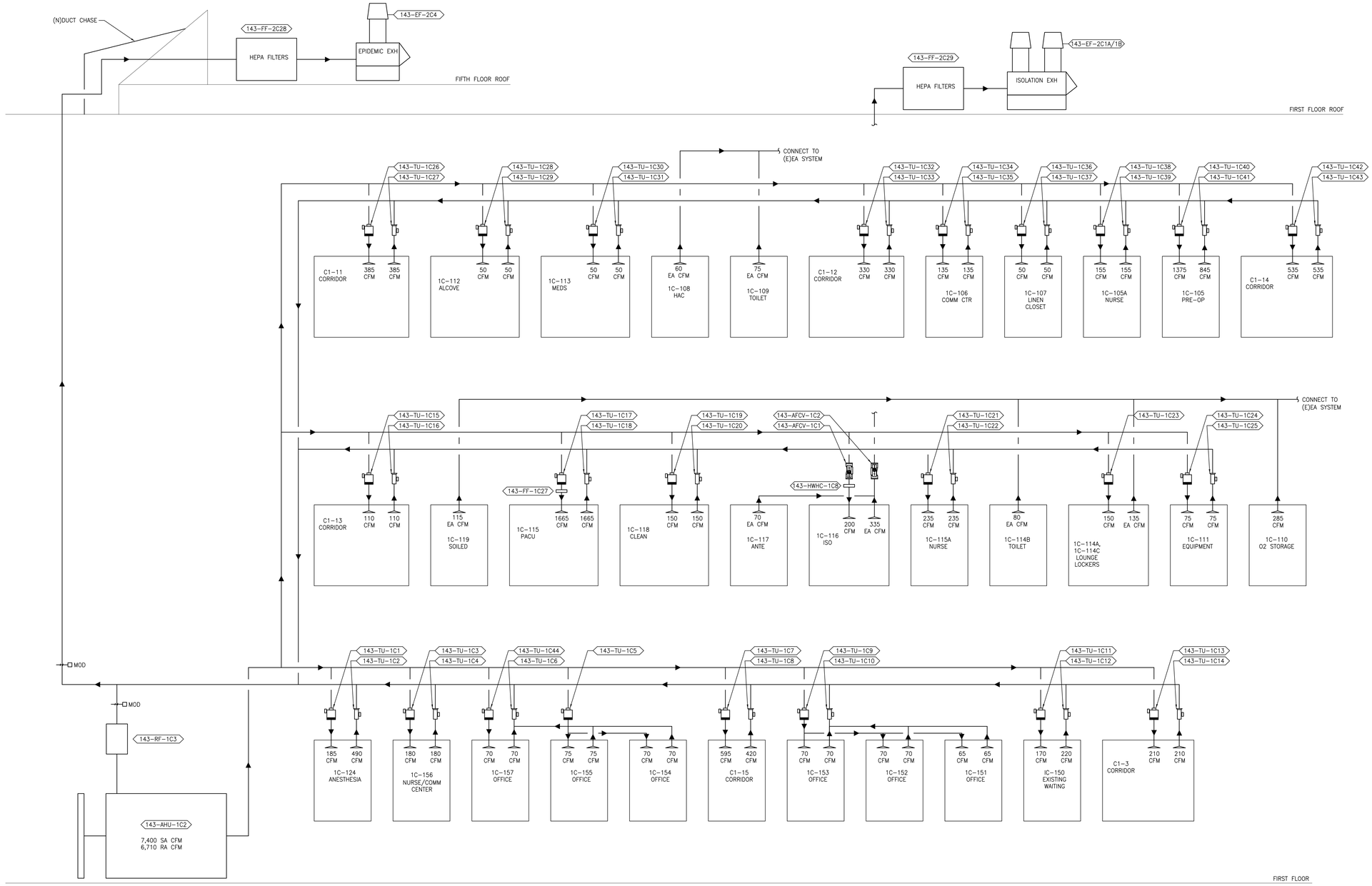
## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MH100 SERIES FOR ACTUAL DUCT LAYOUT.




1 AIRFLOW DIAGRAM – PHASE 3 & 4
SCALE: N.T.S.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

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PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL AIR FLOW DIAGRAM PHASE 4 | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Project Number** 658-19-103 | **Building Number** 143 |
| | | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** M-603 |
| | | **Issue Date** 2025-12-30 / **Checked** BH / **Drawn** CS | |

VA FORM 08-6231

three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
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## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MP100 SERIES FOR ACTUAL DUCT LAYOUT.
3. FLOW DIAGRAM DOES NOT SHOW ALL DEVICES AND ACCESSORIES REQUIRED. PROVIDE AS SHOWN ON DETAILS, AND AS SPECIFIED.

PENTHOUSE
FOURTH FLOOR
THIRD FLOOR
SECOND FLOOR
FIRST FLOOR
BASEMENT
SUB BASEMENT

3" CHS
3" CHR
143-CC-J1
143-AHU-J1
143-CC-4H2
143-CC-4H1
143-AHU-4H2
143-AHU-4H1
2" CHS
2" CHR
2-1/2" CHS
2-1/2" CHR
143-P-4J3
143-P-4J4
143-AHU-4J1
143-AHU-4J2
143-CC-4J1
143-CC-4J2
DPS
143-CC-3H2
143-CC-3H1
143-AHU-3H2
143-AHU-3H1
5" CHR
5" CHS
143-P-3J3
143-P-3J4
143-AHU-3J1
143-AHU-3J2
143-CC-3J1
143-CC-3J2
143-CC-2H2
143-CC-2H1
143-CC-2J3
143-AHU-2H2
143-AHU-2H1
143-AHU-2J3
4" CHS
4" CHR
143-P-2J3
143-P-2J4
143-AHU-2J1
143-AHU-2J2
143-CC-2J1
143-CC-2J2
1-1/2" CHS
1-1/2" CHR
143-AHU-1EF2
143-AHU-1EF1
143-CC-1EF2
143-CC-1EF1
6" CHR
6" CHS
143-AHU-1FG1
143-CC-1FG1
2A-AHU-1A1
2A-CC-1A1
143-AHU-1B1
143-CC-1B1
143-CC-BF1
143-AHU-BF1
143-CC-BG1
143-AHU-BG1
143-CC-BE1
143-AHU-BE1
143-FCU-BD1
1"
143-CC-BB1
143-AHU-BB1
143-CC-BB2
143-AHU-BB2
143-CC-BB3
143-AHU-BB3
143-CC-BD1
143-AHU-BD1
8" CHS
8" CHR
CHS F1
CHR F1
CHS G1
CHR G1
CHS C1
CHR C1
CHR B1
CHS B1
CHR D1
CHS D1
10" CHS
10" CHR
SCOPE AREA
143-P-1
143-P-2
143-ET-SD1
143-AS-SD1
143-P-1C1
143-P-1C2
TO/FROM UTILITY TUNNEL
43-CWCC-1C1
43-CWCC-1C1A
143-RTU-2D1
43-CWCC-1C2
43-CWCC-1C2A
43-CWCC-1C2B
43-CWCC-1C3
43-CWCC-1C3A
43-CWCC-1C3B
143-AHU-1C1
143-AHU-1C2
4" 3"
NO
NC
43-ACCH-1D1
43-ACCH-1D1A
43-ACCH-1D1B
AT GRADE OUTSIDE AT BASEMENT LEVEL

1 CHILLED WATER FLOW DIAGRAM
SCALE: N.T.S.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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MLR PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055075
PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL CHILLED WATER FLOW DIAGRAM | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Project Number** 658-19-103 | **Building Number** 143 |
| | | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** M-604 |
| | | **Issue Date** 2025-12-30 / **Checked** DJ / **Drawn** JAK | |

VA FORM 08-6231

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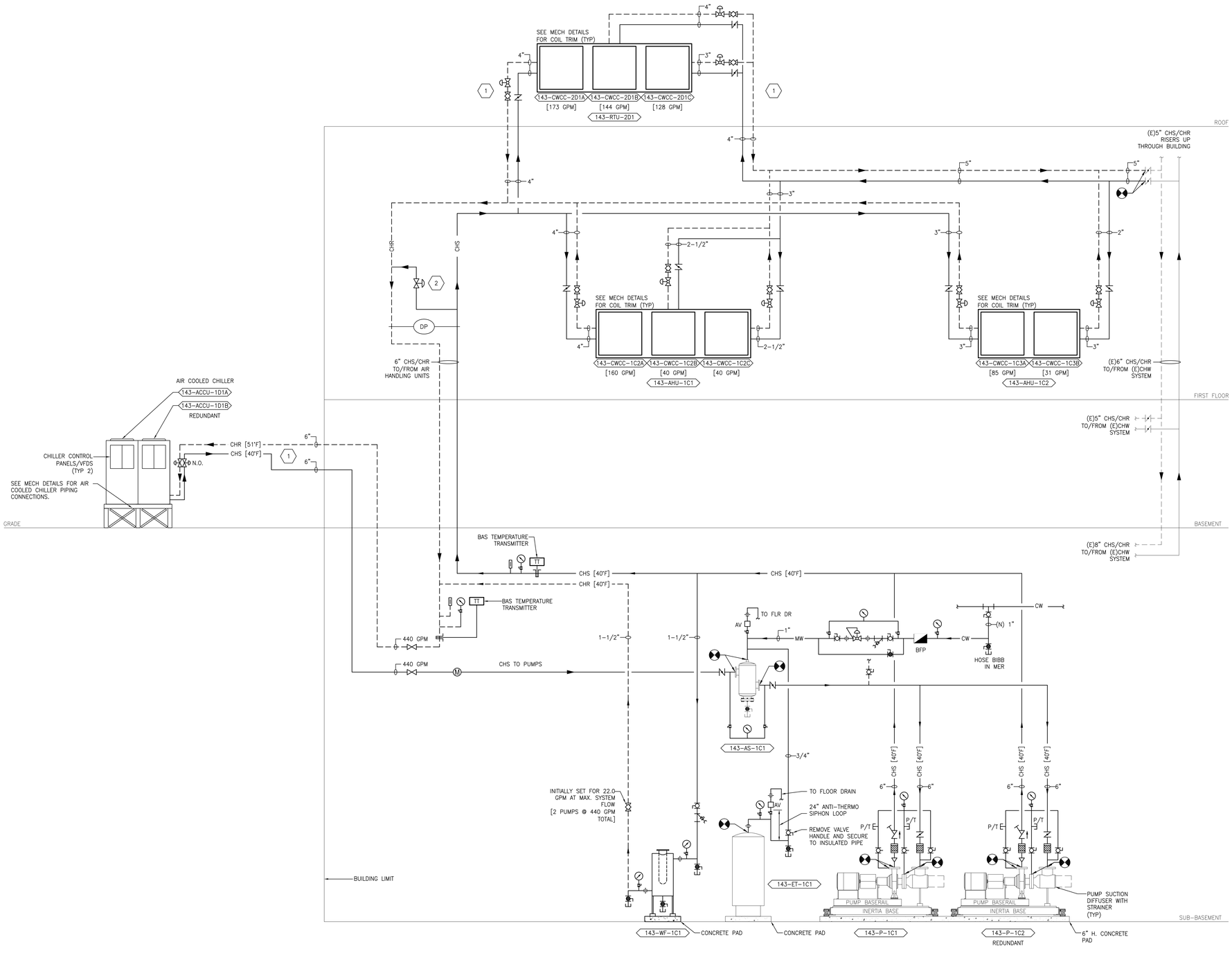
three inches = one foot
one and one half inches = one foot
one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot

## GENERAL SHEET NOTES:

1. SEE DRAWING M-001 FOR ABBREVIATIONS, NOTES, & SYMBOLS.
2. DRAWING IS DIAGRAMMATIC. REFER TO MP100 SERIES FOR ACTUAL DUCT LAYOUT.
3. FLOW DIAGRAM DOES NOT SHOW ALL DEVICES AND ACCESSORIES REQUIRED. PROVIDE AS SHOWN ON DETAILS, AND AS SPECIFIED.

## SHEET KEYNOTES:

1. PROVIDE ELECTRIC HEAT TRACE ON EXTERIOR CHILLER WATER PIPING. COORDINATE INSTALLATION AND POWER WITH ELECTRICAL CONTRACTOR.
2. PROVIDE MINIMUM FLOW BYPASS VALVE SIZED AT 147 GPM PER BASIS OF DESIGN PUMP SELECTION.




1 CHILLED WATER FLOW DIAGRAM
SCALE: N.T.S.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL CHILLED WATER FLOW DIAGRAM |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-605 |

VA FORM 08-6231

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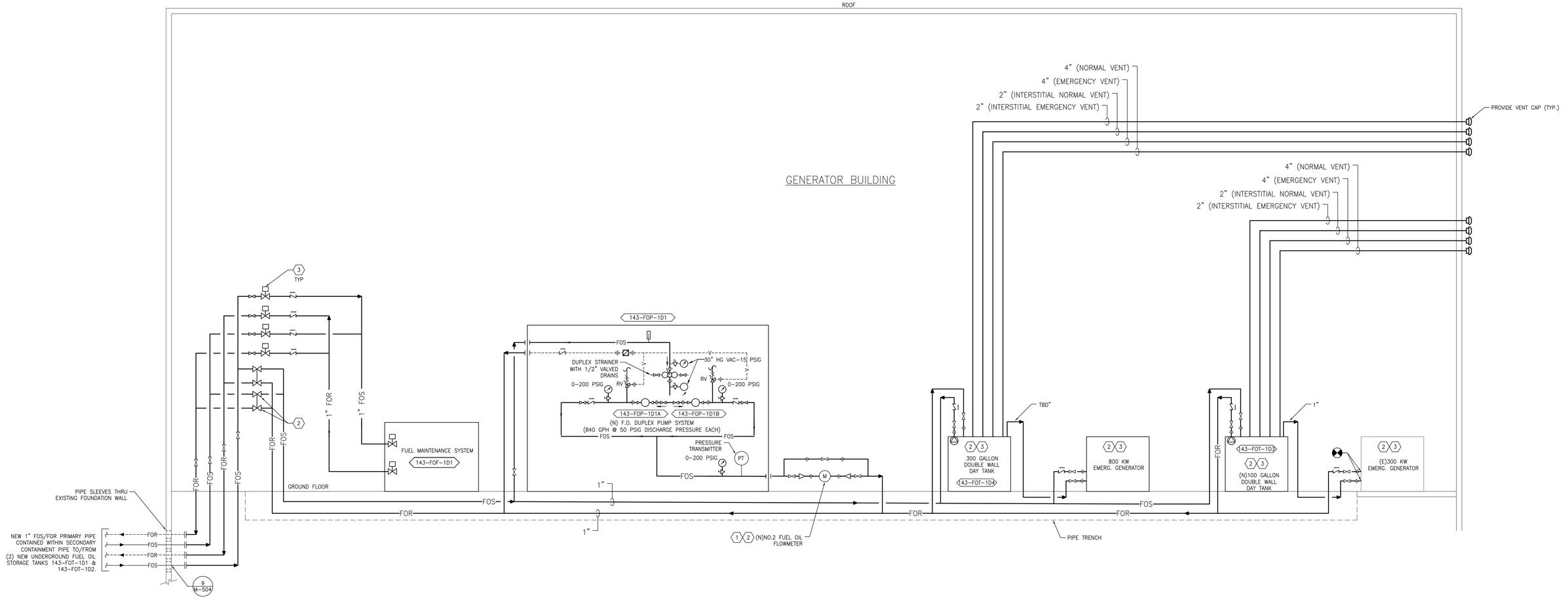
## GENERAL SHEET NOTES:

1. REFER TO DRAWING M-001 FOR MECHANICAL SYMBOLS, ABBREVIATIONS, AND GENERAL NOTES.

## SHEET KEYNOTES:

1. ALL FUEL OIL PIPING NOT LOCATED INSIDE A PIPE TRENCH MUST BE DOUBLE-WALL WITH LEAK MONITORING.
   BASIS OF DESIGN:
   PIPING: TRICON STEEL CON PLUS.
   LEAK MONITORING: PREFERRED-MFG MODEL HD-A2C.
2. VALVES WITH LINKED HANDLES.
3. SOLENOID CONTROL VALVES CONTROLLED BY FOF-1




1 NO. 2 FUEL OIL FLOW DIAGRAM – PUMPS – GENERATORS
SCALE: NTS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

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| | | |

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL NO. 2 FUEL OIL FLOW DIAGRAM - GENERATORS |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-606 |

VA FORM 08-6231

three inches = one foot
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one inch = one foot
three quarters inch = one foot
one half inch = one foot
three eighths inch = one foot
one quarter inch = one foot
one eighth inch = one foot
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## AIR FILTER SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | MERV RATING | CARTRIDGES | | | AIR FLOW | APD | | HOUSING TYPE | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | QUANTITY # | SIZE W x H x D IN | ARRANGEMENT | CFM | INITIAL IN | CHANGEOVER IN | | | |
| 143-PF-2D1A | OR ROOF | 143-RTU-2D1 | PREFILTER | 8 | 9<br>3<br>3 | 24 x 24 x 2<br>24 x 12 x 2<br>12 x 24 x 2 | BY MFG | 22000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-PF-2D1B | OR ROOF | 143-RTU-2D1 | PREFILTER | 11 | 9<br>3<br>3 | 24 x 24 x 4<br>24 x 12 x 4<br>12 x 24 x 4 | BY MFG | 22000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-AF-2D1A | OR ROOF | 143-RTU-2D1 | AFTER PREFILTER | 8 | 9<br>3<br>3 | 24 x 24 x 2<br>24 x 12 x 2<br>12 x 24 x 2 | BY MFG | 22000 | 0.33 | 0.66 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-AF-2D1B | OR ROOF | 143-RTU-2D1 | AFTER FILTER | 14 | 9<br>3<br>3 | 24 x 24 x 4<br>24 x 12 x 4<br>12 x 24 x 4 | BY MFG | 22000 | 0.33 | 0.66 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-PF-1C2A | MECH ROOM | 143-AHU-1C1 | PREFILTER | 8 | 12 | 24 x 20 x 2 | 4 WIDE BY 3 HIGH | 16000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-PF-1C2B | MECH ROOM | 143-AHU-1C1 | PREFILTER | 11 | 12 | 24 x 20 x 4 | 4 WIDE BY 3 HIGH | 16000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-AF-1C2 | MECH ROOM | 143-AHU-1C1 | AFTER FILTER | 14 | 12 | 24 x 20 x 12 | 4 WIDE BY 3 HIGH | 16000 | 0.33 | 0.66 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-PF-1C3A | MECH ROOM | 143-AHU-1C2 | PREFILTER | 8 | 6 | 24 x 20 x 2 | 3 WIDE BY 2 HIGH | 8000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-PF-1C3B | MECH ROOM | 143-AHU-1C2 | PREFILTER | 11 | 6 | 24 x 20 x 4 | 3 WIDE BY 2 HIGH | 8000 | 0.22 | 0.44 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-AF-1C3 | MECH ROOM | 143-AHU-1C2 | AFTER FILTER | 14 | 6 | 24 x 20 x 12 | 3 WIDE BY 2 HIGH | 8000 | 0.33 | 0.66 | SIDE ACCESS | - | AHU MANUFACTURER |
| 143-FF-1D1 | GROUND FLR | 1D-123 HYBRID OR | FINAL FILTER | 17 | 2<br>2 | 24 x 12 x 12<br>24 x 24 x 12 | 2 WIDE BY 1 HIGH<br>2 WIDE BY 1 HIGH | 4905 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D2 | GROUND FLR | 1D-125 OR-2 | FINAL FILTER | 17 | 1 / 1 | 12 x 24 x 12<br>30 x 24 x 12 | 2 WIDE BY 1 HIGH | 3100 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D3 | GROUND FLR | 1D-126 OR-3 | FINAL FILTER | 17 | 1 / 1 | 12 x 24 x 12<br>30 x 24 x 12 | 2 WIDE BY 1 HIGH | 3210 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D4 | GROUND FLR | C1-18 CORRIDOR | FINAL FILTER | 17 | 1 | 24 x 24 x 12 | 1 WIDE BY 1 HIGH | 1250 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C5 | GROUND FLR | 1C-120 CLEAN CORE | FINAL FILTER | 17 | 1 / 1 | 12 x 12 x 12<br>24 x 24 x 12 | 2 WIDE BY 1 HIGH | 2635 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D6 | GROUND FLR | 1D-120 IT | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 115 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D7 | GROUND FLR | 1D-121 EQUIPMENT | FINAL FILTER | 17 | 1 | 24 x 24 x 12 | 1 WIDE BY 1 HIGH | 1345 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D8 | GROUND FLR | 1D-122 PUMP | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 365 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D9 | GROUND FLR | C1-19 CORRIDOR | FINAL FILTER | 17 | 1 | 24 x 12 x 12 | 1 WIDE BY 1 HIGH | 575 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D10 | GROUND FLR | 1D-124 CONTROL ROOM | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 220 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1D11 | GROUND FLR | C1-9 CORRIDOR | FINAL FILTER | 17 | 1 | 24 x 24 x 12 | 1 WIDE BY 1 HIGH | 1640 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-2D12 | ROOF | 143-D2-RTU EPIDEMIC EA | FINAL FILTER | 17 | 12 | 24 x 24 x 12 | 3 WIDE BY 4 HIGH | 2000 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: BF1-4H-3W-2GGF-304-D3 |
| 143-FF-1C13 | GROUND FLR | 1C-128 OR-4 | FINAL FILTER | 17 | 2 | 24 x 24 x 12 | 2 WIDE BY 1 HIGH | 3275 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C14 | GROUND FLR | 1C-126 OR-5 | FINAL FILTER | 17 | 2 | 24 x 24 x 12 | 2 WIDE BY 1 HIGH | 3440 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C15 | GROUND FLR | 1C-130 OR-6 | FINAL FILTER | 17 | 2 | 24 x 24 x 12 | 2 WIDE BY 1 HIGH | 3050 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C16 | GROUND FLR | 1C-129 OR-7 | FINAL FILTER | 17 | 2 | 24 x 24 x 12 | 2 WIDE BY 1 HIGH | 3045 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C17 | GROUND FLR | C1-17 CORRIDOR | FINAL FILTER | 17 | 1 | 24 x 12 x 12 | 1 WIDE BY 1 HIGH | 860 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C18 | GROUND FLR | 1C-131 EQUIPMENT STORAGE | FINAL FILTER | 17 | 1 | 24 x 12 x 12 | 1 WIDE BY 1 HIGH | 800 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C19 | GROUND FLR | C1-16 CORRIDOR | FINAL FILTER | 17 | 1 | 24 x 12 x 12 | 1 WIDE BY 1 HIGH | 860 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C20 | GROUND FLR | 1C-127 ANESTHESIA WORKRM | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 265 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C21 | GROUND FLR | 1C-X ELECTRICAL | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 50 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C22 | GROUND FLR | 1C-125 ORTHO STORAGE | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 90 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C23 | GROUND FLR | 1C-121 IT | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 50 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C24 | GROUND FLR | 1C-123 LINEN STORAGE | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 410 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-1C25 | GROUND FLR | C1-10 CORRIDOR | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 410 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-2C26 | ROOF | 143-C1-AHU EPIDEMIC EA | FINAL FILTER | 17 | 9 | 24 x 24 x 12 | 3 WIDE BY 3 HIGH | 16000 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: BF1-3H-3W-2GGF-304-D3 |
| 143-FF-1C27 | GROUND FLR | 1C-115 PACU | FINAL FILTER | 17 | 1 | 24 x 24 x 12 | 1 WIDE BY 1 HIGH | 1865 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |
| 143-FF-2C28 | ROOF | 143-C2-AHU EPIDEMIC EA | FINAL FILTER | 17 | 4 | 24 x 24 x 12 | 2 WIDE BY 2 HIGH | 8000 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: BF1-2H-2W-2GGF-304-D3 |
| 143-FF-2C29 | ROOF | | | | | | | | | | | | |
| 143-FF-1C30 | GROUND FLR | 1C-144 SOILED | FINAL FILTER | 17 | 1 | 12 x 12 x 12 | 1 WIDE BY 1 HIGH | 135 | 0.48 | 0.96 | SIDE ACCESS | - | AAF: ALPHA 2000 |

**NOTE:**
1. ALL FILTERS SHALL BE REPLACED *AFTER* VA ACCEPTANCE AND PRIOR TO OCCUPANCY.

## HVAC DESIGN DATA

| DESIGN CONDITIONS | SUMMER | | | | | WINTER | | | | | LOWEST AVERAGE ANNUAL DEWPOINT | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | TEMP | | WET BULB TEMP | | % HUMIDITY | TEMP | | DEWPOINT TEMP | | % HUMIDITY | | |
| | °F | [°C] | °F | [°C] | | °F | [°C] | °F | [°C] | | °F | [°C] |
| OUTDOOR DESIGN CONDITIONS | 96.1 | [ 36 ] | 72.8 | [ 23 ] | 46 | 8.4 | [ -13 ] | -3.4 | [ -20 ] | NA | -5 | [ -21 ] |
| INDOOR AREA DESIGN CONDITIONS | | | | | | | | | | | | |
| PRE-OPERATIVE HOLDING | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| GENERAL OPERATING ROOM | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 75 | [ 24 ] | 56.5 | [ 14 ] | 30 | | |
| HYBRID OR PUMP ROOM | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 75 | [ 24 ] | 56.5 | [ 14 ] | 30 | | |
| ROBOTICS OPERATING ROOM | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 75 | [ 24 ] | 56.5 | [ 14 ] | 30 | | |
| HYBRID OPERATING ROOM | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 75 | [ 24 ] | 56.5 | [ 14 ] | 30 | | |
| HYBRID OR CONTROL ROOM | 75 | N/A | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| STERILE STORAGE | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 72 | [ 22 ] | 54.4 | [ 12 ] | 30 | | |
| PHASE 1 RECOVERY PACU PATIENT BAY | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| AIRBORNE INFECTION ISOLATION (AII) PACU ROOM | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| SURGICAL SOILED UTILITY | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | | |
| CLEAN CORE | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| GAS CYLINDER STORAGE ROOM | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | | |
| SEMI-RESTRICTED CORRIDOR | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| CORRIDORS | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| LOCKER ROOM (WITHOUT TOILETS) | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| LOUNGE | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| MEDICATION ROOM | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| OFFICES | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| TOILETS - PUBLIC (INTERIOR) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | | |
| WAITING ROOM | 75 | [ 24 ] | 65.3 | [ 19 ] | 60 | 70 | [ 21 ] | 52.9 | [ 12 ] | 30 | | |
| CLEAN STORAGE ROOM | 66 | [ 19 ] | 57.5 | [ 14 ] | 60 | 72 | [ 22 ] | 54.4 | [ 12 ] | 30 | | |
| ELECTRICAL CLOSET WITHOUT HEAT GAIN | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | | |
| ELECTRICAL CLOSET WITH HEAT GAIN | 86 | [ 30 ] | N/A | N/A | N/A | 40 | [ 4 ] | N/A | N/A | N/A | | |
| HOUSEKEEPING AID CLOSET (HAC) | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | N/A | | |
| AIR HANDLING EQUIPMENT ROOMS | 84 | [ 29 ] | N/A | N/A | N/A | 50 | [ 10 ] | N/A | N/A | N/A | | |
| NURSE STATION | 72 | [ 22 ] | 62.7 | [ 17 ] | 60 | 82 | [ 28 ] | 61.5 | [ 16 ] | 30 | | |
| TELECOMMUNICATIONS ROOM | 80.6 | [ 27 ] | 70.2 | [ 21 ] | 60 | 64 | [ 18 ] | 48.6 | [ 9 ] | 30 | | |

## CLEAN STEAM GENERATOR SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | TYPE | SOURCE OF MAKEUP | WATER | | PRODUCE D STEAM | PRODUCE D STEAM | STEAM | | CONTROL VALVE | TRAP | | | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | FLOW GPM | EWT °F | LBS/HR | PSIG | ENT PSIG | ENT PSIG | LBS/HR | TRAP # | # TRAPS | CAPACITY LBS/HR | | |
| 143-CSG-2D1 | ROOF | 143-RTU-1D2 | HUMIDIFICATION | STEAM-STEAM | RO | 0.6 | 50 | 300 | 15 | 15 | 10 | 185 | - | - | 370 | 120V/1, 3 AMPS | DRISTEEM, 50C |
| 143-CSG-1C1 | MECH RM | 143-AHU-1C1 | HUMIDIFICATION | STEAM-STEAM | RO | 0.6 | 50 | 300 | 15 | 15 | 10 | 202 | - | - | 404 | 120V/1, 3 AMPS | DRISTEEM, 50C |
| 143-CSG-1C2 | MECH RM | 143-AHU-1C2 | HUMIDIFICATION | STEAM-STEAM | RO | 0.6 | 50 | 300 | 15 | 15 | 10 | 101 | - | - | 202 | 120V/1, 3 AMPS | DRISTEEM, 50C |

## STEAM HEATING COIL SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | APPLICATION | AIR FLOW | MAX FACE VELOCITY | APD | TEMPERATURES | | TOTAL MIN CAPACITY | STEAM | | | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | EAT | LAT | | ENT CONT VALVE | ENT COIL | FLOW | STEAM TRAP | |
| | | | | | CFM | FPM | IN WG | °F | °F | MBH | PSIG | PSIG | LBS/HR | MARK | LBS/HR | |
| 143-SHC-2D1 | ROOF | SURGERY SUITE | 143-RTU-2D1 | OA PREHEAT | 22,000 | 444 | 0.03 | 8.4 | 57.4 | 1,164 | 15 | 10 | 1,216 | - | 2433 | ---- |
| 143-SHC-2C2 | ROOF | SURGERY SUITE | 143-AHU-1C1 | OA PREHEAT | 16,000 | 416 | 0.03 | 8.4 | 58.3 | 861 | 15 | 10 | 899 | - | 1798 | ---- |
| 143-SHC-2C3 | ROOF | SURGERY SUITE | 143-AHU-1C2 | OA PREHEAT | 8,000 | 383 | 0.02 | 8.4 | 60.8 | 452 | 15 | 10 | 473 | - | 946 | ---- |

## PUMP SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | TYPE | CIRCULATING FLUID | | | | | MIN % EFF | ELECTRICAL MOTOR | | | | | | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | FLUID | FLOW GPM | HEAD FT | NPSH AVAILAB FT | TEMPERATURE °F | | NOMINAL POWER HP | PHASE | VOLT | MAX RPM | SPEED CONTROL | BASIS OF DESIGN OR APPROVED EQUAL | |
| 143-P -1C1 | SUB-BASEMENT MECHANICAL ROOM SD-100 | BUILDING 143 | CHILLED WATER | END SUCTION | WATER | 750 | 70 | 10.1 | 42 | 82.1 | 20 | 3 | 480 | 1800 | VFD | BELL & GOSSETT; e-1510-4BD | EXISTING TO REMAN |
| 143-P-1C2 | SUB-BASEMENT MECHANICAL ROOM SD-100 | BUILDING 143 | CHILLED WATER | END SUCTION | WATER | 750 | 70 | 10.1 | 42 | 82.1 | 20 | 3 | 480 | 1800 | VFD | BELL & GOSSETT; e-1510-4BD | EXISTING TO REMAIN; REDUNDANT |

## AIR HANDLING UNIT SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | TYPE | MAX AIR FLOW | MIN OA AIR FLOW | RETURN AIR FLOW | SUPPLY FAN MARK | RETURN FAN MARK | PREFILTER MARK | AFTER FILTER MARK | FINAL FILTER MARK | OA PREHEAT COIL MARK | COOLING COIL MARK | HUMIDIFIER MARK | REMARKS | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | CFM | CFM | CFM | | | | | | | | | | |
| 143-RTU-2D1 | ROOF | SURGERY SUITE | ROOF TOP | 22,000 | 4,685 | 17,315 | 143-SF-1A THRU D | 143-RF-1 A THRU D | 143-PF-1A/B | 143-AF-1A/B | AT TERMINAL UNITS | 143-SHC-1 | 143-CWCC-1A-C | 143-SH-1 | SEE NOTES 1-3 | YORK XTO-96x96 |
| 143-AHU-1C1 | MECH ROOM | SURGERY SUITE | COMMERCIAL | 16,000 | 3,400 | 12,600 | 143-SF-2A THRU D | 143-RF-2 | 143-PF-2A/B | 143-AF-2 | AT TERMINAL UNITS | 143-SHC-2 | 143-CWCC-2A-C | 143-SH-2 | SEE NOTES 1-3 | YORK XTI-84x90 |
| 143-AHU-1C2 | MECH ROOM | SURGERY SUITE | COMMERCIAL | 8,000 | 2,240 | 5,760 | 143-SF-3A THRU D | 143-RF-3 | 143-PF-3A/B | 143-AF-3 | AT TERMINAL UNITS | 143-SHC-3 | 143-CWCC-3A-B | 143-SH-3 | SEE NOTES 1-3 | YORK XTI-60x72 |

**NOTE:**
1. PROVIDE WITH A SINGLE POINT POWER CONNECTION. UNIT SHALL BE PROVIDED WITH A FACTORY DISCONNECT SWITCH.
2. SEE DRAWINGS P1-M-121 FOR GENERAL LAYOUT AND OTHER UNIT REQUIREMENTS.
3. PROVIDE WITH SUPPLY AIR DISCHARGE SMOKE DAMPERS.
4. DURING PANDEMIC MODE, MIN OA MATCHES MAX AIR FLOW. ALL RETURN AIR WILL BE EXHAUSTED.

## CHILLED WATER COOLING COIL SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | AIR FLOW | MAX FACE VELOCITY | APD | EAT | | LAT | | TOTAL CAPACITY | SENSIBLE CAPACITY | CHILLED WATER | | | | REMARKS | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | Db | Wb | Db | Wb | | | FLOW | EWT | LWT | WPD | | |
| | | | | CFM | FPM | IN WG | °F | °F | °F | °F | MBH | MBH | GPM | °F | °F | FT | | |
| 143-CWCC-2D1A | ROOF | SURGERY SUITE | 143-RTU-2D1 | 22,000 | 444 | 0.59 | 74.6 | 63.2 | 50.5 | 49.2 | 870 | 586 | 172.7 | 40 | 50 | 9.1 | (N)CHILLER | YORK |
| 143-CWCC-2D1B | ROOF | SURGERY SUITE | 143-RTU-2D1 | 22,000 | 449 | 0.44 | 96.1 | 75.4 | 66.3 | 63 | 1,013 | 732 | 143.7 | 43 | 57 | 1.7 | (E)CHILLERS | YORK |
| 143-CWCC-2D1C | ROOF | SURGERY SUITE | 143-RTU-2D1 | 22,000 | 444 | 0.26 | 103 | 76.1 | 71.6 | 65.4 | 904 | 769 | 128.3 | 43 | 57 | 1.3 | (E)CHILLERS | YORK |
| 143-CWCC-1C2A | MECH RM | SURGERY SUITE | 143-AHU-1C1 | 16,000 | 407 | 0.52 | 78.4 | 66.5 | 50.7 | 49.4 | 801 | 492 | 160 | 40 | 50 | 9.6 | (N)CHILLER | YORK |
| 143-CWCC-1C2B | MECH RM | SURGERY SUITE | 143-AHU-1C1 | 16,000 | 407 | 0.22 | 96.1 | 75.4 | 71.8 | 66.6 | 540 | 434 | 40 | 43 | 57 | 3.0 | (E)CHILLERS | YORK |
| 143-CWCC-1C2C | MECH RM | SURGERY SUITE | 143-AHU-1C1 | 16,000 | 407 | 0.17 | 103 | 76.1 | 83.8 | 70.8 | 341 | 492 | 40 | 43 | 57 | 1.1 | (E)CHILLERS | YORK |
| 143-CWCC-1C3A | MECH RM | SURGERY SUITE | 143-AHU-1C2 | 8,000 | 384 | 0.41 | 82.8 | 68.3 | 51.6 | 50.7 | 427 | 277 | 84.9 | 40 | 50 | 4.6 | (N)CHILLER | YORK |
| 143-CWCC-1C3B | MECH RM | SURGERY SUITE | 143-AHU-1C2 | 8,000 | 384 | 0.22 | 103 | 76.1 | 78.9 | 69.4 | 214 | 214 | 30.5 | 43 | 57 | 2.4 | (E)CHILLERS | YORK |

**NOTE:**
1. THE COOLING COIL FIN SPACING SHALL NOT EXCEED 132 FINS PER FOOT [400 FINS PER METER].

## AIR COOLED CHILLER SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | TYPE | CAPACITY | # OF COMP | MAX kW/TON | PROPYLENE GLYCOL | MAX IPLV (kW/TON) | EVAPORATOR | | | | | CONDENSER | ELECTRICAL | | | | | | | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | | | | | FLOW | EWT | LWT | MAX WPD | FOULING FACTOR | AMBIENT OA TEMP | COMPRESSOR MOTOR | | | CONDENSER FAN MOTORS | | | | |
| | | | | | | | | | | | | | | | kW | PHASE | VOLT | # FANS | POWER kW | PHASE | VOLT | |
| | | | | TONS | | | % | | GPM | °F | °F | FT | | °F | | | | | | | | |
| 143-ACCU-1D1A | BLDG 143 | SURG SUITE | SCROLL | 204 | 6 | 1.17 | 0% | 18.53 | 440 | 40 | 51 | 20 | 0.0001 | 95 | 217.5 | 3 | 460 | 12 | 20.16 | 3 | 460 | YORK; YLAA0230HJ46XI |
| 143-ACCU-1D1B | BLDG 143 | SURG SUITE | SCROLL | 204 | 6 | 1.17 | 0% | 18.53 | 440 | 40 | 51 | 20 | 0.0001 | 95 | 217.5 | 3 | 460 | 12 | 20.16 | 3 | 460 | YORK; YLAA0230HJ46XI |

**NOTES**
1. PROVIDE WITH FREEZE PROTECTION.
2. SEE SPECIFICATIONS FOR OTHER APPLICABLE ENGINEERING REQUIREMENTS.
3. "MAX KW/TON" AND "MIN COP" SPECIFIED ARE AT DESIGN CONDITIONS INDICATED. KW/TON INCLUDES CONDENSER FANS.

## STEAM HUMIDIFER SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SEVICE | HUMIDIFIER TYPE | AIR FLOW | EAT | | LAT | | SOURCE | STEAM | | TRAP | | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | | | | Db | Wb | DEWPOINT | DEWPOINT | | PRESS ENT VALVE | PRESS ENT HUMIDIFIER | FLOW | MARK | CAPACITY | |
| | | | | CFM | °F | °F | °F | °F | | PSIG | PSIG | LBS/HR | | LBS/HR | |
| 143-SH-2D1 | ROOF | 143-RTU-2D1 | UNIT-MOUNTED DISPERSION TUBE | 22,000 | 55 | 42.1 | 25.1 | 36.72 | CLEAN STEAM | 15 | 10 | 185 | - | 370 | YORK |
| 143-SH-1C1 | MECH ROOM | 143-AHU-1C1 | UNIT-MOUNTED DISPERSION TUBE | 16,000 | 55 | 40 | 16.6 | 36.72 | CLEAN STEAM | 15 | 10 | 202 | - | 404 | YORK |
| 143-SH-1C2 | MECH ROOM | 143-AHU-1C2 | UNIT-MOUNTED DISPERSION TUBE | 8,000 | 55 | 42.1 | 25.1 | 41.4 | CLEAN STEAM | 15 | 10 | 101 | - | 202 | YORK |

**NOTES:**

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

MILLER-REMICK
A Service-Disabled Veteran-Owned Small Business
(856) 429-6000 ★ Miller-Remick.com
M-R PROJECT NO. DVPA0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANCIAL SCHEDULES | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: M-701 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

VA FORM 08-6231

P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\M-701.dwg 12-29-25 01:58:40 PM jromano

## VARIABLE AIR VOLUME TERMINAL UNIT SCHEDULE

| MARK | ROOM/AREA SERVED PER EACH BLDG | SYSTEM AIR HANDLING | TYPE | UNIT SIZE | AIR FLOW OCCUPIED CFM | AIR FLOW UNOCCUPIED CFM | STATIC MIN. IN. WC | HOT WATER REHEAT COIL AIR CFM | HOT WATER REHEAT COIL CAP. MBH | HOT WATER REHEAT COIL EAT COOLING °F | HOT WATER REHEAT COIL EAT HEATING °F | HOT WATER REHEAT COIL EWT °F | HOT WATER REHEAT COIL LAT °F | HOT WATER REHEAT COIL APD IN. WC | HOT WATER REHEAT COIL LWT °F | HOT WATER REHEAT COIL WATER GPM | HOT WATER REHEAT COIL WPD FT. WC | ROWS | BASIS OF DESIGN OR APPROVED EQUAL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-TU-1D1 | C1-18 CORRIDOR | 143-RTU-2D1 | SAV | 12 | 1205 | 1202 | - | 1205 | 22.2 | 55.0 | 68.0 | 190.0 | 85.0 | 0.05 | 175.0 | 3.1 | 0.30 | 1 | PRICE SDV | - |
| 143-TU-1D2 | C1-18 CORRIDOR | 143-RTU-2D1 | RAV | 14 | 1670 | 1670 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D3 | 1C-120 CLEAN CORE | 143-RTU-2D1 | SAV | 16 | 2635 | 2597 | - | 2635 | 48.6 | 55.0 | 68.0 | 190.0 | 85.0 | 0.05 | 175.0 | 6.7 | 0.23 | 2 | PRICE SDV | - |
| 143-TU-1D4 | 1C-120 CLEAN CORE | 143-RTU-2D1 | RAV | 16 | 3285 | 3285 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D5 | 1D-120 IT | 143-RTU-2D1 | SAV | 4 | 115 | 115 | - | - | - | - | - | - | - | - | - | - | - | - | PRICE SDV | - |
| 143-TU-1D6 | 1D-121 EQUIP | 143-RTU-2D1 | SAV | 12 | 1345 | 108 | - | - | - | - | - | - | - | - | - | - | - | - | PRICE SDV | - |
| 143-TU-1D7 | 1D-121 EQUIP | 143-RTU-2D1 | RAV | 12 | 1145 | 1145 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D8 | 1D-122 PUMP | 143-RTU-2D1 | SAV | 6 | 365 | 39 | - | 365 | 6.7 | 55.0 | 68.0 | 190.0 | 85.0 | 0.01 | 175.0 | 0.9 | 0.70 | 1 | PRICE SDV | - |
| 143-TU-1D9 | 1D-122 PUMP | 143-RTU-2D1 | RAV | 6 | 265 | 265 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D10 | C1-19 CORRIDOR | 143-RTU-2D1 | SAV | 8 | 575 | 575 | - | 575 | 16.8 | 55.0 | 68.0 | 190.0 | 95.0 | 0.13 | 175.0 | 2.3 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1D11 | C1-19 CORRIDOR | 143-RTU-2D1 | RAV | 8 | 475 | 475 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D12 | 1D-124 CONTROL | 143-RTU-2D1 | SAV | 6 | 225 | 225 | - | 225 | 4.2 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.6 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1D13 | 1D-124 CONTROL | 143-RTU-2D1 | RAV | 8 | 375 | 375 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1D14 | C1-9 CORRIDOR | 143-RTU-2D1 | SAV | 12 | 1640 | 1472 | - | 1640 | 42.7 | 55.0 | 68.0 | 190.0 | 92.0 | 0.02 | 175.0 | 5.9 | 0.64 | 1 | PRICE SDV | - |
| 143-TU-1D15 | C1-9 CORRIDOR | 143-RTU-2D1 | RAV | 16 | 2190 | 2190 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C1 | 1C-127 ANESTHESIA WORK RM | 143-AHU-1C1 | SAV | 6 | 300 | 295 | - | 300 | 5.5 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.8 | 0.30 | 1 | PRICE SDV | - |
| 143-TU-1C2 | 1C-127 ANESTHESIA WORK RM | 143-AHU-1C1 | RAV | 6 | 300 | 295 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C3 | 1C-X EXIST ELEC | 143-AHU-1C1 | SAV | 4 | 50 | 50 | - | - | - | - | - | - | - | - | - | - | - | - | PRICE SDV | - |
| 143-TU-1C4 | 1C-125 ORTHO STORAGE | 143-AHU-1C1 | SAV | 4 | 110 | 110 | - | 110 | 2.0 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.3 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1C5 | 1C-125 ORTHO STORAGE | 143-AHU-1C1 | RAV | 6 | 260 | 260 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C6 | 1C-121 IT | 143-AHU-1C1 | SAV | 4 | 50 | 50 | - | - | - | - | - | - | - | - | - | - | - | - | PRICE SDV | - |
| 143-TU-1C7 | 1C-123 LINEN STORAGE | 143-AHU-1C1 | SAV | 4 | 50 | 20 | - | 50 | 0.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.1 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C8 | C1-10 CORRIDOR | 143-AHU-1C1 | SAV | 10 | 735 | 735 | - | 735 | 13.6 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 1.9 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1C9 | C1-10 CORRIDOR | 143-AHU-1C1 | RAV | 12 | 1095 | 1095 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C10 | C1-17 CORRIDOR | 143-AHU-1C1 | SAV | 4 | 210 | 208 | - | 210 | 3.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.5 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C11 | C1-17 CORRIDOR | 143-AHU-1C1 | RAV | 6 | 250 | 250 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C12 | 1C-131 EQUIPMENT STORAGE | 143-AHU-1C1 | SAV | 6 | 350 | 208 | - | 350 | 6.5 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.9 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C13 | 1C-131 EQUIPMENT STORAGE | 143-AHU-1C1 | RAV | 8 | 365 | 365 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C14 | C1-16 CORRIDOR | 143-AHU-1C1 | SAV | 6 | 460 | 459 | - | 460 | 8.5 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 1.2 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C15 | C1-16 CORRIDOR | 143-AHU-1C1 | RAV | 8 | 530 | 530 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C16 | 1C-142 SOILED | 143-AHU-1C1 | SAV | 4 | 135 | 135 | - | 135 | 2.5 | 55.0 | 68.0 | 190.0 | 85.0 | 0.10 | 175.0 | 0.3 | 0.64 | 1 | PRICE SDV | - |

**NOTES:**

## HOT WATER HEATING COIL SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | AIR FLOW CFM | FACE SIZE WIDTH IN | FACE SIZE HEIGHT IN | APD IN WG | TEMPERATURES EAT °F | TEMPERATURES LAT °F | TOTAL MIN CAPACITY MBH | HOT WATER FLOW GPM | HOT WATER EWT °F | HOT WATER LWT °F | HOT WATER WPD FT | ROWS | % GLYCOL | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-HWC-1D1 | SURGICAL SUITE | 1D-123 HYBRD OR | 143-SVV-1010 | 4905 | 42 | 30 | 0.31 | 50 | 80 | 162 | 21.6 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1D2 | SURGICAL SUITE | 1D-125 OR-2 | 143-SVV-1020 | 3100 | 32 | 24 | 0.33 | 50 | 80 | 102 | 13.6 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1D3 | SURGICAL SUITE | 1-D126 OR-3 | 143-SVV-1030 | 3210 | 32 | 24 | 0.36 | 50 | 80 | 106 | 14.1 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1C4 | SURGICAL SUITE | 1C-128 OR-4 | 143-SVV-1040 | 3275 | 32 | 24 | 0.21 | 50 | 80 | 108 | 14.9 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1C5 | SURGICAL SUITE | 1C-126 OR-5 | 143-SVV-1050 | 3440 | 32 | 24 | 0.23 | 50 | 80 | 114 | 15.8 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1C6 | SURGICAL SUITE | 1C-130 OR-6 | 143-SVV-1060 | 3050 | 30 | 24 | 0.2 | 50 | 80 | 101 | 14.0 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1C7 | SURGICAL SUITE | 1C-129 OR-7 | 143-SVV-1070 | 3045 | 30 | 24 | 0.2 | 50 | 80 | 100 | 14.0 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |
| 143-HWC-1C8 | SURGICAL SUITE | 1C-116 ISOLATION | 143-SVV-1070 | 210 | 12 | 6 | 0.1 | 55 | 80 | 6 | 0.8 | 190 | 175 | 2.0 | 2.0 | 0 | - | AEROFIN W |

**NOTES:**

## VARIABLE AIR VOLUME TERMINAL UNIT SCHEDULE (CONTINUED)

| MARK | ROOM/AREA SERVED PER EACH BLDG | SYSTEM AIR HANDLING | TYPE | UNIT SIZE | AIR FLOW OCCUPIED CFM | AIR FLOW UNOCCUPIED CFM | STATIC MIN. IN. WC | HOT WATER REHEAT COIL AIR CFM | HOT WATER REHEAT COIL CAP. MBH | HOT WATER REHEAT COIL EAT COOLING °F | HOT WATER REHEAT COIL EAT HEATING °F | HOT WATER REHEAT COIL EWT °F | HOT WATER REHEAT COIL LAT °F | HOT WATER REHEAT COIL APD IN. WC | HOT WATER REHEAT COIL LWT °F | HOT WATER REHEAT COIL WATER GPM | HOT WATER REHEAT COIL WPD FT. WC | ROWS | BASIS OF DESIGN OR APPROVED EQUAL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-TU-1C1 | 1C-124 ANESTHESIA | 143-AHU-1C2 | SAV | 4 | 185 | 149 | - | 185 | 3.4 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.5 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C2 | 1C-124 ANESTHESIA | 143-AHU-1C2 | RAV | 8 | 480 | 480 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C3 | 1C-156 NURSE/COMM CNTR | 143-AHU-1C2 | SAV | 4 | 180 | 160 | - | 180 | 3.3 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.5 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C4 | 1C-156 NURSE/COMM CNTR | 143-AHU-1C2 | RAV | 6 | 180 | 160 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C5 | 1C-155, 1C-154 OFFICES | 143-AHU-1C2 | SAV | 6 | 145 | 130 | - | 145 | 2.7 | 55.0 | 68.0 | 190.0 | 85.0 | 0.10 | 175.0 | 0.4 | 0.64 | 1 | PRICE SDV | - |
| 143-TU-1C6 | 1C-157, 1C-155, 1C-154 OFFICES | 143-AHU-1C2 | RAV | 6 | 215 | 215 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C7 | C1-15 CORRIDOR | 143-AHU-1C2 | SAV | 8 | 595 | 363 | - | 595 | 11.0 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 1.5 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C8 | C1-15 CORRIDOR | 143-AHU-1C2 | RAV | 8 | 420 | 420 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C9 | 1C-153, 1C-152, 1C-151 OFFICES | 143-AHU-1C2 | SAV | 6 | 205 | 205 | - | 205 | 3.8 | 55.0 | 68.0 | 190.0 | 85.0 | 0.10 | 175.0 | 0.5 | 0.64 | 1 | PRICE SDV | - |
| 143-TU-1C10 | 1C-153, 1C-152, 1C-151 OFFICES | 143-AHU-1C2 | RAV | 6 | 205 | 205 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C11 | 1C-150 EXISTING WAITING | 143-AHU-1C2 | SAV | 4 | 170 | 167 | - | 170 | 3.1 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.4 | 0.20 | 1 | PRICE SDV | - |
| 143-TU-1C12 | 1C-150 EXISTING WAITING | 143-AHU-1C2 | RAV | 6 | 220 | 220 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C13 | C1-3 CORRIDOR | 143-AHU-1C2 | SAV | 6 | 210 | 210 | - | 210 | 3.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.04 | 175.0 | 0.5 | 0.10 | 1 | PRICE SDV | - |
| 143-TU-1C14 | C1-3 CORRIDOR | 143-AHU-1C2 | RAV | 6 | 210 | 210 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C15 | C1-13 CORRIDOR | 143-AHU-1C2 | SAV | 4 | 110 | 110 | - | 110 | 2.0 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.3 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1C16 | C1-13 CORRIDOR | 143-AHU-1C2 | RAV | 4 | 110 | 110 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C17 | 1C-115 PACU | 143-AHU-1C2 | SAV | 12 | 1665 | 1467 | - | 1665 | 30.7 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 4.2 | 0.28 | 2 | PRICE SDV | - |
| 143-TU-1C18 | 1C-115 PACU | 143-AHU-1C2 | RAV | 14 | 1665 | 1467 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C19 | 1C-118 CLEAN | 143-AHU-1C2 | SAV | 4 | 150 | 83 | - | 150 | 2.8 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.4 | 0.20 | 1 | PRICE SDV | - |
| 143-TU-1C20 | 1C-118 CLEAN | 143-AHU-1C2 | RAV | 6 | 150 | 83 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C21 | 1C-115A NURSE | 143-AHU-1C2 | SAV | 6 | 235 | 212 | - | 235 | 4.3 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.6 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1C22 | 1C-115A NURSE | 143-AHU-1C2 | RAV | 6 | 235 | 212 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C23 | 1C-114A, 1C-114C LOUNGE/ LOCKERS | 143-AHU-1C2 | SAV | 4 | 150 | 131 | - | 150 | 2.8 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.4 | 0.20 | 1 | PRICE SDV | - |
| 143-TU-1C24 | 1C-111 EQUIPMENT | 143-AHU-1C2 | SAV | 4 | 75 | 70 | - | 75 | 1.4 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.2 | 0.28 | 1 | PRICE SDV | - |
| 143-TU-1C25 | 1C-111 EQUIPMENT | 143-AHU-1C2 | RAV | 6 | 75 | 70 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C26 | C1-11 CORRIDOR | 143-AHU-1C2 | SAV | 6 | 385 | 385 | - | 385 | 7.1 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 1.0 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C27 | C1-11 CORRIDOR | 143-AHU-1C2 | RAV | 6 | 385 | 385 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C28 | 1C-112 ALCOVE | 143-AHU-1C2 | SAV | 4 | 50 | 39 | - | 50 | 0.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.1 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C29 | 1C-112 ALCOVE | 143-AHU-1C2 | RAV | 6 | 50 | 39 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C30 | 1C-113 MEDS | 143-AHU-1C2 | SAV | 4 | 50 | 37 | - | 50 | 0.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.1 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C31 | 1C-113 MEDS | 143-AHU-1C2 | RAV | 6 | 50 | 37 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C32 | C1-12 CORRIDOR | 143-AHU-1C2 | SAV | 6 | 330 | 330 | - | 330 | 6.1 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.8 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C33 | C1-12 CORRIDOR | 143-AHU-1C2 | RAV | 6 | 330 | 330 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C34 | 1C-106 COMM CTR | 143-AHU-1C2 | SAV | 4 | 135 | 90 | - | 135 | 2.5 | 55.0 | 68.0 | 190.0 | 85.0 | 0.10 | 175.0 | 0.3 | 0.64 | 1 | PRICE SDV | - |
| 143-TU-1C35 | 1C-106 COMM CTR | 143-AHU-1C2 | RAV | 6 | 135 | 90 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C36 | 1C-107 LINEN CLOSET | 143-AHU-1C2 | SAV | 4 | 50 | 31 | - | 50 | 0.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 0.1 | 0.32 | 1 | PRICE SDV | - |
| 143-TU-1C37 | 1C-107 LINEN CLOSET | 143-AHU-1C2 | RAV | 6 | 50 | 31 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C38 | 1C-105A NURSE | 143-AHU-1C2 | SAV | 4 | 155 | 153 | - | 155 | 2.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.4 | 0.20 | 1 | PRICE SDV | - |
| 143-TU-1C39 | 1C-105A NURSE | 143-AHU-1C2 | RAV | 6 | 155 | 153 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C40 | 1C-105 PRE-OP | 143-AHU-1C2 | SAV | 12 | 1375 | 1279 | - | 1375 | 25.4 | 55.0 | 68.0 | 190.0 | 85.0 | 0.02 | 175.0 | 3.5 | 0.19 | 2 | PRICE SDV | - |
| 143-TU-1C41 | 1C-105 PRE-OP | 143-AHU-1C2 | RAV | 10 | 845 | 845 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C42 | C1-14 CORRIDOR | 143-AHU-1C2 | SAV | 8 | 535 | 535 | - | 535 | 9.9 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 1.4 | 0.24 | 1 | PRICE SDV | - |
| 143-TU-1C43 | C1-14 CORRIDOR | 143-AHU-1C2 | RAV | 8 | 535 | 535 | 1.5 | - | - | - | - | - | - | - | - | - | - | - | PRICE LDV | - |
| 143-TU-1C44 | 1C-157 OFFICE | 143-AHU-1C2 | SAV | 4 | 70 | 60 | - | 70 | 1.3 | 55.0 | 68.0 | 190.0 | 85.0 | 0.13 | 175.0 | 0.2 | 0.28 | 1 | PRICE SDV | - |

**NOTES:**

1. -

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER

12.30.2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| **Drawing Title** MECHANICAL SCHEDULES | **Phase** BID DOCUMENTS | **Project Title** RENOVATE SURGICAL SUITE PHASE II | **VA Drawing Number** 2617 |
| **Approved by:** Jon Manninen, COR | FULLY SPRINKLERED | **Project Number** 658-19-103 | **Building Number** 143 |
| | | **Location** SALEM VA MEDICAL CENTER | **Drawing Number** M-702 |
| | | **Issue Date** 2025-12-30 / **Checked** DJ / **Drawn** JAK | |

VA FORM 08-6231

P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\M-702.dwg 12-29-25 02:01:40 PM jromano

## AIR DEVICE SCHEDULE (SUPPLY)

| MARK | TYPE | AIR FLOW MIN CFM | AIR FLOW MAX CFM | MAX APD IN WG | MOUNTING | PANEL / FRAME IN x IN | NECK SIZE IN | NC | DAMPER | FINISH | BASIS OF DESIGN OR APPROVED EQUAL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| SD-1 | SUPPLY DIFFUSER | 50 | 105 | 0.025 | CEILING | 24x24 | 6ø | N/A | NONE | WHITE | PRICE SPD | PLAQUE FACE |
| SD-2 | SUPPLY DIFFUSER | 110 | 175 | 0.025 | CEILING | 24x25 | 8ø | N/A | NONE | WHITE | PRICE SPD | PLAQUE FACE |
| SD-3 | SUPPLY DIFFUSER | 180 | 290 | 0.026 | CEILING | 24x26 | 10ø | N/A | NONE | WHITE | PRICE SPD | PLAQUE FACE |
| SD-4 | SUPPLY DIFFUSER | 295 | 395 | 0.027 | CEILING | 24x27 | 12ø | 11 | NONE | WHITE | PRICE SPD | PLAQUE FACE |
| SD-5 | SUPPLY DIFFUSER | 405 | 535 | 0.028 | CEILING | 24x28 | 14ø | 13 | NONE | WHITE | PRICE SPD | PLAQUE FACE |
| SD-6 | SUPPLY DIFFUSER | 540 | 605 | 0.028 | CEILING | 24x29 | 15ø | 13 | NONE | WHITE | PRICE SPD | PLAQUE FACE |

**NOTES:**
1. SEE FLOOR PLAN FOR THROW PATTERN.
2. SEE DETAIL FOR DAMPER IN BRANCH DUCT SERVING EACH DIFFUSER.
3. PROVIDE SQUARE TO ROUND ADAPTER.
4. INSTALL ALL AIR DEVICES IN ACCORDANCE WITH MANUFACTURERS RECOMMENDATIONS.
5. MAX NC RATING BASED ON 10dB ROOM ABSORBTION.
6. REFER TO ARCHITECTURAL REFLECTED CEILING PLAN FOR CEILING TYPE.
7. THROW VALUE IN FEET FOR 100 FPM TERMINAL VELOCITY.
8. PROVIDE FLEXIBLE DUCT WHERE SHOWN ON DRAWINGS AT DIFFUSER NECK SIZE,
9. LAMINAR DIFFUSERS TO HAVE FACE PRESSURE TAPS, FACE DOP LEAK TESTING INJECTION AND SAMPLE PORTS. 2" MEDIA

## EXPANSION TANK SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | TYPE | APPROX SYSTEM VOLUME GAL | SYSTEM TEMPERATURE MIN °F | SYSTEM TEMPERATURE MAX °F | INITIAL PRESSURE IN TANK PSIG | MAX OPERATING PRESSURE PSIG | FILL PRESSURE AT TANK: RELIEF VALVE PSIG | FILL PRESSURE AT TANK: AT TANK PSIG | MIN VOLUME TANK GAL | MIN BLADDER VOLUME GAL | PIPE SIZE TO TANK IN | COLD WATER FILL SIZE IN | REMARKS | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-ET 1C1 | MECHANICAL RM SD-100 | CHILLED WATER | VERTICAL DIAPHRAGM | - | 39 | 54 | - | - | - | - | - | - | - | - | EXISTING TO REMAIN | BELL & GOSSETT |

## AIR SEPARATOR SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | TYPE | AIR SEPARATOR SIZE IN IN | AIR SEPARATOR FLOW GPM | AIR SEPARATOR WPD FT | AIR SEPARATOR BUILT-IN STRAINER REQ'D | REMARKS | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|
| 143-AS-1C1 | MECHANICAL ROOM SD-100 | CHILLED WATER | FULL FLOW TANGENTIAL | - | - | - | YES | EXISTING TO REMAIN | BELL & GOSSETT |

## VENTURI VALVE SCHEDULE

| MARK | AREA SERVED | TYPE | SIZE IN Ø | OCCUPIED AIR FLOW CFM | UNOCCUPIED AIR FLOW CFM | APD AT MAX AIR FLOW IN WG | CONTROL TYPE | APPLICATION | BASIS OF DESIGN | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| 143-AFCV-1D1 | 1D-123 HYBRID OR | CV - 2 POSITION | 14 | 4905 | 2453 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1D2 | 1D-123 HYBRID OR | CV - 2 POSITION | 14 | 4405 | 2203 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1D3 | 1D-125 OR-2 | CV - 2 POSITION | 14 | 3100 | 1550 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1D4 | 1D-125 OR-2 | CV - 2 POSITION | 14 | 2600 | 1300 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1D5 | 1D-126 OR -3 | CV - 2 POSITION | 14 | 3210 | 1605 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1D6 | 1D-126 OR -3 | CV - 2 POSITION | 14 | 2710 | 1355 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C1 | 1C-128 OR-4 | CV - 2 POSITION | 14 | 3275 | 1638 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C2 | 1C-128 OR-4 | CV - 2 POSITION | 14 | 2775 | 1388 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C3 | 1C-126 OR-5 | CV - 2 POSITION | 14 | 3440 | 1720 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C4 | 1C-126 OR-5 | CV - 2 POSITION | 14 | 2740 | 1370 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C5 | 1C-130 OR-6 | CV - 2 POSITION | 14 | 3050 | 1525 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C6 | 1C-130 OR-6 | CV - 2 POSITION | 14 | 3050 | 1525 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C7 | 1C-129 OR-7 | CV - 2 POSITION | 14 | 3045 | 1523 | 0.6 | TWO POSITION | OPERATING ROOM SUPPLY | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C8 | 1C-129 OR-7 | CV - 2 POSITION | 14 | 3045 | 1523 | 0.6 | TWO POSITION | OPERATING ROOM RETURN | PHOENIX DUAL | 1,2,3 |
| 143-AFCV-1C1 | 1C-116 ISOLATION RM | CV - 2 POSITION | 8 | 200 | 100 | 0.6 | TWO POSITION | ISOLATION ROOM SUPPLY | PHOENIX SINGLE | 1,2,3 |
| 143-AFCV-1C2 | 1C-116 ISOLATION RM | CV - 2 POSITION | 8 | 335 | 230 | 0.6 | TWO POSITION | ISOLATION ROOM EXHAUST | PHOENIX SINGLE | 1,2,3 |

**NOTES:**
1. AIR VALVE BODY IS TO HAVE FACTORY APPLIED INSULATION, APPROXIMATELY 0.5 INCHES THICK. INSULATION TO BE MOISTURE BARRER.
2. LISTED CFM IS TO BE USED FOR VALVE SELECTION. ACTUAL OPERATING CFM IS TO BE DETERMINED BY ACTIVE PRESSURE CONTROL.
   T&B CONTRACTOR IS TO CALIBRATE VALVE BUT DO NOT SET OPERATING POSITION.
3. VALVE TO BE CONSTANT VOLUME WITH FIXED POSITION, NO CONTROLLER AND NO CONNECTION TO BAS NETWORK.

## FAN SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | SYSTEM AND/OR SERVICE | AIR FLOW CFM | EXTERNAL SP IN | TSP IN | FAN TYPE | FAN DIAMETER IN | FAN DRIVE | FAN MAX RPM | MOTOR ELECTRICAL: NOMINAL POWER BHP | NOMINAL POWER HP | PHASE | VOLT | RPM | ENCLOSURE | SPEED CONTROL | BASIS OF DESIGN OR APPROVED EQUAL | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| **SUPPLY** | | | | | | | | | | | | | | | | | | | |
| 143-SF-2D1A | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 3.00 | 8.69 | PLENUM | 22.2 | DIRECT | 2,515 | 13.79 | 20 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-2D1B | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 3.00 | 8.69 | PLENUM | 22.2 | DIRECT | 2,515 | 13.79 | 20 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-2D1C | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 3.00 | 8.69 | PLENUM | 22.2 | DIRECT | 2,515 | 13.79 | 20 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-2D1D | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 3.00 | 8.69 | PLENUM | 22.2 | DIRECT | 2,515 | 13.79 | 20 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C2A | MECH ROOM | SURGERY SUITE | 143-AHU-1C1 | 4,000 | 3.4 | 8.04 | PLENUM | 16.5 | DIRECT | 3,223 | 8.67 | 15 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C2B | MECH ROOM | SURGERY SUITE | 143-AHU-1C1 | 4,000 | 3.4 | 8.04 | PLENUM | 16.5 | DIRECT | 3,223 | 8.67 | 15 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C2C | MECH ROOM | SURGERY SUITE | 143-AHU-1C1 | 4,000 | 3.4 | 8.04 | PLENUM | 16.5 | DIRECT | 3,223 | 8.67 | 15 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C2D | MECH ROOM | SURGERY SUITE | 143-AHU-1C1 | 4,000 | 3.4 | 8.04 | PLENUM | 16.5 | DIRECT | 3,223 | 8.67 | 15 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C3A | MECH ROOM | SURGERY SUITE | 143-AHU-1C2 | 2,000 | 3 | 7.53 | PLENUM | 13.5 | DIRECT | 3,826 | 4.08 | 7.5 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C3B | MECH ROOM | SURGERY SUITE | 143-AHU-1C2 | 2,000 | 3 | 7.53 | PLENUM | 13.5 | DIRECT | 3,826 | 4.08 | 7.5 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C3C | MECH ROOM | SURGERY SUITE | 143-AHU-1C2 | 2,000 | 3 | 7.53 | PLENUM | 13.5 | DIRECT | 3,826 | 4.08 | 7.5 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-SF-1C3D | MECH ROOM | SURGERY SUITE | 143-AHU-1C2 | 2,000 | 3 | 7.53 | PLENUM | 13.5 | DIRECT | 3,826 | 4.08 | 7.5 | 3 | 480 | 3600 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| **RETURN** | | | | | | | | | | | | | | | | | | | |
| 143-RF-2D1A | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 2.00 | 3.79 | PLENUM | 20 | DIRECT | 1,974 | 5.52 | 10 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-RF-2D1B | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 2.00 | 3.79 | PLENUM | 20 | DIRECT | 1,974 | 5.52 | 10 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-RF-2D1C | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 2.00 | 3.79 | PLENUM | 20 | DIRECT | 1,974 | 5.52 | 10 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-RF-2D1D | ROOFTOP UNIT | SURGERY SUITE | 143-RTU-2D1 | 5,500 | 2.00 | 3.79 | PLENUM | 20 | DIRECT | 1,974 | 5.52 | 10 | 3 | 480 | 1800 | TEFC | VFD | YORK | 1,2,3,4,5,6 |
| 143-RF-1C2 | MECH ROOM | SURGERY SUITE | 143-AHU-1C1 | 16,000 | 2 | - | IN-LINE | 32.4 | DIRECT | 1,770 | 9.1 | 10 | 3 | 480 | 1800 | TEFC | VFD | GREENHECK: AX-80-275-0622 | - |
| 143-RF-1C3 | MECH ROOM | SURGERY SUITE | 143-AHU-1C2 | 8,000 | 2 | - | IN-LINE | 24.5 | DIRECT | 1,621 | 4.2 | 7.5 | 3 | 480 | 1800 | TEFC | VFD | GREENHECK: SQ-20-M2-VG | - |
| **EXHAUST** | | | | | | | | | | | | | | | | | | | |
| 143-EF-2C1A | 1ST FLOOR ROOF | PACU ISO | ISO EXHAUST | 405 | 1.25 | 1.26 | HIGH PLUME MIXED FLOW | - | BELT | 2,833 | 0.67 | 3/4 | 3 | 480 | - | - | VFD | GREENHECK VK-H-10-4 | 7,8 |
| 143-EF-2C1B | 1ST FLOOR ROOF | PACU ISO | ISO EXHAUST | 405 | 1.25 | 1.26 | HIGH PLUME MIXED FLOW | - | BELT | 2,834 | 0.67 | 3/4 | 3 | 480 | - | - | VFD | GREENHECK VK-H-10-4 | 7,8 |
| 143-EF-2D2 | HIGH ROOF | SURGERY SUITE | PANDEMIC | 22000 | 5 | 5.15 | HIGH PLUME MIXED FLOW | - | DIRECT | 1052 | 28.9 | 30 | 3 | 480 | - | - | VFD | GREENHECK VK-MH-40-70-36 | 7,8 |
| 143-EF-2C3 | HIGH ROOF | SURGERY SUITE | PANDEMIC | 16000 | 5 | 5.15 | HIGH PLUME MIXED FLOW | - | DIRECT | 1314 | 20.9 | 25 | 3 | 480 | - | - | VFD | GREENHECK VK-MH-33-70-30 | 7,8 |
| 143-EF-2C4 | HIGH ROOF | SURGERY SUITE | PANDEMIC | 8000 | 5 | 5.15 | HIGH PLUME MIXED FLOW | - | DIRECT | 1700 | 11 | 15 | 3 | 480 | - | - | VFD | GREENHECK VK-MH-24-70-20 | 7,8 |

**NOTE:**
1. FAN CAPACITY LISTED IS CAPACITY REQUIREMENT AND IS NOT THE ACTUAL SYSTEM CAPACITY. SCHEDULED CAPACITY INCLUDES DUCT LEAKAGE ALLOWANCE, SAFETY FACTOR AND FUTURE CAPACITY REQUIREMENTS.
2. FAN IS TO BE PROVIDED WITH PIEZOMETERIC FLOW SENSOR ON INLET BELL TO BE CONNECTED TO BAS PROVIDED TRANSMITTER.
3. VFD IS TO BE PROVIDED BY FAN MANUFACTURER. IF FAN IS SINGLE PHASE, MOTOR IS TO BE ECM TYPE WITH MODULATING SPEED BASED ON SIGNAL FROM BAS.
4. FANS TO OPERATE IN PARALLEL. PROVIDE EACH FAN WITH INLET ELECTRIC MOTOR OPERATED ISOLATION DAMPER, TWO POSITION, SPRING CLOSED AND END POSITION SWITCH TO VERIFY DAMPERS ARE OPEN.
5. FANS ARE TO OPERATE IN PARALLEL. SUBMITTAL TO SHOW THAT OPERATING FANS ARE IN STABLE PART OF FAN CURVE WITH ONE FAN OUT OF OPERATION.
6. FANS IN AIR HANDLING UNITS ARE TO BE SIZED BASED ON THE LISTED EXTERNAL PLUS AHU INTERNAL STATIC PRESSURE DROP INCLUDING CLEAN FILTER PRESSURE DROP, PLUS 1.0 INCHES DIRTY FILTER ALLOWANCE.
7. PROVIDE FAN WITH ROOF CURB AND BIRD SCREEN.
8. FAN PROVIDED WITH ECM CONTROLLER THAT ALLOWS MANUAL SPEED CONTROL AND DRY CONTACT START STOP

## AIR DEVICE SCHEDULE (RETURN AND EXHAUST)

| MARK | TYPE | AIR FLOW MIN CFM | AIR FLOW MAX CFM | MAX APD IN WG | MOUNTING | PANEL / FRAME SIZE IN x IN | NECK SIZE IN x IN | MAX NC | DAMPER | FINISH | BASIS OF DESIGN OR APPROVED EQUAL |
|---|---|---|---|---|---|---|---|---|---|---|---|
| RG-1 | RETURN GRILLE | 50 | 105 | 0.060 | CEILING | 24 x 24 | 6 x 6 | N/A | NONE | | PRICE PDR |
| RG-2 | RETURN GRILLE | 110 | 175 | 0.090 | CEILING | 24 x 24 | 8ø | 13 | NONE | | PRICE PDR |
| RG-3 | RETURN GRILLE | 135 | 215 | 0.090 | CEILING | 24 x 24 | 8 x 8 | 14 | NONE | | PRICE PDR |
| RG-4 | RETURN GRILLE | 220 | 270 | 0.090 | CEILING | 24 x 24 | 10ø | 16 | NONE | | PRICE PDR |
| RG-5 | RETURN GRILLE | 230 | 330 | 0.090 | CEILING | 24 x 24 | 10 x 10 | 17 | NONE | | PRICE PDR |
| RG-6 | RETURN GRILLE | 240 | 395 | 0.090 | CEILING | 24 x 24 | 12ø | 17 | NONE | | PRICE PDR |
| RG-7 | RETURN GRILLE | 250 | 475 | 0.090 | CEILING | 24 x 24 | 12 x 12 | 20 | NONE | | PRICE PDR |
| RG-8 | RETURN GRILLE | 320 | 730 | 0.090 | CEILING | 24 x 24 | 14ø | 21 | NONE | | PRICE PDR |
| RG-9 | RETURN GRILLE | 825 | 1145 | 0.090 | CEILING | 24 x 24 | 18 x 18 | 24 | NONE | | PRICE PDR |
| RG-10 | RETURN GRILLE | 675 | 745 | 0.045 | SIDE WALL | 14 x 22 | 12 x 20 | 14 | NONE | | PRICE 600 |
| RG-11 | RETURN GRILLE | 1000 | 1100 | 0.045 | SIDE WALL | 18 x 26 | 16 x 24 | 15 | NONE | | PRICE 600 |
| EG-1 | EXHAUST GRILL | 60 | 115 | 0.012 | CEILING | 24 x 24 | 6 x 6 | N/A | NONE | | PRICE PDR |
| EG-2 | EXHAUST GRILL | 280 | 340 | 0.016 | CEILING | 24 x 24 | 10 x 10 | N/A | NONE | | PRICE PDR |

**NOTE:**

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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(856) 429-4000 ★ Miller-Remick.com
M-R PROJECT NO: 0499-0136

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PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
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Lic. NO. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | | | |
|---|---|---|---|
| Drawing Title: MECHANICAL SCHEDULES | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Project Number: 658-19-103 | Building Number: 143 |
| | | Location: SALEM VA MEDICAL CENTER | Drawing Number: M-703 |
| | | Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | |

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VA FORM 08-6231

## FUEL OIL TANK SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | FUEL OIL TYPE NO | NOM CAPACITY GAL | DIMENSIONS | ALARM SYSTEM OR LEAK DETECTION | LEVEL INDICATOR | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|
| 143-FOT-1D1 | YARD | GENERATOR | NO. 2 | 8000 | 26'-8"L X 8'ø | VEEDER-ROOT | VEEDER-ROOT | SEE NOTES 1-3 | HIGHLAND TANK |
| 143-FOT-1D2 | YARD | GENERATOR | NO. 2 | 8000 | 26'-8"L X 8'ø | VEEDER-ROOT | VEEDER-ROOT | SEE NOTES 1-3 | HIGHLAND TANK |
| 143-FOT-1D3 | GENERATOR BLDG | GENERATOR | NO. 2 | 100 | 29.5"W x 35"L x 45.5"H | VEEDER-ROOT | VEEDER-ROOT | SEE NOTE 1 | PREFERRED MANUFACTURING; DT-100 |
| 143-FOT-1D4 | GENERATOR BLDG | GENERATOR | NO. 2 | 260 | 29.5"W x 85"L x 45.5"H | VEEDER-ROOT | VEEDER-ROOT | SEE NOTE 1 | PREFERRED MANUFACTURING; DT-260 |

**NOTES:**

1. THE CONSTRUCTION SHALL BE PER SPECIFICATIONS.
2. UNDERGROUND SEE DETAILS.
3. PROVIDE VEEDER-ROOT OVERFILL AUDIO/VISUAL ALARM AND ACKNOWLEDGEMENT DEVICE AND BACNET GATEWAY.

## FUEL OIL FILTER SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | CAPACITY GPH | OIL GRADE | OIL TEMP *F | MOTOR HP | MOTOR PHASE | MOTOR VOLT | BASIS OF DESIGN (OR APPROVED EQUAL) | REMARKS |
|---|---|---|---|---|---|---|---|---|---|---|
| 143-FOF-1C1 | GENERATOR BUILDING | NEW (2)8,000 GALLON FUEL OIL TANKS | 83 | 2 | 50 | 1/2 | 1 | 120 | PREFERRED MFG: PF-503 | NOTES: 1-2 |

**NOTES:**

1. PROVIDE FILTER UNIT AND ACCESSORIES IN A FACTORY PRE-PIPED AND PRE-WIRED SKID PACKAGE.
2. PROVIDE CONTROL PANEL.

## FUEL OIL PUMP SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | TYPE | CAPACITY GPM GPH | DISCHARGE PRESS PSIG | OIL GRADE | OIL TEMP *F | VISCOSITY RANGE (SSU) | SUCTION LIFT IN HG | MOTOR HP | MOTOR PHASE | MOTOR VOLT | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-FOP-1D1A | GENERATOR ROOM | GENERATOR | POSITIVE DISPLACEMENT | 375 | 50 | NO. 2 | 50 | 45 | 9 | 1/2 | 3 | 460 | | PREFERRED UTILITIES : MODEL ATPSF-201 |
| 143-FOP-1D1B | GENERATOR ROOM | GENERATOR | POSITIVE DISPLACEMENT | 375 | 50 | NO. 2 | 50 | 45 | 9 | 1/2 | 3 | 460 | | PREFERRED UTILITIES : MODEL ATPSF-201 |

## SPLIT SYSTEM INDOOR EVAPORATOR SCHEDULE

| MARK | LOCATION | AREA AND/OR BLDG SERVED | TYPE | TOTAL SUPPLY AIR FLOW CFM | MIN. OUTSIDE AIR FLOW CFM | EXT STATIC PRESSURE IN | COOLING CAPACITY MIN TOTAL CAPACITY MBH | MIN SENS CAPACITY MBH | EAT Db *F | EAT Wb *F | OSA DESIGN TEMP *F | HEATING CAPACITY MIN. HEAT CAPACITY MBH | EAT DB *F | LAT Db *F | OSA DESIGN TEMP *F | AIR FILTER MARK NO | ELECTRICAL DATA FAN TYPE | FAN CONTROL | MCA | PHASE | VOLT | SUPPLEMENTAL HEAT Kw | SUPPLEMENTAL HEAT PHASE | SUPPLEMENTAL HEAT VOLT | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-ACU-1D1 | GENERATOR ROOM | GENERATOR ROOM | WALL MOUNTED | 920 | 0 | - | 30 | - | 80 | 67 | - | [ 34 ] | 70 | - | - | - | LINE FLOW | - | 0.5 | 1 | 208/230 | - | - | - | SEE NOTES 1-2 | TRANE ; MODEL TPKFYP |

**NOTES:**

1. PROVIDE WITH MANUFACTURER'S INTEGRAL CONDENSATE PUMP.
2. WALL MOUNTED UNIT. PROVIDE WALL BRACKET.

## AIR COOLED CONDENSING UNIT SCHEDULE

| MARK | LOCATION | SYSTEM AND/OR SERVICE | CAPACITY BTUH | SUCTION TEMP *F | AMBIENT AIR *F | MAX COND TEMP *F | COMPRESSOR MOTOR # COMP | COMPRESSOR MOTOR POWER kW | COMPRESSOR MOTOR PHASE | COMPRESSOR MOTOR VOLT | CONDENSER FAN MOTOR # COND. FAN | CONDENSER FAN MOTOR POWER HP | CONDENSER FAN MOTOR PHASE | CONDENSER FAN MOTOR VOLT | REMARKS | BASIS OF DESIGN |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 143-CU-1 | GENERATOR ROOM GRADE | 143-ACU-1 | 36,000 | - | 95 | 115 | [ 2 ] | 0.074 | 1 | 208 | 2 | 0.25 | 1 | 208 | SEE NOTES 1-3 | TRANE MODEL TUMYP0361AK43 |

**NOTES:**

1. PROVIDE WALL MOUNTED THERMOSTAT CONTROLLER.
2. PROVIDE EQUIPMENT RAILS FOR ROOFTOP MOUNTING.
3. PROVIDE LOW AMBIENT KIT.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

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(856) 429-4000 ★ Miller-Remick.com

M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
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12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

**Drawing Title:** MECHANCIAL SCHEDULES

**Approved by:** Jon Manninen, COR

**Phase:** BID DOCUMENTS

FULLY SPRINKLERED

**Project Title:** RENOVATE SURGICAL SUITE PHASE II

**Project Number:** 658-19-103

**Location:** SALEM VA MEDICAL CENTER

**Issue Date:** 2025-12-30

**Checked:** DJ

**Drawn:** JAK

**VA Drawing Number:** 2617

**Building Number:** 143

**Drawing Number:** M-704

VA FORM 08-6231

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## CONTROLS ABBREVIATIONS

NOTE: NOT ALL ABBREVIATIONS MAY BE USED ON THE PROJECT.

| | |
|---|---|
| ACD | AUTOMATIC CONTROL DAMPER, MODULATING |
| ACV | AUTOMATIC CONTROL VALVE |
| AHU | AIR-HANDLING UNIT |
| AHUDAT | AIR HANDLING UNIT DISCHARGE AIR TEMPERATURE SENSOR |
| AHUHRCLAT | AIR HANDLING UNIT HEAT RECOVERY COIL LEAVING AIR TEMPERATURE SENSOR |
| AHUHRCV | AIR HANDLING UNIT HEAT RECOVERY COIL VALVE |
| ALM | ALARM |
| ATC | AUTOMATIC TEMPERATURE CONTROL |
| AVS | AIR VOLUME STATION |
| AS | AIR SEPARATOR |
| BAS | BUILDING AUTOMATION SYSTEM |
| BCV | BASEBOARD CONTROL VALVE |
| BDD | BACKDRAFT DAMPER |
| BIBO | BAG-IN BAG-OUT |
| BMS | BUILDING MANAGEMENT SYSTEM |
| BV | BYPASS VALVE |
| CCLAT | COOLING COIL LEAVING AIR TEMPERATURE SENSOR |
| CFDPS | CHARCOAL FILTER DIFFERENTIAL PRESSURE SWITCH |
| CHCLAT | CHILLED WATER COIL LEAVING AIR TEMPERATURE SENSOR |
| CHWRT | CHILLED WATER RETURN TEMPERATURE SENSOR |
| CHWST | CHILLED WATER SUPPLY TEMPERATURE SENSOR |
| CHWV | CHILLED WATER VALVE |
| DAT | DISCHARGE AIR TEMPERATURE SENSOR |
| DATL | DISCHARGE AIR TEMPERATURE LIMITER |
| DDC | DIRECT DIGITAL CONTROL |
| DDCFP | DIRECT DIGITAL CONTROL FIELD PANEL |
| DP | DIFFERENTIAL PRESSURE SENSOR |
| DPS | DIFFERENTIAL PRESSURE SWITCH |
| DPV | DIFFERENTIAL PRESSURE BYPASS VALVE |
| DSP | DISCHARGE STATIC PRESSURE SENSOR |
| DWDI | DOUBLE WIDTH DOUBLE INLET |
| EAD | EXHAUST AIR DAMPER |
| ECC | EVIRONMENTAL CONTROL CENTER (BAS) |
| EHRCEAT | EXHAUST HEAT RECOVERY COIL ENTERING AIR TEMPERATURE SENSOR |
| EHRCLAT | EXHAUST HEAT RECOVERY COIL LEAVING AIR TEMPERATURE SENSOR |
| EHRCV | EXHAUST HEAT RECOVERY COIL VALVE |
| ES | END SWITCH |
| ESP | EXHAUST STATIC PRESSURE SENSOR |
| ET | EXPANSION TANK |
| FA | FAULT ALARM |
| FDPS | FILTER DIFFERENTIAL PRESSURE SWITCH |
| FID | FAN ISOLATION DAMPER |
| FMT | FLOW METER TRANSMITTER |
| FZ | FREEZESTAT |
| GAPC | GENESIS AIR PHOTOCATALYSIS COMPOUND (COMMERCIAL GRADE) |
| GFP | GLYGOL FILL PUMP |
| H | HUMIDITY SENSOR |
| HALM | HOOD ALARM |
| HHL | HIGH HUMIDITY LIMIT SENSOR |
| HIV | HUMIDIFIER ISOLATION VALVE |
| HLH | HIGH/LOW HUMIDITY LIMIT SENSOR |
| HOA | HAND-OFF-AUTOMATIC SWITCH |
| HRCEAT | HEAT RECOVERY COIL ENTERING AIR TEMPERATURE SENSOR |
| HRCLAT | HEAT RECOVERY COIL LEAVING AIR TEMPERATURE SENSOR |
| HRLT | HEAT RECOVERY LOOP TEMPERATURE SENSOR |
| HRCV | HEAT RECOVERY COIL VALVE |
| HRGP | HEAT RECOVERY GLYCOL PUMP |
| HRL3WV | HEAT RECOVERY LOOP 3-WAY VALVE |
| HS | HAND SWITCH |
| HSPS | HIGH STATIC PRESSURE SWITCH |
| HSS | HOOD SASH SWITCH |
| HV | HUMIDIFIER VALVE |
| HWST | HOT WATER SUPPLY TEMPERATURE SENSOR |
| HWRT | HOT WATER RETURN TEMPERATURE SENSOR |
| HWV | HOT WATER VALVE |
| IFBD | INTEGRAL FACE & BYPASS DAMPER |
| IGV | INLET GUIDE VANES |
| LAT | LEAVING AIR TEMPERATURE SENSOR |
| LSPS | LOW STATIC PRESSURE SWITCH |
| LS | LIGHT SWITCH INTERFACE |
| MUWV | MAKEUP WATER VALVE |
| NC | NORMALLY CLOSED |
| NO | NORMALLY OPEN |
| OAD | OUTSIDE AIR DAMPER |
| OADBT | OUTSIDE AIR DRY BULB TEMPERATURE SENSOR |
| OAWBT | OUTSIDE AIR WET BULB TEMPERATURE SENSOR |
| OAT | OUTSIDE AIR TEMPERATURE SENSOR |
| PCHWRT | PRIMARY CHILLED WATER RETURN TEMPERATURE SENSOR |
| PCHWST | PRIMARY CHILLED WATER SUPPLY TEMPERATURE SENSOR |
| PCP | POPULATED CATALYST PANEL (GAPC – PCP COMPOUND) |
| RAD | RETURN AIR DAMPER |
| RAT | RETURN AIR TEMPERATURE SENSOR |
| RCV | REHEAT CONTROL VALVE |
| RH | RELATIVE HUMIDITY |
| RI | RUN INDICATOR |
| RID | RETURN ISOLATION DAMPER |
| RT | ROOM TEMPERATURE SENSOR |
| SAV | SUPPLY AIR VALVE |
| SCHWRT | SECONDARY CHILLED WATER RETURN TEMPERATURE |
| SCHWST | SECONDARY CHILLED WATER SUPPLY TEMPERATURE |
| SCLAT | STEAM COIL LEAVING AIR TEMPERATURE SENSOR |
| SCV | STEAM COIL VALVE |
| SD | SMOKE DAMPER |
| SDET | SMOKE DETECTOR |
| SPS | STATIC PRESSURE SENSOR |
| SSID | SUPPLY SMOKE ISOLATION DAMPER |
| SSP | SUCTION STATIC PRESSURE SENSOR |
| SST | START-STOP |
| T | TEMPERATURE SENSOR |
| TCV | TEMPERATURE CONTROL VALVE |
| TT | TEMPERATURE TRANSMITTER |
| VFD | VARIABLE FREQUENCY DRIVE |
| VFDS | VARIABLE FREQUENCY DRIVE SPEED |
| VFDSO | VARIABLE FREQUENCY DRIVE SPEED OUTPUT |
| VS | VIBRATION SWITCH |
| WC | WATER COLUMN |
| WFS | WATER FLOW SWITCH |
| WS | WALL SWITCH |

## CONTROLS SYMBOLS

- CONTROL DEVICE PROVIDED BY ELECTRICAL OR PLUMBING CONTRACTOR BUT INTERFACED TO DDC SYSTEM BY ATC CONTRACTOR
- ATC CONTRACTOR INTERFACE TO MANUFACTURER'S HARDWARE
- AFMD — AIRFLOW MEASURING DEVICE
- AT CO2 — CARBON DIOXIDE TRANSMITTER
- AT CO — CARBON MONOXIDE TRANSMITTER
- AT OC — OCCUPANCY SENSOR
- ATC — ATC CONTRACTOR PROVIDE DDC POINT AND HARDWARE
- CT — CURRENT TRANSMITTER
- COMM — BACNET COMMUNICATIONS
- DAT — DISCHARGE AIR TEMPERATURE SENSOR
- DPS — DIFFERENTIAL PRESSURE SWITCH
- DPSE — ROOM DIFFERENTIAL PRESSURE MONITOR
- DPT — DIFFERENTIAL PRESSURE TRANSMITTER
- EAV — EXHAUST AIR VALVE
- ECC — ENGINEERING CONTROL CENTER (BAS)
- EPT — ELECTRONIC TO PNEUMATIC TRANSDUCER
- FT — FLOW TRANSMITTER
- FC — FLOW CONTROLLER. SEE SEQUENCE OF OPERATION.
- FS — FLOW SWITCH
- FSH — FLOW SWITCH HIGH
- FSL — FLOW SWITCH LOW
- FT — FLOW TRANSMITTER
- HC — HUMIDITY CONTROL
- HHL — HIGH HUMIDITY LIMIT SENSOR

## CONTROLS SYMBOLS (CONTINUED)

- HS — HAND SWITCH (HAND-OFF-AUTO SWITCH) or HUMIDITY SWITCH
- HVAC — HVAC CONTROL PANEL
- IT — CURRENT TRANSMITTER
- KC — TIME CLOCK CONTROLLING EQUIPMENT ON A SCHEDULE
- KR — LOCAL RECORDING TIME CLOCK (RUNTIME)
- LC — LEVEL CONTROLLER
- LS — LIMIT SWITCH
- LT — LEVEL TRANSMITTER
- M — ROOM HUMIDISTAT (MOISTURE)/TRANSMITTER (WALL MOUNTED)
- M — MOTOR ACTUATED DAMPER /OR VALVE
- MT — MOISTURE (HUMIDITY) TRANSMITTER
- PC — PRESSURE CONTROLLER (SEE SEQUENCE OF OPERATIONS)
- PC — PNEUMATIC CONTROLLER (SEE SEQUENCE OF OPERATIONS)
- PDS — PRESSURE DIFFERENTIAL SWITCH
- PDT — PRESSURE DIFFERENTIAL TRANSMITTER
- PSH — PRESSURE SWITCH HIGH
- PSL — PRESSURE SWITCH LOW
- PT — PRESSURE TRANSMITTER
- S20 — INSTRUMENT OR CONTROL AIR @ 20 PSIG
- SAV — SUPPLY AIR VALVE
- SC — SPEED CONTROLLER (SEE SEQUENCE OF OPERATION)
- SCV — STEAM COIL CONTROL VALVE
- SD — SMOKE DETECTOR
- SDET — SMOKE DETECTOR

## CONTROLS SYMBOLS (CONTINUED)

- SPS — STATIC PRESSURE SENSOR
- SST — START-STOP
- T — ROOM THERMOSTAT/TRANSMITTER – WALL MOUNTED
- TC — TEMPERATURE CONTROLLER. SEE SEQUENCE OF OPERATIONS.
- TSH — TEMPERATURE SWITCH HIGH
- TSL — TEMPERATURE SWITCH LOW (FREEZESTAT)
- TT — TEMPERATURE TRANSMITTER – AVERAGING TYPE
- TT — TEMPERATURE TRANSMITTER – SINGLE POINT
- TT — TEMPERATURE TRANSMITTER – EMERSION TYPE
- VFD — VARIABLE FREQUENCY DRIVE
- VS — VIBRATION SWITCH
- VSMC — VARIABLE SPEED MOTOR CONTROLLER
- ZC — VALVE OR DAMPER POSITION CONTROLLER
- ZS — END SWITCH
- MOTOR STARTER
- O — UNOCCUPIED MODE OVERRIDE PUSH BUTTON
- FAN OR PUMP
- PIEZOMETER RING IF SHOWN
- OUTDOOR AIR TEMPERATURE SENSOR – COMBINATION HUMIDITY SENSOR IF SHOWN ON THE DRAWINGS. WITH SOLAR SHADING
- PNEUMATIC OPERATED DAMPER/OR VALVE
- TEMPERATURE SENSING ELEMENT FOR TRANSMITTING TEMPERATURE TO EMCS (PROVIDE 12 INCHES [200mm] MINIMUM LENGTH IN DUCT WHEN SPACE PERMITS.)
- A — SENSOR WITH AVERAGING ELEMENT TO TRANSMIT TEMPERATURE TO EMCS

## CONTROLS GENERAL NOTES

1. THE SEQUENCE OF OPERATIONS OUTLINED SHALL BE PERFORMED BY NEW DIRECT DIGITAL CONTROL, STAND ALONE FIELD PANELS, (DDCFP) COMPATIBLE WITH THE EXISTING BUILDING MANAGEMENT SYSTEM (BMS)/ BUILDING AUTOMATION SYSTEM (BAS). NAMES FOR ALL POINTS AND VARIABLES SHALL BE COORDINATED WITH THE VAMC PRIOR TO SHOP DRAWINGS AND PROGRAMMING.
2. THE SEQUENCES OF OPERATIONS OUTLINED SHALL BE PERFORMED IN A STAGGERED/STAGED MANNER DURING INITIAL START-UP AND DURING THE TRANSITION FROM NORMAL TO EMERGENCY POWER OPERATION TO PREVENT OVERLOADING OF THE BUILDING/SITE ELECTRICAL POWER SYSTEMS. DESIGN INTENT IS FOR MECHANICAL EQUIPMENT TO OPERATE CONTINUOUSLY AND TO BE INDEXED/STARTED IN ORDER OF PRIORITY BEGINNING WITH BUILDING CENTRAL HYDRONIC SYSTEMS (REHEAT/HEATING HOT WATER HEAT EXCHANGERS, PUMPS; AND CHILLED WATER CHILLERS, PUMPS, ETC.), CENTRAL STATION AIR HANDLING EQUIPMENT (INTERLOCKED SUPPLY, RETURN, AND EXHAUST EQUIPMENT), AND ROOM/OCCUPIED SPACE AIR AND HYDRONIC TERMINAL EQUIPMENT. REFER TO CONTROLS DIAGRAMS FOR EACH SPECIFIC SYSTEM/COMPONENT FOR ADDITIONAL INFORMATION, SAFETIES, ETC. SUBMIT THE START/RESTART SEQUENCE WITHIN THE SPECIFIED BAS CONTROLS SUBMITTAL FOR REVIEW BY OWNER AND DESIGN ENGINEERS.
3. EQUIPMENT SHALL BE CAPABLE OF MANUAL OPERATION THROUGH HOA SWITCHES IN STARTERS. THE POSITIONS OF ALL CONTROL VALVES CONTROLLED BY THE BMS SHALL BE CAPABLE OF MANUAL POSITIONING VIA THE CONTROL CENTER TERMINAL OR BY LAPTOP COMPUTER PLUGGED IN TO THE DIRECT DIGITAL CONTROL FIELD CONTROL PANEL.
4. ALL VALVE AND DAMPER ACTUATORS SHALL BE ELECTRIC/ELECTRONIC.
5. REFER TO MECHANICAL DIAGRAMS AND PLANS FOR SENSOR LOCATIONS. COORDINATE TO INSURE PROPER UPSTREAM AND DOWNSTREAM DIAMETERS FOR EACH SENSOR.
6. PROVIDE CURRENT SENSOR ACROSS EACH FAN OR PUMP (WITH THE EXCEPTION OF VFD'S) TO PROVE STATUS AT BMS. SENSORS SHALL BE CAPABLE OF DIFFERENTIATING BETWEEN NORMAL OPERATION AND FREE WHEELING (I.E. FAN BELT BREAKAGE).
7. BMS CONTROLLERS AND ALL OTHER CONTROL DEVICES AND SENSORS THAT REQUIRE POWER SHALL BE POWERED FROM EMERGENCY POWER PANEL.
8. WHEN SYSTEMS ARE OFF, ALL VALVES AND DAMPERS SHALL GO TO THEIR FAIL-SAFE POSITIONS. FAIL-SAFE POSITIONS ARE POSITIONS THAT DEVICES WILL GO TO WHEN DE-ENERGIZED: NO = NORMALLY OPEN, NC = NORMALLY CLOSED, FL = FAIL LAST.
9. ALL SETPOINTS AND TIME DELAYS MENTIONED AS PART OF THIS SYSTEM SHALL BE ADJUSTABLE AT THE BMS FRONT END BY AN OPERATOR WITHOUT ANY HARDWARE OR SOFTWARE REVISIONS.
10. ALL 2-POSITION DAMPERS SHALL HAVE END SWITCHES (ES). EACH END SWITCH SHALL PROVE DAMPER OPEN AND CLOSED. WHEN COMMANDED TO OPEN OR CLOSE, AFTER 30 SECONDS (ADJ.), THE END SWITCH DOES NOT SWITCH, THE CONTROL SYSTEM SHALL SEND AN ALARM TO THE BMS FRONT END.
11. ALL 2-POSITION VALVES SHALL HAVE LIMIT SWITCHES (LS). EACH LIMIT SWITCH SHALL PROVE VALVE OPEN AND CLOSED. WHEN COMMANDED TO OPEN OR CLOSE, AFTER 30 SECONDS (ADJ.), THE LIMIT SWITCH DOES NOT SWITCH, THE CONTROL SYSTEM SHALL SEND AN ALARM TO THE BMS FRONT END.

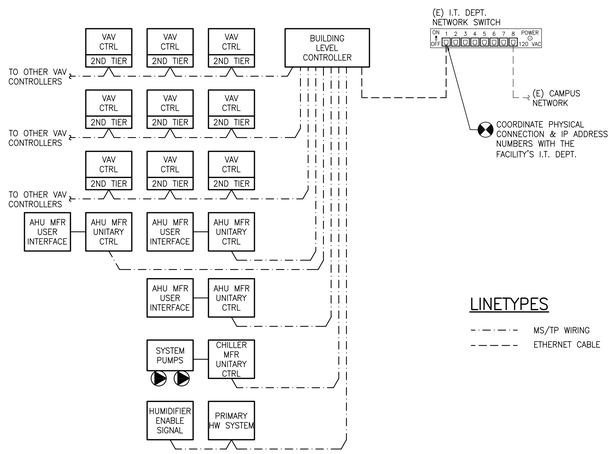
## COORDINATION NOTES

1. COORDINATE WITH TESTING, ADJUSTING, AND BALANCING (TAB) CONTRACTOR FOR ALL OPERATIONAL MODES. REVIEW AND CONFIRM ALL HVAC EQUIPMENT AND CONTROL SEQUENCES TO ALLOW FOR SYSTEM TESTING AND VERIFICATION OF EACH MODE DESCRIBED HEREIN.
2. ALL TEMPERATURE SENSOR, AIRFLOW MEASURING DEVICES, AND MOISTURE TRANSMITTERS, SHALL BE CALIBRATED IN COORDINATION WITH THE TAB CONTRACTOR.
3. VFD STARTUP SHALL MATCH THE MOTOR NAMEPLATE FLA, WHICH MAY EXCEED 60 HZ.
4. REFER TO ELECTRICAL PLANS AND COORDINATE NORMAL POWER SOURCE FOR NEW HVAC EQUIPMENT AND EMERGENCY POWER SOURCE FOR ALL BAS CONTROL PANEL(S) AS REQUIRED TO SUPPORT FINAL BUILDING AUTOMATION SYSTEMS BY BAS CONTRACTOR. DESIGN INTENT IS FOR BAS CONTROL TO REMAIN OPERATIONAL WHILE HVAC SYSTEMS SHUT DOWN DURING LOSS OF NORMAL POWER (EMERGENCY POWER OPERATION) AND FOR HVAC EQUIPMENT TO RE-START AUTOMATICALLY DURING TRANSITION BACK TO NORMAL POWER.
5. BUILDING AUTOMATION SYSTEMS AND CONTROLS SHALL BE FULLY COORDINATED BETWEEN THE CONTROLS CONTRACTOR, AHU MANUFACTURER, AND AHU COMPONENT MANUFACTURERS. CONTROL DEVICES, HARDWARE, ACTUATORS, CONTROLLERS, ETC. SHALL BE COORDINATED AND SUPPLIED BY THE CONTROLS CONTRACTOR TO AHU MANUFACTURER UNLESS THE SPECIFIC PIECE OF EQUIPMENT OR COMPONENT WITHIN THE AHU IS SPECIFIED TO INCLUDE ITS OWN CONTROLS (E.G. SUPPLY FAN PIEZOMETERS PROVIDED BY FAN MANUFACTURER, AIRFLOW MONITORING STATIONS WITH INTEGRAL DAMPER ASSEMBLIES, ETC.). CONTROLS CONTRACTOR AND AHU MANUFACTURER SHALL WORK CAREFULLY WITH ALL SUPPLIERS TO ENSURE THAT THE DEVICES INDICATED ON THE CONTROL DIAGRAM AND SEQUENCE OF OPERATION ARE INSTALLED WITHIN THE AHU. CONTROLS CONTRACTOR PROTOCOLS FOR INSTALLATION OF DEVICES AND HARDWARE SHALL BE FOLLOWED CAREFULLY BY AHU MANUFACTURER INSTALLERS. CONTROLS CONTRACTOR SHALL PROVIDE FACTORY TESTING AND OVERSIGHT OF THE COMPLETED CONTROLS PRIOR TO AND DURING THE FACTORY ACCEPTANCE TESTING AS WELL AS DURING THE SITE START-UP EFFORTS. SEE SPECIFICATIONS FOR ADDITIONAL REQUIREMENTS.

## ADDITIONAL CONTROL POINTS

IN ADDITION TO THE CONTROL POINTS INDICATED IN THE INDIVIDUAL CONTROL SEQUENCES, PROVIDE THE FOLLOWING MONITORING POINTS FOR THE ASSOCIATED ELEMENTS INDICATED/FOUND ON THE PLANS:

| OTHER SECTIONS (AS REQUIRED) | POINTS |
|---|---|
| HVAC SYSTEMS | |
| DAY ROOM RANGE HOOD | STATUS/ALARM |
| PLUMBING EQUIPMENT | |
| DOMESTIC WATER CIRC. PUMP | STATUS/ALARM |
| DOMESTIC WATER CIRC. PUMP | BACNET COMMUNICATION |
| ELECTRICAL SYSTEMS | |
| EMERGENCY POWER SYSTEM | ATS STATUS – NORMAL/EMERGENCY |
| FIRE ALARM SYSTEM | GENERAL ALARM |
| HEAT TRACE (#CIRCUITS) | STATUS |
| HEAT TRACE AMPERAGE (ANALOG) | AMP |




1 BAS ARCHITECTURE

### LINETYPES

- MS/TP WIRING
- ETHERNET CABLE

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| Drawing Title | MECHANCIAL CONTROLS |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
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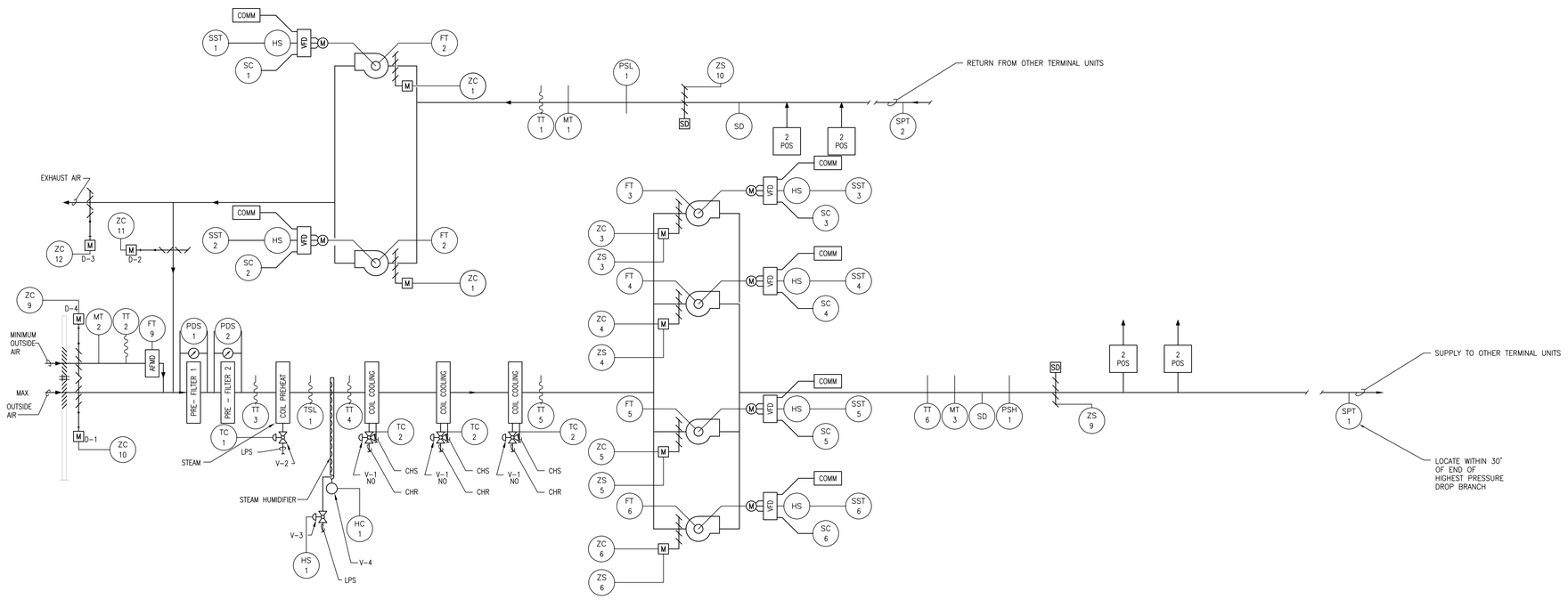


1 MECHANICAL AHU FOR SURGICAL SUITE (VAV) DIAGRAM
SCALE: NTS

## SEQUENCE OF OPERATIONS

GENERAL

1. DESCRIPTION: UNIT IS VAV AIR UNIT WITH SIX PARALLEL OPERATION SUPPLY FANS, TWO PARALLEL OPERATION RETURN FANS, ECONOMIZER CYCLE, MINIMUM AND MAXIMUM ECONOMIZER DAMPERS, STEAM PREHEAT COIL, CHILLED WATER COOLING COIL, AND PLANT STEAM HUMIDIFIER.
2. THE AHU SHALL OPERATE IN THE OCCUPIED MODE WHEN ANY OF THE CONNECTED TERMINALS SERVING THE OPERATING ROOMS, CLEAN CORE, SEMI-RESTRICTED CORRIDOR, PACU, OR PREOP SPACES ARE IN OCCUPIED MODE. THESE TERMINALS MAY BE IN THEIR NORMAL SCHEDULED OCCUPIED MODE OR OFF HOUR OCCUPIED MODE CAUSED BY OPERATOR OR SPACE OCCUPANCY SENSORS.
3. UNIT IS TO BE STARTED AND STOPPED THROUGH AN H-O-A SELECTOR SWITCH LOCATED AT THE UNIT. IN THE AUTOMATIC POSITION, THE OCCUPIED/UNOCCUPIED OPERATION SHALL INDEX THE UNIT TO RUN IN A MODE OF OPERATION. IN THE HAND POSITION, THE UNIT SHALL RUN IN THE OCCUPIED MODE.
4. UNIT IS TO BE KEPT IN THE AUTOMATIC POSITION. "HAND" AND "OFF" POSITIONS ARE TO BE USED TEMPORARILY FOR MAINTENANCE FUNCTIONS ONLY.
5. WHEN THE H-O-A SELECTOR IS IN THE HAND OR AUTO POSITION, THE CONTROLS SHALL BE FUNCTIONAL. IN THE OFF POSITION ALL FANS SHALL BE DE-ENERGIZED AND ALL CONTROLLED DEVICES SHALL ASSUME THEIR "NORMAL" POSITION.

UNIT START UP

1. ON START UP THE SUPPLY AND RETURN AIR SMOKE DAMPERS SHALL BE COMMANDED TO OPEN.
2. UPON THE DAMPER END SWITCHES PROVING THE DAMPERS HAVE OPENED, THE SUPPLY AND RETURN FANS SHALL BE ENABLED TO RUN.
3. ON BEING ENABLED TO RUN THE SUPPLY AND RETURN FAN AUTOMATIC ISOLATION DAMPERS SHALL BE COMMANDED TO OPEN. WHEN ALL DAMPER END SWITCHES PROVE THE DAMPERS ARE ALL OPEN, THE SUPPLY AND RETURN FANS SHALL BE COMMANDED TO START.
4. IF A DAMPER END SWITCH DOES NOT CLOSE WITHIN TWO MINUTES AFTER THE DAMPERS ARE COMMANDED TO OPEN, THE OTHER FANS, WITH OPEN DAMPERS, SHALL START UP.
5. ON COMMAND TO START, THE SUPPLY FANS SHALL ACCELERATE AT THE RATE OF ZERO SPEED TO MAXIMUM SPEED OVER A PERIOD OF TWO MINUTES. WHEN THE DUCT PRESSURE CONTROL IS SATISFIED, THE FANS SHALL BE UNDER THAT CONTROL.
6. THE RETURN FANS SHALL START THE SAME TIME AS THE SUPPLY FANS AND SHALL ACCELERATE AT THE SAME RATE AS THE SUPPLY FANS UNTIL THE SUPPLY FAN CONTROL CHANGES OVER TO THE NORMAL SPEED CONTROL. AT THIS TIME THE RETURN FAN SPEED CONTROL SHAL BE AS DESCRIBED BELOW.
7. AFTER THE SUPPLY FANS HAVE ESTABLISHED DUCT STATIC PRESSURE CONTROL, THE MINIMUM OUTSIDE AIR CONTROL, THE ECONOMIZER CONTROL, STEAM PREHEAT COIL CONTROL, COOLING COIL CONTROL AND THE HUMIDIFIER CONTROL SHALL INITIATE.

SUPPLY FAN SPEED CONTROL:

1. FAN SPEED SHALL BE CONTROLLED TO MAINTAIN SUPPLY DUCT STATIC PRESSURE SETPOINT AS SENSED BY SPT-1. INITIAL SETPOINT IS TO BE 1.3 INCHES W.G. FINAL OPERATING DUCT STATIC PRESSURE SETPOINT IS TO BE ESTABLISHED BY THE TEST AND BALANCE CONTRACTOR.
2. CONTROL SPEED SIGNAL TO ALL SIX FAN VFDS IS TO BE THE SAME VALUE.

RETURN FAN SPEED CONTROL

1. FAN SPEED SHALL BE CONTROLLED TO MAINTAIN RETURN DUCT STATIC PRESSURE SETPOINT AS SENSED BY SPT-2. INITIAL SETPOINT IS TO BE 1.0 INCHES W.G. FINAL OPERATING DUCT STATIC PRESSURE SETPOINT IS TO BE ESTABLISHED BY THE TEST AND BALANCE CONTRACTOR.
2. CONTROL SPEED SIGNAL TO BOTH FAN VFDS IS TO BE THE SAME VALUE.

OUTSIDE AIR/ECONOMIZER CONTROL

1. OUTSIDE AIR TEMPERATURE AND HUMIDITY IS TO BE SENSED BY TT-2 AND MT-2. THESE VALUES ARE TO BE USED TO ESTABLISH OUTSIDE AIR ENTHALPY. WHEN OUTDOOR AIR ENTHALPY IS BELOW 25 BTU/LB AND OUTSIDE AIR IS BELOW 75 DEGREES, THE ECONOMIZER OPERATION IS TO BE ENABLED.
2. WHEN THE ECONOMIZER MODE IS NOT ENABLED, RELIEF DAMPER D-3 SHALL BE FULLY OPEN AND THE CONTROL SHALL MAINTAIN A MINIMUM FLOW OF OUTSIDE AIR. ON A FALL BELOW MINIMUM OUTSIDE AIR FLOW SETPOINT, AS SENSED BY FT-9, MINIMUM AIR DAMPERS D-4 ARE TO BE MODULATED OPEN AND THE RETURN AIR DAMPERS D-2 ARE TO BE MODULATED CLOSED TO MAINTAIN MINIMUM OUTSIDE AIRFLOW QUANTITY. ON A RISE ABOVE AIR FLOW SETPOINT, THE REVERSE ACTION SHALL OCCURE. DURING UNOCCUPIED PERIODS, MINIMUM OUTSIDE AIR DAMPERS D-4 ARE TO BE CLOSED.
3. WHEN ECONOMIZER OPERATION IS ENABLED, THE MINIMUM OUTSIDE DAMPER D-4 POSITION SHALL CONTINUE TO BE MODULATED TO MAINTAIN MINIMUM OUTSIDE AIRFLOW. THE ECONOMIZER AND EXHAUST AIR DAMPERS ARE TO BE MODULATED OPEN, IN UNISON, AND THE RETURN AIR DAMPERS SHALL BE SIMULTANEOUSLY MODULATED CLOSED ON A RISE IN MIXED AIR TEMPERATURE, AS SENSED BY TT-3, ABOVE THE SETPOINT OF (COOLING COIL DISCHARGE SETPOINT MINUS 1 DEGREE). ON A FALL BELOW THE MIXED AIR TEMPERATURE SETPOINT, THE REVERSE ACTION SHALL OCCUR.

TEMPERATURE CONTROL

1. STEAM PREHEAT COIL: THE CONTROL VALVE SHALL BE MODULATED TO MAINTAIN COIL LEAVING TEMPERATURE, AS SENSED BY TT-4 TO A SETPOINT OF 48 DEGREES.
2. CHILLED WATER COOLING COIL: THE THREE WAY MIXING CONTROL VALVE SHALL BE MODULATED TO MAINTAIN A COIL LEAVING SETPOINT. SETPOINT SHALL BE 50 DEGREES DURING OCCUPIED PERIODS AND 60 DEGREES DURING UNOCCUPIED PERIODS.

HUMIDITY CONTROL

1. THE RETURN AIR RELATIVE HUMIDITY, SENSED BY MT-1, SHALL BE USED TO RESET THE SETPOINT OF HUMIDIFIER CONTROL
2. WHEN THE RETURN AIR HUMIDITY IS 33 PERCENT RH OR ABOVE THE HUMIDIFIER IS INACTIVE.
3. WHEN THE RETURN AIR HUMIDITY FALLS TO 31 PERCENT RH, THE HUMIDIFIER CONTROL SHALL BECOME ACTIVE AND THE SETPOINT SHALL BE 30 GRAINS/$LB_{DA}$.
4. THE RETURN AIR RELATIVE HUMIDITY SHALL LINEARLY RESET THE HUMIDIFIER CONTROL SETPOINT TO 36 GRAINS/$LB_{DA}$ WHEN THE RETURN AIR HUMIDITY FALLS TO 25 PERCENT RH.
5. THE HUMIDIFIER CONTROL BLOCK VALVE SHALL OPEN AND THE HUMIDIFIER CONTROL VALVE SHALL BE MODULATED TO MAINTAIN THE CONTROL SETPOINT WHEN THE HUMIDIFIER IS ACTIVE.

SAFETIES

1. UNIT RUN ENABLE SAFETIES INCLUDE:
2. SUPPLY AIR SMOKE DAMPER END SWITCH
3. RETURN AIR SMOKE DAMPER END SWITCH
4. SUPPLY AIR HIGH PRESSURE SWITCH, PSH-1. SETPOINT – +4.0 INCHES W.G. SWITCH SHALL HAVE MANUAL ONLY RESET.
5. RETURN AIR LOW PRESSURE SWITCH, PSL-1. SETPOINT – -4.0 INCHES W.G. SWITCH SHALL HAVE MANUAL ONLY RESET.
6. GENERAL SMOKE/FIRE CONDITION SIGNAL FROM SMOKE DETECTION SYSTEM
7. FREEZESTAT, TSL-1. SETPOINT – 35 DEGREES.
8. ALL FAN VFD START CONTACTS ARE TO BE HARD WIRED THROUGH THE UNIT SAFETIES TO PREVENT RUN CONDITION IN BOTH THE "AUTO" AND "HAND" POSITIONS OF THE UNIT H-O-A SELECTOR SWITCH.
9. IN ADDITION TO THE OVERALL UNIT SAFETIES, EACH SUPPLY FAN AND RETURN FAN ISOLATION DAMPER SHALL BE HARD WIRED INTO INDIVIDUAL FAN RUN ENABLE CIRCUITS IN BOTH THE "HAND" AND "AUTO" POSITION OF THE INDIVIDUAL FAN H-O-A SELECTOR SWITCH.

ALARM AND FAULT HANDLING

1. ALARM CONDITION SHALL BE ANNUNCIATED AT THE ECC WHEN ANY OF THE UNIT SAFETIES; PSH-1, PSL-1, SMOKE DAMPER END SWITCHES AND FREEZESTAT ARE NOT SATISFIED.
2. THE SMOKE DAMPER END SWITCH ALARMS SHALL BE REPRESSED FOR ONE MINUTE AFTER UNIT START TO ALLOW TIME FOR THE DAMPERS TO OPEN. IF DAMPERS DO NOT OPEN, UNIT START UP SHALL BE TERMINATED.
3. SUPPLY AND RETURN FAN ISOLATION DAMPER END SWITCHES SHALL BE ANNUNCIATED AT THE ECC UPON FAILURE TO DETECT OPEN CONDITION. THE DAMPER END SWITCH ALARMS SHALL BE REPRESSED FOR ONE MINUTE AFTER FAN RUN INITIATION TO ALLOW DAMPERS TO OPEN. THIS CONDITION SHALL SHUT DOWN THE ASSOCIATED FAN ONLY AND THE OTHER REMAINING FANS SHALL BE ALLOWED TO PROVIDE THE REQUIRED CAPACITY.
4. IF THE HUMIDIFIER IS ACTIVE AND THE DISCHARGE HUMIDITY RISES ABOVE 90 PERCENT, THE HUMIDIFIER SHALL BE SHUT DOWN AND ALARM SHALL BE ANNUNCIATED AT THE ECC.
5. THE ALARMS INDICATED IN THE POINTS LIST SHALL BE ANNUNCIATED AT THE ECC.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

ENGINEER OF RECORD

MILLER-REMICK
A Service-Disabled Veteran-Owned Small Business
(856) 429-4000 ★ Miller-Remick.com
M-R PROJECT NO. CMR-0236

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL AHU FOR SURGICAL SUITE (VAV) DIAGRAM |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-802 |

VA FORM 08-6231

P:\0499_DVA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\M-802-7.dwg 12-29-25 02:00:53 PM eremick

## POINTS SUMMARY AHU

| SYSTEM COMPONENT | Point ID | Applies To Both Base and Deductive Alternate Bids | Applies to Deductive Alternate Bid Only | Applies to Base Bid Only | Device Fail Position (O)-Open, (C)- Closed | System Outputs: Binary: Electric | System Outputs: Binary: Pneumatic | System Outputs: Analog: Electric | System Outputs: Analog: Pneumatic | System Inputs: Binary: Status | System Inputs: Analog: Temperature | System Inputs: Analog: Pressure | System Inputs: Analog: Humidity | System Inputs: Analog: Flow | System Inputs: Analog: Current | System Inputs: Analog: Data | System Software/Control: Alarm Processing: Equipment Status | System Software/Control: Alarm Processing: Low Limit | System Software/Control: Alarm Processing: High Limit | System Software/Control: Application/Function: Schedule Start/Stop | System Software/Control: Application/Function: Lead/Lag | System Software/Control: Application/Function: Temperature Control | System Software/Control: Application/Function: Enthalpy Economizer | System Software/Control: Application/Function: Pressure/Volume Control | System Software/Control: Application/Function: Humidity Control | System Software/Control: Application/Function: Humidity Reset | System Software/Control: Application/Function: Value | Digital Communications: Operating Speed | Digital Communications: Status | Digital Communications: Fault | Digital Communications: Amps (Actual and Precent of Full Capacity Amps) | Graphic Display | Trending | | Remark |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| HOA On/Off | HS | ● | | | | ● | | | | ● | | | | | | | | | | ● | | | | | | | | | | | | | | | |
| Return Air Temperature | TT-1 | ● | | | | | | | | | ● | | | | | | | | 78 | | | | | | | | | | | | | | ● | ● | |
| Return Air Humidity | MT-1 | ● | | | | | | | | | | | ● | | | | | 20 | 60 | | | | | | | ● | | | | | | | ● | | |
| Outside Air Temperature | TT-2 | ● | | | | | | | | | ● | | | | | | | | | | | | ● | | | | | | | | | ● | | |
| Outside Air Humidity | MT-2 | ● | | | | | | | | | | | ● | | | | | | | | | | ● | | | | | | | | | ● | | |
| Mixed Air Temperature | TT-3 | ● | | | | | | | | | ● | | | | | | | 45 | | | | | | | | | | | | | | ● | | |
| Preheat Coil Leaving Temperature | TT-4 | ● | | | | | | | | | ● | | | | | | | 45 | 80 | | | | | | | | | | | | | ● | | |
| Cooling Coil Leaving Temperature | TT-5 | ● | | | | | | | | | ● | | | | | | | | 62 | | | | | | | | | | | | | ● | | |
| Supply Air Temperature | TT-6 | ● | | | | | | | | | ● | | | | | | | | 66 | | | | | | | | | | | | | ● | ● | |
| Supply Air Humidity | MT-3 | ● | | | | | | | | | | | ● | | | | | | 85 | | | | | | | ● | | | | | | | ● | ● | |
| Supply Static Pressure | SPT-1 | ● | | | | | | | | | | ● | | | | | | | | | | | | | ● | | | | | | | ● | | |
| Return Static Pressure | SPT-2 | ● | | | | | | | | | | ● | | | | | | | | | | | | | ● | | | | | | | ● | | |
| Return Fan On/Off | SST-1 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Return Fan On/Off | SST-2 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Return Air Flow | FT-1 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Return Air Flow | FT-2 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Return Fan Speed | SC-1 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Return Fan Speed | SC-2 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Return Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Return Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Return Fan Isolation Damper | ZC-1 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Return Fan Isolation Damper | ZC-2 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Return Fan Damper End Switch | ZS-1 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Return Fan Damper End Switch | ZS-2 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan On/Off | SST-3 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Fan On/Off | SST-4 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Fan On/Off | SST-5 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Fan On/Off | SST-6 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Fan On/Off | SST-7 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Fan On/Off | SST-8 | ● | | | | ● | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | |
| Supply Air Flow | FT-3 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Air Flow | FT-4 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Air Flow | FT-5 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Air Flow | FT-6 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Air Flow | FT-7 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Air Flow | FT-8 | ● | | | | | | | | | | | | ● | | | | | | | | | | | | | | | | | | ● | | |
| Supply Fan Speed | SC-3 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan Speed | SC-4 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan Speed | SC-5 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan Speed | SC-6 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan Speed | SC-7 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan Speed | SC-8 | ● | | | | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan VFD | COMM | ● | | | | | | | | | | | | | | | ● | | | | | | | | | | | | ● | ● | ● | ● | ● | | |
| Supply Fan Isolation Damper | ZC-3 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Isolation Damper | ZC-4 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Isolation Damper | ZC-5 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Isolation Damper | ZC-6 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Isolation Damper | ZC-7 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Isolation Damper | ZC-8 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Fan Damper End Switch | ZS-3 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan Damper End Switch | ZS-4 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan Damper End Switch | ZS-5 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan Damper End Switch | ZS-6 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan Damper End Switch | ZS-7 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply Fan Damper End Switch | ZS-8 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Supply High Pressure | PSH-1 | ● | | | | | | | | | | | | | | | | | ● | | | | | | | | | | | | | ● | | |
| Return Low Pressure | PSL-1 | ● | | | | | | | | | | | | | | | | ● | | | | | | | | | | | | | | ● | | |
| Minimum Outside Air Damper | ZC-9 | ● | | | C | | | ● | | | | | | | | | | | | | | | | | ● | | | | | | | ● | | |
| Economizer Outside Air Damper | ZC-10 | ● | | | C | | | ● | | | | | | | | | | | | | | | | ● | ● | | | | | | | ● | | |
| Return Air Smoke Damper | SD | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Return Air End Switch | ZS-10 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Return Air Economizer Damper | ZC-11 | ● | | | O | | | ● | | | | | | | | | | | | | | | | ● | ● | | | | | | | ● | | |
| Supply Air Smoke Damper | SD | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | | | | | | | | | |
| Supply Air End Switch | ZS-9 | ● | | | | | | | | ● | | | | | | | ● | | | | | | | | | | | | | | | ● | | |
| Exhaust Air Damper | ZC-12 | ● | | | C | | | ● | | | | | | | | | | | | | | | | ● | ● | | | | | | | ● | | |
| Preheat Steam Control Valve | TC-1 | ● | | | C | | | ● | | | | | | | | | | | | | | | | ● | | | | | | | | ● | | |
| Humidifier Capacity Control Valve | HC-1 | ● | | | C | | | ● | | | | | | | | | | | | | | | | | | ● | | | | | | ● | | |
| Humidifier Block Control Valve | HS-1 | ● | | | C | ● | | | | | | | | | | | | | | | | | | | | ● | | | | | | ● | | |
| Cooling Coil Chilled Water Valve | TC-2 | ● | | | C | | | ● | | | | | | | | | | | | | | | | ● | | | | | | | | ● | | |
| Pre Filter Pressure Drop | PDS-1 | ● | | | | | | | | ● | | | | | | | | | 0.5 | | | | | | | | | | | | | ● | | |
| Intermediate Filter Pressure Drop | PDS-2 | ● | | | | | | | | ● | | | | | | | | | 0.7 | | | | | | | | | | | | | ● | | |

POINTS INDICATED FOR TRENDING SHALL BE SET UP WITH 30 MINUTE INTERVALS BETWEEN READINGS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|
| | | |

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

MILLER•REMICK
A Service-Disabled Veteran-Owned Small Business
(856) 429-4000 ▪ Miller-Remick.com
M-R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER

COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

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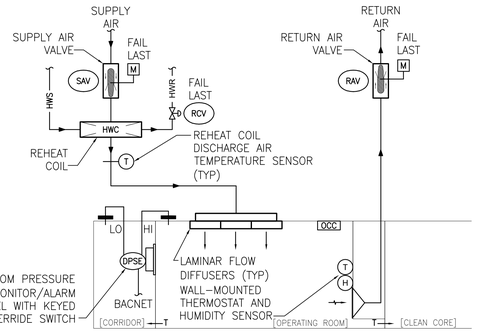
|---|---|
| Drawing Title | MECHANICAL<br>AHU FOR SURGICAL SUITE (VAV)<br>POINTS SUMMARY |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE<br>PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-803 |

VA FORM 08-6231

P:\0499_DWA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\M-802-7.dwg 12-29-25 02:01:22 PM sremick



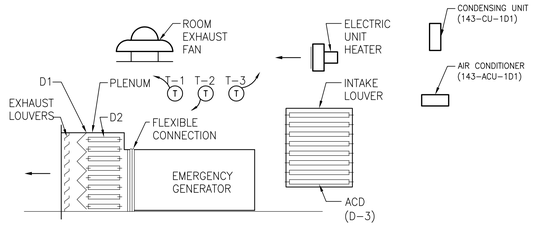

A. THE SUPPLY AND RETURN AIR FLOW CONTROL VALVES SHALL BE PRESSURE INDEPENDENT AND SHALL MAINTAIN CONSTANT AIR FLOW BASED UPON THE ENGAGED OPERATIONAL SEQUENCE FOR THE THIRD FLOOR NEW SURGICAL OPERATING ROOM SUITES (OCC/UNOCC).

B. AIR QUANTITIES FOR ALL AIRFLOW CONTROL VALVES SHALL BE FULLY ADJUSTABLE WITHIN THE BAS. SUPPLY AND EXHAUST AIRFLOW, ROOM TEMPERATURE, ROOM HUMIDITY AND ROOM PRESSURE SHALL BE MEASURED, MONITORED AND ALARMED WITHIN THE BAS FOR EACH SPACE.

C. POSITIVE PRESSURE/STERILE OPERATING ROOMS SHALL INCLUDE ACTIVE ROOM PRESSURE CONTROL/MONITORING WITH LOCAL AND BAS MEASUREMENT, MONITORING AND ALARM. LOCAL ROOM PRESSURE MONITOR PANELS (TSI OR APPROVED EQUAL–SEE PLANS FOR LOCATIONS) SHALL INCLUDE KEY SWITCH TO ALLOW LOCAL AND REMOTE/BAS ALARMS TO BE DISABLED WHEN SPACES ARE BEING CLEANED/AFTER–HOURS/DURING PERIODS OF NON–USE/UNOCCUPIED. BAS CONTINUES TO MONITOR PRESSURE, ONLY THE ALARMS ARE DISENGAGED TO ALLOW FOR ROOM CLEANING, DOOR OPENINGS, ETC. BASED ON VAMC STANDARD OPERATING PROCEDURES/QUALIFIED PERSONNEL.

D. THE OPERATING ROOM TEMPERATURE SENSORS SHALL PROVIDE REHEAT COIL CONTROL AS DESCRIBED IN AIRFLOW CONTROL VALVE SEQUENCES. OCCUPIED TEMPERATURE SETPOINT SHALL BE 66°F IF NOT RESET AT ECC OR LOCALLY AT THE SPACE THERMOSTAT. UNOCCUPIED SETPOINT SHALL BE 64°F. HUMIDITY SENSORS ARE FOR FEEDBACK MONITORING AS INDIVIDUAL OPERATING ROOM HUMIDITY CONTROL IS NOT REQUIRED/PROVIDED IN DESIGN.

## 1 OPERATING ROOMS (O.R.) CONTROL DIAGRAM
SCALE: NTS




NOTES:

1. EMERGENCY GENERATOR SHALL BE INTERLOCKED WITH D3. WHEN EMERGENCY GENERATOR IS ENERGIZED D3 SHALL OPEN. WHEN EMERGENCY GENERATOR IS DE–ENERGIZED, D3 SHALL CLOSE, PROVIDED ROOM EXHAUST FAN IS OFF.

2. ROOM EXHAUST FAN SHALL BE INTERLOCKED WITH D3 & ROOM THERMOSTAT T1. WHEN ROOM THERMOSTAT RISES ABOVE 85°F [29°C], ROOM EXHAUST FAN SHALL RUN AND D3 SHALL OPEN. WHEN ROOM THERMOSTAT DROPS BELOW 80°F [27 C], ROOM EXHAUST FAN SHALL STOP AND D3 SHALL CLOSE, PROVIDED EMERGENCY GENERATOR IS DE–ENERGIZED.

3. POWER–OPERATED, OPPOSED BLADE DAMPERS D1 & D2 SHALL BE INTERLOCKED WITH ROOM THERMOSTAT T2 SET AT 60°F [16°C]. ON A RISE IN ROOM TEMPERATURE ABOVE 60°F [16°C], D1 SHALL MODULATE OPEN AND D2 SHALL MODULATE CLOSED. ON A DROP IN ROOM TEMPERATURE BELOW 60°F [16°C], D1 SHALL MODULATE CLOSED AND D2 SHALL MODULATE OPEN.

4. ELECTRIC UNIT HEATERs SHALL BE INTERLOCKED WITH ROOM THERMOSTAT T3 SET AT 45°F [7.2°C]. ON A DROP IN ROOM TEMPERATURE BELOW 43°F [6.1°C], ELECTRIC UNIT HEATER SHALL BE ENERGIZED AND ON A RISE IN ROOM TEMPERATURE ABOVE 47°F [8.3°C].

5. WHEN ROOM THERMOSTAT RISES ABOVE 85°F [29°C], THE AIR CONDITIONING UNIT 143–ACU–1D1 SHALL RUN. WHEN ROOM THERMOSTAT DROPS BELOW 80°F [27°C], ROOM 143–ACU–1D1 SHALL STOP.

6. ALL DAMPERS TO HAVE END SWITCHES. GENERATOR SHALL START WHEN END SWITCHES HAVE CONFIRMED THAT THE DAMPERS HAVE FULLY OPENED. IF DAMPERS FAIL TO OPEN AN AUDIBLE ALARM SHALL SOUND AND THE GENERATOR START CIRCUIT SHALL BE DISABLED.

## 2 EMERGENCY GENERATOR ROOM CONTROLS
SCALE: NTS

### DIFFERENTIAL PRESSURE CONTROL WITH MONITOR/ALARM SEQUENCE OF OPERATION (OPERATING ROOMS)

GENERAL

A. DIFFERENTIAL PRESSURE BETWEEN POSITIVE PRESSURE OPERATING ROOMS AND ADJACENT SURGICAL SUITE SEMI–RESTRICTED CORRIDOR, SHALL BE A MINIMUM OF –0.01 IN. WC (–2.5 Pa). INITIAL SETPOINT: –0.025 IN. WC. (ADJ). COORDINATE TIGHT SEAL OF OPERATING ROOM ENVELOPE WITH ALL TRADES. THE AIRFLOW PRESSURIZATION CASCADE SHALL BE AS INDICATED ON DRAWING M7.00 AND SHALL BE VERIFIED AS "POSITIVE" FOR NORMAL OCCUPIED MODE/UNOCCUPIED MODES WITHOUT ALARMS (KEY–SWITCH ENGAGED TO DISABLE ALARMS). BAS SHALL MONITOR PRESSURE DIFFERENTIALS FOR ALL MODES.

B. A ROOM DIFFERENTIAL PRESSURE MONITOR (TSI OR APPROVED EQUAL), WITH KEYED OVERRIDE SWITCH, SHALL COONTINUOUSLY MONITOR/ALARM THE ROOM DIFFERENTIAL PRESSURE AT EACH OPERATING ROOM AS REFERENCED TO ITS ADJACENT SPACE(S). AN ADDITIONAL SUMMARY MONITORING & ALARM PANEL SHALL BE PROVIDED AT THE MAIN SURGICAL SUITE NURSE STATION (3A–108) TO DISPLAY STATUS AND ALARMS FOR ALL SEVEN (3 PRESENT, 3 FUTURE) OPERATING ROOMS. ALL DIFFERENTIAL PRESSURE MONITORS AND KEYED OVERRIDE SWITCHES SHALL BE ON "EMERGENCY" POWER.

C. WHEN THE ROOM DIFFERENTIAL PRESSURE IS WITHIN ITS ACCEPTABLE PRESSURE ALARM PARAMETERS (MINIMUM DIFFERENTIAL OF 0.01 IN. WC./2.5 Pa, ADJUSTABLE), AS SET AT THE ROOM DIFFERENTIAL PRESSURE MONITOR/ALARM, THE GREEN "NORMAL" LIGHT SHALL BE ILLUMINATED. WHEN THE ROOM DIFFERENTIAL PRESSURE IS NOT WITHIN ITS ACCEPTABLE PRESSURE ALARM PARAMETERS, THE RED "ALARM" LIGHT SHALL BE ILLUMINATED AND THE ROOM DIFFERENTIAL PRESSURE MONITOR/ALARM SHALL PRODUCE A LOCAL AUDIBLE ALARM. AN ALARM SHALL ALSO BE GENERATED AT THE BAS WORKSTATION(S).

D. SUMMARY PANEL AT NURSE STATION/CONTROL SHALL BE SIMILAR IN APPEARANCE AND FUNCTION TO INDIVIDUAL SPACE MONITORS, BUT PROVIDE OVERALL SUMMARY SCREEN INDICATING FULL STATUS OF THE OPERATING ROOMS/ASSOCIATED KEY SWITCHES AT ONE GLANCE. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.

E. EACH MONITOR SHALL HAVE A KEYED OVERRIDE SWITCH THAT ALLOWS THE MONITOR TO BE TAKEN OUT OF SERVICE FOR SPECIFIC/USER–DEFINED PERIODS WHEN SPACES ARE NOT IN USE FOR SURGICAL/STERILE OPERATIONS (CLEANING/MAINTENANCE/AFTER–HOURS, ETC.). WHEN THE KEYED OVERRIDE SWITCH IS PLACED IN THE "NEUTRAL" POSITION THE ROOM MONITOR DISPLAY INDICATOR SHALL CHANGE FROM "POSITIVE" TO "NEUTRAL". WHEN THE KEYED OVERRIDE SWITCH IS PLACED IN THE "POSITIVE" POSITION ROOM MONITOR DISPLAY SHALL CHANGE FROM "NEUTRAL" TO "POSITIVE". NO HVAC AIRFLOW CHANGES SHALL OCCUR, JUST DISABLING OF LOCAL AND REMOTE/BAS ALARMS BASED UPON VAMC PROTOCOL/AUTHORIZED PERSONNEL USAGE OF KEY SWITCH.

F. ROOM DIFFERENTIAL PRESSURE IS TO BE ACTIVELY CONTROLLED BY THE ROOM DIFFERENTIAL CONTROLLER/MONITOR. THE SUPPLY AIR QUANTITY TO THE SPACE IS TO BE CONSTANT AND THE SETPOINT FOR THE OCCUPIED AND UNOCCUPIED MODES SHALL BE AS DEFINED IN THE AIR VALVE SCHEDULE.

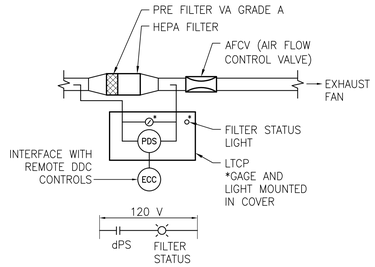
G. THE RETURN AIR VALVE SETPOINT ON START UP OR CHANGE OF OCCUPIED/UNOCCUPIED STATUS IS TO BE AS INDICATED ON THE AIR VALE SCHEDULE. AFTER THE INITIAL AIRFLOW SETPOINTS HAVE BEEN ESTABLISHED, THE ROOM DIFFERENTIAL PRESSURE CONTROLLER/MONITOR SHALL RESET THE RETURN AIR VALVE SETPOINT TO CONTROL THE SPACE PRESSURE TO 0.015 INCHES W.G. POSITIVE RELATIVE TO THE ADJACENT SPACES. THE RESETTING OF THE CFM SETPOINT SHALL BE LIMITED TO + 20% AND – 10% OF THE SCHEDULED VALUE FOR THE OCCUPIED OR UNOCCUPIED MODE.

OCCUPANCY SENSOR OVERRIDES:

A. THE OCCUPANCY SENSOR SHALL OVERRIDE THE SPACE OCCUPIED/UNOCCUPIED MODE FOR CONTROLLING THE AIR VALVE FLOW SETPOINT.

B. IF THE ROOM IS SCHEDULED TO BE OCCUPIED AND OCCUPANCY IS NOT SENSED FOR 10 MINUTES, THE SUPPLY AND RETURN AIR VALVE FLOW SETPOINTS SHALL BE SET TO THE UNOCCUPIED MODE VALUES.

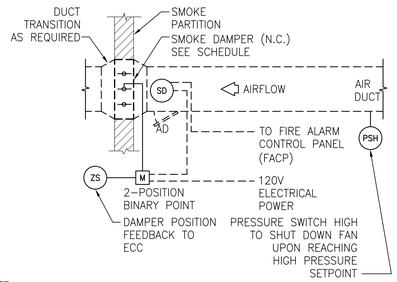
C. UPON OCCUPANCY BEING SENSED, THE OCCUPANCY SCHEDULE SHALL BE OVERRIDEN AND THE SUPPLY AND RETURN AIR VALVES FLOW AND THE SPACE TEMPERATURE SETPOINTS SHALL BE SET TO THE OCCUPIED MODE VALUES.




SEQUENCE OF OPERATION:

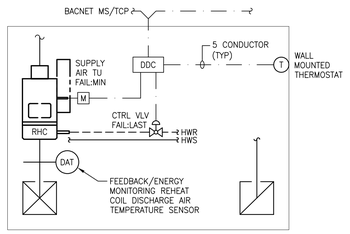
1. WHEN FILTER PRESSURE DROP RISES TO 2" [7 KPA] OF WATER COLUMN,
FILTER STATUS LIGHT (RED) SHALL BE ENERGIZED.

## 3 HEPA FILTER CONTROLS – EXHAUST SYSTEMS
SCALE: NTS




NOTE:
UPON DETECTION OF SMOKE BY THE SMOKE DETECTOR, THE SMOKE DAMPER SHALL CLOSE & SEND AN ALARM TO THE ECC.

## 5 SMOKE DAMPER CONTROL DIAGRAM
SCALE: NTS




ZONE TEMPERATURE SETPOINT ADJUST

1. SPACE OCCUPANTS SHALL BE ABLE TO MODIFY ZONE TEMPERATURE HEATING AND COOLING SETPOINTS AT THE ZONE SENSOR. PROVIDE TWO DEGREE UP/DOWN (ADJUSTABLE). CONFIRM RANGE WITH VAMC.

   HEATING SPACE TEMPERATURE SETPOINT (OCCUPIED): 69.0°F
   COOLING SPACE TEMPERATURE SETPOINT (OCCUPIED): 72.5°F

   HEATING SPACE TEMPERATURE SETPOINT (UNOCCUPIED): 62.0°F
   COOLING SPACE TEMPERATURE SETPOINT (UNOCCUPIED): 78.0°F

NORMAL/OCCUPIED MODE

1. THE VARIABLE VOLUME SUPPLY AIR TERMINAL UNIT (TU) SHALL PROVIDE THE REQUIRED MAXIMUM AND MINIMUM SCHEDULED AIRFLOWS CONTINUOUSLY REGARDLESS OF VARIATIONS IN DUCT OR ROOM STATIC PRESSURE. THE BAS SHALL ACTIVELY MEASURE, MONITOR, AND ALARM AIRFLOW.
2. THE ROOM TEMPERATURE SENSOR SHALL, THROUGH THE DDC, MODULATE THE TERMINAL UNIT CONTROL DAMPER AND DUCT–MOUNTED REHEAT COIL TWO WAY VALVE TO MAINTAIN ROOM TEMPERATURE SETPOINT. THE BAS SHALL ACTIVELY MEASURE, MONITOR, AND ALARM SPACE TEMPERATURE.
3. ON A RISE IN SPACE TEMPERATURE, ABOVE SETPOINT, THE REHEAT COIL CONTROL VALVE SHALL MODULATE CLOSED. ON A FURTHER RISE IN SPACE TEMPERATURE, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MAXIMUM POSITION. ON A FALL IN SPACE TEMPERATURE, THE REVERSE SHALL OCCUR.
4. ON A FALL IN SPACE TEMPERATURE, BELOW SETPOINT, THE TERMINAL UNIT DAMPER SHALL MODULATE TO ITS MINIMUM POSITION. ON A FURTHER FALL IN SPACE TEMPERATURE, THE REHEAT COIL CONTROL VALVE SHALL MODULATE OPEN. ON A RISE IN SPACE TEMPERATURE, THE REVERSE SHALL OCCUR.

UNOCCUPIED MODE

1. WHEN INITIATED BY TIME OF DAY OR DAY OF THE WEEK AS COORDINATED WITH THE VAMC, THE BAS SHALL MODULATE THE SUPPLY AIR TERMINAL UNITS TO THEIR SCHEDULED MINIMUM/UNOCCUPIED AIRFLOWS AND SPACE TEMPERATURES.

WARM–UP MODE

1. BASED UPON TIME OF DAY/DAY OF WEEK SCHEDULE COORDINATED/REVIEWED BY VAMC, THE BAS SHALL INITIATE A WARM UP CYCLE TO ENGAGE OCCUPIED MODE IN ADVANCE OF TYPICAL OCCUPIED HOURS (TIME ADJUSTABLE). BAS SHALL ENGAGE OCCUPIED MODE, INDEXING THE AIR HANDLING UNITS, AS WELL AS ALL AIR TERMINAL UNITS TO THE SCHEDULED OCCUPIED AIRFLOWS AND MODULATE REHEAT COIL CONTROL VALVES TO ACHIEVE OCCUPIED SPACE TEMPERATURES.

PANDEMIC MODE

1. SUPPLY TU'S OPEN TO THEIR SCHEDULED AIRFLOW AT A CONSTANT VOLUME REGARDLESS OF VARIATIONS IN THE DUCT PRESSURE.

LOSS OF NORMAL POWER/FAILURE OR "OFF" MODE

1. WHENEVER THE AIR HANDLING UNIT (AHU) IS "OFF" FOR ANY REASON, THE AIR TERMINAL UNITS SHALL MODULATE TO THEIR UNOCCUPIED POSITIONS. THE REHEAT COIL CONTROL VALVE SHALL CONTINUE TO MODULATE TO MAINTAIN THE TEMPERATURE SETPOINT OF ITS RESPECTIVE MODE OF OPERATION.
2. ONCE NORMAL POWER HAS BEEN RESTORED, THE SYSTEM SHALL RESUME NORMAL OPERATION AUTOMATICALLY, THROUGH THE BAS.

## 4 AIR TERMINAL UNIT W/ RHC DIAGRAM
SCALE: NTS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
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CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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COMMONWEALTH OF VIRGINIA
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PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

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|---|---|---|---|
| Drawing Title: MECHANICAL CONTROLS DIAGRAMS | Phase: BID DOCUMENTS | Project Title: RENOVATE SURGICAL SUITE PHASE II<br>Project Number: 658-19-103 | VA Drawing Number: 2617 |
| Approved by: Jon Manninen, COR | FULLY SPRINKLERED | Location: SALEM VA MEDICAL CENTER<br>Issue Date: 2025-12-30 / Checked: DJ / Drawn: JAK | Building Number: 143<br>Drawing Number: M-804 |

VA FORM 08-6231

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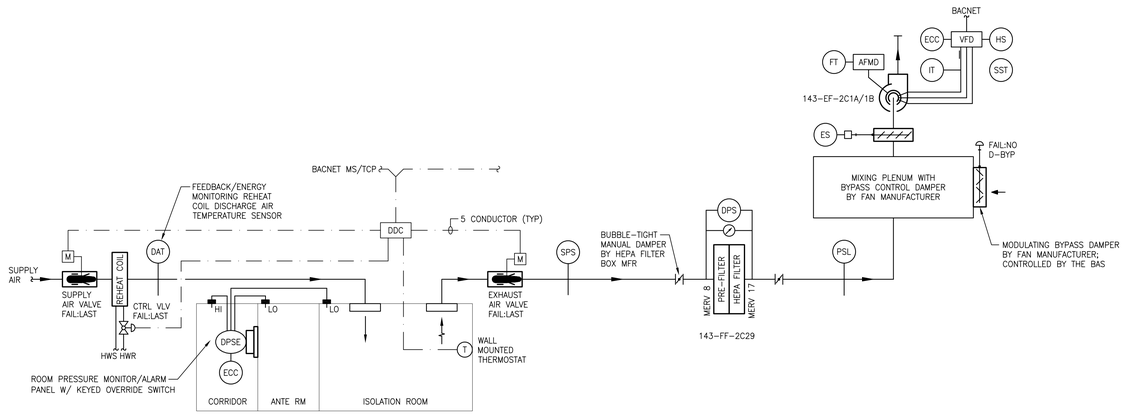


# SEQUENCE OF OPERATION

1. GENERAL

1.1 EXHAUST FANS ARE NORMALLY STARTED AND STOPPED REMOTELY VIA THE BAS (BUILDING AUTOMATION SYSTEM) AND OPERATE AS FULLY REDUNDANT (N+1) ACTIVE/STANDBY FANS. THE SYSTEM WILL NORMALLY OPERATE 24 HOURS/DAY. H-O-A SWITCH SHALL BE KEPT IN THE "AUTO" POSITION. "HAND" AND "OFF" POSITIONS SHALL BE USED ONLY FOR MAINTENANCE. WHEN THE UNIT IS "OFF", D-1 SHALL BE FULLY OPEN.

2. AIR FLOW CONTROL

2.1 WHEN THE EXHAUST FAN IS ENABLED, BYPASS DAMPER, D-BYP, IS FULL OPEN. THE EXHAUST FAN SHALL RAMP UP TO MINIMUM SPEED AS DETERMINED BY THE APPROVED FAN CURVES WHILE THE FAN INLET ISOLATION DAMPER OPENS. IF THE DAMPER END SWITCH IS NOT MADE AFTER 60-SEC (ADJ), THE FAN SHALL SHUT DOWN AND THE BAS SHALL RECEIVE AN ALARM. THE RETURN FAN CONTINUES TO OPERATE.

2.2 ONCE THE FAN INLET DAMPER END SWITCH IS PROVEN FULLY OPEN, THE EXHAUST FAN SHALL RAMP UP TO MAINTAIN THE STATIC PRESSURE SETPOINT AT SPS, WHILE THE BYPASS DAMPER MODULATES CLOSED.

2.3 THE VARIABLE FREQUENCY DRIVE SHALL MODULATE TO MAINTAIN THE SYSTEM STATIC PRESSURE SETPOINT. THE SYSTEM SHALL VARY AIRFLOW EXHAUSTED FROM FILTER LOADING.

2.4 THE AIR FLOW MEASURING DEVICE (PIEZOMETER) AT THE INLET OF THE EXHAUST FAN IS TO PROVIDE SYSTEM CAPACITY FEEDBACK TO THE BAS AS A MONITORING POINT ONLY. COORDINATE FINAL FAN SYSTEM AIRFLOW WITH TESTING, ADJUSTING, AND BALANCING (TAB) CONTRACTOR.

3. FAILURE MODE

3.1 WHEN THE EXHAUST FAN IS COMMANDED TO RUN AND AFTER A 60 SECOND DELAY, THE CONTROL SYSTEM SENSES NO STATUS VIA THE CURRENT TRANSFORMER RELAY, THE INLET EXHAUST AIR DAMPER SHALL CLOSE AND SEND AN ALARM TO THE BAS. THE BAS SHALL ALSO CLOSE THE SUPPLY AND EXHAUST AIRFLOW CONTROL VALVES AND ASSOCIATED REHEAT COIL CONTROL VALVE SERVED BY THE EXHAUST FAN SYSTEM, IN ORDER TO MAINTAIN NEUTRAL AIR PRESSURE WITHIN THE THREE SPACES SERVED BY THE FAILED EXHAUST FAN.

4. HEPA FILTER CONTROL

4.1 MEASURE DIFFERENTIAL PRESSURE ACROSS FILTER BANK. WHEN FILTER PRESSURE DROP ACROSS ANY THE FILTER BOX RISES ABOVE THE SCHEDULED LIMIT, AN ALARM SHALL BE GENERATED AT THE BAS. INITIAL SETPOINT SHALL BE -1.6" WG.

5. INTERLOCKS AND SAFETIES

5.1 LOW STATIC PRESSURE SWITCH (PSL) SHALL PREVENT THE EXHAUST FAN FROM DEVELOPING OVER -3.0" OF STATIC PRESSURE (FIELD ADJUSTABLE) IF STATIC PRESSURE EXCEEDS -3.0" THE EXHAUST FAN SHALL BE DE-ENERGIZED. PSL SHALL BE HARDWIRED TO THE EXHAUST FAN AND SHALL SHUT DOWN THE FAN IN HAND, AUTO, OR BYPASS MODE. PSL WILL REQUIRE MANUAL RESET AT THE DEVICE.

6. ZONE TEMPERATURE SETPOINT ADJUST

6.1 SPACE OCCUPANTS SHALL BE ABLE TO MODIFY ZONE TEMPERATURE HEATING AND COOLING SET POINTS AT THE ZONE SENSOR. PROVIDE TWO DEGREE UP/DOWN (ADJUSTABLE). CONFIRM RANGE WITH VAMC.

NORMAL/UNOCCUPIED MODE

6.2 THE SUPPLY AIR CONTROL VALVE SHALL SUPPLY A CONSTANT VOLUME OF AIR CONTINUOUSLY REGARDLESS OF VARIATIONS IN DUCT OR ROOM STATIC PRESSURE. THE BAS SHALL ACTIVELY MEASURE, MONITOR, AND ALARM SUPPLY AIRFLOW AND USE FOR OFFSET CONTROL OF EXHAUST.

6.3 THE ROOM TEMPERATURE SENSOR SHALL, THROUGH THE DDC, MODULATE THE DUCT-MOUNTED REHEAT COIL TWO-WAY CONTROL VALVE TO MAINTAIN ROOM TEMPERATURE SET POINT.

6.4 THE EXHAUST AIR CONTROL VALVE SHALL TRACK THE SUPPLY AIRFLOW PLUS/MINUS THE CFM DIFFERENTIAL SCHEDULED.

KEYED OVERRIDE/OCCUPIED MODE

6.5 WHEN INITIATED BY KEYED SWITCH OVERRIDE AT THE ROOM PRESSURE MONITOR, THE SUPPLY AIR CONTROL VALVES SHALL INCREASE TO ITS SCHEDULED MAXIMUM AIRFLOW.

6.6 THE EXHAUST AIR CONTROL VALVE SHALL TRACK THE SUPPLY AIRFLOW PLUS OR MINUS THE CFM DIFFERENTIAL SCHEDULED.

LOSS OF NORMAL POWER/FAILURE MODE

6.7 UPON SENSING A LOSS OF POWER OR FAILURE OF THE AIR HANDLING UNIT, THE AIR VALVES AND REHEAT COIL CONTROL VALVE SHALL CONTINUE TO OPERATE TO MAINTAIN ALL REQUIRED SETPOINTS.

7. DIFFERENTIAL PRESSURE MONITOR/ALARM

7.1 DIFFERENTIAL PRESSURE BETWEEN NEGATIVE ISOLATION PATIENT ROOM AND ADJACENT CORRIDOR SHALL BE A MINIMUM OF -0.02 IN. WC. INITIAL SETPOINT: -0.05 IN. WC. (ADJ). COORDINATE TIGHT SEAL OF ISOLATION ROOM ENVELOPE WITH ALL TRADES. SPACES SHALL MEET ISOLATION ROOM CRITERIA (WITH LOCAL AND BAS ALARMS DESCRIBED HEREIN) AND SHALL BE VERIFIED AS "NEGATIVE" FOR NORMAL OCCUPIED MODE/UNOCCUPIED MODES WITHOUT ALARMS (KEYED SWITCH ENGAGED TO DISABLE ALARMS). BAS SHALL MONITOR PRESSURE DIFFERENTIALS.

7.2 WHEN THE ROOM DIFFERENTIAL PRESSURE IS WITHIN ITS ACCEPTABLE PRESSURE ALARM PARAMETERS (MINIMUM DIFFERENTIAL OF 0.01 IN. WC. (ADJ.), AS SET AT THE ROOM DIFFERENTIAL PRESSURE MONITOR/ALARM, THE GREEN "NORMAL" LIGHT SHALL BE ILLUMINATED. WHEN THE ROOM DIFFERENTIAL PRESSURE IS NOT WITHIN ITS ACCEPTABLE PRESSURE ALARM PARAMETERS, THE RED "ALARM" LIGHT SHALL BE ILLUMINATED AND THE ROOM DIFFERENTIAL PRESSURE MONITOR/ALARM SHALL PRODUCE A LOCAL AUDIBLE ALARM.

7.3 ISOLATION ROOM SHALL BE MONITORED AND ALARMED AT THE BAS. THE BAS CONTRACTOR SHALL ASSIST IN THE VERIFICATION AND DOCUMENTATION OF ALL OPERATION MODES TO ENSURE THE PROPER NEGATIVE PRESSURE IS MAINTAINED.

7.4 ISOLATION ROOM SHALL HAVE ITS OWN MONITOR MOUNTED AS INDICATED ON PLANS (TSI OR APPROVED EQUAL) AND SHALL BE ON "EMERGENCY" POWER. MONITOR SHALL HAVE A LCD DISPLAY WITH KEYPAD, ROOM ALARM STATUS INDICATOR, VISUAL, AND AUDIBLE ALARM ANNUNCIATOR, ALARM RELAY OUTPUT, ANALOG PRESSURE OUTPUT (MA OR VDC TO BE MONITORED BY BAS), SERIAL ASYNCHRONOUS COMMUNICATIONS PORT UTILIZING ASCII OR BACNET MS/TP PROTOCOL, ALARM SILENCE VIA KEYPAD, TEST VIA KEYPAD, AND ISOLATED POWER SUPPLY. SEE SPECIFICATIONS FOR ADDITIONAL INFORMATION.

7.5 SENSOR AND MONITOR SHALL HAVE DEMONSTRATED SUCCESSFUL OPERATION FOR A MINIMUM OF 3 YEARS IN ROOM MONITORING APPLICATIONS.

7.6 EACH MONITOR SHALL HAVE A KEYED OVERRIDE SWITCH THAT ALLOWS THE MONITOR TO BE TAKEN OUT OF SERVICE WHEN SPACES ARE NOT IN USE FOR ISOLATION. WHEN THE KEYED OVERRIDE SWITCH IS PLACED IN THE "NEUTRAL" POSITION THE ROOM MONITOR DISPLAY INDICATOR SHALL CHANGE FROM "NEGATIVE" TO "NEUTRAL". WHEN THE KEYED OVERRIDE SWITCH IS PLACED IN THE "NEGATIVE" POSITION ROOM MONITOR DISPLAY SHALL CHANGE FROM "NEUTRAL" TO "NEGATIVE". NO HVAC AIRFLOW CHANGES SHALL OCCUR, JUST DISABLING OF LOCAL AND REMOTE/BAS ALARMS.

7.7 EACH MONITOR SHALL HAVE A KEYED OVERRIDE SWITCH THAT ALLOWS THE USER TO ENGAGE AN "OCCUPIED" MODE OF OPERATION, WHICH SHALL INCREASE THE AIRFLOW TO THE SPECIAL PROCEDURE ROOM, AS SCHEDULED.

7.8 ROOM PRESSURE SENSOR AND MONITOR SHALL BE PROVIDED AS A PACKAGE FROM A SINGLE MANUFACTURER.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
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CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
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| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

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Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

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| Drawing Title | MECHANICAL ISOLATION PATIENT ROOM AND EXHAUST FAN CONTROL DIAGRAM |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
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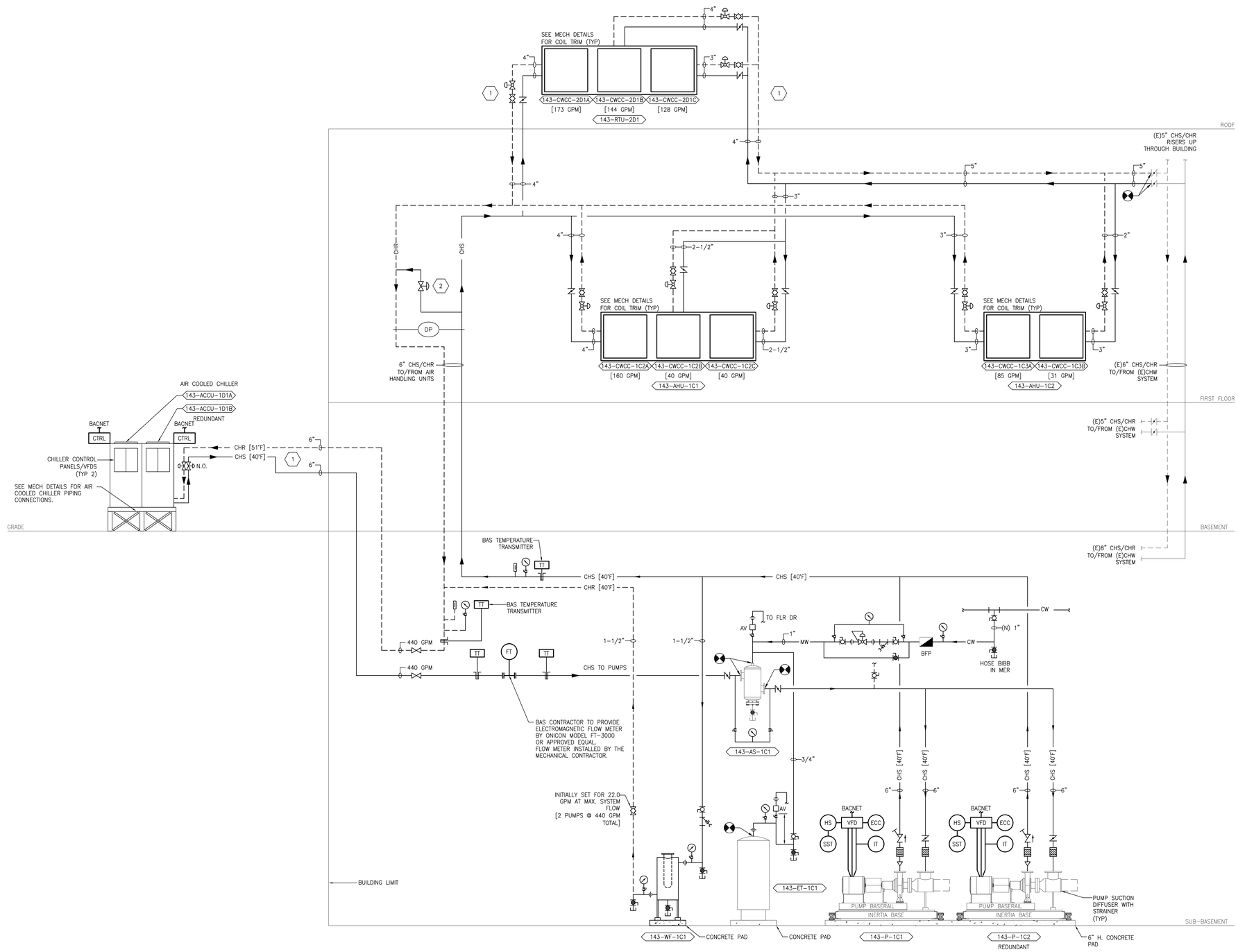



1 AIR COOLED CHILLER AND CHILLED WATER CONTROL DIAGRAM
SCALE: NTS

## SHEET KEYNOTES:

1. THE MINIMUM FLOW RATE IS 60 GPM, BASED ON THE "BASIS OF DESIGN" CHILLER MANUFACTURER ON THE MECHANICAL SCHEDULE DRAWINGS.
2. BUILDING SYSTEM AUTOMATION (BAS) COMPONENTS AND DEVICES ARE SHOWN TO ENSURE THAT THE MECHANICAL CONTRACTOR PROVIDES ALL WELD-O-LETS, TREAD-O-LETS, AND PETE'S PLUGS REQUIRED FOR THE BAS CONTRACTOR, THE TAB CONTRACTOR, AND ALL MANUFACTURER START-UP TECHNICIANS.

1. GENERAL

1.1 THE VARIABLE FLOW CHILLED WATER DISTRIBUTION IS A PRIMARY SYSTEM. THE PUMPS SHALL BE PROVIDED WITH THE CHILLERS AND SHALL HAVE ALL THE INTEGRAL COMPONENTS TO CONTROL THE MANUFACTURER'S VARIABLE SPEED PUMPS AND REFRIGERANT CIRCUITS.

1.2 CHILLER AND PUMPS SHALL BE STARTED AND STOPPED AUTOMATICALLY BY THE BUILDING AUTOMATION SYSTEM (BAS).

1.3 ALL SETPOINTS DESCRIBED HEREIN SHALL BE ADJUSTABLE THROUGH THE BAS, INCLUDING THOSE WITHIN THE CHILLER CONTROLLER. THE BAS SHALL BE CAPABLE OF INTEGRATING THE CHILLERS AND BACNET REMOTE COMMUNICATIONS AT THE FRONTEND WORKSTATION. THE CHILLER'S MICROPROCESSOR SHALL MAINTAIN ALL INTERNAL SAFETY FEATURES.

1.4 PROVIDE ON/OFF RUN STATUS INDICATION FOR EACH PUMP THROUGH THE VARIABLE FREQUENCY DRIVE (VFD).

1.5 THE CHILLED WATER PUMPS SHALL BE CONTROLLED BY THEIR RESPECTIVE VFD'S AND BE CAPABLE OF MANUAL OPERATION.

1.6 PUMPS AND CHILLERS ARE FULLY REDUNDANT (N+1) AND OPERATE IN PARALLEL OR IN AN ACTIVE/STANDBY MODE OF OPERATION AS DESCRIBED. HARDWIRE INTERLOCK OR SOFTWARE LATCH IS NOT PERMITTED.

1.7 COORDINATE CONTROL OF THE CHILLERS AND PUMPS BASED ON FLOW AND COOLING REQUIREMENTS (FULL RANGE-MIN. TO PEAK) AS COORDINATED WITH TESTING, ADJUSTING, AND BALANCING (TAB) CONTRACTOR AND THE FACTORY-CERTIFIED STARTUP TECHNICIANS IN ACCORDANCE WITH DRAWINGS AND SPECIFICATIONS.

2. CHILLERS

2.1 THE NEW CHILLERS SHALL BE ENABLED WHENEVER THERE IS A CALL FOR COOLING BY ANY AIR HANDLING UNIT.

2.2 A LEAD CHILLER SHALL BE DESIGNATED BY BAS. IF THE CHILLED WATER RETURN TEMPERATURE REACHES 56°F (ADJ) THE SECOND (LAG) CHILLER SHALL START. EACH CHILLER HAS IT'S INTERNAL SEQUENCE TO RUN EACH CIRCUIT PER LOAD DEMAND AND PRIMARY LOOP RETURN TEMPERATURE.

2.3 IF THE DELTA T (°F) OF THE CHILLED WATER HEADERS DROP BELOW THE MANUFACTURER'S REQUIREMENTS, THE BAS SHALL DE-ENERGIZE THE CHILLER.

2.4 CHILLERS TO ACTIVATE INTERNAL EVAPORATOR IMMERSION FREEZE PROTECTION HEATER ON A WATER TEMPERATURE DROP BELOW 38°F.

2.5 MODULAR CHILLER CONTROLLER: THE CHILLER MANUFACTURER SHALL OPERATE AND CONTROL ITS INTERNAL CONTROL COMPONENTS, COMPRESSORS, AND FREEZE PROTECTION CONTROL LOGIC. THE BACNET CONTROLLER SHALL ALLOW FOR SETPOINT ADJUSTMENTS AT THE ECC. ALLOWABLE SETPOINT ADJUSTMENTS SHALL BE LIMITED TO PREVENT DAMAGING THE CHILLER.

3. SECONDARY PUMPS

3.1 THE CHILLED WATER PUMPS SHALL BE INDIVIDUALLY CONTROLLED THROUGH THE BAS TO PROVIDE AUTOMATIC CONTROL BASED ON SYSTEM DIFFERENTIAL PRESSURE. A LEAD PUMP SHALL BE DESIGNATED BY THE BAS. THE BAS SHALL ALTERNATE THE LEAD PUMP BASED ON RUN HOURS, AS SENSED BY THE VFD FEEDBACK. UPON SENSING PUMP FAILURE OF THE LEAD PUMP, THE BAS SHALL START THE LAG PUMP AUTOMATICALLY. THE LEAD PUMP SHALL REPORT AS FAILED WITHIN THE BAS AND SHALL BE SHUT DOWN.

3.2 THE CHILLERS SHALL REMAIN ACTIVE UNTIL THE RETURN WATER TEMPERATURE OF THE PRIMARY LOOP FALLS BELOW THE MANUFACTURE'S DELTA T (°F).

3.3 ON A CALL FOR COOLING BY ANY AIR HANDLING UNIT, AS SENSED BY ANY COOLING COIL CONTROL VALVE ACTUATOR FEEDBACK, THE BAS SHALL ENABLE THE LEAD PUMP TO RAMP UP TO SATISFY THE SYSTEM DIFFERENTIAL SETPOINT. WHEN THE BAS HAS PROVEN THAT COOLING IS NOT REQUIRED, THE ACTIVE PUMP SHALL BE SHUT DOWN.

3.4 THE PUMPS VARIABLE FREQUENCY DRIVE (VFD) SHALL MAINTAIN THE REQUIRED SYSTEM DIFFERENTIAL PRESSURE. THE DIFFERENTIAL PRESSURE SETPOINT SHALL SATISFY THE SYSTEM REQUIRED FLOW AND THE MINIMUM WATER FLOW REQUIRED FOR THE PUMPS, AS DETERMINED BY THE TESTING, ADJUSTING, AND BALANCING CONTRACTOR.

3.5 PUMP ADDITION AND SUBTRACTION: THE PUMPS SHALL RAMP UP AND DOWN IN UNISON, WITH ONE (1) PUMP ON CONSTANT STANDBY. WHEN TWO (2) OPERATING PUMPS REACH THEIR MINIMUM SPEED, AS DETERMINED BY THE INSTALLED PUMP CURVE, ONE (1) PUMP SHALL BE DE-ENERGIZED AND THE ACTIVE PUMP SHALL RAMP UP TO MAINTAIN DIFFERENTIAL PRESSURE SETPOINT. WHEN THE ACTIVE PUMP REACHES 95% VFD SPEED FOR 30-SECONDS (ADJ), AS DETERMINED BY THE VFD FEEDBACK; SIMULTANEOUSLY, THE ACTIVE PUMP SHALL RAMP DOWN TO 50% (ADJ) VFD SPEED WHILE THE LAG PUMP RAMPS UP TO 50% (ADJ) VFD SPEED. WHEN BOTH PUMPS REACH 50% VFD SPEED, THE TWO (2) PUMPS SHALL RAMP UP AND DOWN IN UNISON TO MAINTAIN DIFFERENTIAL PRESSURE SETPOINT.

- IF 50% MATCHING SPEED OVERSHOOTS THE DIFFERENTIAL PRESSURE SETPOINT, THEN LOWER THE INITIAL MATCHING VFD SPEED.
- IF 50% MATCHING SPEED FALLS BELOW THE DIFFERENTIAL PRESSURE SETPOINT, THEN RAISE THE INITIAL MATCHING VFD SPEED.

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

**CONSULTANTS**

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

**ENGINEER OF RECORD**

MILLER•REMICK
A Service-Disabled Veteran-Owned Small Business
(856) 429-4000 • Miller-Remick.com
M-R PROJECT NO: 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 0402055070
PROFESSIONAL ENGINEER
12-30-2025

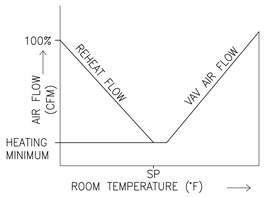
Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL AIR COOLED CHILLER AND CHILLED WATER CONTROL DIAGRAM |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-806 |

VA FORM 08-6231

P:\0499_DWA\0499-0136 Salem Renovate Surgical Suite\4_Drawings\1_Cad\1_Current Submittal Phase\6_Mechanical\M-802-7.dwg 12-29-25 02:03:01 PM sremick

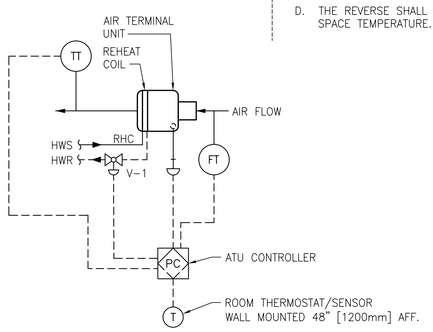



VAV BOX CONTROL SEQUENCE
NO DEADBAND

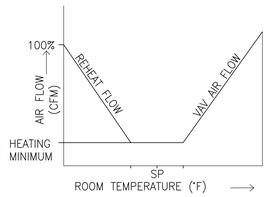
A. UPON FALL IN SPACE TEMPERATURE, THE VAV DAMPER WILL MODULATE TO MINIMUM POSITION.

B. UPON FURTHER DROP IN SPACE TEMPERATURE, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

C. THE REVERSE SHALL OCCUR ON RISE IN SPACE TEMPERATURE.




NO SUPPLEMENTAL HEATING



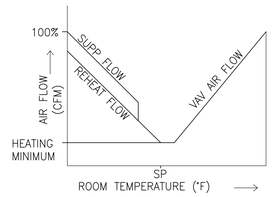

VAV BOX CONTROL SEQUENCE
W/DEADBAND

A. SETPOINTS SHALL BE SET AS FOLLOWS:
- COOLING 75°F (ADJ)
- HEATING 70°F (ADJ)
- DEADBAND OF 5°F BETWEEN HEATING AND COOLING SET POINT WILL BE MAINTAINED.

B. UPON FALL IN SPACE TEMPERATURE, THE VAV DAMPER WILL MODULATE TO MINIMUM POSITION.

C. UPON FURTHER DROP IN SPACE TEMPERATURE, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

D. THE REVERSE SHALL OCCUR ON RISE IN SPACE TEMPERATURE.



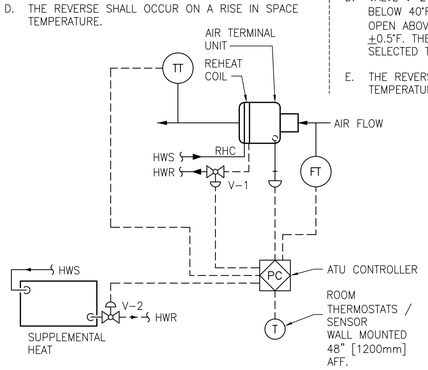

VAV BOX CONTROL SEQUENCE
NO DEADBAND

A. UPON FALL IN SPACE TEMPERATURE, THE VAV DAMPER WILL MODULATE TO MINIMUM POSITION.

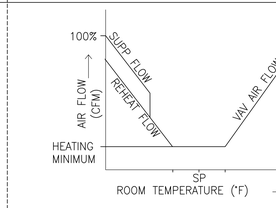
B. UPON FURTHER DROP IN SPACE TEMPERATURE, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

C. VALVE V-2 SHALL BE ENABLED WHEN OUTSIDE AIR FALLS BELOW 40°F (ADJ) AND VALVE V-1 HAS BEEN MODULATED OPEN ABOVE 30% (ADJ). VALVE V-2 SHALL MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

D. THE REVERSE SHALL OCCUR ON A RISE IN SPACE TEMPERATURE.




WITH SUPPLEMENTAL HEATING




VAV BOX CONTROL SEQUENCE
W/DEADBAND

A. SETPOINTS SHALL BE SET AS FOLLOWS:
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B. UPON FALL IN SPACE TEMPERATURE THE VAV DAMPER WILL MODULATE TO MINIMUM POSITION.

C. UPON FURTHER DROP IN SPACE TEMPERATURE VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

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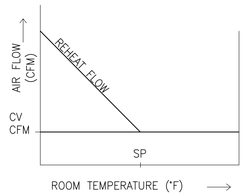
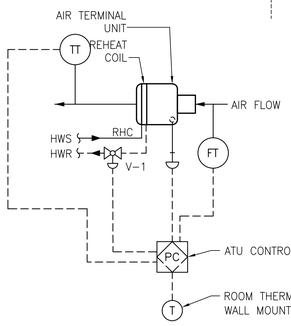
E. THE REVERSE SHALL OCCUR ON THE RISE IN SPACE TEMPERATURE.




CV BOX CONTROL SEQUENCE
NO DEADBAND

A. UPON FALL IN SPACE TEMPERATURE BELOW SETPOINT, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

B. THE REVERSE SHALL OCCUR ON A RISE IN SPACE TEMPERATURE.

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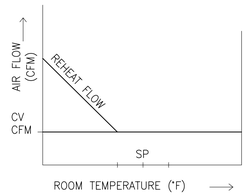


CV BOX CONTROL SEQUENCE
W/DEADBAND

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- HEATING 70°F (ADJ)
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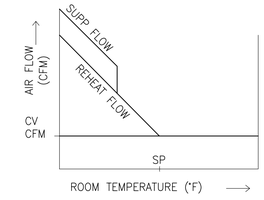
B. UPON FALL IN SPACE TEMPERATURE BELOW SETPOINT, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

C. THE REVERSE SHALL OCCUR ON A RISE IN SPACE TEMPERATURE.

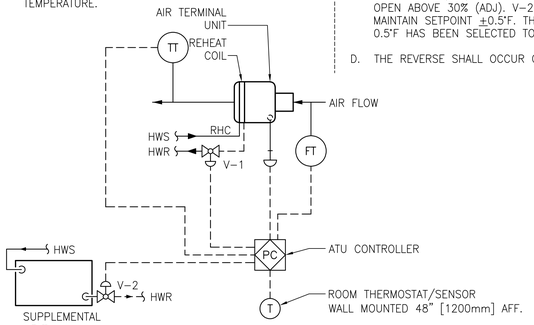



CV BOX CONTROL SEQUENCE
NO DEADBAND

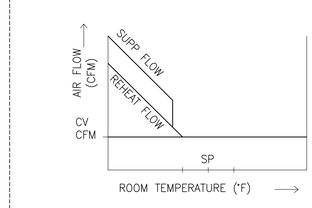
A. UPON FALL IN SPACE TEMPERATURE BELOW SETPOINT, VALVE V-1 WILL MODULATE TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

B. VALVE V-2 SHALL BE ENABLED WHEN OUTSIDE AIR FALLS BELOW 40°F (ADJ) AND VALVE V-1 HAS BEEN MODULATED OPEN ABOVE 30% (ADJ). V-2 SHALL THEN BE MODULATED TO MAINTAIN SETPOINT ±0.5°F. THE ADJUSTABLE TOLERANCE OF ±0.5° HAS BEEN SELECTED TO PREVENT VALVE HUNTING.

C. THE REVERSE SHALL OCCUR ON A RISE IN SPACE TEMPERATURE.




WITH SUPPLEMENTAL HEATING

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D. THE REVERSE SHALL OCCUR ON RISE IN SPACE TEMPERATURE.

1 VARIABLE VOLUME AIR TERMINAL UNIT CONTROL DIAGRAM
SCALE: NTS

2 CONSTANT VOLUME AIR TERMINAL UNIT CONTROL DIAGRAM
SCALE: NTS

PRINTS OF THIS DRAWING SHALL NOT BE USED FOR ANY PURPOSE WHATSOEVER WITHOUT THE SEAL AND SIGNATURE OF THE PROFESSIONAL ENGINEER.

| NO. | DESCRIPTION | DATE |
|---|---|---|

CONSULTANTS

| Architecture: | Civil Engineering: | Fire & Life Safety: |
|---|---|---|
| PF&A DESIGN, P.C. | LABELLA ASSOCIATES | STANTON ENGINEERING SERVICES, LLC |

ENGINEER OF RECORD

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M & R PROJECT NO. 0499-0136

MILLER-REMICK LLC
PROFESSIONAL ENGINEER
COMMONWEALTH OF VIRGINIA
ANTHONY DANIEL PACARO JR.
Lic. No. 402055079
PROFESSIONAL ENGINEER
12-30-2025

Office of Construction and Facilities Management

VA U.S. Department of Veterans Affairs

| | |
|---|---|
| Drawing Title | MECHANICAL AIR TERMINAL UNIT CONTROL DIAGRAM |
| Approved by: | Jon Manninen, COR |
| Phase | BID DOCUMENTS |
| | FULLY SPRINKLERED |
| Project Title | RENOVATE SURGICAL SUITE PHASE II |
| Project Number | 658-19-103 |
| Location | SALEM VA MEDICAL CENTER |
| Issue Date | 2025-12-30 |
| Checked | DJ |
| Drawn | JAK |
| VA Drawing Number | 2617 |
| Building Number | 143 |
| Drawing Number | M-807 |

VA FORM 08-6231

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